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Z E R O S Y N T A X

Experiencers
and
Cascades

David Pesetsky

Zero Syntax

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Zero Syntax

David Pesetsky

Experiencers and Cascades

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For Janis, Ben, and Jonathan

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Preface

As its subtitle suggests, this book deals with two main topics.

First, it develops a simple view of syntactic linking regularities (which determine the mapping of arguments onto syntactic positions) and defends it in the challenging domain of Experiencer predicates.

Second, it develops a theory of syntactic constituency that involves two parallel modes of structural organization (of which the novel mode is the “Cascade Syntax” of the title). This theory is an essential part of the simple view of syntactic linking, but it also has an empirical sweep that extends beyond the considerations that justify its inclusion in this book. I present the second topic as an outgrowth of the first, but I might just as well have presented the first topic as an outgrowth of the second. Were this book a postmodern novel, I might have been able to write it so that the same text could be read Cascade-end first or Experiencer-end first with no loss of intelligibility. Unfortunately, perhaps, I have not been able to do this, so I have chosen one of the two possible orders of presentation: Experiencer-end first. This reflects the order in which the research was carried out.

But what does the main title mean? The main title names the glue that holds the book together: the supposition that phonologically null (*zero*) morphology is present in structure, detectable through its syntactic and morphological consequences, and important to the discovery and explanation of the phenomena dealt with in the book. Thus, the title is properly read as a compound noun, and quite surely not as an adjective followed by a noun: *ZERO Syntax*, not *Zero SYNTAX*.

There are also certain topics with which this book does not deal, or with which it does not deal directly.

I have adopted certain ideas here more as a matter of convenience than for reasons motivated by the discussion itself. These include most notably

the hypothesis that subjects of IP originate within VP (the VP-Internal Subject Hypothesis (Kitagawa 1986, among others)) and the hypothesis that the head of nominal phrases is D (the DP Hypothesis (Abney 1987)), as well as the syntactic *koine* often referred to as “Government-Binding Theory” (discussed at greater length in chapter 1). When choice of theory is more important, I have attempted to be as explicit as possible.

During the latter stages of research, a number of manuscripts came my way whose contents should have had a greater impact on this work than they have. In particular, Bouchard (1992) has developed an alternative account of some of the material discussed in the earlier chapters of this book that should be compared to my proposals. I have not done so here. Ruwet (1993) takes issue with certain aspects of the prefinal draft of this book, mostly on empirical grounds that do not seem to me correct. To avoid extensive delay in the completion of this book, I have decided to go to press without taking proper account of these criticisms—if only because the time required to think carefully about them would no doubt bring new papers and manuscripts to my desk, requiring further thought, and so on ad infinitum. Likewise, although I have benefited from Beth Levin and Malka Rappaport Hovav’s research in many ways, I have not taken serious advantage of their recently completed book on unaccusativity, (Levin and Rappaport Hovav, in press), except insofar as its contents appeared in earlier published form. I drew the line somewhere around 1992, except for certain scattered observations that thrust themselves at me since then.

Certain topics lie next door to masses of supplemental literature that I have decided not to deal with for the present. For example, the zero morphology that is the backbone of this book should, by rights, be discussed in the context of current debates over the nature of word formation (even including the reality of morphemes themselves). No such discussions are included here. Such omissions, of course, are unfortunate, but once again arose from pragmatic decisions necessary to the completion of what would otherwise have been an endless project. As it happens, I strongly suspect that most of my proposals can be adapted to fit reasonable conceptions of morphology other than the fairly traditional approach to word structure adopted here, but obviously the alternatives need to be developed before one can be sure.

Likewise, I have not devoted much attention to the growing literature that debates the concept “thematic role” in a general fashion. This is not because the literature is irrelevant or uninteresting, but because (perhaps

surprisingly) I do not see a direct connection between this debate and the proposals I make here. In order to state my proposals, I need to discuss the nature and classification of arguments. For this, I employ the familiar notion “ θ -role,” as well as a repertoire of names for θ -roles whose lineage includes Gruber 1965 and Chomsky 1981. Where necessary (and it is quite necessary), I supplement and clarify some of the traditional notions of θ -theory. Where not necessary, I have been conservative. I do not see any respect in which the content of this book is tied to my use of terms like *Agent* and *Causer*, as opposed to other classification schemes (e.g., Dowty 1981). Proposals are often less tied to their putative foundations than they seem.

Finally, I should add a few words in the same vein concerning the relation of this work to Chomsky’s (1993) minimalist program. Chomsky’s recent paper proposes a refreshing and exciting housecleaning of the linguistic apparatus that has been widely assumed in research that followed in the footsteps of his (1981) *Lectures on Government and Binding (LGB)*. As far as I can tell, however, very little in the present work is directly affected by this housecleaning. If correct, the minimalist program may situate the present proposals in an interesting and theoretically parsimonious context, but it deals mainly with issues orthogonal to those discussed here.

On the other hand, this book quite consciously fails to incorporate one important and influential proposal from Chomsky’s paper: the suggestion (based on earlier work by Koopman (1992)) that objective Case, like nominative Case, is uniformly licensed in a specifier-head configuration with an Agreement category. Throughout this book, I assume that Case is assigned or checked much as in *LGB*—under government and adjacency. Still, as Akira Watanabe (personal communication) has pointed out, much of the discussion can be reconstructed with structures that incorporate various Agreement categories whose specifiers are the locus of Case assignment. This even includes the Cascade structures of chapters 6 and 7, if certain novel proposals concerning the location of Agreement are made. Nonetheless, I have stayed away from this alternative here, since it would complicate the exposition rather than improve it. On the other hand, other aspects of Chomsky’s recent proposals about Case are important here. In particular, Chomsky’s view that morphological features (including Case features) are checked, rather than assigned, in the syntax is put to good use in section 6.2.2. I view this as a proposal that can easily be detached from Chomsky’s specific views concerning the syntactic locus of Case.

My proposals also depart from Chomsky's (1993 and related work) in a deeper way. Our hunches about research strategy are different. Work like Chomsky's limits the apparatus of syntactic theory to exactly what is necessary in order to incorporate a set of insights concerning constituency, movement, binding, and other properties that have been gleaned over the last decade. This "tight-fitting" theory is of interest because it helps reveal the true shape of the discoveries made in this research. This, in turn, is important because this "true shape" seems to be in accord with certain "minimal" views of the interfaces between syntax and other cognitive/motor systems. On the other hand, this last aspect of the minimalist program has scientific bite only to the extent that the proposals concerning the shape of the theory are correct.

If my proposals are on the right track (particularly the "Dual System" of Cascade and Layered Syntax presented in the final chapters), the syntactic picture drawn in the minimalist program and similar schemes is seriously incomplete. Whether this conclusion alters our ability to accept Chomsky's general view of interface conditions on syntax is, at present, an open question. Perhaps there will be no conflict or even no interaction whatsoever between these proposals. It is hard to tell: the issues are not yet sharply enough in focus to even pose the right questions. At this point, the only real question that can be posed about these issues concerns strategy: when is it helpful in linguistics to venture a high-level "theory of everything," and when is it unprofitable, because important theoretical questions crucial to the shape of the higher-level enterprise are still not well answered? If the work reported here is on the right track, the high-level "theory of everything" should wait at least long enough for the dust to settle on fundamental questions of constituent structure, but nothing in the letter or spirit of this book is inimical to the idea that there are deeper, more general properties of syntax to be discovered than those discussed here.

One question I have often been asked after presentations of this research concerns the quantity and nature of the zero morphemes available to languages: "When will it all stop? Can you posit a zero morpheme whenever it suits the analysis?" My answer is that we must wait and see, and, yes, investigate analyses with zero morphemes wherever such an investigation seems fruitful. In this book, I believe that I have given evidence in favor of the existence, in English and a few other languages, of a set of phonologically null morphemes that express a few simple semantic notions and serve a few syntactic functions. As with any phonologically

null element (from traces to phrases), the existence and behavior of zero morphemes poses a puzzle for acquisition. The easiest solution would root zero morphemes in Universal Grammar in some fashion, either by positing a repertoire of zero morphemes invariant across languages, or by furnishing the child with a Universal Grammar-driven means of detecting language-specific occurrences of zero morphemes. Certain possibilities of the latter sort are considered in this book. On the other hand, I think it is premature even to speculate about what, if any, substantive limitations are placed by Universal Grammar on the set of zero morphemes—nor do I think that the work reported here suffers from this gap. After all, one purpose of this work is to discover zero morphology, which, as we will see, requires close reasoning and discussion of facts from a variety of domains. Only after much more work of this sort has been accomplished will it be possible to see what shape, if any, the general theory of zero morphology will assume. First things first.

Acknowledgments

Almost none of the material presented in this book has been published previously, but earlier reports on this research have circulated under various titles. A fragment of the semantic analysis of psychological predicates was reported as “Binding Problems with Experiencer Verbs” (Pesetsky 1987a). A fuller version of the empirical material, with less of a theory, was circulated under the title “Psych Predicates, Universal Alignment, and Lexical Decomposition” (Pesetsky 1987b). For important assistance with this stage of the work, I wish to thank F. Roger Higgins and Angelika Kratzer, as well as audiences at the University of Massachusetts at Amherst (particularly Noriko Kawasaki and Anne Vainikka). Luigi Burzio was very generous in replying to endless queries from me concerning Italian, as was Alessandro Zucchi. Noriko Kawasaki, Kiyoshi Kurata, Michiko Terada, and Hajime Hoji were just as generous in supplying information concerning Japanese.

The morphological portions of this book, along with the idea that the Target/Subject Matter restriction might be a syntactic consequence of morphology, represent research carried out from 1988 to 1990, when a manuscript incorporating these results, called *Zero Syntax (version 1.0)*, was circulated. This material was presented in lectures at MIT and at the 1990 Girona International Summer School of Linguistics. I wish to thank the audiences at these lectures, particularly Eulàlia Bonet, Chris Collins, and Sabine Iatridou at MIT, and Marcel den Dikken, René Mulder, Lea Nash, and Jan-Wouter Zwart in Girona, for crucial ideas, observations, and critical comments. Discussions with Barry Schein, Alec Marantz, and Hagit Borer were important in clarifying and directing my research during this time, as were comments by Teun Hoekstra and Esther Torrego.

Chapter 4, which makes thematic sense out of my morphological proposals, owes its explanatory parts to a conversation with Richard Kayne

in March 1991 (about absorption of external θ -roles) and to Kayne's lectures on reflexive clitics at the 1990 Girona International Summer School of Linguistics.

Finally, the work on Cascade and Layered Syntax (reported in chapters 5 and 6), which ties together the strands developed in the earlier chapters, was developed largely during the spring of 1992. It was first tried out in a series of lectures at the Institute for Research in Cognitive Science at the University of Pennsylvania and was polished in classes at MIT, the Free University of Amsterdam, and the 1993 Linguistic Institute of the Linguistic Society of America (Ohio State). I am particularly grateful to Kevin Broihier, Chris Collins, Danny Fox, John Frampton, Erich Groat, Masatoshi Koizumi, Geoff Poole, Tanya Reinhart, William Snyder, Akira Watanabe, and Ken Wexler, among many others, for probing questions and helpful suggestions.

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Chapter 1

Introduction: The Lexicon

1.1 Preliminaries

Although linguists struggle to make sense of the grammatical patterns of human languages, children take a mere two years or less to discover most of the grammar and much of the basic vocabulary of their native language. Modern linguistics relies on the hunch (due in its modern form to Chomsky) that the child is so successful because a basic linguistic system and the ability to use this system are somehow enshrined in the child's biology. The linguist labors to describe this system in a manner accessible to conscious thought, but the child simply *has* the system. Linguistic experience, combined with the properties of the innate system, yields the native language that the child comes to know and use.

It follows from the hunch just described that hypotheses about language should put as small a burden as possible on the child's linguistic experience and as great a burden as possible on the biologically given system, which we call *Universal Grammar (UG)*. Of course, the role of experience is not zero, or else every detail of language would be fixed along genetic lines. Nonetheless, given the facts that linguistics tries to explain, the null hypothesis should place the role of experience as close to zero as possible. In practical terms, this means that the linguist's null hypothesis should start with no role for experience. Those linguistic facts that can only be ascribed to experience can then be characterized cautiously. Everything else should arise from the interaction of these experience-induced facts with UG.

One area where experience is clearly relevant is the *lexicon*. Children simply do acquire lexical entries. As a bare minimum, these lexical entries consist of arbitrary pairs of sounds and meanings. Knowledge of these pairings is obviously a result of the child's linguistic experience. Nonethe-

less, the lexicon is no place to abandon a cautious approach to the role of experience in language acquisition. As with any aspect of language, proposals about the lexicon should proceed from the null hypothesis that *nothing* is acquired through experience, progressing with cautious and conservative steps toward an understanding of exactly what is acquired through experience and how.

In fact, traditional discussions of the lexicon often presuppose more content than the mere pairing of sound with meaning—and not without reason. For one thing, lexical items also impose various requirements on the syntactic *structures* in which they find themselves. These other requirements form part of the speaker's knowledge of those lexical items. This is clearest in the case of argument-taking categories: verbs, adjectives, nominalizations like *destruction* and *eagerness*, as well as nouns like *father*. For argument-taking lexical items, these include three crucial bits of information: semantic selection (*s-selection*), selection for syntactic categories (*c-selection*), and assignment of arguments to syntactic positions (*linking*).

S-selection is quite basic. It is nothing more than the consequences for argument structure of a predicate's lexical semantics. This includes selection for independent categories like *proposition* or *thing* as well as selection for relational categories like *Agent* or *Patient*. Thus, the fact that we utter but do not eat propositions entails that a verb like *say* allows a propositional argument whereas *eat* does not. The semantics of wonderment and belief entail that *wonder* takes an interrogative argument whereas *believe* does not. Likewise, the semantics of all these verbs tell us the semantic relations borne by their arguments (Agents, Experiencers, etc.). One could not conceive of a verb that meant 'say' that did not take an Agent argument.

- (1) a. Suo said that the world is round.
b. *Sue ate that the world is round.
- (2) a. Bill wonders whether the world is round.
b. *Bill believes whether the world is round.

Linking is a bit less basic, involving the relation between the semantic categories of a predicate's arguments and their syntactic positions. Linking tells us that the propositional argument of *believe* in (3) is its object and not its subject, that the Agent of *throw* is a subject, not an object in (4), and that *like* and *please* differ in the placement of their Experiencer argument in (5).

- (3) a. Sue believed that the world is round.
b. *That the world is round believed Sue.
- (4) a. Bill threw the ball.
b. *The ball threw Bill. [with *Bill* as Agent]
- (5) a. Mary liked the play.
b. *The play liked Mary.
c. The play pleased Mary.
d. *Mary pleased the play.

C-selection, in addition to *s-selection*, is relevant to the facts in (1) and (2). *Say*, but not *eat*, *c-selects* a CP complement; *wonder* and *believe* differ in whether the subordinate Comp triggers *wh*-movement. Furthermore, *c-selection* tells us that the interrogative argument of *ask* may be a DP, whereas the interrogative argument of *wonder* may not in (6), and determines that the semantically close verbs *like* and *enjoy* differ in their ability to take an infinitival argument in (7).

- (6) a. John asked the time.
b. *John wondered the time.
- (7) a. She liked the concerto.
b. She liked hearing the concerto.
c. She liked to hear the concerto.
d. She enjoyed the concerto.
e. She enjoyed hearing the concerto.
f. *She enjoyed to hear the concerto.

Much work over the last decade has attempted to discover relationships among *s-selection*, *c-selection*, and linking. Since *s-selection* is so rooted in the irreducible pairing of sound and meaning, there have been persistent hopes and speculations that the lexical entries of predicates need not specify their *c-selection* and linking properties directly (Grimshaw 1979, 1981, Pesetsky 1982, Chomsky 1986b). If these speculations are correct, then theories that ascribe separate status to the *s-selection*, *c-selection*, and linking properties of lexical items are insufficiently cautious theories and should be reconsidered. If we pursue these speculations, the lexical entry of a predicate does not contain explicit information concerning *c-selection* or linking. Most instances of either must be explained as consequences of *s-selection* aided by principles of UG that map semantic categories onto syntactic categories and syntactic positions. Since *s-selection*

itself is an aspect of lexical semantics, we will have the best theory of the lexicon that we can hope for: children learn pairings of sound and meaning; UG does the rest.

Thus, for example, the fact that *eat* s-selects things, not propositions, along with the fact in UG that CPs do not denote things, is sufficient to explain (1). No particular c-selectional facts need be learned by the child. Likewise, if UG in general requires Agents to be linked to subject position, then the linking facts in (4) follow from the s-selectional properties of the main verb and the linking principles supplied by UG. No specific linking facts need be learned by the child. This reasoning has the (probably) correct consequence that the facts in (1) and (2) have cross-linguistic validity. We would not expect a verb meaning 'eat' to select a proposition, nor would we expect a verb meaning 'believe' to select an interrogative complement in any language.

The same reasoning can apply to morphologically complex lexical items, once UG is provided with the ability to project the argument structure of complex words from their component parts. Following Randall (1981), I will call this property *morphological inheritance*. Thus, an Agent-Patient verb like *destroy* will pass on its s-selectional properties to the process reading of *destruction* in accordance with mechanisms of morphological inheritance given by UG plus language-specific facts about the suffix *-ion*. For any nominalized verb a similar procedure can be expected to obtain.

Of course, no one supposes that pairings of sound and meaning can possibly exhaust the content of the lexicon. Once we leave c-selection, linking, and morphological inheritance, it is clear that language acquisition involves certain types of facts that can only be the consequence of experience. Thus, children learn facts about declension class, agreement class, and conjugation class, affixal status, availability as a host for affixation, selection for quirky Case, and other facts that attach themselves like barnacles to lexical entries. What aspects of lexical semantics coupled with what principles of UG could predict (5), (6), or (7), the fact that the past tense of *go* is *went*, that *-ion* suffixed to *destroy* yields *destruction*, or the fact that the Russian verb *vladet'* 'command' governs the instrumental Case? Our model of the lexicon should allow such facts to be acquired, but should do so grudgingly, limiting the acquisition of idiosyncratic facts to those that seem absolutely inescapably idiosyncratic, but explaining away c-selection and linking as by-products of s-selection and syntax.

Content with these methodological preliminaries, we might be tempted to rest on our speculative laurels and consider the description of lexical learning accomplished. This would be an error, since our picture could easily be wrong. Since the stakes are so high—an entire view of the logical problem of language acquisition—we must work hard to determine whether our view of the lexicon is tenable. Indeed, once we move away from cheap victories for UG like *believe* and *destruction* and obvious cases of experience-based acquisition like conjugation class and quirky Case, many problems and questions arise.

This book has a simple goal: to show that certain observations that seem to situate the lexicon far from our minimalist hopes actually resolve themselves in a satisfactory and exciting way once we dig deeper into the nature of things. In other words, this book seeks to reduce the number of problems with the *a priori* most attractive view of lexical acquisition.

Suppose we discover some pattern of c-selection, linking, or morphological inheritance that neither looks like declensional class or quirky Case nor is derivable from lexical semantics and current views of UG. There are two kinds of responses we might make. First, we might abandon the view that the lexicon is maximally simple to accommodate the case at hand, positing some new mechanism employed by the child in acquiring the pattern under discussion. Alternatively, we might modify our characterization of the problem in such a way that the problem disappears. We do this by changing our view of UG.

Of course, it is sound method to begin with some set of well-supported “off the rack” ideas about UG as a background to discussion. In linguistics, such sets of ideas are typically referred to as *frameworks*. Such frameworks are helpful because they provide links between different proposals and allow the linguist to focus on one problem without having to worry about every adjacent problem. Nonetheless, they are also pernicious, since as often as not, the solution to a problem that has not been solved lies in an idea that cannot quite be bought “off the rack.” This book takes a framework as theoretical backdrop, but out of necessity develops a few new ideas not “off the rack” to reach its conclusions—the most notable case being the “Dual System” (including a Cascade Syntax) developed in chapters 6 and 7.

The framework with which I begin is the body of work that flows from the results reported in Chomsky 1981. This is usually called *Government-Binding Theory* (GB) or, more generally, *principles-and-parameters syntax*

(P&P). Where the choice among competing proposals in this literature is important, it will become clear which path I take as the discussion progresses.

A central novelty of this work is its approach to cross-linguistic variation. GB/P&P research has discovered principles of grammar that cross-cut the traditional division of syntax into constructions typically named according to their key morpheme or semantic content, for example, *passive*, *reflexive*, or *wh-question*. The articulations of GB/P&P theories lie elsewhere, so that some properties of the passive construction are held in common with properties of reflexive anaphora, others with *wh-questions*. Syntactic differences among languages are attributed to the role of experience in choosing values for certain open parameters in the theory. Since the articulations of the theory largely crosscut traditional construction boundaries, the setting of a parameter for any articulation can be expected to have consequences that crosscut constructions. Although work on these questions is in its infancy, the approach appears to be productive. Surprising linkages between constructions of the sort expected in this approach are beginning to be discovered. To pick one recent example, Kayne (1990, 1991) shows that a single parameter links the relative positions of infinitive and clitic pronoun in Romance languages to the possibility of infinitival yes-no questions, and suggests that this is a direct consequence of Chomsky's (1986b) theory of pronoun and anaphor binding.

The same property of GB/P&P research that predicts surprising linkages in cross-linguistic variation yields surprising linkages for the linguist attempting to explain language. A slight change in theory, or even a relatively minor change in the analysis of some structure, can be expected to have surprising consequences over a range of linguistic constructions. This book presents some relatively banal proposals concerning a few constructions that pose serious problems for the "minimalist" view of linking and c-selection just discussed. I will try to show that these proposals, set against the backdrop of GB/P&P theories, provide a key to a large number of thorny problems concerning c-selection, linking, and morphological inheritance.

1.2 Zero Syntax

In this book (and in Pesetsky, in preparation), I propose that a series of problems concerning the lexicon can be solved by positing a phonologically zero derivational affix.

Phonologically zero *inflectional* morphemes are nothing new. It has always been common to posit a phonologically zero pluralizer in plural *sheep*, a zero participial morpheme for *come* in *I have come*. Furthermore, in theories of the French verb like those of Emonds (1976) and Pollock (1989), finite verbs move from V to Infl because of the affixal property of the finite morpheme occupying Infl.

- (8) Nous [_{Infl} *parl*_i [_{Infl} -*ons*]] [_{VP} à peine t_i français].

+ affix

we speak 1pl barely French

'We barely speak French.'

When the same sort of movement is attested without the phonological expression of Infl, we can only conclude that Infl contains a phonologically null affix.¹

- (9) Il [_{Infl} *parl*_i [_{Infl} $\emptyset$]] [_{VP} à peine t_i français].

he speak 3sg

'He barely speaks French.'

Phonologically zero derivational affixes are less well established. They are obvious mechanisms to propose for relating noun-verb pairs like *support*_V/*support*_N or nominal and adjectival uses of *immortal*. I will discuss such cases in detail, in order to motivate some less obvious candidates for zero derivational morphology.

Where else might one posit phonologically zero derivational morphemes? There are a variety of circumstances in which such a proposal might be reasonable.

- Suppose a verb with a particular meaning is bimorphemic in Japanese, yet superficially monomorphemic in English. It would not be shocking if we were to learn that the relevant form is bimorphemic in both languages, except that one of the morphemes is phonologically null in English.

- (10) a. Sono sirase-ga Tanaka-o kanasim-sase-ta.
that news-NOM Tanaka-ACC be sad-CAUSE-past

'That news saddened Tanaka.'

- b. The news [*CAUS* [depressed _V]] Bill.

- One might posit a null morpheme in a more general circumstance. Suppose the same phonological form is used in two senses, one more complex than the other. One might attribute the greater complexity in the second sense to vagueness, or else one might posit a phonologically null

morpheme that adds the complexity. Thus, consider the use of adjectives like *nervous* or *happy* when predicated directly of animate beings (*Mary is nervous*) and when predicated of the behavior of animate beings (*Her manner is nervous*). In the latter use, *nervous* means something like *suggests nervousness on her part*. This added complexity might be contributed by a morpheme, which we could call *SUG*, though it has no phonological form.

(11) Her manner was [[nervous_A] *SUG*_A].

- Next consider a construction involving two DP objects, where the second object has no visible source for abstract Case. Suppose that the second object is asymmetrically c-commanded by the first with no visible heads creating the asymmetry. Suppose further that this construction has a near synonym in which one of the objects is introduced with a preposition. It might be plausible to propose that the second DP in this double object construction is in fact the object of a phonologically null preposition. It might not be implausible (though not motivated at present) to propose that this preposition is affixed by head movement to the main verb of the construction.

(12) They [_P_i [give_v]_V] Bill [_P t_i] books.

- Now consider a construction involving an embedded finite clause, in which the embedded clause is not introduced by a phonologically overt complementizer. Following Stowell (1981) and Kayne (1984a:chap. 5), we might propose that these constructions actually are introduced by a complementizer—namely, a phonologically null complementizer. Once again, it might not be implausible (though unmotivated for the moment) to affix this complementizer to the higher verb.

(13) They [Comp_i [announced_v]_V] [_{CP} t_i [_{IP} the train was about to leave]].

In later work (Pesetsky, in preparation), I will show that the affixation in (13) accounts for gaps in the distribution of this zero complementizer in subject and adjunct sentences.

- Finally, the same analysis might be accorded to an embedded infinitival clause like that in (14).

(14) They [Comp_i [considered_v]_V] [_{CP} t_i [_{IP} Bill to be happy]].

In Pesetsky, in preparation, I will also show that affixation of the null complementizer to *consider* in (14) explains the fact that *consider* governs

the embedded subject. This will capture the effects of Chomsky's (1981) proposal of CP-deletion.

Of course, it is one thing to demonstrate circumstances in which zero morphemes are "not implausible," and quite another to demonstrate their existence. This will be my task here and in Pesetsky, in preparation. For now, by way of introduction, note that each construction mentioned above shows a peculiar morphological gap: nominalization of the expected sort cannot take place.

- (15) a. *the book's annoyance of Bill
 b. *her manner's nervousness
 c. *their gift of Bill (of) books
 d. *their announcement the train was about to leave
 e. *their consideration (of) Bill to be happy

Under the analyses that I will suggest, the examples in (15) all show a nominalizing suffix attached to a zero-derived form.

- (16) a. *the book's [[[annoy_v] CAUS_v] ance_N] of Bill
 b. *her manner's [[[nervous_A] SUG_A] ness_N]
 c. *their [[P_i [gif-_v]_v] -t_N] of [_{PP} Bill [_{t_i} _P] books]
 d. *their [[Comp_i [announce_v]_v] ment_N] [_{CP} _{t_i} [_{IP} the train was about to leave]]
 e. *their [[Comp_i [consider_v]_v] ation_N] (of) [_{CP} _{t_i} [_{IP} Bill to be happy]]

Furthermore, these examples all contrast with similar examples that do not involve zero derivation and show no difficulty undergoing nominalization. Noncausative psychological nominalizations, predication by *nervous* of human experiencers, dative verbs with overt *to*, proposition-taking nominals with overt complementizers—all lack the slightest hint of evidence for zero morphology, and all avoid the problems that star the examples in (15)–(16).

- (17) a. Bill's [[annoy_v] ance_N] at the book
 b. her [[nervous_A] ness_N]
 c. their [[gif-_v] -t_N] of [_{PP} books to Bill]
 d. their [[announce_v] ment_N] [_{CP} that [_{IP} the train was about to leave]]
 e. la sua [[suppos_v] izione_N] [_{Comp} di [_{IP} PRO essere
 his/her consideration Comp to be
 felice]] [Italian]
 happy

In the consistent impossibility of nominalization in (15) and its acceptability in (17), we have a good hint that the examples displayed in (15) have something in common. The plausibility of zero affixation in each case invites us to explore the possibility that zero affixation is actually found in these cases. In the course of this book, I will show that zero affixation provides a crucial clue that helps us resolve problems for the minimalist lexicon hypothesis. I will also suggest a general picture of affixation in which the stars in (16) are expected.

In chapters 6 and 7, I will make a second, equally banal proposal. This proposal, which I will call the *Dual System*, posits two types of structural organization in predicate-headed phrases like VP. One, *Layered Syntax*, is based on simple principles that map selectional relations into structural relations, yielding structures like those traditional in syntax through the 1970s. The other, *Cascade Syntax*, creates quite different structures from a minor modification of these principles: in essence, permitting selection of specifiers of sisters as well as sisters. When this proposal is combined with the first, not only the problems that arose in the first half of the book, but a variety of others as well, find a solution. These include

1. why prepositions sometimes seem to be overlooked for c-command relations;
2. why the structure induced by prepositions is overlooked by coordination;
3. why heavy NP shift acts like both upward and downward movement;
4. why there are limitations on the occurrence of *to* in the double object alternation.

If it is on the right track, this book will advance the central program of modern linguistics: to characterize UG and the fruits of linguistic experience, putting as much burden as is plausible on the former and as little as possible on the latter. More than that, the sort of results reported here fall within a large field of results that validate linguistics itself. The only doctrine of linguistics is the conviction that the facts of language pose puzzles that can be solved—that language has a shape, forms a system. The very structure of this work demonstrates the point: it starts by noting problems with the lexicon, solves these problems with the help of a novel syntactic proposal, and then discovers that the novel syntactic proposal solves other so far intractable problems. This is the sort of structure one expects from an extended investigation of an orderly, law-governed system like human language.

Chapter 2

Linking Problems with Experiencer Predicates

2.1 Linking

In this chapter, I will first discuss general problems for theories of linking. Specific linking problems with Experiencer predicates will then lead me to consider some solutions that are attractive yet, I believe, ultimately flawed. In the next chapter, these problems will lead to proposals concerning zero morphology.

The child who has acquired a verb such as *push* in *John pushed Bill* knows that the Agent (John) is “linked” to subject position and the Patient (Bill) is linked to object position. Must the child learn linking facts on an item-by-item basis? The strongest and most attractive answer is, of course, no. For active Agent-Patient verbs, it is quite tempting to argue that UG maps Agent uniformly onto subject, and Patient onto object.² Can we extend this sort of reduction to other lexical items? The optimistic hypothesis that we can was a mainstay of much work in Generative Semantics and has been recently revived in its strongest form by Baker (1988:46) as the *Uniformity of Theta Assignment Hypothesis*.

(18) *Uniformity of Theta Assignment Hypothesis (UTAH)*

Identical thematic relationships between items are represented by identical structural relationships between those items at the level of D-Structure.

A slightly weaker condition was earlier proposed by Perlmutter and Postal (1984) as their *Universal Alignment Hypothesis*.

(19) *Universal Alignment Hypothesis (UAH)*

There exist principles of UG which predict the initial relation borne by *each* [argument] in a given clause from the meaning of the clause.³

The UAH as stated in (19) is weaker than the UTAH in that it does not require *identical* syntactic linking patterns in cases of semantic identity, but merely requires *predictable* linking patterns.⁴ Exploiting the happy coincidence of initials, I will refer to the family of hypotheses that includes the UTAH and the UAH as the *U(T)AH*. U(T)AH hypotheses are important because they provide plausible (albeit partial) explanations for the rapidity of language acquisition. It is a truism that the more regular the correspondences are between sound, structure, and meaning, the less there is for the child to learn.⁵

The U(T)AH is worth investigating because it is a very obvious and optimistic proposal. It is also worth investigating because there are many apparent counterexamples that could bring it down. The purpose of this study is to explore two very different ways of explaining apparent counterexamples to the U(T)AH. I wish to show that both methods are fruitful, and that only rather extensive investigation can decide whether these apparent counterexamples are genuine, or yield to the first method of explaining apparent counterexamples, or yield to the second. The specific domain of investigation will be Experiencer (or “psychological”) predicates, but we can first examine the general questions on the basis of somewhat less controversial examples.

First consider Baker’s UTAH, the strongest form of the U(T)AH. One sort of case that puts it to the test is an instance where one and the same type of argument appears to be assigned to more than one grammatical relation. A familiar example of this is the inchoative/causative alternation seen in well-known pairs like these:

- (20) a. The ice melted.
b. Bill melted the ice.
- (21) a. The door opened.
b. Sue opened the door.
- (22) a. The ship sank.
b. Bill sank the ship.

In the (a) examples, the Patient or Theme argument is found in subject position; in the (b) examples, it is found in object position. How can these data be reconciled with the UTAH? There are three general types of solutions.

1. *Finer-grained syntax* There is more to the syntax than meets the eye. Closer examination of the syntax will show that the apparent syntactic

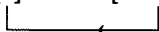
identity between the subject of the (a) sentence and the subject of the (b) sentence is false. The superficial subjects of the (a) sentences are D-Structure objects. There is no problem for the UTAH.

2. *Finer-grained semantics* There is more to the semantics than meets the eye. Closer examination of the semantics will show that the apparent semantic identity between the subject of the (a) sentence and the object of the (b) sentence is false. Once more, there is no problem for the UTAH.

3. *Abandonment of the UTAH* The UTAH is wrong, and weaker hypotheses (e.g., the UAH, or the assumption of total arbitrariness) should be examined.

A “type 1” approach leads to the best-known and probably correct solution for the data in (20)–(22): an *unaccusative* analysis for the (a) examples. This analysis holds that certain subjects are underlying objects, as illustrated in (23).

(23) a. [e] melted [the ice].



b. Bill melted [the ice].

For the problem raised by (20)–(22), the literature contains abundant evidence for the hypothesis sketched in (23). Indeed, this evidence itself provides support for the derivation of certain linking patterns from UG. Were it not for UG principles, there would be no reason for the child to assume that the underlying structure for a sentence like *The boat sank* differs significantly from its PF representation. Nonetheless, even if we suppose that the UTAH in its strongest form is correct, the fact that a type 1 approach proved fruitful for this particular problem does not mean that *every* problem for the U(T)AH will have a solution along these lines.

Let us look at type 2 solutions. Both of the U(T)AH hypotheses hinge crucially on a theory of lexical meaning. This is central to the notion “identity” invoked by Baker’s UTAH, which should probably be replaced by some notion of “similarity” with respect to some property. Not all detectable semantic distinctions are of syntactic relevance. Two predicates might show subtle or even glaring semantic differences that are irrelevant to the syntax. Clauses containing these predicates will count as “equivalent in meaning” and will display “equivalent thematic relationships” for the U(T)AH, despite their semantic differences.

For example, though there are doubtless relevant and identifiable distinctions between *shout* and *whisper*, *amble* and *scamper*,⁶ or even *shout* and *holler*, these distinctions probably play no role at all in syntax/

semantics interactions. Both *shout* and *whisper*, for example, have properties characteristic of agentive predicates; their Agent argument is a subject. Both verbs share selectional properties characteristic of manner-of-speaking verbs (e.g., nonextractability of adjuncts from their complements). Both verbs resist deletion of the complementizer of a complement *that*-clause. There are very palpable differences between the meanings of *shout* and *whisper*; these differences are simply of no consequence for syntax.

The relationship between semantics and syntax in this case is similar to the relationship between phonetics and phonology. For example, the side of the mouth chosen to articulate lateral consonants is a gross and obvious physiological fact—but one of no known phonological relevance. The exact degree of lip pursing during production of a rounded vowel may be quite observable, but is similarly of no phonological relevance. For the phonetics-phonology interface, the theoretical instantiation of these observations is feature theory, which gives a particular sort of coarse phonological grain to the continuum of phonetic reality. For the semantics-syntax interface, there may be a number of analogous theoretical constructs. One is θ -theory, where many observable semantic distinctions are ignored in favor of a coarse, grainy classification.

That said, however, the nature of the graininess and the identity of the grains is an empirical question. For example, it seems that the distinction between “verbs of loud speech” and “verbs of quiet speech” (e.g., *holler* and *shout* vs. *whisper* and *murmur*) is syntactically irrelevant, but the distinction between “verbs of manner of speaking” (including both *holler* and *whisper*) and “verbs of content of speaking” (e.g., *say*, *propose*) is not irrelevant. Verbs of the latter class in English do not resist adjunct extraction and allow complementizer deletion.⁷

Let us return to the pairs in (20)–(22). Suppose one were able to argue for a type 2 solution to this problem. The holder of this view would claim that the alleged semantic identity between the subjects of the (a) sentences and the objects of the (b) sentences is incorrect and based on an excessively coarse-grained semantic analysis. Suppose, for example, that the subject of (20a) were to turn out to be a Theme_a, and the object of (20b) were to turn out to be a Theme_b, where Theme_a and Theme_b were semantically distinct in some fashion. Whether the distinction between Theme_a and Theme_b is relevant to the syntax would then be an empirical matter. If the distinction were relevant to the syntax, we would then propose that UG includes the following statements:

- (24) a. Agent is mapped to subject.
 b. Theme_a is mapped to subject.
 c. Theme_b is mapped to object.

We would want to ask *why* these principles are the way they are, of course (see Levin and Rappaport 1988c). Nonetheless, independent of this question, such principles would yield an alternative analysis of the data in (20)–(22) that does not conflict with the UTAH. The difficulty with this approach applied to the data in (20)–(22), of course, is the existence of strong evidence for the Unaccusative Hypothesis and the nonexistence of any evidence whatsoever for a systematic distinction between Theme_a and Theme_b. For this reason, there has been no attempt (of which I am aware) to deal with the data in (20)–(22) from the perspective of finer-grained semantics.⁸ A priori, however, either approach might have turned out to be correct, and both approaches preserve the UTAH.

Now let us consider the weaker version of the U(T)AH, Perlmutter and Postal's UAH. This condition allows a variety of solutions to the problem posed by (20)–(22), so long as the initial relation borne by every argument is predictable. For example, the following rules would cover (20)–(22), respect the UAH, but violate the UTAH (see Chomsky 1972:171, (55)):

- (25) a. Agent (Causer) is assigned to subject.
 b. Theme (Patient) is assigned to subject, unless another argument is assigned to subject by rule (25a), in which case Theme (Patient) is assigned to object.

The linking theory outlined by Bresnan and Kanerva (1989) is a theory of this general sort, respecting the UAH, but not the UTAH.⁹

Even if we grant that the U(T)AH can be successfully defended for cases like (20)–(22), we must still ask whether the defense is well founded. Are there cases that force us toward a type 3 solution—abandonment of the UTAH? Rosen (1984) has argued that there are such cases, on the basis of cross-linguistic as well as language-internal linking patterns. If the UAH is an aid in language acquisition, then the rules governing linking are surely a property of UG and should not be susceptible to random cross-language variation. With this conclusion in mind, Rosen notes that there is cross-linguistic variability in the unaccusativity/unergativity of a number of predicates, including verbs glossed as 'bleed', 'die', 'suffer', and 'be hungry'. More troubling still, she also notes variability internal to Italian both with verbs of "bodily function" like 'snore' and 'blush' and with motion verbs like 'fall' and 'walk'.

- (26) a. Mario è arrossito. [unaccusative]
 Mario blushed (Rosen's (79a))
 b. Mario ha russato. [unergative]
 Mario snored (Rosen's (78a))
- (27) a. Luigi è caduto apposta. [unaccusative]
 Luigi fell on purpose (Rosen's (76a))
 b. Ugo ha camminato meglio ieri. [unergative]
 Ugo walked better yesterday (after Rosen's (86a))

Rosen concludes that the UAH is simply incorrect.¹⁰ It is entirely possible, however, that some of these apparent deviations from principles believed to govern unaccusativity should be explained as sketched above, by taking a finer-grained (or simply different) view of the semantics of the relevant verbs. For motion verbs, for example, Levin and Rappaport (1988a) have successfully pursued the finer-grained semantics approach and shown that the seemingly unpredictable distribution of motion verbs is actually predictable from two generalizations. The first concerns whether the verb's meaning "includes a specification of inherent direction" (which explains (27a–b)).¹¹ The second concerns whether the verb's meaning "specifies a direct external cause" (which explains other problematic cases). Further research along these lines might well bring order to the apparent chaos of 'snore' versus 'blush', as Levin and Rappaport note, but so far nothing has been established about these cases.¹²

Nonetheless, even if there *are* domains within which linking is unpredictable, such as predicates of involuntary bodily function, Rosen's own data suggest that weaker forms of the U(T)AH can be maintained.

(28) *UTAH (weaker version)*

Identical thematic relationships between *certain* categories of items are represented by identical structural relationships between those items at the level of D-Structure.

(29) *UAH (weaker version)*

For *certain* types of arguments, there exist principles of UG that predict the initial relation that they bear in a given clause from the meaning of the clause.

The addition of the word *certain* in the weaker versions of the U(T)AH does not void them of content. The absence of totally predictable linking does not mean that the child cannot take advantage of what predictability is available. For example, none of Rosen's examples of apparent capri-

ciousness (apart from motion verbs) involves agentive predicates. Rules for the linking of Agent appear to apply without exceptions, even though some other argument types might display fluctuations. If the assignment of semantic relations to grammatical functions were idiosyncratic and chaotic, we would not expect to find this sort of regularity. I will speak of arguments such as Agents whose linking patterns are predictable as *falling under* the U(T)AH, meaning by this that Agent arguments are among those whose syntactic representation is always the same (UTAH) or is always predictable (UAH).¹³ As for arguments that do not fall under the U(T)AH (if such arguments exist), these might need special stipulations. Even so, the language learner might not have to learn such stipulations for every argument: in the case of the subjects of involuntary bodily function verbs meaning 'blush' or 'sneeze', we would then assume that an unaccusative analysis is the default, but that some expected unaccusatives may be idiosyncratically assigned to the unergative class. We might then still claim that Theme arguments fall under the U(T)AH as the unmarked case. More research would be needed to establish this, so I will leave the matter open (see Postal 1986:31).

Obviously, the original versions of the U(T)AH are the strongest and most interesting. The stronger versions, if true, require less of the language learner than the weaker versions do. On the other hand, even if the stronger versions are false, the U(T)AH in any form is still of great importance. Requiring a little of the language learner is still better than requiring a lot.

What does this suggest for the investigation of unaccusative verbs? Rosen's proposal is that *nothing* falls under the U(T)AH in the sense described above. Baker's and Perlmutter and Postal's proposals are that *all* argument types fall under the U(T)AH. Even if Baker and Perlmutter and Postal are wrong, the null hypothesis of the researcher faced with any pattern of linking should be that *the pattern does fall under the U(T)AH*. Our initial assumption in any given instance should be that linking is predictable, until facts prove otherwise. If we do not take this as the null hypothesis, then we will not only be "making life difficult" for the language learner, we will also be learning nothing about the subject. The two general methodologies available for solving problems with the U(T)AH are the finer-grained syntax approach and the finer-grained semantic approach sketched above.

With the preceding discussion as guide, we are now in a position to tackle a particularly well known and controversial problem: the linking of

Experiencer predicates. Like Rosen's examples, these seem at first blush like counterexamples to both the UTAH and the UAH. I will suggest that this particular linking problem provides an excellent example of the points sketched above. The familiar linking problem with Experiencer verbs cannot be uniformly solved either by an unaccusative analysis or by refining the semantics. The linking problem is actually two problems: for one set of predicates, the problem is solvable in unaccusative terms; for another set of predicates, the problem is not solvable in this way, but does yield to a finer semantic analysis.

Experiencer predicates present a problem for the U(T)AH because of pairs like those in (30)–(41).

- (30) a. Bill was very angry at the article in the *Times*.
b. The article in the *Times* angered/enraged Bill.
- (31) a. The paleontologist liked/loved/adored the fossil.
b. The fossil pleased/delighted/overjoyed the paleontologist.
- (32) a. Bill disliked/hated/detested John's house.
b. John's house displeased/irritated/infuriated Bill.
- (33) a. Bill was satisfied/content with the Chinese dinner.
b. The Chinese dinner satisfied/contented Bill.
- (34) a. Sue resented Bill's remarks.
b. Bill's remarks embittered Sue.
- (35) a. Mary rejoiced at the French victory.
b. The French victory cheered/exhilarated Mary.
- (36) a. John worried about the television set.
b. The television set worried John.
- (37) a. Bill was furious about/fumed about the article in the *Times*.
b. The article in the *Times* infuriated Bill.
- (38) a. Sue's remarks puzzled us.
b. We puzzled over Sue's remarks.
- (39) a. Sue grieved over/at the court decision.
b. The court decision grieved Sue.
- (40) a. John is bored with the problem of lexical entries.
b. The problem of lexical entries bores John.
- (41) a. Bill fears/is afraid of ghosts.
b. Ghosts frighten Bill.

Pairs such as these suggest at first sight that for Experiencer predicates, linking is arbitrary. This is the case because, on the most straightforward and obvious analysis of (30)–(41), all the examples display the same two θ -roles. Belletti and Rizzi (1988) call these roles *Experiencer* and *Theme*. An analysis of the (a)–(b) contrast in (30)–(41) that assumes arbitrariness is sketched in (42), where the underscored argument must c-command all other arguments.

- (42) a. predicate (Exper, Theme)
 b. predicate (Exper, Theme)

In the (a) examples, Experiencer is the subject, and Theme is the object; in the (b) examples, Theme is the subject, and Experiencer is the object. I will henceforth call the predicates in the (a) examples the *Subject Experiencer* (*SubjExp*) class, and the predicates in the (b) examples the *Object Experiencer* (*ObjExp*) class. If the “straightforward” view were correct, then the existence of these two classes would show that these Experiencer verbs refute the U(T)AH hypotheses in its strongest forms. The arguments in (30)–(41) would not fall under the U(T)AH.¹⁴

2.2 An Unaccusative Solution to the Experiencer-Object Problem

Belletti and Rizzi (1988; henceforth *B&R*) have argued at length that the problem seen in (30)–(41) is simply a species of the problem seen in (20)–(22). They thus propose a solution along the lines of the type 1 possibilities discussed above: a finer-grained syntax in the form of an unaccusative analysis for the *ObjExp* predicates. According to them, although the S-Structure representations for *SubjExp* and *ObjExp* class predicates differ sharply, their D-Structure representations are identical in most respects. If *B&R* are correct, then these predicates pose few problems for the U(T)AH.¹⁵

In particular, *B&R* propose the linking principle in (43).

- (43) *Linking Principle for Experiencer Verbs*

Given a θ -grid [Experiencer, Theme], the Experiencer is projected to a higher position than the Theme. (*B&R*’s (119))

The Experiencer position is sometimes the traditional subject position (with *SubjExp* predicates) and sometimes a VP-internal position higher than direct object (with *ObjExp* predicates). The Theme position is always the underlying direct object position.

How are predicates of the *ObjExp* class derived on such an analysis? Recall that the *ObjExp* class shows Theme in S-Structure subject position.

B&R argue that this is the result of movement, as with more traditional unaccusatives.

(44) Theme_i [_{VP} [_V, V e_i] Exper]

B&R's proposal is entirely in line with our discussion of linking. It represents a type 1 solution to the problems for the U(T)AH that are raised by the SubjExp ~ ObjExp alternation. It is also the best, most thoroughly worked out discussion of these predicates in a structure-based theory of grammatical relations.

Nonetheless, we will see that this solution is as wrong as it is right. In some ways ObjExp constructions behave as B&R's solution predicts, but in other ways they behave in a completely contrary fashion. When we chase down the solutions to these paradoxes in the next few chapters, we will discover an entirely new view of the VP, which brings dividends of its own and helps us solve the problems with which we started.

In a movement-based analysis of the sort pioneered by Burzio (1981), unaccusative verbs have two properties. First, they do not have a designated external (or subject) argument, allowing for an athematic subject position. Second, they license movement or a movement-like relation between an internal argument position and the normal position for external arguments. We may divide arguments for the unaccusativity of ObjExp verbs into those that point to an athematic subject position and those that point to a movement-like relation between internal and external positions. In Italian and in English (as well as Dutch, among other languages), there are indeed ObjExp verbs that have both properties associated with unaccusativity, but only a proper subset of the ObjExp verbs fall into this "pure" category.

Another (larger) group of ObjExp verbs are not unaccusative in the straightforward fashion that allows B&R's solution to the linking problem. These verbs, I will suggest, do license a movement-like relation between their internal argument position and subject position, but do not show the other unaccusative property—an athematic subject position. The solution to this unaccusativity paradox, coupled with a type 2 analysis in terms of a finer-grained semantics, not only will put our minds at ease about ObjExp verbs (a minor result) but also will reveal properties of the mapping between argument structures and S-Structure representations that have remained unexplored in previous work.

Thus, some of my conclusions will be conservative and some will be more radical. Among the conservative conclusions will be the results con-

cerning “pure” unaccusative ObjExp verbs. ObjExp predicates in languages like Italian and Dutch fall into two very obvious subclasses with respect to compound past tenses: some take the auxiliary ‘be’ normally associated with unaccusatives (e.g., Italian *piacere* ‘please’). For these verbs, my unaccusative analysis will be unsurprising. Others take the auxiliary ‘have’, normally associated with unergatives (e.g., Italian *preoccupare* ‘worry’). For these verbs, therefore, an unaccusative analysis like B&R’s *would* be surprising. I will argue that the “surprising” part of B&R’s analysis is the incorrect part.¹⁶ These are the verbs for which a finer-grained semantics will be important. These verbs will also turn out to be unaccusative in the partial sense mentioned above, but finer-grained semantics, and not unaccusativity, will be relevant to the solution for their linking problems.

In particular, I will argue that the label “Theme” as applied to the non-Experiencer in (30)–(41) incorrectly lumps together a number of distinct θ -roles. Once these roles are distinguished, the problem for the U(T)AH disappears. (The same distinctions are found in English, where there is no auxiliary selection to guide us.) In its stead, however, a new problem appears; this problem will lead to a “bimorphemic” analysis of the non-unaccusative ObjExp verbs (somewhat reminiscent of work in Generative Semantics (Akatsuka 1976)), involving zero causative morphology. The special properties of this zero causative morpheme will solve the problem that motivated its existence in the first place. Finally, the zero causative morpheme will also help resolve the unaccusativity paradox that originally led us into the discussion.

The initial discussion of B&R’s proposal will introduce points and examples that will become important as the analysis progresses. Though this preliminary discussion may appear negative, it will lead to points and observations that are extremely important later. The critique of B&R’s solution is the vehicle for discovering the solution presented in this book.

Let us now consider B&R’s unaccusative solution more carefully. I will focus on English and Italian data, and will begin with B&R’s arguments that the subject of ObjExp predicates is athematic.

2.3 Verbal Passivization

B&R claim that the nature of passive forms from ObjExp verbs argues for the absence of a thematic external argument. I will suggest on several fronts that this conclusion is incorrect.

A characteristic property of unaccusative verbs in many languages, identified by Perlmutter and Postal (1984) and discussed by Perlmutter and Zaenen (1984), Marantz (1984), and others, is their incompatibility with passive morphology. This property is explained within Relational Grammar as a consequence of the *I-Advancement Exclusiveness Law*. Working within a phrase-structural theory of passive advancement, Marantz (pp. 144–49) explains this incompatibility in a different way, as a consequence of two claims.

- (45) a. Passive morphology absorbs the *external* (underscored) θ -role.
- b. Vacuous dethematization is impossible.

More recently, Baker, Johnson, and Roberts (1989), developing ideas by Jaeggli (1986b), suggest that the passive affix does not *deprive* a verb of its external argument role, but actually functions as an argument nominal and *receives* the external argument role. Under this theory, if a passivized verb has no external argument role, the passive affix stands in violation of the θ -Criterion.

Whether or not the “external argument” must be syntactically external to all projections of its predicate, as suggested by Williams (1981), is a separate issue, related to recent work on the VP-Internal Subject Hypothesis (which I have adopted here). Clearly, one argument may be marked as special, since it is this argument that is absorbed by or assigned to the passive affix. When this special argument is missing, as with unaccusative verbs, passive is impossible. Whatever the proper account of this phenomenon, it is of great interest that many English ObjExp verbs seem to passivize freely.

- (46) a. Bill was angered by Mary’s conduct.
- b. The paleontologist was pleased by the discovery of the fossil.
- c. Bill was irritated by the loud noises coming from next door.
- d. Bill would not be satisfied by halfway measures.
- e. Sue was embittered by her experiences with discrimination.
- f. Mary was cheered by the French victory.
- g. John was worried by my remarks.
- h. Harry was puzzled by Sue’s curious behavior.
- i. Harry was grieved by the court’s decision.
- j. Sue was bored by her work on lexical entries.
- k. Bill was frightened by strange noises.

The same is true of Italian ObjExp verbs that take *avere* ‘have’ in the active. The following are B&R’s examples (1988:(47)):

- (47) a. Gianni è disgustato dalla corruzione di questo paese.
 Gianni is disgusted by-the corruption of this country
 b. Gianni è affascinato da questa prospettiva.
 Gianni is fascinated by this prospect

Examples like those in (46)–(47) suggest *prima facie* that these ObjExp predicates are not unaccusative, contrary to B&R. B&R argue that appearances are misleading. They claim that apparent passives of this sort are adjectival, and not verbal. Although they do not explain why this should make a difference, examples like those in (48) provide a precedent for the claim that adjectival passives are formed from unaccusative verbs. Some of these examples are based on Bresnan (1978:8–9); others were provided by Beth Levin (personal communication).

- (48) a. elapsed time
 b. departed travelers
 c. newly arrived packages
 d. newly appeared book
 e. capsized boat
 f. a fallen leaf
 g. collapsed lung
 h. blistered paint
 i. a failed writer
 j. a deceased celebrity
 k. a stalled machine
 l. well-rested children
 m. a risen Christ
 n. a stuck window
 o. drifted snow
 p. a lapsed Catholic

On the other hand, the generalization is far from clear. Many seemingly unaccusative verbs in English do not form adjectival passives, as (49) shows.¹⁷

- (49) a. *an (already) occurred event
 b. *(recently) left travelers
 c. *(newly) come packages
 d. *(recently) grown interest
 e. *a (recently) surfaced problem
 f. *(recently) descended balloon
 g. *(recently) peeled skin

- h. *(often) stunk paint
- i. *a (recently) succeeded writer
- j. *a (recently) died celebrity
- k. *a (frequently) paused machine
- l. *(well-)slept children [?]
- m. *(brightly) glistened paint [?]
- n. *a (visibly) trembled orator [?]
- o. *the (regularly) twinkled star [?]
- p. *the (already) stumbled horse [?]

In chapter 4, I will suggest that (49) is the general case (as expected) and will provide a reason why examples like (48) exist; but for now it is sufficient to note the absence of any general possibility for adjectival passives of unaccusatives.

In other cases, adjectivizing a passive provides no refuge at all from the ban on forming passives from unaccusatives (at least in English). We can see this quite clearly. Consider first the contrasts in (50)–(52). These contrasts were attributed by Perlmutter and Postal (1984:102–3) to unaccusativity in the (b) sentences. The additional examples in (53), also Perlmutter and Postal's (p. 101), received the same analysis.

- (50) a. The closet was slid into by Ted.
b. *The closet was slid into by the soap.
- (51) a. The desk was sat on by the gorilla.
b. *The desk was sat on by the lamp.
- (52) a. The house was leaned against by the athlete.
b. *The house was leaned against by the ladder.
- (53) a. *The package was accumulated on by dust.
b. *The room was burst in by the bubble.
c. *The dome was collapsed under by the model.
d. *The bridge was existed under by trolls.
e. *The bed was fallen on by dust.

Pseudopassives do participate in the adjectival passive construction, albeit somewhat colloquially, as seen in (54).¹⁸

- (54) a. a much talked-about solution
b. an often referred-to article
c. a much looked-at painting
d. an often relied-on result

- e. a much struggled-against vice
- f. a man more sinned-against than sinning

On the other hand, adjectival pseudopassives from the verbs in (50)–(52) have only the sense of the (a) examples, and adjectival pseudopassives from the examples in (53) are entirely impossible.¹⁹

- (55) a. an often slid-into closet [by people, not soap]
- b. a much sat-on desk [≠ a cluttered desk]
- c. an often leaned-against house [by people, not ladders]
- (56) a. *a much accumulated-upon object
- b. *an often burst-in cloud chamber
- c. *a frequently collapsed-under dome
- d. *a recently existed-under bridge
- e. *a much fallen-on bed [except if fallen on by, say, actors]

B&R do present two arguments from Italian that are intended to support the adjectival analysis of ObjExp passives (including both *preoccupare*-class and *piacere*-class). These arguments, however, do not hold up to closer scrutiny.

2.3.1 Reduced Relatives

B&R's first argument concerns cliticization in Italian reduced relatives. They note that clitic pronouns may attach themselves to passive participles in reduced relatives—but only if the participle is verbal. Thus, attachment to the verbal participle in (57) is acceptable, but attachment to the adjectival participle in (58) is not.

- (57) a. [_{DP} la notizia che *gli* è stata comunicata]
 the news that to him was communicated
- b. [_{DP} la notizia comunicatagli]
- (58) a. [_{DP} la notizia che *gli* è ignota]
 the news that to him was unknown
- b. *[_{DP} la notizia ignotagli]

B&R then note that a passive *by*-phrase may not cliticize to a passive participle of an ObjExp verb, as shown in (59). From this, they conclude that the passive of an ObjExp verb is an adjectival passive.

- (59) a. (?)[_{DP} la sola persona che *ne* è affascinata]
 the only person who by it is fascinated
- b. *[_{DP} la sola persona affascinata*ne*]

In fact, the examples in (59) are irrelevant, since a passive *by*-phrase quite generally cannot cliticize to a passive participle functioning as a reduced relative, as shown in (60)–(63). This fact is independent of verb class.

- (60) a. [DP la sola persona che ne è stata uccisa]
 the only person that by it was killed
 b. *la sola persona uccisane
- (61) a. [DP la sola persona che ne è stata colpita]
 the only person that by it was struck
 b. *la sola persona colpitane
- (62) a. [DP la sola persona che ne è stata toccata]
 the only person that by it was touched
 b. *la sola persona toccatane
- (63) a. # [DP la sola persona che ne è stata arrestata]
 the only person that by it was arrested
 b. *la sola persona arrestatane

I know of no explanation for the prohibition seen in (60)–(63). However, the general impossibility of reduced relatives with *ne*-cliticization means that there is no argument from reduced relatives in favor of adjectival status for ObjExp participles.²⁰

2.3.2 Passive Auxiliary Choice

B&R's other argument that passives of Italian ObjExp verbs are all adjectival concerns the choice of auxiliary. They note that Italian allows passives with the verb *venire* 'come' replacing *essere* 'be'. They argue that although (64a) "is ambiguous between the adjectival interpretation (the door is in the state of being closed at five) and the verbal interpretation (somewhat marked with present tense, but still possible) 'Somebody closes the door at five'; [(64b)] is not ambiguous, only the verbal interpretation is allowed" (p. 310). From this, they conclude that *venire* is compatible only with verbal passives and hence forms a test for such passives.

- (64) a. La porta è chiusa alle cinque. [stative or eventive]
 the door is closed at five
 b. La porta viene chiusa alle cinque. [only eventive]
 the door comes closed at five

Now consider (65a). *Apprezzare* ‘appreciate’ is a SubjExp predicate: its Experiencer is in surface subject position. By contrast, *affascinare* ‘fascinate’ and *preoccupare* ‘worry’ in (65b–c) are ObjExp predicates: their Experiencers are in surface object positions.

- (65) a. Gianni viene apprezzato dai suoi concittadini.
 Gianni comes appreciated by his fellow citizens
 b. *Gianni viene affascinato da questa prospettiva.
 Gianni comes fascinated by this perspective
 c. *Gianni viene preoccupato da tutti.
 Gianni comes worried by everybody

(65a) shows that *venire* is compatible with the SubjExp predicate *apprezzare*. (65b–c) show that *venire* is *not* compatible with ObjExp predicates of the *preoccupare*-group, including *affascinare* and *preoccupare* itself. Based on these observations, B&R conclude that ObjExp predicates yield only adjectival passives.

Once again, however, I believe that the situation has been misanalyzed. B&R are correct in claiming that *venire* is compatible only with “eventive” predicates. Furthermore, adjectival passives are noneventive. In fact, adjectives are in general quite impossible with *venire*. Nonetheless, the impossibility of *venire* does not diagnose adjectival passives; it merely diagnoses noneventiveness, a property shared by adjectival passives, some verbal passives, and other forms as well.²¹ Thus, as Sandro Zucchi (personal communication) points out, the SubjExp predicate in (65a) is acceptable with *venire* only if appreciating Gianni is somehow an event or, better, an action. For example, (65a) suggests that *signs* of appreciation are being given, for example, a cheer, a slap on the back, a broad smile, or some display of this sort. Likewise, *venire*-passives with ObjExp predicates become progressively more acceptable as the predicate becomes more and more eventive. Indeed, most of my informants find (66a–c) (minimally modified from B&R’s examples) to be quite acceptable.

- (66) a. (?)Il pubblico venne affascinato dalla conclusione di quel
 the public came fascinated by the conclusion of that
 concerto.
 concert(o)
 b. Gianni venne spaventato da questa prospettiva alle cinque.
 Gianni came frightened by this perspective at five
 c. ?Gianni venne terrorizzato da questa prospettiva (alle cinque).
 Gianni came terrified by this perspective at five²²

(66a) uses the same verb as (65b), but invites a more eventive interpretation. (Present versus past tense also seems to make a difference, for some reason.) Additionally, Denis Delfitto (personal communication) notes that examples like those in (66) become perfect if an adverb such as *spesso* 'often' is added. Thus, neither of B&R's two arguments in support of adjectival status for these Italian passives can be maintained.

2.3.3 Interim Conclusions

For Italian, I have defused B&R's arguments that all ObjExp passives are adjectival. I am not aware of arguments from Italian that press the stronger claim—that these passives can be verbal. In English and Dutch, however, such arguments are available. These arguments are of interest for two related reasons.

First, in other respects English and Dutch ObjExp verbs do not look too different from their Italian counterparts. Italian, English, and Dutch ObjExp verbs display similar binding phenomena (considered in section 2.5.1). Italian and Dutch verbs show a similar distribution of auxiliaries (see section 2.5.2). Other facts considered by B&R and in this study are similar.

Second, if there are universal linking patterns, we do not expect semantically similar ObjExp verbs in Italian, English, and Dutch to differ systematically in unaccusativity. None of these languages produces verbal passives from unaccusatives. Thus, a demonstration of unequivocally verbal behavior in the ObjExp passives of any of these languages has significance for the analysis of ObjExp predicates in all of these languages. In the next sections, we will look first at ObjExp passives in English and then briefly at Dutch, arguing in each case that these passives can be verbal.

2.3.4 English: Stativity and Passives

In this section, I draw on work by Grimshaw (1987, 1991:114ff.), except that I reach conclusions quite different from hers. Grimshaw argues in favor of B&R's proposal concerning the passive of ObjExp predicates. In support of this proposal, she claims that English ObjExp passives are always stative. Since stativity is a property of the adjectival passive, she takes this observation as evidence that B&R's adjectival analysis of Experiencer predicate passives is correct. The argument is not conclusive, of course, since the data are merely consistent with the analysis; they do not force it. In fact, there are also empirical problems. I believe that the relevant data, when considered more carefully, lead to a conclusion opposite from Grimshaw's.

Grimshaw's premise is correct: adjectival passives are stative. This can be shown with the aid of adverbs like *much* and *very*. These adverbs, though restricted in distribution, can modify an adjectival passive, but not a verbal one.²³

- (67) a. This idea was much discussed in the '70s.
 b. The invasion was much condemned by the press.
 c. John is much maligned.
 d. The much awaited performance lived up to expectations.
 e. Our much battered car finally made it over the hill.
 f. *We much discussed this idea in the '70s.
 g. This idea was (*much) considered important in the '70s.
- (68) a. This edition is very abridged.
 b. The circle was very elongated.
 c. His reply was very balanced.
 d. The tree limb was very bloated.
 e. Sue was very hurried.
 f. France is a very industrialized country.
 g. *The censors very abridged this edition.
 h. This idea was (*very) considered important in the '70s.

Progressive aspect is generally incompatible with stative predicates. As predicted, progressive aspect is incompatible with a passive form modified by *very* or by *much* (as (69) shows). These data can be explained if adjectival status for a passive entails stativity.²⁴

- (69) a. The book was still being (*very) abridged when the order came through to publish it in its entirety.
 b. This idea was being (*much) talked about in the '70s.

Now let us turn to data involving ObjExp predicates. It is clear enough that some ObjExp predicates are often most comfortable as statives, even in the active. Thus, for example, the ObjExp verb *depress* resists the progressive form with the meaning of iterated action, and is odd in the punctual use of the simple past tense. This oddness is, unsurprisingly, recapitulated in the passive. The question marks in (71) cannot be taken as evidence that the passive of *depress* is adjectival, given the comparable question marks for (70).

- (70) a. ??Odd noises were continually depressing Sue.
 b. ??Bill was sitting around happy as a lark, when an unexpected groan from the next room suddenly depressed him.

- (71) a. ??Sue was continually being depressed by odd noises.
 b. ??Bill was sitting around happy as a lark, when suddenly he was depressed by an unexpected groan from the next room.

Now consider ObjExp predicates that are not exclusively stative. *Scare* is quite acceptable in the progressive or punctual past. Crucially, there is no problem with the passive forms.

- (72) a. Odd noises were continually scaring Sue.
 b. Bill was sitting around calm as could be, when an unexpected groan from the next room suddenly scared him.
- (73) a. Sue was continually being scared by odd noises.
 b. Bill was sitting around calm as could be, when he was suddenly scared by an unexpected groan from the next room.

Like *scare* are *terrify*, *alarm*, *startle*, *dismay*, *shock*, and *surprise*, among others. Like *depress* are *worry* and *bore*. It is quite likely that the relevant distinction has to do with the nature of the onset of the emotion referenced by the ObjExp verb. I conjecture that emotions that typically come on suddenly and consciously (e.g., frights and surprises) allow the iterative progressive, whereas emotions that typically grow imperceptibly (e.g., boredom and depression) do not, but I have not investigated these matters carefully. The crucial point is that the actives and the passives do not contrast in any relevant fashion.

Grimshaw (1991:114) presents a rather different emblematic example of the progressive in ObjExp verbs. I accept the star on (74b).

- (74) a. The situation was depressing Mary.
 b. *Mary was being depressed by the situation.

The present discussion invites new questions about these examples. Given (70)–(71), what is surprising is not the *unacceptability* of the passive (74b), but the *acceptability* of the active (74a). The acceptability of (74a) recalls the surprising acceptability of the progressive with certain SubjExp statives.

- (75) a. Karen is finally understanding this proof.
 b. Donald is finding your accusations ludicrous.
 c. I think Bill is really liking this performance.
 d. Sue is truly hating the sea-urchin sushi.
 e. Harry is clearly fearing an outbreak of the flu.²⁵

Examples (75a–c) are modeled on examples presented by Baker (1989: 489, (191)–(193)), who notes that predicates like these function elsewhere as statives. He remarks further that the progressive-aspect predicates found here

appear to assert the existence of a judgment of some sort concerning an individual entity or a set of entities ... imply[ing] that the judgment is an intermediate one based on only part of the available evidence. Sentence [(75a)] would typically be used if Karen was only partly done going through the proof, [(75c)] would be appropriate at an intermediate point in the performance, and [(75b)] would be used if Donald had heard only some of the accusations. (pp. 489–90)

Clearly, this is a use of the progressive distinct from (though related to) the “iterative” use seen in (70)–(73). (74a) shows this use, not the “iterative” one. If someone says that “the situation is depressing Mary,” we naturally infer that this person is making a judgment (in this context; see below) about some situation that has not played itself out at the time of the utterance.

Now none of the verbs in (75) is unaccusative in B&R’s analysis. In particular, verbs of this sort (B&R’s *temere*-class) are said to yield verbal passives, and fail B&R’s other tests for unaccusativity. Nonetheless, the passive progressive of these verbs in English yields judgments ranging from odd to unacceptable.

- (76) a. ??This proof is finally being understood by Karen.
 b. *Your accusations are being found ludicrous by Donald.
 c. *I think this performance is really being liked by Bill.
 d. *The sea-urchin sushi is truly being hated by Sue.
 e. *An outbreak of the flu is clearly being feared by Harry.

Nonprogressive passives are stilted, but unproblematic.

- (77) a. This proof is understood by Karen.
 b. Your accusations were found ludicrous by Donald.
 c. I think this performance was really liked by Bill.
 d. The sea-urchin sushi was truly hated by Sue.
 e. An outbreak of the flu is feared by Harry.

I suggest that the data in (76) and (74a) are of a piece: progressive forms of stative predicates require a particular interpretation that is for some reason incompatible with the passive. Crucially, unaccusativity is not diagnosed.²⁶ If my interpretation of the data in (70)–(73) is right, then just the opposite is true: we have an argument *against* the unaccusativity of

ObjExp predicates, since they form fully verbal passives that, under the right conditions, also participate in the progressive.

Other contrasts make the same point concerning Grimshaw's claim. Assuming that the availability of *much*, as in (67), can be used as a test for adjectival passives, let us apply it to ObjExp passives. It turns out that the passive of an ObjExp verb like *scare* or *frighten* is incompatible with the progressive aspect only when it is modified by *much* (or similar adverbs). When not modified by *much*, these verbs are acceptable in the progressive. This contrast is expected if only the disambiguating power of an adverb like *much* can force ObjExp passives to be adjectival. Thus, (78a) *does* show an adjectival passive derived from *frighten*, but this form differs in a predictable manner from the verbal passive seen in (78b).

- (78) a. Bill was (much) frightened by my remark.
 b. In those days, Bill was often being (*much) frightened by one thing or another when I would come home from work.
- (79) a. Sue was (extremely) annoyed by Bill's behavior.
 b. In those days, Sue was often being (*extremely) annoyed by Bill's behavior.²⁷

Finally, consider cases where the true adjectival passive takes an idiosyncratic preposition. Here, the contrast between the uncontroversial adjective and the passive with the *by*-phrase is sharp and robust.^{28,29}

- (80) a. Sue was continually being scared by sudden noises.
 b. *Sue was continually being scared of sudden noises.
- (81) a. Bill was often being enraged by totally innocent remarks.
 b. *Bill was often being enraged at totally innocent remarks.
- (82) a. Sue was continually being annoyed by mysterious sounds from the cellar.
 b. *Sue was continually being annoyed with mysterious sounds from the cellar.
- (83) a. John was always being deeply impressed by things that left the rest of us cold.
 b. *John was always being deeply impressed with things that left the rest of us cold.

If the *-ed* marking is a sign of some sort of passive, then the (b) examples here show true adjectival passives. These adjectival passives do not behave at all like the verbal passives seen in the (a) examples.

Thus, the types of facts raised by Grimshaw not only undermine B&R's arguments *for* the adjectival analysis of ObjExp passives, but also actively argue *against* such an analysis.

2.3.5 Raising in Dutch

In Dutch, the phenomenon of V-raising (henceforth *VR*) shows quite straightforwardly that Dutch ObjExp passives may be verbal. My discussion of VR is taken from Den Besten 1989:196ff., as are most of the examples.³⁰ I am grateful to René Mulder (personal communication) for bringing this argument to my attention, and for supplying the crucial punchline concerning Experiencer verbs.

VR is a phenomenon that reverses the expected order of a matrix verb and an embedded verb. (I will limit myself to embedded *dat*-clauses, in order to avoid the complications of matrix verb-second word order.) VR is obligatory in some contexts, but is optional in the perfect tense. In the perfect, the participle may invert with the auxiliaries *hebben* 'have' and *zijn* 'be'.

- (84) a. dat hij gelachen heeft [no VR]
 that he laughed has
 'that he has laughed'
 b. dat hij heeft gelachen

- (85) a. dat zij gearriveerd is
 that she arrived is
 'that she has arrived'
 b. dat zij is gearriveerd

(86a–b) show that VR is also optional in passives with auxiliary *worden* 'become'. (87a–b) show that VR (unlike the choice of *venire* or *essere* in Italian) is insensitive to stativity.

- (86) a. dat hem de P.C. Hooft-prijs toegekend werd
 that to him the P.C. Hooft prize awarded was
 'that the P.C. Hooft prize was awarded to him'
 b. dat hem de P.C. Hooft-prijs werd toegekend
- (87) a. dat Jan zijn vader nooit echt gekend heeft
 that Jan his father never really known has
 'that Jan has never really known his father'
 b. dat Jan zijn vader nooit echt heeft gekend

Finally, nothing analogous to VR (again, regardless of stativity) applies to adjectives. (88) and (89) were supplied by René Mulder. The adjective *verliefd* in (90) (from Den Besten 1989:199, (82)) is participial in form, but nonetheless does not allow VR.

- (88) a. dat Jan de hele dag druk bezig is
 that Jan the whole day very busy is
 ‘that Jan is very busy the whole day’
 b. *dat Jan de hele dag druk is bezig
- (89) a. dat Jan de hele dag boos was
 that Jan the whole day angry was
 ‘that Jan was angry the whole day’
 b. *dat Jan de hele dag was boos
- (90) a. dat hij op haar verliefd is
 that he with her in love is
 ‘that he is in love with her’
 b. *dat hij op haar is verliefd

Den Besten argues that there is a systematic difference between verbal and adjectival passives in Dutch, using VR as one of a number of crucial tests. Verbal and adjectival passives differ in Dutch (as in English) in a variety of ways. Only verbal passives allow a nonstative interpretation. Only verbal passives allow a *by*-phrase (using the preposition *door*).³¹ Finally, only verbal passives use the auxiliary *worden* ‘become’. This auxiliary, however, is normally deleted in the perfect. Thus, Dutch speakers generally accept only the version of (91) without the participle *geworden*.

- (91) Zij is gisteren gearresteerd (%geworden).
 she is yesterday arrested become
 ‘She has been (or ‘was’) arrested yesterday.’

As a consequence, (92a) is ambiguous between a verbal, nonstative, perfect reading (when the hearer assumes a deleted *geworden*) and an adjectival, stative, present tense reading (when the hearer does not assume deleted *geworden*). The ambiguity is resolved in favor of a verbal reading by the addition of a *door*-phrase, as in (92b).

- (92) a. Dit zwembad is gesloten. [ambiguous]
 ‘This swimming pool is closed.’ [no deleted *worden*]
 ‘This swimming pool has been/was closed.’ [assuming deleted
 worden]

- b. Dit zwembad is door de gemeente
 this swimming pool is by the authorities
 gesloten. [unambiguous]
 closed

‘This swimming pool has been/was closed by the authorities.’

(93) (supplied by René Mulder) without *jarenlang* ‘for years’ is also ambiguous between a verbal, nonstative, past tense reading (when the hearer assumes a deleted *geworden*) and an adjectival, stative, present tense reading (when the hearer does not assume deleted *geworden*). The inclusion of *jarenlang* eliminates the stative possibility, leaving only the verbal passive.

- (93) dat Jan (jarenlang) getrouwd is
 that Jan for-years married is
 ‘that Jan has been married (for years)’
 ‘that Jan is married (*for years)’

Crucially, if VR reverses the order of *getrouwd* and *is* (as well as the deleted *geworden*, presumably), only the verbal, nonstative, past tense reading is available.

- (94) dat Jan (jarenlang) is getrouwd

Likewise, Den Besten (1989:199) considers (95)

- (95) dat hij er niet van overtuigd is
 that he it not of { convinced is }
 { been-convinced has }

and notes,

This example is ambiguous. Either it is a *zijn*-passive or it is a *worden*-passive, and then it can be related to [(96)] . . . :

- (96) [%]dat hij er niet van overtuigd is geworden³²
 that he it not of convinced is become
 [i.e., convinced has been]

Under its stative reading, [(96)] does not allow an agent phrase, nor does it allow an application of Verb Raising, most probably because the past participle of a *zijn*-passive is an adjective and not a verb. Under its dynamic reading, however, [(96)] allows both an agent phrase and an application of Verb Raising. Compare [(97)]:

- (97) dat hij er niet door zijn broer van is overtuigd
 that he it not by his brother of is convinced

I have reproduced here a well-known argument from Dutch grammar in favor of the verbal status of the past participle in *worden*-passives as well as in favor of the adjectival status of the “past participle” in *zijn*-passives.

Now let us turn to the passives of ObjExp verbs. Taking my lead from B&R, I will stick for now to ObjExp verbs that take the auxiliary *hebben* ‘have’, returning to ObjExp verbs with *zijn* in section 2.5.2. As in English, verbal passives from certain ObjExp verbs are slightly reduced in acceptability for some speakers. This is reported to be the case for *irriteeren* ‘irritate’ and *ergern* ‘annoy’ in the examples below, though not for *boeien* ‘fascinate’. Nonetheless, even the worse passives are reported to be “pretty much acceptable.” Furthermore, both the (a) and (b) examples contain a *door*-phrase, which may already force a verbal reading on the passive. Additionally, the (b) examples show VR, which quite clearly forces a verbal reading, as we have seen.

- (98) a. (?)dat ik door dat boek nogal geirriteerd werd
 that I by that book rather irritated became
 b. (?)dat ik door dat boek nogal werd geirriteerd
 ‘that I got rather irritated by that book’
- (99) a. (?)dat ik door die opmerking nogal geergerd werd
 that I by that remark rather annoyed became
 b. (?)dat ik door die opmerking nogal werd geergerd
 ‘that I got rather annoyed by that remark’
- (100) a. dat ik door het college geboeid werd
 that I by the classes fascinated became
 b. dat ik door het college werd geboeid
 ‘that I got fascinated by the classes’

None of these examples yield the sharp judgments of unacceptability found when adjectives are subjected to VR in (88)–(90). Nor do they yield the types of judgments reported for passives of unaccusatives discussed by Perlmutter (1978) and Perlmutter and Zaenen (1984:190–91).

- (101) a. *Er wordt (door veel patienten) in dat ziekenhuis
 there were by many patients in that hospital
 gestorven.
 died
 b. *Er werd door veel toeristen in de hotelbrand gebleven.
 there were by many tourists in the hotel fire remained

I conclude that the passives of Dutch ObjExp verbs do not have to be adjectival. Since Dutch does not form verbal passives from unaccusatives, it follows that Dutch ObjExp verbs are not unaccusative.

2.4 Athematic Subjects: Other Discussion

We have seen that the analysis of ObjExp passives as adjectival is dubious in Italian and incorrect for English and Dutch. If ObjExp verbs can form verbal passives in defiance of standard properties of verbs with athematic subjects, we must ask what, if any, other evidence might bear on the matter.

2.4.1 Free Inversion

One of the best-known and best-studied properties of unaccusative verbs in Italian cannot serve as a test for the unaccusativity of ObjExp verbs. We expect the underlying object of an unaccusative verb in Italian to be able to remain in object position at S-Structure. Free inversion with ObjExp verbs is impossible, however, as B&R note.

- (102) *Preoccupano le tue idee Gianni.
 worry your ideas Gianni

B&R relate the impossibility of (102) to a general condition (p. 340) requiring that noneventive clauses “involve a nonvacuous predication at S-structure (with a referential subject)” and note that the same word order, *verb Theme Experiencer*, is impossible even with SubjExp predicates.

- (103) *Teme le tue idee Gianni.
 worry your ideas Gianni

B&R are correct in claiming that (102) and (103) can fall under the same generalization if the unaccusative analysis for (102) is correct. Indeed, observations that point to a generalization quite close to B&R’s have been made independently by Diesing (1989) and by Kratzer (1989) on the basis of rather different German and English data. It is important to note, however, that the ungrammaticality of (102), coupled with the unexpected possibility of verbal passivization just observed, remove from discussion two of the better-established diagnostics for unaccusativity. The facts in (102)–(103) do not intrinsically cast doubt on the analysis as the passive facts do, but they do make unavailable data that might have significantly supported the analysis.

2.4.2 Arbitrary pro

B&R present one other argument directed at the athematic status of the subject of ObjExp verbs. This argument turns out to be spurious. The details of the refutation are included here because of their intrinsic interest, although I view the athematic proposal as sufficiently refuted at this point. This subsection may thus be regarded as an appendix, which may be skipped without cost to the general discussion.

Developing unpublished work by Alfredo Hurtado (and following Jaeggli (1986a)), B&R (p. 299) use as a diagnostic of unaccusativity a “kind of arbitrary interpretation [of third person plural pro] in which the plural specification does not imply semantic plurality: there is simply no commitment as to the real number of the argument in question.” As a consequence, the following dialogue is felicitous, despite the continuation of plural pro with singular *Gianni* (B&R’s examples (20)–(21)).

- (104) pro ti stanno chiamando. Deve essere Gianni.
 they you are-3pl calling (it) must be Gianni
 ‘Somebody is calling you. It must be Gianni.’

B&R make the claim in (105).

- (105) “[A]rb interpretation can be assigned to deep subject *pro*’s only; it is incompatible with unaccusative structure.” (p. 300)

This claim is illustrated by the contrast between the unergative and transitive examples in (106) and the unaccusative examples in (107) (B&R’s examples (22)–(23)). All the relevant verbs are third person plural.

- (106) a. pro hanno telefonato a casa mia.
 ‘Somebody telephoned my place.’
 b. pro mi hanno mandato un telegramma.
 ‘Somebody sent me a telegram.’
 c. pro hanno arrestato Gianni.
 ‘Somebody arrested Gianni.’
 d. pro hanno visto Gianni in giardino.
 ‘Somebody saw Gianni in the garden.’
- (107) a. *pro sono arrivati a casa mia.
 ‘Somebody arrived at my place.’
 b. *pro mi sono sembrati matti.
 ‘Somebody seemed to me crazy.’

- c. *pro sono stati arrestati dalla polizia.
'Somebody has been arrested by the police.'
- d. *pro sono stati visti in giardino.
'Somebody has been seen in the garden.'

B&R next note the impossibility of the relevant interpretation for third person plural subjects of ObjExp verbs and conclude that these subjects are underlying objects, in keeping with the unaccusative analysis. ((108a–c) are B&R's (24b), (26b), and (25c), respectively.)

- (108) a. *Evidentemente, in questo paese per anni pro
evidently in this country for years
hanno preoccupato il governo.
worried the government
'Evidently, in this country somebody worried the government
for years.'
- b. *pro hanno colpito il giornalista per la gentilezza.
'Somebody struck the journalist with his kindness.'
- c. ??Qui pro hanno sempre entusiasmato/commosso gli americani.
'Here, someone always excited/moved the American people.'

But are the diagnostic and the generalization correct? First, a sharp distinction must be made between what B&R call the "existential" use of the third person plural and the "generic" use (alluded to in their footnote 6). Once this is done, a rather different picture of the facts seems to emerge.³³

Let us consider the "existential" usage. The term *corporate* is perhaps more illuminating than "existential" for the first use of the third person plural. The corporate usage can be seen in English examples like (109a–g) and their Italian equivalents (110a–g).

- (109) a. They robbed Mrs. Johnson.
- b. ?They rob someone different every night.
- c. They're making us fill out our income tax forms early this year.
- d. They came for Charley.
- e. They accepted our check at the supermarket.
- f. They punched me at the supermarket.
- g. They sell cigarettes on Melrose. (Jaeggli 1986a)
- (110) a. Hanno rapinato la Signora Rossi.
- b. Rapinano una persona diversa tutte le sere.
- c. Quest'anno ci fanno fare la dichiarazione dei redditi prima.

- d. Sono venuti a cercare Gianni.
- e. Al negozio hanno accettato il nostro assegno.
- f. Al mercato mi hanno malmenato.
- g. Al mercato vendono sigarette.

The English plural pronoun in (109) seems to pick out some socially designated group of people, prototypically governments, bosses, criminals, or shopkeepers. As B&R note, the referent of the pronoun need not be plural: the acceptor in (109e) could be a single person. This single person must, however, be acting as the representative of some larger group. The parallel Italian examples (including (106)) apparently have the same flavor, making the “somebody” of B&R’s glosses somewhat misleading.

Now consider again the contrast in (106)–(107) with the corporate reading pointed to by the glosses. The data really admit two descriptions: the verbs in (106) differ from those in (107) in unaccusativity, but they also differ in agentivity.³⁴ In many cases, the two notions coincide: unaccusative verbs are usually nonagentive, and many unergative or transitive verbs are agentive. Nonetheless, as noted in section 2.1, there are cases where they do not coincide. Thus, (111) must be considered as a possible alternative to (105).

- (111) “Corporate” interpretation can be assigned to agentive third person pronouns (pro in Italian) only; it is incompatible with nonagentive θ -roles.

In fact, the formulation in (111) appears to be correct, and (105) false. First consider nonagentive underlying subjects. The claim in (105) predicts that these should allow the corporate reading, all things being equal. By contrast, (111) predicts that they should not. In fact, they do not. Contrast (109f) with (112). (109f) is felicitous if I’ve been banned from the supermarket, and the supermarket’s hired guard punched me. (112) is not felicitous if I punched him.³⁵

- (112) a. *They received a punch in the nose at the supermarket.
 b. *Al mercato hanno preso un pugno sul naso.
- (113) a. *They received a phone call yesterday.
 b. *Ieri hanno ricevuto una telefonata.

Note the auxiliary *avere* in (112)–(113). The main verbs here are not unaccusative.

Next consider agentive underlying objects. As noted above, certain verbs of motion are unaccusatives even when used agentively. Theory (105) predicts straightforwardly that corporate third person plural pronouns should be impossible with such verbs. All things being equal, theory (111) predicts that they should be possible. Once again, theory (111) seems to be correct. Corporate third person plural pronouns seem to be completely possible in these environments, under the circumstances indicated in square brackets.³⁶

- (114) a. Sono venuti a riparare il lavandino.
 ‘Somebody came to fix the sink.’
 [One repairman came from the shop.]
 b. Sono venuti a cercare Gianni.
 ‘Somebody came looking for Gianni.’
 [One policeman came to the door.]
 c. Sono andati a cercarlo a casa di sua madre.
 ‘Somebody went looking for him at his mother’s house.’
 [One person went.]³⁷

If corporate *they* diagnoses *agency*, not unaccusativity, in both English and Italian, its impossibility as the subject of ObjExp predicates tells us little about the underlying grammatical relations of these verbs. It is merely sufficient to note that the subject of ObjExp predicates is typically not an Agent, and (111) predicts that corporate *pro* will be impossible in examples like (108a–c).³⁸

The other use of third person plural relevant to B&R is a “generic” use. This appears to have a different distribution than the corporate use, but similarly fails to diagnose unaccusatives. Relevant examples are as follows:

- (115) a. In Japan, they drive on the left.
 b. In America, they’re required to fill out income tax forms every year.
 c. In Canada, they wilt if the temperature goes above 60°F.
- (116) a. In Giappone, viaggiano sulla sinistra.
 b. In America, sono ?costretti/obbligati a fare la dichiarazione dei redditi tutti gli anni.³⁹
 c. In Canada, ?avvizziscono/?soffocano quando la temperatura supera i 15 gradi.

Plainly, deep objects can be third person plural generics, as the (b) and (c) examples of (115)–(116) illustrate. A similar observation is made by Cinque (1988:545).

In fact, it is likely that the correct characterization of third person plural generics is once again thematic. B&R note (somewhat indirectly) that this interpretation is available for the Experiencer subject of SubjExp predicates.⁴⁰

(117) Evidentemente, in questo paese per anni pro hanno temuto il terremoto.

‘Evidently, in this country people feared the earthquake for years.’

By contrast, periphrastic counterparts to sentences with ObjExp verbs resist generic third person plural pronouns.

- (118) a. *In France, they worry you.
 b. *In France, they make you worried.
 c. *In Francia, ti preoccupano.
 d. *In Francia, ti rendono preoccupato.

If we now examine the good examples of third person plural generics, they include Agents (as in (115a)), Patients (as in (115b–c)), and Experiencers, but exclude subjects of *make* and *please* (which B&R would probably call Causer and Theme, but which we will shortly identify as both being Causer arguments) and certain direct object third person plural generics (**In totalitarian countries, they are known by the police*).

Apparently, third person plural generic uses of *pro* involve arguments that are affected by the event in which they occur, either by being changed or by being conscious participants in the event. (This generalization is similar to the generalizations concerning direct object *pro* discussed by Rizzi (1986a).) In any case, it is clear that the generalization does not single out grammatical functions like subject and object, but is semantic in nature. Third person plural pronouns thus provide no argument for the unaccusativity of ObjExp predicates.

2.5 The Relation between Subject and Object Position

2.5.1 “Backward Binding” and ObjExp Predicates

B&R’s structure for the ObjExp verbs does not merely involve an athematic subject position; it involves a reversal of c-command relations among the arguments between D-Structure and S-Structure, as seen in (44), repeated here.

(119) Theme_i [_{VP} [_V V e_i] Exper]

B&R argue that the reversal of c-command relations provides an explanation for the well-known binding peculiarities of ObjExp verbs: anaphors contained within the subject of such verbs may be bound by the object (Akatsuka 1976, Giorgi 1984, Pesetsky 1987a), in violation of the usual c-command condition on bound anaphora.

- (120) a. Questi pettegolezzi su di sè_i preoccupano Gianni_i più di ogni altra cosa. (B&R (57a))
 b. These rumors about himself_i worry Gianni_i more than anything else.
- (121) a. I propri_i sostenitori preoccupano Gianni_i.
 'His own supporters worry Gianni.'
 b. Each other's supporters worried Freud and Jung.
- (122) Each other_i's remarks annoyed John and Mary.

If the surface subject in these examples is c-commanded by the Experiencer object at D-Structure, then D-Structure application of Principle A of the binding theory, or any one of a variety of plausible mechanisms (e.g., reconstruction or Barss's (1986) Chain Binding) can explain the unexpected grammaticality of these examples.

I will ultimately propose an account of these phenomena very close to B&R's. Indeed, I will adopt their idea that Principle A of the binding theory may be satisfied earlier than S-Structure or LF, and I will adopt an analysis in which movement applies to the non-Experiencer in a way not too different from B&R's proposal (despite the fact that the subject will be argued to be thematic, rather than athematic).

Nonetheless, here too there are complications and subtleties that do not follow naturally from B&R's analysis. The phenomenon seen in (120)–(122) for English extends beyond the domain of ObjExp predicates to constructions that clearly do not involve an athematic subject. In particular, periphrastic causatives show what looks like the same phenomenon.⁴¹

- (123) a. ?Questi pettegolezzi su di sè_i hanno reso Gianni_i felice.
 'These rumors about himself have made Gianni happy.'
 b. Questi pettegolezzi su di sè_i hanno persuaso Gianni_i a partire.
 'These rumors about himself persuaded Gianni to leave.'
- (124) a. Each other's remarks made John and Mary angry.
 b. Pictures of each other make us happy.

- c. These stories about herself made Mary nervous.
 - d. Pictures of himself give John the creeps.
- (125) a. Each other's criticisms forced John and Mary to confront their problems.
- b. Pictures of each other caused John and Mary to start crying.
 - c. Those rumors about himself made John behave more carefully.
 - d. Pictures of herself used to make Sue blush.
- (126) a. ?Each other's stupid remarks eventually killed John and Mary.
- b. ?Each other's criticisms harmed John and Mary.
 - c. ?Those pictures of himself ultimately destroyed Bill.
 - d. ?Rumors about herself always plunge Mary into a deep depression.

The examples in (126) contradict usual descriptions of binding possibilities, but should be compared with similar agentive examples, which are clearly worse.

- (127) a. *Each other's stupid friends eventually killed John and Mary.
- b. *Each other's parents harmed John and Mary.
 - c. *Each other's teachers insulted John and Mary.
 - d. *Each other's swimming coaches plunged John and Mary into the pool.

Now let us look carefully at (123a) and (124a–d). If we attempt to explain the binding possibilities in these examples in the way B&R explain the parallel ObjExp verb examples, we soon run into trouble. Suppose the structure of these examples is unaccusative in a way that will explain the binding problems in a simple manner.

- (128) [Each other_i's remarks]_j [made e_j [John and Mary]_i angry].

The question of Case assignment immediately arises. B&R face the problem in structures like (119) of explaining why the first object (the "Theme") must move to subject position, whereas the second object stays in place. If the verb assigns only structural accusative Case, then we might expect either (1) both objects to remain in situ (if two accusative Cases can be assigned), or (2) one or the other of the two objects, freely chosen, to move (if only one accusative Case can be assigned). In either instance, we also have a violation of Burzio's Generalization to contend with: if ObjExp verbs are unaccusative, we expect problems in accusative Case assignment.

B&R opt for the only solution available, given their *LGB*-based view of Case theory (which I will also accept in this book). They restrict Burzio's Generalization to structural Case. By excluding inherent Case from the jurisdiction of Burzio's Generalization, B&R leave open the possibility that an unaccusative verb may assign *inherent* accusative Case. Inherent Case is defined as Case linked to θ -role assignment (Chomsky 1980, 1986b, Pesetsky 1982). It is for this reason that ObjExp verbs are allowed to specify that they assign inherent accusative Case to their Experiencer argument.

But now turn to the (flat) structure in (128). By parity of reasoning, we must assume that *make* is assigning inherent accusative Case to *John and Mary*; structural accusative Case would violate Burzio's Generalization, and would in any case lead to the wrong predictions about movement sketched above. But *make* does not necessarily assign any θ -role to *John and Mary*, even when backward binding is found. In (129), for example, *John and Mary* receives its θ -role from *angry* in the lowest clause.

(129) [Each other_i's remarks]_j [made e_j [John and Mary]_i seem t_i to be angry].

Suppose that *John and Mary angry* in (128) forms a small clause in the sort of syntactic structure B&R would probably assume, as in (130).

(130) [Each other_i's remarks]_j [made e_j [John and Mary_i angry]].

In this structure, *John and Mary* does not c-command *each other's remarks* even at D-Structure. Thus, if the small clause analysis is correct and a c-command condition governs anaphoric binding, there is an additional reason for rejecting a B&R-like solution to the problem of (123a) and (124).

Now consider the Japanese constructions that correspond to English sentences with ObjExp predicates. Akatsuka (1969, 1976) notes facts very similar to those just discussed.⁴² Examples (131a–c) are hers; (131d) is due to Hoji and Saito (1983).

(131) a. [Zibun_i-ga gan kamo sirenai koto]-ga Hiroshi_i-o
 refl-NOM cancer may have fact-NOM Hiroshi-ACC
 nayam-ase-ta.⁴³
 worry-CAUSE-past
 'The fact that himself_i may have cancer worried Hiroshi_i.'

- b. [*Zibun_i*-ni-mo yotugi-ga umareta koto]-ga *Hideyosi_i*-o
 refl-DAT-prt heir-NOM was born fact-NOM Hideyoshi-ACC
 itaku yorokob-ase-ta.
 very pleased-CAUSE-past
 'The fact that an heir was born to himself_i pleased Hideyoshi_i very much.'
- c. [*Koibito*-ga *zibun_i*-o uragitta koto]-ga *Hiroshi_i*-o
 girl friend-NOM refl-ACC betrayed fact-NOM Hiroshi-ACC
 hungai-s-sase-ta.
 resentment-do-CAUSE-past
 'The fact that his girl friend had betrayed himself_i infuriated Hiroshi_i.'
- d. [*John*-ga *zibun_i*-no kuruma-o kowasita koto]-ga
 John-NOM refl-GEN car-ACC broke fact-NOM
Mary_i-o odorok-ase-ta.
 Mary-ACC surprised-make-past
 'The fact that John broke herself_i's car surprised Mary_i.'

The antecedent of *zibun*, like the antecedent of an English reflexive, must normally occupy a c-commanding position.

- (132) *[*John_i*-no hahaoya-ga] *zibun_i*-o aisiteiru.
 John-GEN mother-NOM himself-ACC loves

The first thing to note is that the matrix verbs of (131), though given simple translations like 'worry', 'please', and 'infuriate', are actually morphologically complex causative verbs (as the glosses make clear). If one accepts the analysis of Japanese causatives first proposed by Kuroda (1965), under which they are syntactically complex forms that have undergone VR, the examples in (131) are already more like the English periphrastic examples in (124) than the single-verb examples in (120)–(122). This point will become more important in section 3.4. Although apparently most Japanese translations of English or Italian ObjExp verbs are causatives of some sort, there are idiomatic uses of simplex verbs that more closely resemble the English ObjExp class. The following examples are attributed by Akatsuka (1976:97) to S.-Y. Kuroda (personal communication).⁴⁴

- (133) a. [*Zibun_i*-ga *Marii*-ni karakawareta koto]-ga *Zyon_i*-o
 refl-NOM Mary-by made fun of fact-NOM John-ACC
 zetuboo-e oiyatta.
 despair-to drove
 'That himself_i was made fun of by Mary drove John_i to despair.'

- b. *Marii-ga zibun_i-o hinan sita koto]-ga Zyon_i-o*
 Mary-NOM refl-ACC accused fact-NOM John-ACC
utinomesita.
 bowled over
 'That Mary accused himself_i bowled John_i over.'

Of interest here is the fact that, although the Japanese data resemble the better-known English and Italian data, the Japanese verbs fail an important and heretofore reliable test for unaccusativity developed by Miyagawa (1989). Miyagawa looks at "floated" phrases consisting of a numeric quantifier and a classifier (henceforth *NQs*). He shows that they must be syntactically quite close to the noun phrase with which they are associated. (This association is indicated by italic type in the examples below.) Miyagawa argues that mutual c-command is the relevant relation. On the assumption that Japanese has the sort of articulated phrase structure found in standard analyses of English, the examples in (135) show a failure of mutual c-command between the italicized DP and the italicized NQ.

- (134) *Taroo-ga hon-o 3-satu kat-ta.*
 Taro-NOM book-ACC 3-cl buy-past
 'Taro bought three books.'
- (135) a. **[Tomodati-no kuruma]-ga 2-ri nusum-are-ta.*
 friend-GEN car-NOM 2-cl steal-PASS-past
 b. **Gakusei-ga hon-o 2-ri kat-ta.*
 student-NOM book-ACC 2-cl buy-past
 c. *?*Kodomo-ga [kono kagi]-de 2-ri doa-o ake-ta.*
 child-NOM this key-by 2-cl door-ACC open-past
 d. *?*Gakusei-ga [zibun-no kane]-de 2-ri denwa-si-ta.*
 student-NOM self-GEN money-with 2-cl telephone-past

Given (135), the following data are at first sight quite surprising:

- (136) a. *Doa-ga [kono kagi]-de 2-tu ai-ta.*
 door-NOM this key-by 2-cl open-past
 'Two doors were opened by this key.'
- b. *Kinoo, tekihei-ga [ano hasi]-o 2-3-nin*
 yesterday enemy soldier-NOM that bridge-ACC 2-3-cl
 watat-ta.
 cross-past
 'Yesterday two or three enemy soldiers crossed that bridge.'

- c. *Gakusei-ga ofisu-ni 2-ri ki-ta.*
 student-NOM office-to 2-cl come-past
 'Two students came to the office.'
- (137) a. *Kuruma-ga doroboo-ni 3-dai nusum-are-ta.*
 car-NOM thief-by 3-cl steal-PASS-past
 'Three cars were stolen by the thief.'
- b. Kinoo, *gakusei-ga [ano otoko]-ni 2-ri koros-are-ta.*
 yesterday student-NOM that man-by 2-cl kill-PASS-past
 'Yesterday two students were killed by that man.'

Miyagawa notes, however, that the class of verbs seen in examples like (136a–c) looks very much like the class of unaccusatives, including various verbs of motion and the (superficially) intransitive member of pairs meaning 'rise'-'raise', 'collapse'-'destroy', 'be cut'-'cut', 'close' (intrans.)-'close' (trans.). On this assumption, although the floated NQ in (136) does not c-command the surface subject, it does c-command the D-Structure position of the surface subject, represented at S-Structure by a trace.

- (138) *Doa_i-ga [kono kagi]-de e_i 2-tu aita.*
- └──────────┘

Obviously, exactly the same explanation can be provided for the passive examples in (137), if passive in Japanese involves movement from direct object position.

- (139) *Kuruma_i-ga doroboo-ni e_i 3-dai nusum-are-ta.*
- └──────────┘

We can now ask, internal to Japanese, whether the Experiencer verbs, both causatives and noncausatives, are unaccusative by this test. First let us look at the causatives. Unexpectedly, even NQs adjacent to the subject are rather unacceptable, but there is a clear contrast between these cases and nonadjacent cases that argues against an unaccusative analysis.⁴⁵

- (140) a. [3-tu-no omosiroi koto]-ga Hanako-o
 3-cl-GEN interesting thing-NOM Hanako-ACC
 yorokob-ase-ta.
 happy-CAUSE-past
 'Three interesting things made Hanako happy.'
- b. ??[*Omosiroi koto*]-ga 3-tu Hanako-o yorokob-ase-ta.
- c. *[*Omosiroi koto*]-ga Hanako-o 3-tu yorokob-ase-ta.

- (141) a. [3-tu-no iyana koto]-ga Hanako-o
 3-cl-GEN terrible thing-NOM Hanako-ACC
 kanasim-ase-ta.
 sad-CAUSE-past
 ‘Three terrible things saddened Hanako.’
 b. ??[*Iyana koto*]-ga 3-tu Hanako-o kanasim-ase-ta.
 c. *[*Iyana koto*]-ga Hanako-o 3-tu kanasim-ase-ta.

Next, the noncausatives. Here the judgments are crisper, since there is little or no problem with examples in which the NQ is adjacent to the subject.

- (142) a. [2-tu-no ii sirase]-ga Zyon-o zetuboo-e oiyat-ta.
 2-cl-GEN good news-NOM John-ACC despair-to drive-past
 ‘Two pieces of good news drove John to despair.’
 b. [*Ii sirase*]-ga 2-tu Zyon-o zetuboo-e oiyat-ta.
 c. *[*Ii sirase*]-ga Zyon-o 2-tu zetuboo-e oiyat-ta.
 d. *[*Ii sirase*]-ga Zyon-o zetuboo-e 2-tu oiyat-ta.
- (143) a. [2-satu-no Faulkner-no’ hon]-ga Zyon-o
 2-cl-GEN Faulkner-GEN book-NOM John-ACC
 utinome-si-ta.
 bowl over-past
 ‘Two of Faulkner’s books bowled John over.’
 b. ?*Faulkner-no hon-ga 2-satu Zyon-o utinome-si-ta.*
 c. **Faulkner-no hon-ga Zyon-o 2-satu utinome-si-ta.*

At this point, one might object that the various instances of backward binding could have different explanations. This is logically possible, but I think it is unlikely, given the similarities in the semantics of these cases, all of which (as we will see) involve some notion of causation. Compare the noncausal examples in (144).

- (144) a. *Each other_i’s relatives considered [John and Mary]_i angry.
 b. *Each other_i’s rocks hit [John and Mary]_i on the head.
 c. *Articles about himself_i in the *Times* attacked John_i.
 d. *That picture of himself_i flatters John_i.

The relevant generalization so far seems to be as follows:

- (145) A Causer argument of a predicate π may behave as if
 c-commanded by an argumental DP governed by π .

This generalization covers the cases we have looked at. On the other hand, it is unprincipled: (145) bears no relation to anything else we know about binding phenomena.

By contrast, B&R's solution, insofar as it works, is principled. It has the admirable property that it reduces an odd binding phenomenon to a natural interaction of movement with more familiar binding phenomena. The choice between an unprincipled description of the facts and a principled but inadequate explanation of the facts is an unpleasant one. Fortunately, by the end of our long discussion of ObjExp predicates, we will be able to adopt B&R's explanation of backward binding in the context of a new and more nuanced view of the syntax of ObjExp predicates. For now, however, we are stuck with conflicting evidence. B&R's analysis of backward binding requires movement from internal to external position in ObjExp constructions. This movement, under normal assumptions, would require the subject position to be athematic, a description already argued to be incorrect. Furthermore, the analysis does not extend to clearly related cases. Yet the unaccusative analysis is the only one on the table that has the potential to explain backward binding.

For the moment, let us assume that the case for the unaccusativity of ObjExp verbs like *annoy* or *preoccupare* has not been proven, even though we have no better alternative as yet. This will require us to return to the linking problems posed by these verbs and seek a fresh solution. We will return to the binding problems later.

2.5.2 Other Arguments

Before closing this discussion, let us consider auxiliary selection and its relation to Case theory. I noted above that the unaccusative analysis of verbs like *preoccupare* contradicts usual assumptions about auxiliary selection in Italian. *Preoccupare*-class verbs take the auxiliary *avere* in the compound past tense, and not the *essere* found with all verbs tagged as unaccusative in previous literature. Additionally, I noted that there *are* also *essere* ObjExp verbs, which generally assign some inherent Case, as discussed by B&R.

- (146) a. A Gianni è sempre piaciuta la musica.
 to Gianni is always pleased the music
 'The music always pleased Gianni.'
 b. La musica è sempre piaciuta a Gianni.

If we assume that the traditional interpretation of auxiliary selection is correct, then we will naturally conclude that ObjExp verbs like *piacere* (unlike *preoccupare*) are unaccusative. We might very well accept for these verbs the analysis proposed by B&R, which posits the following representation at D-Structure:

(147) [e] è [_{VP} [_{V'} *piacuta* la musica] a Gianni]

As we expect, these verbs form neither adjectival nor verbal passives, unlike *avere* ObjExp verbs.

- (148) a. A Gianni piace questo libro.
to Gianni pleases this book
'This book pleases Gianni.'
- b. *Questo libro è stato piaciuto (da Gianni).
this book was pleased by Gianni
- c. *(A) Gianni è stato piaciuto (da questo libro).
to Gianni was pleased by this book
- (149) a. Ai bambini non manca energia.
to the kids not lacks energy
'The kids don't lack energy.'
(see Perlmutter 1984:293, (4b))
- b. *Energia non è mancato dagli bambini.
energy not is lacked by the kids
- c. *Ai bambini è stato mancato (da energia).
to the kids was lacked by energy

Additionally, Guglielmo Cinque (personal communication) has noted that a postverbal nominative argument of *piacere*, like postverbal arguments of unaccusative verbs, allows *ne*-cliticization. By contrast, postverbal Causer arguments of verbs like *preoccupare* pattern with postverbal arguments of transitive verbs in disallowing *ne*-cliticization.

- (150) a. ?Ne sono piaciuti a Maria [solo due ____].
of them pleased to Maria only two
- b. *Ne hanno preoccupato Gianni [solo due ____].
of them worried Gianni only two
- c. *Ne hanno interessato Maria [due ____].
of them interested Maria two

Much the same distinction among Experiencer verbs has been proposed for Dutch by Hoekstra (1984) and Everaert (1986). Dutch has a class of

Experiencer verbs that are conjugated with *hebben* ‘to have’ and form verbal passives, and another class of Experiencer verbs that are conjugated with *zijn* ‘to be’ and do not (Hoekstra 1984:185–86).⁴⁶

- (151) a. Die fout *is* mij opgevallen.
that mistake is me struck
b. *Ik ben/werd door die fout opgevallen.
I am by the mistake struck
- (152) a. Die fout *heeft* mij getroffen.
that mistake has me struck
b. Ik ben/werd door die fout getroffen.

Assuming this discussion to be correct, I will not claim that verbs with B&R-style analyses fail to exist. I will merely claim that the B&R-style analysis is wrong for the “surprising” cases of *avere* verbs, and right for the unsurprising cases of *essere* verbs.

Indeed, B&R are also not wrong in supposing that unaccusative verbs that assign accusative Case (anti-etymologically) exist. We have already seen a Japanese instance of this sort in (136b). The (a) sentences of (153)–(154) quite plausibly display English examples of accusative-assigning unaccusative verbs—at least if the failure of passive is any guide.⁴⁷

- (153) a. Smith’s name escaped us for some reason.
b. *We were escaped by Smith’s name for some reason.
c. We didn’t remember Smith’s name for some reason.
d. Smith’s name was not remembered by us for some reason.
- (154) a. The correct generalization eluded Pāṇini.
b. *Pāṇini was eluded by the correct generalization.
c. Pāṇini missed the correct generalization.
d. The correct generalization was missed by Pāṇini.
(see Perlmutter and Postal 1984:115)

The (a) examples in (155)–(157) show plausible examples of unaccusative verbs assigning structural dative, like *piacere*.⁴⁸ Crucially, as the (b) examples in (153)–(157) demonstrate, these verbs also form neither verbal nor adjectival passives at all (in contrast to their non-unaccusative congeners in the (c–d) sentences).⁴⁹

- (155) a. The play didn’t appeal to Mary.
b. *Mary wasn’t appealed to by the play.
c. Mary didn’t care for the play.
d. The play wasn’t cared for by Mary.

- (156) a. This mattered to John.
 b. *John was mattered to by this.
 c. John cared about this.
 d. This was cared about by John.
- (157) a. The same idea occurred to Mary.
 b. *Mary was occurred to by the same idea.
 c. Mary thought of the same idea.
 d. The same idea was thought of by Mary.
 (see Perlmutter and Postal 1984:104–5)

Backward binding is possible with these predicates.

- (158) a. Each other's names escaped Tom and Sue.
 b. The solutions to each other's problems eluded the two scientists until they compared notes.
- (159) a. Each other's remarks appealed to John and Mary.
 b. Each other's welfare mattered to the students.
 c. The problems with each other's ideas had never even occurred to Heisenberg and Bohr.

These instances of backward binding might very well be explained by B&R's view of binding, if the structures of these sentences are of the type they propose, with *to* disregarded for c-command purposes in (160b).⁵⁰

- (160) a. [Each other's names]_i [_{VP} [_{V'} escaped *t*_i] Tom and Sue].
 b. Each other's remarks]_i [_{VP} [_{V'} appealed *t*_i] to Tom and Sue].

In any case, the existence of verbal passives of ObjExp predicates provides a positive argument against the unaccusative analysis of these predicates. I draw the conclusion that we cannot save the U(T)AH for this class by recourse to a simple unaccusative analysis, even though there are examples that fall into this basket. We need new suggestions—hypotheses whose consequences are more nuanced than the proposals we have examined. This will be the topic of the following chapters.

Chapter 3

U(T)AH and the Zero Morpheme *CAUS*

Having rejected the simplest solution to the U(T)AH problem posed by ObjExp verbs, we are back where we started. Let us approach the problem from another angle. Examples (30)–(41) pose questions for the U(T)AH because of the assumption that the θ -role assigned to the object DPs of the SubjExp class is the same role as that assigned to the subject DPs of the ObjExp class. What if this basic assumption were false, and the two roles were distinct after all? That is, suppose that we did not have a Theme argument in both the SubjExp and ObjExp examples but two distinct θ -roles. In such a case, there would be no a priori problem for the null hypothesis. In fact, the claim that these θ -roles are distinct has already been pursued in philosophical literature on emotions and predicates of emotion (see, e.g., Kenny 1963). I raised this theme briefly in Pesetsky 1987a. My discussion here is an elaboration of those remarks.

Let us put aside B&R's use of the term *Theme* in the context of SubjExp and ObjExp predicates. Instead, I want to suggest that the subject argument with the ObjExp class always bears the role *Causer*, whereas the object argument with the SubjExp class always bears one of two entirely distinct roles, which I will rename *Target of Emotion* and *Subject Matter of Emotion*. Both Target and Subject Matter are generally lumped together by philosophers under the cover term "Object of Emotion." This difference is not important for the moment, but must be noted nonetheless. All that is really important now is the existence of a distinction between the θ -role associated with the subject of the ObjExp class (*Causer*) and the θ -roles associated with the object of the SubjExp class (*Target*, *Subject Matter*).

3.1 Causer versus Target

Nonetheless, it will be helpful to spend some time sorting out the semantic nature of these roles—since these distinctions form the empirical basis for the rest of this study. Let us first distinguish Causer from Target. To see this distinction, compare the pair *anger/angry* in (30), repeated here.⁵¹

- (30) a. Bill was very angry at the article in the *Times*. [Target]
 b. The article in the *Times* angered/enraged Bill. [Causer]

The truth conditions of these two sentences are noticeably distinct. For (30a) to be true, Bill must have evaluated the article, and he must have formed a bad opinion of some aspect of it. In other words, Bill must find the article objectionable in some respect. (30b) is rather different. Bill *might* be mad at the article in (30b) as well—the meaning of (30b) is not inconsistent with (30a). Nonetheless, (30b) is appropriate even if Bill thinks the article is splendid. It can be true, for example, if Bill's favorite columnist has written, in Bill's opinion, a great article revealing examples of government corruption. The article does *cause* Bill to be angry, and possibly angry at someone or something, but he is not necessarily angry at the article itself.

Distinctions of this sort between Causer and Target distinguish many pairs of predicates like *anger* and *be angry at*. A Causer argument must simply be causally connected to the emotion described by the predicate and borne by the Experiencer. The Target argument, however, is *evaluated* by the Experiencer as part of what Nissenbaum (1985) calls “the emotional episode.” In general, a negative emotion (like anger) entails a negative evaluation, a positive emotion (like love) entails a positive evaluation. Crucially, throughout the examples in (30)–(41), both verbal and adjectival, Causer is always associated with the *subject* position, and Target is associated with the *object* position. Experiencer takes up the other grammatical relation.

Examples (31) and (33), repeated here, show much the same thing. The (a) examples imply a positive evaluation of the direct object; the (b) examples merely causally link the subject to the emotion denoted by the predicate. For example, (33b) has a purely causal reading in which the dinner eliminated Bill's hunger, but in (33a) Bill expressed a judgment about the *quality* of the dinner.

- (31) a. The paleontologist liked/loved/adored the fossil.
 b. The fossil pleased/delighted/overjoyed the paleontologist.
- (33) a. Bill was satisfied/content with the Chinese dinner.
 b. The Chinese dinner satisfied/contented Bill.

3.2 Causer versus Subject Matter

Let us turn now to the distinction between Causer and Subject Matter, seen clearly in (36).

- (36) a. John worried about the television set.
 b. The television set worried John.

In (36a), whenever John was experiencing the worry described in the example, he was thinking in some way about the television set. Perhaps he was worried that it might catch fire, or that it was perched too precariously and might fall. Whatever the nature of John's specific concern, *the television set* is the Subject Matter of Emotion. In (36b), on the other hand, the DP *the television set* bears the now familiar role of Causer. It is sufficient that the television set causes John to experience worry, but the Subject Matter of his thoughts while experiencing worry could have nothing to do with the television set. There is simply a causal relationship between the set and some state of worry. For example, John could be a detective. Seeing the television set in a suspect's living room sets off a chain of worries: for example, "What would a completely blind man be doing with a fancy color television?" He is not worrying *about* the set. The set merely provokes worries about other matters.

(40) shows something similar. In (40a), John has had some experience with the problem of lexical entries and has lost a prior interest in it. In (40b), it is sufficient that mentioning lexical entries causes John to yawn.

- (40) a. John is bored with the problem of lexical entries.
 b. The problem of lexical entries bores John.

The fact that Causer and Subject Matter are distinct does not mean that no logical relationship exists between sentences with a Causer and sentences with a Subject Matter argument. Thus, consider the following sentences:

- (161) a. John worried about Mary's poor health, but Mary's poor health did not worry John. [contradiction]

- b. Mary's poor health worried John, but John did not worry about Mary's poor health. [noncontradiction]

(161a) is clearly a contradiction, but (161b) is not.⁵² For example, in (161b), the subject of John's worry might be the possibility of an epidemic. While John is worrying, he might have completely forgotten about Mary, the instigator of his train of thought. By contrast, there is no plausible scenario for (161a) that makes it noncontradictory. This is because of a one-way implicational relation between statements about the Subject Matter of emotion and statements about Causers of emotion. The situation is summarized in (162).

- (162) a. If X worried about Y, then Y worried X. [true]
 b. If Y worried X, then X worried about Y. [false]

We can see this in examples like the following: (163a) is a tautology, because of (162a), but (163b) is not:

- (163) a. Because/Whenever John worried about Mary's poor health, Mary's poor health worried John. [tautology]
 b. Because/Whenever Mary's poor health worried John, John worried about Mary's poor health. [nontautology]

Similarly:

- (164) a. Because John was bored with the problem of lexical entries, the problem of lexical entries bored John. [tautology]
 b. Because the problem of lexical entries bored John, John was bored with the problem of lexical entries. [nontautology]
- (165) a. Because Bill feared the ghost, the ghost frightened Bill. [tautology]
 b. Because the ghost frightened Bill, Bill feared the ghost. [nontautology]

Let us see where we stand. We began by noting that apparent doublets like those in (30)–(41) pose questions for the U(T)AH. I have now suggested that these pairs are not true doublets at all. Although all the predicates assign the role Experiencer, the other role is either Causer, in which case it is assigned to the subject, or else it is Target or Subject Matter, in which case it is assigned to an object position.⁵³ We thus rescue the U(T)AH for these predicates. The linking of particular arguments to grammatical relations is completely predictable. We can assume that the assignment for Experiencer predicates is made on the basis of the hierar-

chy in (166), where the highest argument is mapped onto the highest D-Structure position in its clause. Obviously, (166) represents a portion of a larger hierarchy, on which the various θ -roles are arranged.⁵⁴

(166) Causer > Experiencer > Target/Subject Matter

A verb like *anger*, with Causer and Experiencer arguments, will link Causer with some high position in its clause (e.g., subject of VP) and will link Experiencer with a lower position (e.g., direct object).

(167) [_{VP} [_V V Experiencer] Causer]

A verb like *love*, with an Experiencer and a Target, will link Experiencer with the higher position and Target with the lower position. Likewise for the Experiencer and Subject Matter arguments of one use of *worry*:

- (168) a. [_{VP} [_V V Target] Experiencer]
 b. [_{VP} [_V V Subject Matter] Experiencer]

If the VP-Internal Subject Hypothesis is correct, then other principles (e.g., Case theory) will ensure that the Experiencer is promoted to the Spec, IP subject position in normal tensed clauses. This will be true, for example, if V Case-marks the Target/Subject Matter argument, but not Experiencer. If, with some other verb, V Case-marks the Experiencer argument but fails to Case-mark the Target/Subject Matter argument, then it is the Target/Subject Matter argument that will raise. Such is the case with the genuinely unaccusative ObjExp predicates, for example, *appeal to*, as presented in (155a–b), repeated here.

- (169) a. The play didn't appeal to Mary.
 b. *Mary wasn't appealed to by the play.

Such a predicate contrasts with the non-unaccusative *please* in passivization possibilities.

- (170) a. The play didn't please Mary.
 b. Mary wasn't pleased by the play.

As predicted, *appeal to* and *please* contrast semantically as well. The surface subject of *please* is a Causer; the surface subject of *appeal to* is a Target, moved from an underlying direct object position. Thus, if the play didn't *appeal to* Mary, then she must have given it a negative evaluation. On the other hand, if the play failed to *please* Mary, she might rate it highly, as long as for some reason the play fails to bring her pleasure. For example, an excellent play written by Mary's rival could fail to please her

while simultaneously appealing to her.⁵⁵ The correct English translation of Italian *piacere* is thus not *please*, but *appeal to*. Likewise, there is nothing causal about any of the unaccusative ObjExp predicates described in (153)–(157): *escape*, *elude*, *matter to*, *occur to*.

Our finer-grained semantic investigation of these predicates has now allowed us to save the U(T)AH for Experiencer predicates. This, as discussed in chapter 1, is a most desirable result.

3.3 A New Problem: The Target/Subject Matter Restriction

Nonetheless, we are not done yet. Of course, we have not explained backward binding phenomena. Worse still, we now have a brand-new problem: if Causer of Emotion is distinct from either Subject Matter or Target, why can't *Causer*, on the one hand, and *Target/Subject Matter*, on the other hand, cooccur with the same predicate? For example, why are (171a–g) absolutely impossible?

- (171) a. *The article in the *Times* angered Bill at the government.
 b. *The Chinese dinner satisfied Bill with his trip to Beijing.
 c. *The television set worried John about the veracity of Bill's alibi.
 d. *The problem of lexical entries bores John with his life as a linguist.
 e. *Something Bill had said bothered Mary about her future.
 f. *Bill driving at night always worries John about the adequacy of his insurance coverage.
 g. *The distant rumbling frightened Mary of another tornado.

The problem is real and not trivial. For one thing, many SubjExp predicates are morphologically related to the ObjExp predicates. Just because the ObjExp predicates assign an additional Causer role, there is no reason to expect them not to also assign the other roles assigned by the SubjExp predicates. Additionally, the phenomenon seen in (171) is not the idiosyncratic property of a few verbs. A search through the verbs listed in the *Oxford Advanced Learner's Dictionary*, matched against the Experiencer predicates listed by Norrick (1978), reveals almost no examples of the type seen in (171). This suggests that we are dealing here, not with lexical idiosyncrasy, but with an important pattern.⁵⁶ Furthermore, the meaning assigned to the ungrammatical examples of (171) is clear. There is no

semantic incoherence in (171a–g), and, in fact, periphrastic counterparts to these examples are fine.

- (172) a. The article in the *Times* made Bill angry at the government.
 b. The Chinese dinner made Bill satisfied with his trip to Beijing.
 c. The television set made John worry about the veracity of Bill's alibi.
 d. The problem of lexical entries made John bored with his life as a linguist.
 e. ?Something Bill had said made Mary bothered about her future.
 f. Bill driving at night always makes John worry about the adequacy of his insurance coverage.
 g. The distant rumbling made Mary afraid/fearful of another tornado.

Furthermore, in my judgment at least, there is another set of ObjExp expressions that do not show this restriction, or show it only weakly. These are verb-particle constructions like those in (173)–(178).

- (173) a. *The election results really irritated Sue at the media.
 b. The election results really riled Sue up at the media.
 (174) a. *The check calmed Bill about the accident.
 b. The check calmed Bill down about the accident.
 (175) a. *Her remarks really depressed Bill about it.
 b. Her remarks really got Bill down about it.
 (176) a. *The news cheered Sue about her plight.
 b. The news cheered Sue up about her plight.
 (177) a. *The lectures excited Bill about classical music.
 b. The lectures turned Bill on to classical music.
 (178) a. *The article angered Bill at Mary.
 b. The article really pissed Bill off at Mary.

We are thus left with a genuine problem. If Causer of Emotion is distinct from either Subject Matter or Target, why are there no simplex predicates that simultaneously realize the Causer argument and the Target or Subject Matter argument?⁵⁷

The data in (171), however, make possible a serious and reasonable objection to the “finer-grained semantic” proposal. One might claim that

(171) simply disproves the contention that Causer is distinct from Target and Subject Matter. If these roles are not truly distinct, then we might rule out the examples in (171) by a constraint that excludes two Themes in a single clause (using *Theme* here as it is used by B&R). We would then view the verb-particle constructions, like the periphrastic constructions with *make*, as biclausal in some sense, perhaps following Kayne (1984b). This is not fully a straw-man proposal: such a constraint is in fact invoked by B&R (p. 306, fn.) to handle precisely this problem. A restriction against two “Themes” could in turn be a consequence of a broader constraint on lexical entries, which we might call a requirement of *Thematic Diversity*.⁵⁸

(179) *Thematic Diversity*

If α and β are distinct arguments of a predicate P, the thematic role assigned to α must be distinct from the thematic role assigned to β .

This requirement is plausible, explaining (for example) the nonexistence of predicates with two Agents or with two Targets of Emotion (**John is angry at Mary at her family*). It is another question, however, whether Thematic Diversity is the factor at stake in (171).

The issue is not as simple as the existence of semantic distinctions between Causer and Target or between Causer and Subject Matter. Thematic Diversity might explain (171) even if the members of the pairs (Causer, Target) and (Causer, Subject Matter) are semantically distinct. We already saw in section 2.1 that the syntax ignores some very palpable semantic distinctions, like the one between verbs of quiet speech and verbs of noisy speech. I took the view that syntax imposes a coarse grain on information from other systems, such as the semantics, noticing certain distinctions and blurring others. In the present context, one could recognize the semantic distinctions among Causer, Target, and Subject Matter and still propose that Thematic Diversity blurs these distinctions. From the standpoint of this constraint, then, Causer, Target, and Subject Matter would all count as nondistinct.⁵⁹

If this counterproposal were correct, then—if our critique of B&R’s analysis is sound—we would be back where we started, with a severe problem for the U(T)AH. There is, however, some simple evidence that suggests that this counterproposal is on the wrong track. If Causer and Target are nondistinct, and Causer and Subject Matter are nondistinct, then we expect Target and Subject Matter likewise to be nondistinct. Thematic Diversity should prevent the cooccurrence of Target and Sub-

ject Matter just as it would prevent the cooccurrence of Causer with either of these roles. This consequence is incorrect, however: where Target and Subject Matter are both compatible with a given predicate, they do cooccur. The object of *with* in (180a) is a Target because it is evaluated (negatively), whereas the object of *about* is a Subject Matter argument. Likewise for the other examples of (180):⁶⁰

- (180) a. Sue is angry with Bill about the party.
 b. John is irritated at Mary about the mistake.
 c. Bill likes this about his new job: he doesn't have to get up very early.⁶¹
 d. What_i Mary hates t_i about Sue is her stubbornness.⁶²

In the next sections, I will argue that the phenomenon seen in (171) is more widespread and extends to cases in which not only is resort to the notion Theme irrelevant, but there is no evident similarity among the θ -roles involved. These cases have one thing in common: a predicate that can be semantically or morphologically analyzed as a SubjExp predicate embedded in something else. I will argue that this property, not any problem with the proposed thematic analysis of ObjExp verbs, is to blame for (171).

3.4 Zero Morphemes

Let us call the restriction noted at the end of the previous section the *Target/Subject Matter (T/SM) restriction*. This section has two goals, one immediate and the other more ambitious. The immediate goal is to demonstrate that the T/SM restriction on ObjExp predicates is not an artifact of an incorrect analysis. I do this by displaying other constructions that appear to show the same restriction, but are not amenable to the easy solutions that might undermine my conclusions about ObjExp predicates. The more ambitious goal, of course, is to explain the T/SM restriction. In fact, I will show that the T/SM restriction in its various manifestations is a simple effect of a broader generalization that governs the movement of syntactic heads, the *Head Movement Constraint* of Travis (1984) and Baker (1988). Only a demonstration of this sort can show that the unification of the T/SM restriction with other cases is not spurious. Thus do the ambitious and immediate goals depend on each other.

This is where zero morphology enters the picture. In the sections that follow, I will argue that ObjExp predicates like *annoy* are actually

morphologically complex. I will suggest that such predicates consist of a phonologically zero causative morpheme and a bound root (in this case pronounced *annoy*) that is actually a SubjExp predicate. The morphological structure of these predicates is thus similar to that of their Japanese counterparts, except that the causative morpheme is here phonologically zero. I will argue for this analysis by demonstrating that ObjExp predicates (and, in fact, the other constructions that show conditions like the T/SM restriction) obey a morphological restriction on zero derivation discovered by Myers (1984)—perhaps itself a special case of a generalization discovered by Fabb (1988).

I will then show that the T/SM restriction derives from the Head Movement Constraint once an independently motivated characterization has been given of the place of this zero morpheme in argument structure and syntactic structure—a characterization with other consequences that I explore in chapter 6. By the conclusion of this book, I will have partly vindicated the optimistic view of the relation between the lexicon and syntax with which I began. However, I need to start slowly, with a defense of the reality of the T/SM restriction.

3.4.1 Higgins's Cases and the Morpheme *SUG*

Data strongly reminiscent of the T/SM restriction with ObjExp verbs were noted first by Higgins (1973:168–70).⁶³ Crucially, no Causer role is involved in Higgins's examples; these are not ObjExp constructions.

- (181) a. John was proud (of his son).
 b. Bill was angry (at the government).
 c. Sue was nervous (about the exam).
 d. Mary was optimistic (about the future).
 e. Bill was sad (about John).
 f. Tom was fearful (of an earthquake).
 g. John is over-eager (to leave).
- (182) a. John's manner was proud (*of his son).
 b. Bill's remarks were angry (*at the government).
 c. Sue's behavior was nervous (*about the exam).
 d. Mary's expression was optimistic (*about the results).
 e. Bill's words were sad (*about John).
 f. Tom's attitude was fearful (*of an earthquake).
 g. John's behavior is over-eager (*to leave).

There is no general restriction on transitive adjectives predicated of nouns like *manner*. Thus, there is something special about Higgins's cases.

- (183) a. John's manner was full of bluster.
 b. Bill's remarks were suggestive of deeper anxieties.
 c. Sue's behavior was shocking to his friends.
 d. Mary's expression was typical of her.
 e. Bill's words were clear to all.
 f. Tom's attitude was reminiscent of Bill's.

At the same time, Higgins's phenomenon is not limited to SubjExp adjectives. Agentive adjectives also display it.

- (184) a. John was careful (with the electrodes).
 b. Bill was wary (of snap judgments).
 c. Sue was attentive (to every detail).
 d. Mary was stingy (with her prize money).
 e. Bill was reckless (with his money).
 f. Tom was unwilling (to leave).
- (185) a. John's manner was careful (*with the electrodes).
 b. Bill's remarks were wary (*of snap judgments).
 c. Sue's behavior was attentive (*to every detail).
 d. Mary's behavior was stingy (*with her prize money).
 e. Bill's expression was reckless (*with his money).
 f. Tom's attitude was unwilling (*to leave).

Suppose these facts are of a piece with the impossibility of **The article in the Times angered Bill at the government* and other examples of the T/SM restriction with ObjExp predicates. Then we can make an important observation: an explanation for the T/SM restriction in terms of B&R's notion of Theme is untenable. One might account for the example with *anger* by claiming that the syntax considers both Causer and Target as subspecies of the same role *Theme*, but one can hardly say something similar about the roles assigned in (182a) to *John's manner* and *his son*. There is no plausibility to identifying the subject and object θ -roles in (182a), since there is no sense in which behavior that may be described as *angry* can be confused with the object of the anger.

Next we must ask what kind of light Higgins's cases can shed on the distribution and nature of the T/SM restriction. Higgins (p. 168) characterizes his examples as follows:

[A]pparently, whatever can be predicated directly of a person that relates to some aspect of his behavior can, with various restrictions, also be “predicated” of the “entities” referred to by subjects with head nouns such as *aspect*, *behavior*, *look*, or *manner*. . . . [I]t seems to be the case that one cannot use adjective phrases with complements in such sentences.

This observation covers the examples just considered, but is nonetheless not quite accurate. Unlike SubjExp adjectives, ObjExp adjectives may be predicated directly of a person, relating to some aspect of that person’s behavior, yet not lose their complements when predicated of nouns like *manner*.

- (186) a. Bill was annoying to us.
 b. Mary was worrisome to her classmates.
 c. Sue was frightening to the little children.
- (187) a. Bill’s behavior was annoying to us.
 b. Mary’s manner was worrisome to her classmates.
 c. Sue’s expression was frightening to the little children.

Instead, the important factor in (182) and (185) seems to stem from the semantics and morphology of the adjectives themselves. Adjectives like *proud* and *careful*, which have a particular meaning when predicated of a person, have a more complex meaning when predicated of a manner or remark. If we represent the meaning of ‘angry’ predicated of a person as ϕ , then the meaning of ‘angry’ predicated of a manner or remark is $(\phi + x$, where x is some extra component of meaning. *John’s manner was proud* means something like ‘John’s manner suggested that he was proud’. Similarly, *John’s manner was careful* means something like ‘John’s manner suggested that he was being careful’. The locution ‘suggested that he was’ is x .

This difference could be seen as a vagueness in the meaning assigned to *proud* and *careful*. Suppose, however, that there is no vagueness. Instead, consider the possibility that *proud* and *careful* in these examples are morphologically complex. On this view, a phonologically null affix, which I will call *SUG* (for “*suggest*”), adds the component of meaning x that allows it to be predicated of a DP whose head is a word like *manner*.

- (188) a. John’s manner is [proud-*SUG*] (*of his son).
 b. John’s manner was [careful-*SUG*] (*not to offend anyone).

By contrast, the adjectives in (183) and (186) are predicated directly of their subjects, with no special meaning that depends on whether the sub-

ject is a person or on a person's behavior. There is no reason to assign to (183a), for example, an analysis involving *full-SUG*.

3.4.2 English ObjExp Causatives and the Morpheme *CAUS*

At present, the *SUG* hypothesis has been presented, but not supported by any evidence. I will shortly present morphological evidence of its existence. In any case, if the phenomenon in (182a) can be traced back to a morphological complexity in *proud* as used here, we can imagine a similar proposal for non-unaccusative ObjExp verbs like *annoy*. We have already noted that ObjExp predicates like *annoy* differ from SubjExp predicates like *hate* or *angry* in taking a subject marked with the θ -role Causer. One can easily imagine that ObjExp verbs like *annoy* are in fact derived from SubjExp predicates by the addition of a morpheme that adds just this Causer role—that is, a causative morpheme. On this approach, the word pronounced *annoy* is bimorphemic, containing a SubjExp root meaning 'be annoyed' or 'get annoyed' and a causative morpheme that I will call *CAUS*. The impossibility of expressing the Target or Subject Matter in the environment of *CAUS* will then be traced to the same thing that prevents the cooccurrence of Target or Subject Matter with *SUG*. What is more, the examples in (185) give reason to believe that this constraint will generalize to arguments other than Target and Subject Matter.

A bimorphemic analysis of non-unaccusative ObjExp verbs is neither novel nor farfetched. Blurring some details that will shortly be important, a bimorphemic, causative approach to ObjExp verbs was entertained by Chomsky (1965:189, 1972:24). It is transparently correct for many cases in Japanese, as we have already seen. Examples (191)–(192) are Kuroda's (1965), quoted by Akatsuka (1976). The others are modeled on examples of Akatsuka's. Indeed, Akatsuka (1976) herself proposed the extension to English of the analysis transparently motivated in Japanese.

- (189) a. Tanaka-ga sono sirase-o yorokon-da.
 Tanaka-NOM that news-ACC be pleased-past
 'Tanaka was pleased at that news.'
- b. Sono sirase-ga Tanaka-o yorokob-ase-ta.
 that news-NOM Tanaka-ACC be pleased-CAUSE-past
 'That news pleased Tanaka.'
- (190) a. Tanaka-ga sono sirase-o kanasin-da.
 Tanaka-NOM that news-ACC be sad-past
 'Tanaka was sad about that news.'

- b. Sono sirase-ga Tanaka-o kanasim-ase-ta.
that news-NOM Tanaka-ACC be sad-CAUSE-past
'That news saddened Tanaka.'
- (191) a. Taroo-ga sono koto-o nagei-ta.
Taro-NOM that fact-ACC be-grieved-past
'Taro was grieved at that fact.'
- b. Sono koto-ga Taroo-o nagek-ase-ta.
that fact-NOM Taro-ACC be grieved-CAUSE-past
'That fact grieved Taro.'
- (192) a. Taroo-ga ongaku-o tanosin-da.
Taro-NOM music-ACC be amused-past
'Taro was amused at the music.'
- b. Ongaku-ga Taroo-o tanosim-ase-ta.
music-NOM Taro-ACC be amused-CAUSE-past
'The music amused Taro.'

Interestingly, Japanese causative ObjExp verbs, when used agentively, show exactly the same T/SM restriction we have seen in English. This observation is due to Akatsuka (p. 78 ff.), as are the following examples:

- (193) a. Tanaka-wa otoko-no ko-no tanzyoo-o
Tanaka-NOM baby-GEN boy-GEN birth-ACC
yorokon-da.
be pleased-past
'Tanaka was pleased with the birth of a baby boy.'
- b. *Satoo-wa Tanaka-ni otoko-no ko-no tanzyoo-o
Sato-NOM Tanaka-DAT baby-GEN boy-GEN birth-ACC
yorokob-ase-ta.
pleased-CAUSE-past
- (194) a. Tanaka-wa [zibun-ga gan-ni kakatta koto]-o
Tanaka-NOM self-NOM cancer-DAT had fact-ACC
kanasin-da.
be sad-past
'Tanaka was sad about the fact that he had cancer.'
- b. *Satoo-wa Tanaka-ni [zibun-ga gan-ni kakatta
Satoo-NOM Tanaka-DAT self-NOM cancer-DAT had
koto-o] kanasim-ase-ta.
fact-ACC be sad-CAUSE-past

No such prohibition is found with normal causative constructions.

- (195) a. Hiroshi-ga Taroo-o nagut-ta.
 Hiroshi-NOM Taro-ACC hit-past
 'Hiroshi hit Taro.'
- b. Ziroo-ga Hiroshi-ni Taroo-o nagur-ase-ta.
 Ziro-NOM Hiroshi-DAT Taro-ACC hit-CAUSE-past
 'Ziro made Hiroshi hit Taro.'

It must be noted, however, that the effect documented in (194)–(195) appears to be absent or quite weak when the subject of the causative is inanimate and a non-Agent. The numerous speakers from whom I have data on this point have disagreed greatly on nonagentive examples, and their judgments are often murky.⁶⁴

- (196) a. (?)Sono sirase-ga Tanaka-ni otoko-no ko-no
 that news-NOM Tanaka-DAT baby-GEN boy-GEN
 tanzyoo-o yorokob-ase-ta.
 birth-ACC pleased-CAUSE-past
- b. (?)Sono koto-ga Tanaka-ni [zibun-ga gan-ni
 that fact-NOM Tanaka-DAT self-NOM cancer-DAT
 kakatta koto-o] kanasim-ase-ta.
 had fact be sad-CAUSE-past

I will return to an explanation of these Japanese data in section 6.3.1. For now, the data in (193)–(194) may simply be taken as another indication that the T/SM restriction in English is not an artifact of an incorrect thematic analysis, but is a widespread phenomenon that demands an explanation.

3.4.3 Zero Morpheme or Lexical Decomposition?

It is one thing to show that an idea has "initial plausibility," and another thing to show that it is right. With this in mind, let us first clarify the analysis being assumed for cases of *CAUS* and *SUG*. As noted above, the bimorphemic analysis of ObjExp verbs like *annoy* is reminiscent of analyses in Generative Semantics. Words that might be pronounced *annoy* or *angry* (or *kill*) were often analyzed as syntactically multipartite, consisting of several "abstract" morphemes inserted in semantically appropriate positions in phrase markers. These morphemes were combined into one unit by a transformational rule of Predicate Raising that, in McCawley's (1968:75) words, "applies to trees which terminate in semantic matter rather than in lexical matter."

A few years earlier, however, Lakoff (1970; originally written in 1965) had presented a slightly different proposal. He also posited multipartite analyses for semantically causative predicates, and various rules of Predicate Raising. In his analysis, however, the predicates that undergo Predicate Raising do receive PF interpretation. Thus, he analyzed *persuade* (p. 93) as consisting of one verb with features [+V, +PRO, +CAUS-ATIVE] and another with various features including the phonological features that yield the pronunciation *persuade*. Lakoff's feature "+PRO" on the causative morpheme presumably indicates that this morpheme corresponds to a null string at PF, whereas the phonological features on the other morpheme yield the actual sounds of the word *persuade*. In addition, certain morphemes like *persuade* bear features that indicate that they are *bound morphemes* (not Lakoff's terminology)—only licensed if they undergo the rule that raises them to the causative morpheme.

These two views are morphologically quite different, but little attention seems to have been paid to morphology in this context. Instead, McCawley's general approach was developed in the context of ongoing debates over lexical decomposition and the status of Deep Structure. Much of this work did not survive the criticisms aimed in the early 1970s against the Generative Semantics program and its identification of semantic structures with Deep Structure. Nonetheless, the correctness of multipartite syntactic analyses for various words is logically independent of the motivation for these analyses. Thus, for example, there may be reasons for the child to assume that a verb like *annoy* is bimorphemic above and beyond its causative meaning (see Kayne 1984a:157). We will return to these reasons later.

If certain bipartite analyses are correct, we must return to the unfinished task of comparing Lakoff's structures with later approaches. Lakoff postulated a phonologically zero causative morpheme attached to a non-causative root. This root in some cases does not occur as an independent verb. If Lakoff's view had been applied to a causative analysis of verbs like *annoy* (instead of the popular analysis of the time in terms of Psych Movement, adopted by Lakoff (1970) himself), then the phonological string *annoy* would have been seen as the pronunciation of a SubjExp predicate that does not occur except when combined with a phonologically zero causative morpheme. (I will use the symbol $\sqrt{\quad}$, where necessary, to mark roots that are homophonous but not coextensive with words.)

(197) a. *John $\sqrt{\text{annoy}}$ ed with Mary

b. The book [$\sqrt{\text{annoy}}$ $\emptyset_{\text{CAUS}}$]-ed John (*with Mary)

(198) [[“BE-ANNOYED”] “CAUSE”] [output of Predicate Raising]

annoy [late lexical insertion]

3.4.4 Evidence for Bound SubjExp Morphemes: Nominalization

Now consider the nominalizations that are related to causative ObjExp verbs like *annoy*. These nominalizations uniformly lack all causative force (as observed first, perhaps, by Lakoff (1970:126)). The present analysis allows an interesting alternative description of this fact, in morphological

terms. Nouns like *agitation*, *annoyance*, *amusement*, and *surprise* are not nominalizations of the causative predicates *agitate*, *annoy*, *amuse*, and *surprise*. Rather, they are nominalizations of corresponding noncausative predicates. Thus, *annoyance* does not mean 'the process of making annoyed', but 'the state of being annoyed'. *Amusement* does not refer to something amusing someone, but to the state of being amused.

These nouns are not result nominals (which may lack argument structure), but argument-taking nouns (contra Abney (1987:126)). Their singular forms take modifiers typical of nonresult usage and may have the properties of mass nouns, resisting pluralization (see Grimshaw 1987:1991 for discussion of these and other diagnostic tests).

- (199) a. Bill's continual agitation about the exam was silly.
 b. Mary's constant annoyance about/at/with us got on our nerves.
 c. Despite Sue's frequent amusement at the goings-on, she was, deep down, quite upset.
 d. John's constant embarrassment about his looks was unnecessary.
 e. Notwithstanding our continual surprise at the news from the East . . .
- (200) a. Agitation about exams is silly.
 b. Annoyance at one's teacher should be suppressed.
 c. Amusement at the misfortunes of another is a sign of poor character.
- (201) a. *The agitations about exams are silly.
 b. *Annoyances at one's teacher should be suppressed.
 c. *Those amusements at Sue's mistakes were uncalled for.

Some nominalizations of this type do have uses that refer to objects (though not results). These usages are sharply distinct from those seen above and can be put aside.

- (202) a. The broken elevator was at best a minor annoyance.
 b. Sally didn't interfere with Bill's amusements.
 c. Talking to John was a real pleasure.

To all appearances, the nominalizations in (199)–(200) are true nominalizations of predicates meaning 'be agitated', 'be annoyed', and so on, or perhaps 'get agitated', 'get annoyed'. If these nouns are morphologi-

cally derived from morphemes pronounced *agitate*, *annoy*, *amuse*, and *surprise*, how can apparently related nominals have such noncausative semantics?⁶⁵ Our “neo-Lakovian” morphological hypothesis provides the answer. Noncausative derivatives of roots $\sqrt{\textit{amuse}}$ and $\sqrt{\textit{annoy}}$ are no surprise if these roots are noncausative themselves. On this analysis, we are simply dealing with bound roots found in two general environments—causative and nominalization. Notice also that hiding behind the cover term *nominalization* is a significant variety of nominalizing suffixes, including *-ance*, *-ment*, *-ion*, *-al*, and zero. Thus, as in the case of Latinate bound roots, there are a reasonable number of affixes that serve to produce actually occurring words from our bound SubjExp predicates.

If these bound SubjExp predicates actually exist, we must analyze them as belonging to the category V. This is because the nominalizing affixes that attach to these predicates are unambiguously deverbal. This in turn means that *CAUS* is a suffix that attaches to verbs and yields verbs—an observation that will be of great importance later in explaining the T/SM restriction.

In fact, there are a few *unbound* SubjExp verbs that allow affixation of *CAUS*, if the presence of the T/SM restriction is traceable (as I will ultimately suggest) to this zero affix. This class consists of *worry*, *puzzle*, and *grieve* (already seen in (36), (38), and (39), repeated here) and, in a slightly higher style, *delight* (Levin 1989:6).

- (203) a. John worried about the television set.
b. The television set worried John.
- (204) a. We puzzled over Sue’s remarks.
b. ‘Sue’s remarks puzzled us.
- (205) a. Sue grieved over/at the court decision.
b. The court decision grieved Sue.
- (206) a. Bill delighted in his new-found wealth.
b. His new-found wealth delighted Bill.

3.4.5 Evidence for Zero Morphemes: Myers’s Generalization

The facts about nominals related to causative ObjExp verbs are, of course, more striking than I have stressed so far. There is nothing optional about the absence of causative force. There are *no* causative nominalizations related to any of these verbs. Alongside structures of the form in (207a), we do not find structures that we would analyze as (207b).⁶⁶

- (207) a. [[$\sqrt{\text{SubjExp-predicate}}_V$] nominalizer]
 b. *[[[$\sqrt{\text{SubjExp-predicate}}_V$] $\emptyset_{CAUS}$] nominalizer]
- (208) a. *The exam's continual agitation of Bill was silly.
 b. *Our constant annoyance of Mary got on our nerves.
 c. *Despite the goings-on's frequent amusement of Sue, she was, deep down, quite upset.
 d. *His looks' constant embarrassment of John was unnecessary.
 e. *Notwithstanding the news's continual surprise of Sue, she kept her sense of humor.

In fact, if we posit not only a *CAUS* affix for *annoy*, but also a *SUG* affix for Higgins's examples, there are other facts of this type. Forms affixed with *SUG* also may not be nominalized.⁶⁷

- (209) a. [[$\sqrt{\text{SubjExp-predicate}}_A$] nominalizer]
 b. *[[[$\sqrt{\text{SubjExp-predicate}}_A$] $\emptyset_{SUG}$] nominalizer]
- (210) a. his pride
 b. His bearing was proud.
 *his bearing's pride
- (211) a. your anger
 b. Your remarks were angry.
 *your remarks' anger
- (212) a. her optimism
 b. Her expression was optimistic.
 *her expression's optimism
- (213) a. his irritability
 b. His comments were irritable.
 *his comments' irritability
- (214) a. his sadness
 b. His words were sad.
 *his words' sadness
- (215) a. her fearfulness
 b. Her behavior was fearful.
 *her behavior's fearfulness
- (216) a. her annoyance
 b. Her manner was annoyed.
 *her manner's annoyance

Finally, forms affixed with *SUG* also may not be causativized by *CAUS* or by an overt causativizer like *en-*.

- (217) a. $[[\sqrt{\text{SubjExp-predicate}}] \emptyset_{SUG}]$
 b. $*[[[\sqrt{\text{SubjExp-predicate}}] \emptyset_{SUG}] \emptyset_{CAUS}]$
- (218) a. Bill's remarks were angry.
 b. *These events angered Bill's remarks.
- (219) a. Sue's attitude was bitter.
 b. *Bill's remarks embittered Sue's attitude.
- (220) a. Bill's words were sad.
 b. *The recent events saddened Bill's words.
- (221) a. Tom's behavior was fearful.
 b. *The recent events frightened Tom's behavior.

The (b) examples of (217)–(221) may be irrelevant to the point at hand, since affixation of *SUG* yields an adjective, whereas the previous instances of *CAUS* have involved affixation to verbs. If (217)–(221) are subcategorization violations, they still argue for the existence of a causativizing process, but they do not form a part of the argument being developed in this section.

Consider the cases that are ruled out, under the analysis proposed here (with (222c) parenthesized, since it might be independently explained as just discussed).

- (222) a. $*[[[\sqrt{\text{SubjExp-predicate}}_V] \emptyset_{CAUS}] \text{nominalizer}]$
 b. $*[[[\sqrt{\text{SubjExp-predicate}}_A] \emptyset_{SUG}] \text{nominalizer}]$
 c. $(*[[[\sqrt{\text{SubjExp-predicate}}] \emptyset_{SUG}] \emptyset_{CAUS}])$

In each case, a hypothesized zero morpheme is followed by another derivational suffix. As Allen (1978:chap. 4) notes, such cases are at best rare; this leads her to posit a place for zero derivation late in a level-ordered morphology. In fact, in an interesting investigation of the combinatorial possibilities of zero-derived forms, Myers (1984) makes the strongest claim.

(223) *Myers's Generalization*

Zero-derived words do not permit the affixation of further derivational morphemes.

As evidence for (223), Myers notes that wherever a phonological string like *support* is assigned to two syntactic categories (here, verb and noun),

only one of its categorizations (here, the verb) allows the affixation of derivational morphemes. Thus, the verb *support* yields *supportive*, but the noun *support* yields nothing like **supportial_A* or **supportious*. Myers interprets these data as arguing that one member of the pair is always zero-derived from the root represented by the other member. The derived member shows the restriction against further derivational affixation. Myers's study is, unfortunately, flawed by an insufficient number of roots with more than one derivative. Since much derivational morphology is semiproductive in the first place, it is risky to draw conclusions from the absence of **supportial* in a list that contains mainly roots with one derivative.

Nonetheless, Myers's Generalization is worth pursuing further because of its interest in the present context. Under an analysis that posits null *CAUS* and *SUG* morphemes, the facts summarized in (222) are simply cases of this generalization.⁶⁸ In fact, if true, it can now be used as strong support for the existence of zero morphemes. As noted earlier, we should believe in the existence of zero morphemes in the *CAUS* and *SUG* examples to the extent that the characteristic behavior of zero morphemes is observed. If I am right, characteristic behavior is observed.

Finally, there are some systematic exceptions to Myers's Generalization. To the extent that the exceptions do exist, we will see that they apply uniformly across all null morphology including *CAUS* and *SUG*, and thus continue to support the proposed analysis. Equally important, these exceptions point the way toward an account of Myers's Generalization as a special case of broader observations about English morphology due to Fabb (1988).

3.4.6 Exceptions to Myers's Generalization: *-er* and *-able*

Among the derivational suffixes of English, there are at least two genuine exceptions to Myers's Generalization: the agentive nominalizer *-er* and the adjectivizer *-able*.⁶⁹ Thus, though by Myers's tests the verbs *accent*, *document*, *envy*, and *fracture* are denominal (since they generally fail to take deverbal affixes), the forms in (224) are grammatical, and the construction is productive.⁷⁰

(224) *accentable*, *documentable*, *enviable*, *fracturable*

Similarly, given the right context, the *-er* nominals in (225) are fine.⁷¹

(225) *accenter*, *documenter*, *envier*, *fracturer*

The presence of exceptions does not mean that Myers's Generalization is irredeemable. An electronic dictionary search, albeit informal and incomplete, suggests that *-er* and *-able* are the only true exceptions to the generalization. This type of investigation is important, since judgments about lexical matters have an uncertain status, given the listedness of many items and the pervasive (if mysterious) effects of blocking phenomena. Forms like **accetive*, **abuseous*, forbidden by Myers's Generalization, are not clearly worse than forms like **accentious*, **abusion*, which do not violate the generalization, but are presumably blocked by *accetual* and *abuse_N*.

The entries from A to C in a machine-readable version of *Webster's Seventh Collegiate Dictionary* were searched for noun-verb pairs with identical spelling. This section of the dictionary contains 783 such noun-verb pairs (from *abandon* to *cycle*), most of which are examples of zero derivation. Of these 783 headwords, 297 have derivatives listed as such (and of these, 92 have more than one listed derivative). Listing each headword and its derivatives on a single line, I searched for the intersection of lines that contained words ending in unambiguously denominal suffixes and lines that contained words ending in unambiguously deverbal suffixes (taking my lists from Fabb 1988). After irrelevant examples (nonderivatives, derivatives of derivatives) were thrown away, the only remaining lines in the output were lines involving *-able*, *-er*, and the stress-shifting deverbal noun *contract* (which I shall ignore for the present, returning to the issue below) discussed at the end of this chapter.

I also examined derivatives that have their own headword in the dictionary. Of the 783 examples of noun-verb pairs with identical spelling, approximately 418 have derivatives listed in the dictionary entry. Of these, 382 have more than one derivative. This datum is important; the majority of Myers's examples show only one derivative, making the segregation of the data into "deverbal affixes only" and "denominal affixes only" unconvincing. After a large number of erroneously listed words had been weeded out, the total number of derivatives was 1,545.⁷² Since the trapping of "derivatives" was based on orthography, derivatives that involve a spelling change (e.g., *acclaim/acclamation*) were not trapped.⁷³ Again, after discarding irrelevant examples, the only remaining lines in the output were lines involving *-able*, *-er*, or stress-shifting deverbal nouns like *contract*, mentioned above. I did not try to combine the data in the two sets, so counterexamples to Myers's Generalization may indeed be lurking there, but within these two reasonably large data sets it appears that—except for *-er* and *-able*—the generalization is correct. The data can be summarized as follows:

(226) *A–C entries in Webster’s Seventh Collegiate Dictionary*

Total N-V pairs	783
Have derivatives under headword	297
Have more than one derivative under headword	92
Have deverbal and denominal derivative	0 [except <i>-er</i> and <i>-able</i>]
Have separately listed derivatives	≈ 418
Have more than one derivative	≈ 382
Total number of derivatives	≈ 1545
Have deverbal and denominal derivative	0 [except <i>-er</i> and <i>-able</i> plus <i>contraction/contractual</i> (stress-shifting)]

I will return to the exceptional behavior of *-er* and *-able* in section 3.4.8.2. Notice here, however, that if these deverbal affixes are exceptions to Myers’s Generalization in the types of examples that Myers considered, then they should be exceptions to our extension of this generalization to ObjExp verbs with *CAUS*. In fact, this is true.⁷⁴

- (227) a. John is an annoyer of little children.
 b. Someone was finally able to amuse the prince. The amuser of the prince was none other than Bill.
 c. The world is divided into the upsetters and the upset.
 d. Loose false teeth is the leading embarrasser of heads of state.
- (228) a. I can’t help annoying John. He’s so annoyable!
 b. In general, Queen Victoria was not amuseable.
 c. In general, Bill is not upsettable.
 d. Embarrassing Sue is no big achievement. She’s embarrassable by any four-year-old.

Although these words generally have the status of nonce forms, requiring some context,⁷⁵ the contrast with truly impossible words like **accentive* is sharp.

3.4.7 Interim Conclusions and Extensions

This discussion has shown that causative ObjExp verbs are bimorphemic, containing a null causative morpheme *CAUS* affixed to a (usually) bound root.

I demonstrated first that the putative bound root can be licensed by affixes other than *CAUS*; nominalizing affixes of all sorts occur with these bound roots. The evidence for this was the noncausative character of the nominalizations. I argued next that the utter absence of nominalizations from *causative* ObjExp words is part of a larger phenomenon that includes the absence of nominalizations and causativizations from predicates affixed with *SUG*. If this phenomenon falls under Myers's broader generalization concerning zero affixation, then we have strong arguments for the presence of a zero affix with causative meaning in causative ObjExp constructions.

Nominalization restrictions due to Myers's Generalization are not limited to the non-unaccusative ObjExp verbs that we have analyzed as containing a null causative morpheme. Causative verbs like *grow* that have homophonous inchoative counterparts also cannot be nominalized.⁷⁶ Thus, the (c) examples in (229)–(236) can only be related to the (a) examples, and they lack any causative meaning. The (d) examples are therefore ungrammatical.⁷⁷

- (229) a. Tomatoes grow.
- b. Bill grows tomatoes.
- c. the growth of tomatoes
- d. *Bill's growth of tomatoes (Chomsky 1972:25)
- (230) a. The curtain dropped.
- b. The mechanism dropped the curtain.
- c. the drop of the curtain
- d. *the mechanism's drop of the curtain
- (231) a. The money returned.
- b. The thief returned the money.
- c. the return of the money
- d. *the thief's return of the money
- (232) a. His salary shrank.
- b. Inflation shrank his salary.
- c. the shrinkage of his salary
- d. *inflation's shrinkage of his salary
- (233) a. His salary diminished.
- b. Inflation diminished his salary.
- c. the diminishment of his salary
- d. *inflation's diminishment of his salary

- (234) a. The pendulum is swinging.
 b. Gravity is swinging the pendulum.
 c. the swing of the pendulum
 d. *gravity's swing of the pendulum
- (235) a. The string vibrated.
 b. The bow vibrated the string.
 c. the vibration of the string
 d. *the bow's vibration of the string
- (236) a. The activity ceased/stopped.
 b. Bill ceased/stopped the activity.
 c. the cessation/stoppage of the activity
 d. *Bill's cessation/stoppage of the activity⁷⁸

These facts can now be seen as examples of Myers's Generalization. They provide an argument that non-Experiencer causative verbs homophonous with their inchoative counterparts are analyzed by the language learner as bimorphemic, containing a null causative morpheme: $\sqrt{\text{grow}} + \text{CAUS}$, $\sqrt{\text{drop}} + \text{CAUS}$, $\sqrt{\text{return}} + \text{CAUS}$, and so on.⁷⁹

Interestingly, nominalizations that have no homophonous inchoative counterpart do not seem to show the restrictions just examined.⁸⁰

- (237) a. *Tomatoes cultivate.
 b. Bill cultivates tomatoes.
 c. the cultivation of tomatoes
 d. Bill's cultivation of tomatoes
- (238) a. *The town destroyed.
 b. The bomb destroyed the town.
 c. the destruction of the town
 d. the bomb's destruction of the town
- (239) a. *Herculaneum buried.
 b. The volcano buried Herculaneum.
 c. the fortuitous burial of Herculaneum
 d. the volcano's fortuitous burial of Herculaneum
- (240) a. *The monarchy restored.
 b. The emperor restored the monarchy.
 c. the restoration of the monarchy
 d. the emperor's restoration of the monarchy

- (241) a. *Controversy created.
 b. The proposal created controversy.
 c. the creation of controversy
 d. the proposal's creation of controversy
- (242) a. *The room illuminated.
 b. The sun illuminated the room.
 c. the illumination of the room
 d. the sun's illumination of the room
- (243) a. *The activity discontinued/suspended.
 b. Bill discontinued/suspended the activity.
 c. the discontinuation/suspension of the activity
 d. Bill's discontinuation/suspension of the activity

Suppose that the language learner evidently does not posit zero derivation when a non-Experiencer causative verb is *not* homophonous with any non-causative counterpart. Myers's Generalization is thus inapplicable to these causative verbs, since they contain no causative morpheme. Therefore, further affixation (e.g., nominalization) is fine. On the other hand, when there is an alternation between homophonous noncausative and causative verbs, the language learner posits a derivation of the latter from the former by means of a null causative affix.⁸¹ This instance of the null causative morpheme, like all instances of zero derivation, does not tolerate further affixation—hence the impossibility of phrases like Chomsky's **Bill's growth of tomatoes* and a causative reading of *the growth of tomatoes*.⁸²

There is now an asymmetry between causative ObjExp verbs and other causative verbs. Nonexperiencer causative verbs are analyzed with a null causative morpheme only under the circumstances described above. On the other hand, ObjExp verbs like *annoy* are always analyzed as containing a null causative morpheme, even though these verbs have no homophonous SubjExp counterpart. I return to this question in later chapters.

3.4.8 Myers's Generalization as a Special Case of Fabb's Generalization

3.4.8.1 Myers's Explanation Our discussion has assumed the existence of zero derivational morphemes. This runs counter to Myers's (1984) explanation of his generalization. In this section, I will argue against Myers's explanation and develop an alternative account. It is important to understand the reasons for Myers's Generalization in order to be confident that it tests for what it is claimed to.

Myers argues, contrary to what has been assumed here, that “zero derivation” is not the affixation of a phonologically zero morpheme. Instead, he argues that it results from the affixation of inflectional material characteristic of one category to a root belonging to another category. For Myers, the adjective *mortal*, for example, becomes a zero-derived noun simply by affixation of nominal inflection—for example, the plural suffix *-s*.

(244) [[*mortal*_A] *s*_N]

Although inflection in turn may be phonologically null (e.g., singular number on nouns in English), Myers posits no intermediate zero morpheme as the nominalizer: the inflection itself does the job. Since derivational morphology cannot be affixed to already inflected forms, an explanation is provided for Myers’s Generalization. As support for this view, Myers claims that zero derivation in English (unlike Icelandic) never forms adjectives. This result follows from Myers’s theory if the absence of number, tense, or gender morphology in the English adjective (in contrast to languages like Icelandic) entails the nonexistence of adjectival inflection. This consequence seems incorrect, and I return to it shortly.

The present discussion is incompatible with Myers’s proposal for two reasons, involving the existence of non-category-changing zero derivation and the nature of the counterexamples presented by *-er* and *-able*.

Consider first the phenomenon of non-category-changing zero derivation. If the proposed analysis of causative ObjExp verbs is correct, then the bound SubjExp root must be a *verb*. This is because the nominalizations of these SubjExp roots involve the repertoire of deverbal suffixes (*-ion*, *-ment*, etc.), and not the repertoire of deadjectival suffixes (*-ity*, *-ness*). Since *CAUS* forms verbs, we conclude that the grammar contains non-category-changing zero derivation.

If zero derivation involves the affixation of a phonologically zero derivational morpheme, there is nothing surprising here. Overt derivational morphemes also come in non-category-changing and category-changing varieties. In Myers’s theory, however, “non-category-changing zero derivation” is an incoherent notion, since it cannot be distinguished from normal inflection. Attaching verbal inflection for person, number, or tense to a verbal root cannot form causative *annoy* from noncausative $\sqrt{\text{annoy}}$. All it can produce is an inflected form of $\sqrt{\text{annoy}}$. Similarly, no kind of inflection can have the effect of deriving *angry-SUG* from *angry*. (Indeed, as noted earlier, Myers argues that there is no adjective-forming zero

derivation in English.) If *CAUS* and *SUG* exist, and if they obey Myers's Generalization, then Myers's explanation for his generalization must be wrong.

Next consider the affixes *-er* and *-able*. As we have seen, both of these affixes attach to verbs, including verbs zero-derived from nouns. Neither affix behaves like an inflectional morpheme: *-er* is followed by nominal inflection (the plural *-s*), and *-able* may be followed by the derivational affix *-ity*.⁸³ Myers's explanation for his generalization has as a consequence that exceptions like *-er* and *-able* cannot be logically accommodated at all. If verbs like *document_V* are just the result of attaching verbal inflection to a noun, then *-er* and *-able* must be postinflectional. But this is false: *-er* and *-able* attach not to inflected forms (**documented-able*, **documenting-er*), but to bare roots. I see no possible niche for these affixes in Myers's theory.

Finally, note that Myers's proposal does not forbid all phonologically zero morphology—only phonologically zero *derivational* morphology. Phonologically zero *inflection* must exist if the verb *document_V* in *They document atrocities* is held to be the consequence of adding verbal inflection to the noun *document_N*. By contrast, phonologically zero derivation cannot exist, if Myers's Generalization is to be explained in the manner he suggests. Let in just one instance of phonologically zero derivation, and we expect to find counterexamples of a sort that are unattested: classes of zero-derived forms that freely allow subsequent derivational affixation. (This is not the same situation found with *-er* and *-able*, where isolated affixes can freely attach to any zero-derived verb.) This peculiar asymmetry between inflection and derivation suggests that something is being missed.

3.4.8.2 Fabb's Observations This study has pursued a different idea entirely. I have assumed, contrary to Myers, that there are phonologically zero affixes of all sorts—derivational as well as inflectional. If this is the case, we should ask why zero morphemes followed by most derivational suffixes are deviant, and why zero morphemes followed by inflection, *-er*, and *-able* are not deviant. Before we search for an answer to this question, however, it should be clearly noted that we are under no logical compulsion to explain Myers's Generalization, if it is true and can be invoked in service of our search for zero morphology. However, the better we understand the reasons for the generalization, the more sure we can be of arguments that depend on it. Therefore, the discussion in this section will

involve a level of empirical and technical detail that is not fully warranted by its logical position in the general discussion, and the reader who wishes to skip it will not risk any major loss of continuity.

There are two possible approaches to Myers's Generalization. The first approach would assume that there is something special to be said about zero morphology. On this approach, we might draw a parallel between the "specialness" of zero derivation and the specialness of empty categories in sentence-level syntax. Such an approach would assume that phonologically zero morphemes must be licensed and would limit this licensing ability to inflection, *-er*, and *-able* structurally close to the zero morpheme. Thus, any word headed by a zero morpheme followed by a nonlicenser like *-ment* or *-ance* would violate the licensing requirements of the zero morpheme. Early in the research reported here, I pursued this path, but as the conditions involved seemed quite ad hoc, I will not present the results.

I will follow a second approach, which was suggested by Nigel Fabb (personal communication) and which follows directly from the results he reported in Fabb 1988. This approach explains Myers's Generalization, not as a property of zero affixation, but as a property of those morphemes that cannot attach to zero-derived forms.

Fabb was interested in the types of restrictions that limit combinations of (nonzero) suffixes in English. Fabb observed that if there were no restrictions whatsoever (not even categorial selection), the 43 most frequently occurring suffixes should give rise to 1,849 suffix pairs. In fact, however, there are categorial-selectional restrictions: that is, suffixes are subcategorized so as to attach only to forms of a particular syntactic category (e.g., *-ness* to adjectives, *-ion* to verbs). Once these categorial restrictions are taken into account, the number of possible combinations reduces to 663. Various phonological restrictions further lower this number to 614. This is still many more combinations than actually occur. In fact, there are "only about 50 attested pairs of suffixes" (p. 530). Thus, there are further restrictions. Fabb summarizes these restrictions as follows (the text is mostly Fabb's (p. 532); the group designations are mine).

- A. Many suffixes never attach to an already suffixed word.
- B. Some suffixes attach outside only one other particular suffix.
- C. Some suffixes attach outside some but not all of the suffixes that we would expect.
- D. Some suffixes attach to all the suffixes that we would expect, given categorial-selectional (subcategorization) restrictions.

The thrust of Fabb's article was to show that the restrictions in A–D make Allen's (1978) hypothesis of Level Ordering unnecessary. Here, however, it is the restrictions themselves that are of primary interest. Among the morphemes in group A are the abstract-noun-forming suffixes *-ance* and *-ment*. Suppose the structure of occurring noncausative and nonoccurring causative uses of *annoyance* and *amusement* is as in (245) and (246) (cf. (207)).

- (245) a. $[[\sqrt{\text{annoy}}_v] \text{ance}_N]$
 b. $*[[[\sqrt{\text{annoy}}_v] \text{CAUS}_v] \text{ance}_N]$
- (246) a. $[[\sqrt{\text{amuse}}_v] \text{ment}_N]$
 b. $*[[[\sqrt{\text{amuse}}_v] \text{CAUS}_v] \text{ment}_N]$

If these structures are correct, then the impossibility of the causative (b) examples is explained as an instance of the attachment of a group A suffix to an already suffixed form.⁸⁴

Now let us look at nominalizations in *-ion*. Unlike *-ment* and *-ance*, this suffix may attach to previously suffixed bases, but the base must be Latinate (p. 536), a familiar observation that places these suffixes in group C.⁸⁵ Consider examples like noncausative and causative uses of *agitation* and *depression*.

- (247) a. $[[\sqrt{\text{agitat}}_v] \text{ion}_N]$
 b. $*[[[\sqrt{\text{agitat}}_v] \text{CAUS}_v] \text{ion}_N]$
- (248) a. $[[\sqrt{\text{depress}}_v] \text{ion}_N]$
 b. $*[[[\sqrt{\text{depress}}_v] \text{CAUS}_v] \text{ion}_N]$

If *CAUS* is [–Latinate], then the impossibility of the causative examples is once again explained using Fabb's criteria. Conceivably, if *CAUS* is a resource of UG rather than a morpheme particular to any language, it is incapable of being positively specified as belonging to a particular sub-domain of the morphology. This would account for its status as [–Latinate].

Certain unproductive nominalizers relevant to the present discussion are not treated by Fabb. The nominalizers *-ure* (*pleasure*, *discomfort*) and *-dom* (*boredom*) seem to fall in group A, and behave like *-ment* and *-ance*. Similarly with *-th* (*growth*, discussed above). Finally, a number of ObjExp psychological predicates are related to nominals (as well as causatives) via zero derivation. For example:

- (249) awe, charm, cheer, comfort, dismay, pain, pique, quiet, shame, surprise, worry

Suppose the verbal uses of these forms are basic, the nominal uses being derived from the verbal uses by a nominalizer *-NOM*. If *-NOM* belongs to group A, then we may rule out causative uses of nominals like *surprise* the same way we ruled out causative uses of nominals in *-ment* and *-ance*—as an illegal attachment of *-NOM* to a previously suffixed form. If instead the nominal uses of the forms in (249) are basic, the verbal uses being derived from an underlying noncausative noun (e.g., by affixation of a verbalizer), then *-NOM* will still be needed to derive a causative noun, and the assumption that *-NOM* is in group A will once again rule out causative uses of nouns like *surprise*.⁸⁶

As noted earlier, this discussion has a very different cast from Myers's. Myers's approach posits that suffixation after zero derivation is impossible. The approach that I am proposing simply places most suffixes in a category that prevents affixation to a zero-derived form.

Let us restrict discussion for the moment to nominalizations of verbs formed with *CAUS*. Fabb's approach makes a weaker prediction than Myers's. If the appropriate properties are assigned to *CAUS*, then no suffix in group A, B, or C should attach to such a verb. We have seen examples of this for groups A and C. (There are no relevant affixes in group B.) On the other hand, an affix from group D should be able to attach to such a verb, since these affixes are restricted only by categorial selection for the familiar syntactic categories. Fabb lists only two deverbal suffixes in group D: *-able* and *-er*. These are, of course, the only two noun-forming suffixes that can attach to forms that we have analyzed with *CAUS*. Their special behavior in the domain that we have been examining thus constitutes important support for the general approach proposed here.

Nonetheless, there is a problem. Fabb lists one (and only one) other suffix in group D: deadjectival *-ness*. If ungrammatical words that violate Myers's Generalization are actually cases in which morphemes of groups A, B, and C are attached outside a zero morpheme, then there should be no problem adding *-ness*, a group D morpheme, to a zero-derived form. In fact, this seems to be false in the two instances where it can be tested.

Consider first category-changing, adjective-forming zero derivation. This is quite rare: English nouns and verbs simply cannot be made into zero-derived adjectives with the same ease that nouns are made into verbs and verbs into nouns. Thus, we do not find phrases like an *[[accent]_N]_A* *SYSTEM* with main stress on *system* signaling it as an instance of the sequence *Det Adj N*, nor do we find *support SYSTEM* with main stress on *system*. We do, however, find toponyms and organization names bearing what appears to be adjectival stress, with adjectival meaning.

- (250) a. the ²Boston ¹SUBWAY
 b. a ²KGB ¹MOLE
 c. an Oxford ²SNOB

This usage is impossible with nation or state names that have related adjectives with the same interpretation.

- (251) a. the Lòndon police
 b. *the Britain police
 c. the British police
- (252) a. the U.S. ecónomy
 b. *the Amèrica ecónomy
 c. the Amèrican ecónomy
- (253) a. Sòviet inténctions
 b. *Rùssia inténctions
 c. Rùssian inténctions
- (254) a. Connècticut réstaurants
 b. *Hawàii réstaurants
 c. Hawàiiian réstaurants
- (255) a. Geòrgia wáter [only the U.S. state]
 b. Geòrgian wáter [only the country in the Caucasus]

These data suggest that *Boston*, *KGB*, and *Oxford* in (250) are indeed adjectives zero-derived from nouns, since the existence of other adjectives blocks the zero derivation.⁸⁷ (I presume that blocking is category-bound.) If this conclusion is correct, then *London*, *U.S.*, *Soviet*, *Connecticut*, and *Georgia* in (251)–(255) are adjectives zero-derived from nouns. These adjectives (in contrast to other ethnic adjectives) do not allow further affixation of *-ness*.⁸⁸

- (256) a. *USness
 b. *Sovietness
 c. *Connecticutness
 d. *Georgianess [U.S. state]
- (257) a. ?Americanness
 b. ?Russianness
 c. ?Hawaiianness
 d. ?Georgianness [country in the Caucasus]

Finally, of course, I have also posited an instance of non-category-changing adjective-forming zero derivation in the form of *SUG*. If I am

correct about its existence, *SUG* is phonologically null, attaches to adjectives, and forms adjectives. Here too, it is crucial that *-ness* be prevented from attaching to forms in *SUG*, given the impossibility of expressions like (214) and (215), reproduced here, with the structure in (258c).

- (258) a. *his words' sadness
 b. *her behavior's fearfulness
 c. *[[[sad_A] *SUG*_A] ness_N]

If Myers's Generalization is to be explained as a consequence of combinatory generalizations of the sort discovered by Fabb, then *-ness* must be removed from group D and restricted so as not to apply to forms headed by the zero morphemes seen above. Perhaps this leaves us where we began, without a means of linking Myers's Generalization—with its exceptions for *-er* and *-able*—to other known facts. On the other hand, the fact that *-er* and *-able* are the unique deverbal morphemes in group D and the unique exceptions to Myers's Generalization remains a striking observation, which should encourage us to pursue this line of thinking further, despite the problem observed with deadjectival *-ness*.⁸⁹

We can make some sense of this problem if we refine our view of Fabb's discoveries a bit, and slightly enlarge the range of possibilities. At this point, the reader uninterested in the solution to this problem may skip to the next chapter. Nothing that follows affects the logical flow of the rest of the book. This is simply an attempt to ground Myers's Generalization in a version of Fabb's theory that has the correct empirical consequences. Ultimately, this attempt helps shore up the book as a whole, but it does not yield any particular insights.

Fabb's group A contains a large set of morphemes that do not attach to previously suffixed forms. The remaining groups of morphemes also fail in general to attach to previously suffixed forms, but these morphemes treat certain suffixes as *opaque* for the purposes of this restriction. When a morpheme α attaches to a form whose outermost morpheme β is opaque, α cannot see any brackets internal to β (hence the metaphor of opacity). Thus, α cannot tell that it is attaching to a derived form.

Assume that all morphemes bear a unique *identifying mark*—but if and only if they are phonologically realized. This makes sense as a design feature of the system. Except in the case of homophonous forms, the identifying mark of a morpheme is its phonological form (presumably an underlying representation), plus any other features relevant to phonology, like [+Latinate]. When two nonzero morphemes belonging to the same

syntactic category are homophonous, the identifying mark may be supplemented by an integer or some other indicator that maintains the uniqueness of the mark. By making the identifying mark parasitic on the presence of phonological features, we prevent phonologically zero morphemes from bearing an identifying mark (and prevent them from bearing features like [+Linate]).⁹⁰ Now let us organize Fabb's data using the following system:

(259) *A formalism for Fabb's system: Morphological opacity*

- a. A suffix β may attach to a form headed by a suffix α only if α is opaque to β .
- b. Suffix α is *opaque* to suffix β for (259a) iff α satisfies the *opacity index* of β .
- c. The *opacity index* of a morpheme β is
 - i. an identifying mark or variable over identifying marks (e.g., [+Linate] or a wildcard [*]), or
 - ii. a syntactic feature (e.g., N, V, A).

Let us now look at the types of suffixes found in English, to see how this works.

- Group A suffixes have no opacity index. Thus, a suffix in this group never attaches to an already suffixed word.
- The opacity index of a group B suffix contains the identifying mark of the suffix to which it may attach; consequently, these suffixes may attach to unsuffixed forms or to forms suffixed with the mentioned morpheme. A group B suffix (Fabb 1988:534) is *-ic*, which attaches to forms suffixed with *-ist* (*modernistic*, *capitalistic*, *violinistic*), but not to forms suffixed with other morphemes (**conductoric*). Examples like *balsamic* and *idiotic* do not pose a problem, since they are unsuffixed. Since zero morphemes lack an identifying mark, no group B suffix will specify a zero morpheme as a suffix to which it can attach.
- The opacity index of a group C suffix mentions a feature like [+Linate], a property of identifying marks; consequently, such a suffix may attach to unsuffixed forms or to forms suffixed with a [+Linate] morpheme. An example of a group C suffix (Fabb 1988:536) is *-ize*.⁹¹ In *industrialize*, for example, deadjectival *-ize* may attach to the suffixed form *industri-al* because the opacity index of *-ize* is [+Linate]. *Pasteurize* and *Thatcherize* are acceptable, even though their roots are (presumably) not Linate, since *Pasteur* and *Thatcher* are nonsuffixed. Once again, since the feature [+Linate] is a property of identifying marks, no zero

morpheme can be marked [+Linate]. Therefore, no group C suffix can attach to a [+Linate] suffix.

- The opacity index of *-ness* is the wildcard over identifying marks, [*]. Thus, *-ness* can attach to any suffixed form, so long as that form is headed by a phonologically realized morpheme. Unlike *-er* and *-able*, then, *-ness* cannot attach to zero-derived forms. Since opacity indices are identifying marks, including the wildcard, we expect to find morphemes that attach to suffixed forms only if the suffix is phonologically realized (like *-ness*), but we do not expect to find morphemes that attach to suffixed forms only if the suffix is zero. This is probably correct, though I have not investigated the matter seriously.⁹²
- The opacity index of suffixes *-er* and *-able* from Fabb's group D consists of the syntactic features for V. Thus, these suffixes can attach to any verbal base, even those with suffixes.

Inflectional affixes might be regarded as subject to (259a), in which case they must be treated on a par with *-er* and *-able*. Alternatively, we might restrict (259a) to derivational suffixes (as Fabb does), removing inflection from the picture altogether.

The zero affixes *SUG* and *CAUS* can themselves be classified. Probably, *CAUS* is also restriction-free or nearly so, if we analyze alternations like the one in (260) as shown.⁹³

- (260) a. The Soviet Union [industrialize]ed.
 b. Stalin [industrialize]ed the Soviet Union.

SUG, on the other hand, may be restricted in the way *-ness* is restricted. Although *SUG* may attach to morphologically complex adjectives (e.g., *Tom's attitude was fearful*), SubjExp adjectives derived from past participles resist this attachment.

- (261) a. *Bill's remarks were pleased.
 b. *Sue's behavior was agitated.
 c. *Mary's expression was impressed.
 d. *Bill's words were depressed.
 e. *Tom's attitude was frightened.

This can be explained if these adjectives are derived from passive verbs by the addition of a phonologically null adjectivizer.

- (262) [[please]_v ed]_v ∅_A (*-SUG)

In section 4.1.3.4, I will argue on independent grounds for this analysis. Interestingly, the same contrast is found with the overt adverbializing suffix *-ly*, suggesting that this suffix, too, belongs in a class with *-ness*.

- (263) a. proudly
 b. angrily
 c. nervously
 d. optimistically
 e. sadly
 f. fearfully
 g. eagerly
- (264) a. *pleasedly
 b. *agitatedly
 c. *impressedly
 d. *depressedly
 e. *frightenedly

There is one final empirical problem with the analysis. Consider verb-noun pairs like those in (265), taken with a few changes from Marchand 1969 via Myers 1984.

- (265) *abstráct_V-ábstract_N, affíx_V-áffix_N, augmént_V-aúgment_N,
 compóund_V-cómound_N, compréss_V-cómpress_N,
 cóncert_V-cóncert_N, cóndúct_V-cónduct_N*⁹⁴

In many of these cases, both noun and verb allow affixes that should be impossible if zero derivation is involved. For example: deverbal *contraction* alongside denominal *contractual*; deverbal *objective* alongside denominal *objectual*; deverbal *rebellion* alongside denominal *rebellious*; deverbal *áffix* alongside denominal *affixal* (in linguistics jargon). In fact, as discussed by Kiparsky (1982), the nouns in such stress-shifting pairs appear to be deverbal, yet may undergo further zero derivation into verbs. Thus, alongside *protést_V* and *prótest_N* there is a verb *prótest_V*; alongside *abstráct_V* (or *abstráct_A*) and *ábstract_N* there is the verb *ábstract* 'to write an abstract of'; alongside *affíx_V* and *áffix_N* there is the linguist's verb *áffix_V*. As Kiparsky notes, the denominal derivation of the initially stressed verbs is clear from the semantics: *protést* may be predicated of children, but *prótest* only of demonstrators. Likewise, Martin Luther's theses were *affixed* to the cathedral door, but only a morpheme may be *áffixed*. Although these examples appear at first glance to be exceptions to Myers's

Generalization, they actually fare quite well under a theory in which Myers's conclusions are not subsumed by Fabb's observations, but they present problems if we follow this section in reducing Myers's facts to Fabb's.⁹⁵

If Myers's Generalization (with exceptions for *-er* and *-able*) is not explained in terms of Fabb's categories, then these forms need not be seen as problems. These may be instances of derivation, but they are not necessarily instances of *zero* derivation. The deverbal noun-forming morpheme seen in (265) contains those phonological features that govern stress retraction (perhaps an instruction to make the final syllable extrametrical; see Hayes 1981). It is a morpheme that does effect a phonological change and thus may be categorized with the nonzero morphemes in having an identifying mark. (In this it differs from the morpheme that derives, for example, *commánd_N* from *commánd_V*.) Thus, there is no reason for the stress-shifting nominalizer to be relevant to Myers's Generalization, and indeed it is not.

Stress-shifting $V \rightarrow N$ derivation poses problems for an approach based on Fabb's proposals, however. On this approach, the zero-ness of zero derivation plays no particular role. The denominal *-(u)al* of *contractual* is a group A suffix, which (except for the cases at hand) cannot be attached to previously suffixed forms. We might argue, in this context, that stress-shifting $V \rightarrow N$ derivation does not involve addition of any suffix at all. Instead, perhaps, it is an instance of nonconcatenative morphology and is for this reason opaque to the requirements of group A morphology.

Here too, there is a problem. Kiparsky (1982) notes the nonexistence of zero-derived nouns formed from affixed verbs (**a publicize*, **a demonstrate*, **a clarify*). This can be accounted for straightforwardly if $V \rightarrow N$ zero derivation is itself an instance of affixation of a group A suffix. If the $V \rightarrow N$ suffix in question does not count as a suffix for the purposes of further affixation, it surely counts as a suffix when it itself is affixed. I leave these difficulties as unresolved questions.

Let us review what has been achieved and what has not been achieved here. I have suggested that the phenomena covered by Myers's Generalization are part of a more general picture identified by Fabb. I proposed a view of Fabb's discoveries according to which group A behavior is the basic case, complex affixation being the result of exception clauses stated in terms of positive feature specifications. This allows Myers's Generalization, even with the unexpected behavior of *-er*, *-able*, and (in a different

way) *-ness*, to be explained in terms of the parameters by which English suffixes vary even when no zero derivation is involved.

None of this discussion explains why group A behavior should exist in the first place, or why there should be such a phenomenon as opacity. I will leave the matter here, however. I am content, for now, with the more modest achievement of explaining an odd and restricted phenomenon in terms of an odd and less restricted phenomenon. This alone constitutes a step forward, although it is not a situation that should satisfy us indefinitely. In any case, by fitting the true cases of Myers's Generalization into a broader picture, we can assure ourselves that the behavior of ObjExp verbs and Higgins's adjectives under nominalization does diagnose the presence of a zero morpheme, and we can proceed to use that discovery in service of our ultimate goal.

Chapter 4

A Meteorology of Emotions: What *CAUS* Attaches To

4.1 The Nature of $\sqrt{\textit{amuse}}$

Let us now return to the causative psychological predicates with which we began. In analyzing psychological predicates, I have posited a large class of verbs that do not occur independently, such as $\sqrt{\textit{annoy}}$ and $\sqrt{\textit{amuse}}$. I justified the existence of these bound roots in two ways. First, I noted that they also appear to be the semantic basis for nominalizations like *annoyance* and *amusement*. Second, the evidence for *CAUS* in verbs like *annoy* raised the question of what the noncausative part could be: the only candidate was $\sqrt{\textit{annoy}}$ ‘be annoyed’ or ‘become annoyed’ which would combine with *CAUS* and with nominalizers, but not occur by itself.

These bound morphemes rightly raise eyebrows. With a few exceptions like *worry*, the roots behind almost all the semantically causative ObjExp verbs of English are bound. Clearly, some explanations are in order.

This predicament is all the more worrisome because the proposed analyses apparently should extend to structurally similar languages like Italian, French, and Russian. ObjExp verbs in these languages behave much as their counterparts do in English. The status of a morpheme as bound or free does not typically display any cross-linguistic constancy (unless borrowing is at stake, which is not systematically the case here). In this chapter, I attempt to lay these worries to rest by providing a reason for the bound status of $\sqrt{\textit{annoy}}$ and similar roots in English. This discussion will involve a consideration of reflexive counterparts to $\sqrt{\textit{annoy}}$ in other languages and will lead ultimately to a solution for the unaccusative paradoxes of ObjExp predicates as well as to a solution for the T/SM problem.

4.1.1 Gaps in the English Lexicon

Let us begin with a simple observation. The usable vocabulary of non-bound SubjExp verbs in English is startlingly small. This fact stands on its own, entirely apart from any analyses I have put forward in this book. The following two classes almost exhaust the possibilities:

1. There are a few SubjExp verbs such as *like*, *love*, *adore*, *admire*, *hate*, *loathe*, *detest*, *care* that involve nothing besides *evaluation* of a Target of Emotion. If Sue is angry at Mary, she evaluates her negatively, but also blames her. If John is afraid of an object, he evaluates it negatively, but also has a more complex attitude toward it, involving desire for avoidance, suspicions of possible harm, and so on. By contrast, if Sue likes Mary, she merely evaluates her positively, and, if she hates Mary, she merely evaluates her negatively. The differences among verbs such as *like* and *hate* seem to be merely differences of degree (*like* vs. *love* vs. *adore*) and differences that reflect the attribute of the object being evaluated (*like* vs. *admire*).
2. There are a few SubjExp verbs like *worry*, *grieve*, *delight*, *puzzle* (*over*) (see (203)–(206)) that can be affixed with *CAUS*; and others (*fret*, *mourn*, *rage*, *enjoy*) that cannot (**The events fretted Mary*). These verbs denote emotions that are also *activities*. Unlike *like* and *love*, they are not stative (*#John was liking the picture*; *John was worrying about the picture*).⁹⁶

As far as I can tell, there are no freely occurring nonactivity SubjExp verbs involving emotions like anger, annoyance, and satisfaction. No verb in English means ‘be pleased’, ‘be sad’, ‘be amused’ or their inchoative counterparts ‘become pleased’ ‘become sad’, ‘become amused’. In their place, we find adjectives, to which *CAUS* does not attach.⁹⁷

I have already remarked that the gap in the inventory of freely occurring Experiencer verbs is logically independent of my analysis of causative Experiencer predicates. This means that, even if the analyses presented here are wrong in every detail, there is still a peculiarity in the distribution of English verbs of emotion that requires an explanation. To be sure, my analysis imposes a particular character on the gap: the verbal roots are not missing from the language, they are merely bound morphemes. But at least the mere existence of the gap cannot be counted as a demerit of the present account of Experiencer predicates.

4.1.2 Reflexive SubjExp Verbs

Now let us examine some other languages.⁹⁸ At first sight, French displays a pattern reminiscent of English. A T/SM restriction is found with the verbs of the *amoy* class,⁹⁹ and similar restrictions hold on nominalizations. It is thus tempting to provide a similar analysis. The semantically causative verb *étonner* 'amaze', for example, will be analyzed as a SubjExp root $\sqrt{\text{étonner}}$ ¹⁰⁰ 'be amazed' plus a null causative affix *CAUS*. As with English roots like $\sqrt{\text{amaze}}$, $\sqrt{\text{étonner}}$ does not occur independently, unlike *aimer* 'love'.

- (266) a. Le bruit étonne Marie.
 the noise amazes Marie
 b. *Marie étonne (du) bruit.
 Marie is amazed at the noise
 c. Marie aime le bruit.
 Marie loves the noise

We might thus be tempted to conclude that French also resembles English in assigning bound status to SubjExp roots like $\sqrt{\text{étonner}}$. This conclusion would not be entirely correct. As noted by Ruwet (1972:201ff.), many verbs like causative *étonner* have SubjExp counterparts with the reflexive clitic *se*.

- (267) Marie s'étonne du bruit qu'on fait sur cette histoire.
 Marie refl-amazes at the fuss that one makes about this story
 'Marie is amazed at the fuss made about this story.'

The semantics of this construction are noncausative, with the object of the preposition bearing either Target or Subject Matter role. Some evidence for this comes from Ruwet's observation (p. 20, (86), (88)) that the subject in ObjExp (266) and the object in SubjExp (267) constructions show different selectional restrictions.

- (268) $\left\{ \begin{array}{l} \text{Paul} \\ \text{Cette table} \\ \text{Le bruit qu'on fait sur cette histoire} \\ \text{Que Jules soit sorti} \end{array} \right\} \text{ étonne Marie.}$
- $\left\{ \begin{array}{l} \text{Paul} \\ \text{this table} \\ \text{the fuss made about this sotry} \\ \text{that Jules left} \end{array} \right\} \text{ amazes Marie}$

- (269) Marie s'étonne $\left\{ \begin{array}{l} *de \text{ Paul} \\ *de \text{ cette table} \\ \text{du bruit qu'on fait sur cette histoire} \\ (\text{de ce}) \text{ que Jules soit sorti} \end{array} \right\}.$
- Marie refl-amazes $\left\{ \begin{array}{l} \text{Paul} \\ \text{of this table} \\ \text{the fuss made about this story} \\ \text{that Jules left} \end{array} \right\}$

Evidently the object argument of *s'étonner* must be abstract, whereas no such restriction is imposed on the subject of *étonner*. Lea Nash (personal communication) notes very similar nominalization and T/SM data from Russian (as also noted by Merzon and Pjatetskaja (1983:107–58)), similar semantic contrasts in causativity, and similar selectional restrictions.

- (270) $\left\{ \begin{array}{l} \text{Èta kniga} \\ \text{Ira} \\ \text{Èë povedenie} \\ \text{Èë postupok} \end{array} \right\}$ udivljaet Ivana.
- $\left\{ \begin{array}{l} \text{this book} \\ \text{Ira} \\ \text{her conduct} \\ \text{her action} \end{array} \right\}$ -NOM surprises Ivan-ACC
- (271) Ivan udivljaet-sja $\left\{ \begin{array}{l} *ètoj knige \\ *Ire \\ èë povedeniju \\ èë postupke \end{array} \right\}.$ ¹⁰¹
- Ivan-NOM surprise-refl at $\left\{ \begin{array}{l} \text{his book} \\ \text{èë} \\ \text{her conduct} \\ \text{her action} \end{array} \right\}$ -DAT

These data suggest that many of the SubjExp roots that I postulate for semantically causative verbs like *étonner* and *udivljaet'-sja* (the infinitive from *udivljaet-sja*) do occur, but are specified as inherently reflexive. I will return shortly to a possible explanation for this instance of “inherent reflexivity.”

Reflexiva tantum (verbs used only reflexively) are found in French and Russian, for example, *s'évanouir* (**évanouir*) ‘faint, vanish’, *stremit'-sja*

'strive'. This leads to the perhaps surprising conclusion that the direction of derivation for pairs like *s'étonner/étonner*, *udivljat'-sja/udivljat'* proceeds from reflexive to nonreflexive. That is, the nonreflexive verb is the zero-derived causative of the reflexive verb.¹⁰² Obviously, if this proposal is correct, we must explain the disappearance of the reflexive morpheme under causativization.

Like any zero morpheme, the causative morpheme *CAUS* is suspect precisely because it is phonologically null. Since it is null, it can be detected only indirectly. If among the properties of this mysterious zero morpheme is the mysterious disappearance of an otherwise obligatory reflexive morpheme, we certainly hope that this property is mirrored by instances of nonzero derivation. In fact, it is. As alluded to above, the meaning of nominalizations formed from Experiencer stems in French and Russian indicates, as in English, that they are derived only from the noncausative, reflexive forms. Thus, the nominalizations *étonnement* and *udivlenie* have only the noncausative meanings and uses of English nominalizations like *amazement*. This follows as an instance of Myers's Generalization if causative *étonner* and *udivljat'* involve affixation of *CAUS* to *s'étonner* and *udivljat'-sja*. Furthermore, zero morphemes must behave in French and Russian much as they do in English with respect to Fabb's Generalization.¹⁰³

Remarkably, no harm seems to result from the disappearance of the reflexive pronoun in the nominalization. Nothing like **s'étonnement* or **udivlenie-sja* is required or possible. To be sure, we do not expect these forms to occur, since the reflexive has no place to lodge itself inside DP, for morphological or perhaps syntactic reasons. Clitic pronouns attach to Infl and, perhaps, to V. Nonetheless, this is not the end of the story. The absence of a landing site for *se/sja* might have had the consequence that *reflexiva tantum* would lack any nominalization. But there is no such consequence. *Reflexiva tantum* have nominalizations; these nominalizations simply lack the otherwise obligatory overt reflexive.

The disappearance of the reflexive is not a fact unique to Experiencer verbs, but is the general pattern: *reflexiva tantum* like *s'évanouir* and *stremit'-sja* yield nonreflexive nominalizations like *évanouissement* 'fainting fit' and *stremlenie* 'striving'. The point of these examples should be clear: the disappearance of the reflexive under nominalization looks like the same phenomenon that we must posit for causativization, and it provides an important precedent. As a stopgap, let us suppose that *reflexiva*

tantum bear the feature [+ reflexive], which is expressed by the reflexive morpheme. Then we may entertain (272) as an accurate description of the facts under discussion.

- (272) If α bears the feature [+ reflexive], this feature is expressed with a reflexive pronoun iff no derivational morphemes have been attached to α .

I will turn shortly to an explanation for (272).

The disappearance of an obligatory reflexive morpheme under causativization has another precedent—in the behavior of Italian periphrastic causatives. Following Burzio (1981), Zubizarreta (1985) notes the following paradigms (which I have somewhat rearranged):

- (273) a. Le nubi *(si) dissipano.
the clouds refl dissipate
'The clouds dissipate.'
b. Il vento dissipa le nubi.
the wind dissipates the clouds
c. Il vento ha fatto dissipare/*dissiparsi le nubi.
the wind made dissipate/*dissipate-refl the clouds
- (274) a. Le foglie *(si) disperdono.
the leaves refl scatter
b. Il vento disperde le foglie.
the wind scatters the leaves
c. Il vento ha fatto disperdere/*disperdersi le foglie.
- (275) a. La candela *(si) spegne.
the candle refl goes-out
b. Il vento spegne la candela.
the wind puts-out the candle
c. Il vento ha fatto spegnere/*spegnersi la candela.

These examples show that when the noncausative verbs of (273)–(275) are embedded under an overt causative *fare*, the reflexive morpheme is suppressed, just as it is under nominalization and (if I am correct) affixation with *CAUS*. Belletti and Rizzi (1988:304–8) note the same phenomena with Experiencer predicates.

- (276) a. Il presidente si entusiasma (per la partenza dei Marines).
the president refl excites at the departure of the Marines
'The president gets excited at the departure of the Marines.'

- b. Questo ha entusiasmato il presidente.
 this excited the president
- c. Questo ha fatto entusiasmare/*entusiasmarsi il presidente per
 la partenza dei Marines.

Zubizarreta (1985) argues that even in periphrastic causatives like the (c) examples of (273)–(276), the matrix causative verb behaves in some fashion as an affix. It would be tempting in light of this suggestion to relate the *reflexive drop* observed above to (272). In turn, this might shed light on the explanation for (272); but I have not investigated these matters further. *Fare* in (276) would have to be affix enough to trigger (272), but not affix enough to trigger the T/SM effect, given the acceptability of the Subject Matter argument *per la partenza dei Marines* in (276c). There are other difficulties: as Burzio (1981, 1986:409) observes, French and Spanish do not in general drop the reflexive clitic *se* in periphrastic causatives (with certain exceptions noted by Ruwet (1972:108)). The French and Spanish causative constructions must therefore be analyzed as “less morphological” than their Italian counterpart (though possibly still involving some sort of verb incorporation, as argued by Den Dikken (1990) for French), perhaps in the spirit of Zubizarreta’s proposals, which I will not examine here.

In any case, if there is a phenomenon of reflexive drop under morphological derivation, then we are free to replace the mysterious bound status of many postulated SubjExp roots in our description with an equally mysterious, but distinct, hypothesis: these SubjExp roots bear the feature [+ reflexive]. To be sure, some residue of pure bound-ness must remain, in order to accommodate a sprinkling of cases of SubjExp *causativa tantum* in languages like French and Russian. For example, Ruwet (1972:202) notes that the semantically causative *gêner* in the meaning ‘to embarrass’ has no corresponding inchoative **se gêner* ‘become embarrassed’; similarly with *tenter* in the sense ‘allure’. Likewise, not every [+ reflexive] SubjExp verb in languages of this type can legally be affixed with *CAUS*. For example, alongside the Russian *reflexivum tantum gordit’-sja* ‘be proud’ there is no causative verb **gordit’* ‘make proud’.¹⁰⁴ Nonetheless, the analysis of reflexive-causative alternations in languages like French and Russian that I have sketched throws the problem for these languages into a ballpark much more familiar than the one it previously occupied.

We still need an explanation for the prevalence of *reflexiva tantum* SubjExp predicates—but, as we will see, this is a much more tractable

and interesting problem to face than the problem of simple bound forms. We still need to posit some SubjExp bound forms, but a sprinkling of language-specific bound morphemes (as opposed to a cross-linguistic downpour) is not a novelty in morphology. Finally, we need to posit some SubjExp forms that do not causativize, but semiproductivity is also not a novelty (though poorly understood, particularly from an acquisition standpoint).

The problems that we have eased for languages like French and Russian are not, however, eased for English. In English, as things stand, there still remains a true gap among verbs. Though there are, on the proposed analysis, many SubjExp roots that predicate emotional states of their Experiencer arguments, most of these roots are truly bound: they do not occur independently. Now, however, we can see the glimmerings of an explanation, in light of languages like French. Call the unknown reason why many or most French SubjExp verbs are [+reflexive] *fact R*. Suppose that fact R—and indeed the phenomenon of reflexive drop—is also a fact about English. Suppose further that the expression of the feature [+reflexive] is impossible in languages that do not have a reflexive clitic. English is one such language (ignoring the few obligatorily reflexive verbs in English; see note 104).¹⁰⁵ Then a language like English will be unable to realize [+reflexive] SubjExp verbs unless these verbs undergo derivational morphology. The bound property of these verbs will therefore be explained.

Of course, this line of reasoning hinges upon two puzzles that are so far unsolved: fact R and the reasons for reflexive drop. In the next section, I turn to these puzzles.

4.1.3 Why Is $\sqrt{\text{annoy}}$ an Inherent Reflexive?

4.1.3.1 Reflexive Clitics Reflexive clitics in languages like French are found in a variety of constructions, including configurations in which they function as a particular argument of a predicate.

- (277) Marie se voit.
 Marie refl sees
 'Marie sees herself.'

Interestingly, at least in French and Italian, these constructions, though they involve transitive verbs with an external argument, display the auxiliary 'be' that otherwise signals unaccusative advancement of the underlying object to subject position.

- (278) Marie s' est vue.
 Marie refl is seen
 'Marie saw herself.'

Even though perhaps not all unaccusative verbs in French choose *être* 'be', it seems that all verbs that choose *être* are unaccusative. The choice of *être* with reflexive verbs thus leads us to ask where the unaccusativity lies in these constructions.

I will follow a suggestion that has been developed for Romance in recent lectures by Richard Kayne. (Marantz (1984) first proposed this sort of analysis in its most general form.) According to Kayne, reflexive clitics in French and Italian are always *external* arguments, though they are assigned accusative Case. The full DP in constructions like (277)–(278) is thus the underlying object, which moves to Spec,IP and binds the reflexive clitic from its new position. Movement of the full DP is forced by the assignment of accusative Case by the verb to the reflexive clitic.

- (279) Marie_i [_{VP} se_i voit t_i].

These constructions are unaccusative in that they involve A-movement of their object to a Case position. They resemble passive constructions (as Marantz notes) in that they involve assignment of the external θ -role and verbal Case to an element that does not otherwise bear these properties.

Richard Kayne (personal communication) has pointed out one consequence of this analysis. If reflexive clitics are always external arguments, they are predicted to be incompatible with clauses whose main verb lacks an external argument. This is in fact the case, as documented extensively by Rizzi (1986b) (whose observations stem from Kayne 1975 and Burzio 1981:1986); the following data are his. The (a) and (b) examples show the absence of any effect with nonreflexives, and the (c) examples show the absence of any effect with nonclitics. The (d) examples are the important ones.

(280) *Passives*

- a. Gianni è stato affidato [_{PP} a lui].
 Gianni was entrusted to him
- b. Gianni *gli* è stato affidato.
 Gianni to him was entrusted

- c. Gianni è stato affidato [PP a se stesso].
Gianni was entrusted to himself
- d. *Gianni *si* è stato affidato.
Gianni to himself was entrusted

(281) *Unaccusatives*

- a. Il ladro e il poliziotto sono caduti addosso a lui.
the thief and the cop fell on top of him
- b. Il ladro e il poliziotto *gli* sono caduti addosso.
- c. Il ladro e il poliziotto sono caduti l'uno
the thief and the cop fell one
addosso all'altro.
on top of the other
- d. ?*Il ladro e il poliziotto *si* sono caduti addosso.

(282) *Raising predicates*

- a. A Piero, Gianni non sembra fare il suo dovere.
to Piero Gianni not seems to do the his duty
'To Piero, Gianni does not seem to do his duty.'
- b. Gianni non *gli* sembra fare il suo dovere.
- c. A se stesso, Gianni non sembra fare il suo dovere.
to himself Gianni not seems to do the his duty
- d. *Gianni non *si* sembra fare il suo dovere.¹⁰⁶
Gianni not to himself seems to do the his duty

Rizzi explains these examples as the consequence of a condition on Chain Formation that prevents configurations of the following sort, where α occupies an A-position, t is the trace of α , and β c-commands t :

$$(283) \alpha_i \dots \beta_i \dots t_i$$

Rizzi's condition is empirically problematic. The reflexive clitic, for him, has two crucial properties. First, it is necessarily bound by the subject (as represented by the indexing in (283)). Second, because of the nature of the clitic position, it c-commands the trace of a moved subject in the illegal configuration.¹⁰⁷ The English equivalent to sentences like (282d), for Rizzi, is acceptable because English uses a full PP, not a clitic. Rizzi presumes that a reflexive object in a PP does not c-command out of that PP in examples like (284).

- (284) a. Mary_i seems [to herself_i] t_i to have done her duty.
b. John_i appeared [to himself_i] t_i to be capable of doing the job.

However, this seems not to be the case (for reasons I will examine in chapter 6). The object of *to* in (284), at least for Principle C of the binding theory, behaves as if it does c-command into the infinitival complement.

(285) *Mary seems [to him_i] *t* to like John_i.

Since the θ -Criterion entails that the landing site of A-movement is always a non- θ -position, Kayne's hypothesis, which requires the reflexive clitic to bear an external θ -role, automatically accounts for Rizzi's paradigms without creating any false expectations that (284) will be impossible. In an example like (280d), on Kayne's hypothesis, *si* must be the external argument of *è stato affidato* and *Gianni* the dative, with *si* absorbing the Case assigned by the verb to *Gianni*. But, by hypothesis, the passive *è stato affidato* lacks an external argument, so *si* is impossible.

If the reflexive clitic is always an external argument, some account is needed of the phenomena that I earlier associated with the ad hoc feature [$\pm$ reflexive]. These phenomenon can naturally be subsumed under *control*.¹⁰⁸ Control of PRO and other elements across clause boundaries is a familiar notion, but control of reflexives and pronouns internal to a clause is also well attested (though poorly understood). For example, control is found in expressions involving movement of inalienably possessed body parts.

- (286) a. Bill_i shrugged his_i shoulders.
 b. Sue_i nodded her_i head.
 c. Tom_i craned his_i neck.

- (287) a. *Bill shrugged my shoulders.
 b. *Sue nodded his head
 c. *Tom craned her neck.

Certain phrases of possession also show control.

- (288) a. John has only five dollars on him.
 b. John has only five dollars on his person.
 c. Mary didn't have five dollars to her name.
- (289) a. *John has only five dollars on us.
 b. *John has only five dollars on my person.
 c. *Mary didn't have five dollars to your name.

Idioms like those in (290) provide other examples.

- (290) a. John_i lost his_i way.
 b. Sue_i got her_i kicks by decapitating reindeer.
 c. Mary_i has her_i mind on something else.
- (291) a. *John lost our way.
 b. *Sue got his kicks by decapitating reindeer.
 c. *Mary has his mind on something else.

In each of these cases, the controlled element is either an internal argument or a subpart of an internal argument of the verb. Suppose, however, that a verb could also require its external argument (or a subpart) to be controlled by one of its internal arguments.

Controllers are subjects and objects, that is, A-positions. Let us assume that this is a general fact about control, just as it appears to be a fact about binding. Let us further assume that control requires the controlled element to be c-commanded by the controller. This requirement has apparent exceptions, like (292).¹⁰⁹

- (292) [PRO_i having to talk to Sue] would annoy Bob_i.

These seem to be the same environments that license backward binding, an already known “exception” to the c-command property of binding. Let us put aside such cases for now.

I will limit the present discussion to active sentences. Assuming that the controller must occupy a c-commanding A-position, a verb that requires its external argument to be controlled by an internal argument can only be used in a structure that allows movement of the internal argument to an A-position that is higher than its external argument. In languages typologically similar to English, French, Italian (or Japanese), these factors quite straightforwardly limit this sort of control to the reflexive clitic construction. Consequently, verbs that require control of the external argument by an internal argument can occur overtly only in languages that have reflexive clitics.¹¹⁰ This is so because (1) A-movement from VP must target a Case position within the minimal IP that contains this VP, (2) external arguments must either move to a Case position outside VP or else be cliticized, and (3) there is only one Case position in IP outside VP.¹¹¹ Control of the external argument by an internal argument thus mandates a reflexive clitic configuration and can replace the feature [+ reflexive] in lexical description.

The only other possibility might arise in passive sentences, if the external argument can be realized as a *by*-phrase controlled by an internal argument that moves to Spec,IP, as seen in (293).

(293) $DP_i [V + \text{passive}] t_i [\text{by reflexive}_i]$

This would satisfy a lexical entry that requires the external argument to be controlled by an internal argument on very particular assumptions about the nature of the *by*-phrase. Change these assumptions, and the possibility disappears. In any case, as has often been noted, reflexives are not very acceptable as the object of *by* in a passive.

- (294) a. ?*Sue was seen by herself in the mirror.
 b. ?*Bill was taught by himself to ride a bicycle.
 c. ?*Mary was believed by herself to have done her duty.
 d. ?*John was given a book by himself.

Baker, Johnson, and Roberts (1989) explain these effects as an instance of the configuration in (283), with the passive morpheme *en* referentially linked to the object of *by* in some fashion and functioning as β .

In fact, however, the phenomenon appears to be more general. With respect to Principles A and C of the binding theory, a VP-final *by*-phrase generally acts as if c-commanded by preceding material. (These facts are of a piece with many discussed in chapter 6, and can be easily understood in light of the theory of phrase structure presented there.)

- (295) a. The book was given t to her_i by Sue_j 's mother. [$*i = j$]
 b. John was given t her_i by Sue_j . [$*i = j$]

- (296) a. ?These books were given to the $girls_i$ by each $other_i$.
 b. I was assigned t John and $Mary_i$ by each $other_i$'s teachers.

Yet the object of *by* cannot be a reflexive in a passive construction even when its antecedent is a preceding internal argument—that is, in constructions that are not instances of (283).

- (297) a. *These books were given to the $girls_i$ by $themselves_i$.
 b. *I was assigned John_i by $himself_i$.

Danny Fox (personal communication) has pointed out that the ill-formedness of examples like (297a–b) could be attributable to the effects of Principle C, if (with Baker, Johnson, and Roberts (1989)) we suppose that the passive affix *-en* shares an interpretation with the object of *by* and if we further assume that this affix counts as a binder for Principle C.

- (298) a. *These books were $giv-en_i$ to the $girls_i$ by $themselves_i$.
 b. *I was $assign-ed_i$ $John_i$ by $himself_i$.

Nonetheless, this conclusion seems to be unlikely, in view of the lack of Principle C interaction between the passive affix and *subparts* of internal arguments that precede the *by*-phrase.

- (299) a. These books were *giv-en*_i to *Mary*_i's brother by her_i.
 b. I was *assign-ed*_i *John*_i's book by him_i.

I conclude that passive *-en* does not count as an antecedent for Principle C, and that consequently the impossibility of reflexive objects of *by* in (297) must have a different cause.

In fact, a similar effect is found with nonpassive uses of *by* (though not, obviously, in the usage in which *by herself* means 'alone').

- (300) a. ?*Sue bought some [pictures by herself].
 b. ?*Tolstoy liked [stories by himself] best of all.

Clearly, we are dealing here with an effect that is unrelated to the configuration in (283), but has the consequence that configurations like (301) are excluded.

- (301) *[*by* reflexive]

This, in turn, eliminates the possibility in (293).

Thus, the lexical entry for a verb like *s'évanouir* need only specify that it takes two arguments, of which the external is controlled by the internal. The condition in (301) (assuming that it is universal) will rule out any passive use of this verb. Given this, only an active configuration with a reflexive clitic will satisfy this property. For the moment, I will not attempt to specify the semantic function, if any, of the reflexive external argument of such a verb. It may be that either the external or the internal argument is semantically vacuous, after the fashion of an idiom; or perhaps there is a semantic analysis that can make more interesting sense out of the full DP and the controlled clitic. In the case of true inherent reflexives like *s'évanouir* it is, for now, hard to tell, but once we discuss Experiencer predicates, an interesting possibility will open up. I also do not attempt a full and clear analysis of the range of constructions in which a reflexive clitic is possible or obligatory, leaving that for future work.

This analysis has clear and interesting consequences for the reflexive SubjExp verbs with which this chapter began. Following our analysis of *s'évanouir*, we should assume that verbs like French *s'étonner* have (at least) three arguments: the Experiencer, the T/SM argument, and a third, external, argument controlled by the Experiencer.

- (302) *Marie*_i *s'* *étonne* *t_i* *du* [*bruit qu'on fait* . . .].
 ARG *ARG* *ARG*

Unlike verbs like *s'évanouir*, where the presence of the extra reflexive argument might well be idiomatic, these verbs demand a good reason for the reflexive. This is because of the existence of fact R, which *requires* most ObjExp verbs like *s'étonner* to be reflexive. (Where we find something odd yet systematic like fact R, there must be an explanation.) The proposed analysis of reflexive clitics and the nature of obligatory reflexivity commits us to a particular style of explanation.

I can offer informed speculation concerning the explanation, rather than an explanation for which independent evidence is available. What follows might need to be viewed as a "suggestion" rather than an "idea," but since it is at worst a good stopgap (and at best correct), I will proceed.

4.1.3.2 Ambient *it* Why might a class of Experiencer verbs take an argument that has the following two properties: (1) it is always external, and (2) it is always controlled by the Experiencer? We might attempt to explain the first property by looking for some other argument type that is always external.

The only candidate (see section 2.4.2 for a spurious example) of which I am aware is "*weather*" *it*, also called *ambient it*. This element occurs only in subject position of full and small clauses.

- (303) a. It is raining.
 b. I consider [it to be stuffy in that room].
 c. Mary likes [it in Cambridge].
 d. *Mary likes it. [unless, of course, *it* is a pronoun coreferential with some other expression]

Ruwet (1991:chap. 4) has suggested that *ambient it* is not an external argument after all and has proposed that weather verbs like *rain* are unaccusative. If this is so, we might replace the requirement that *ambient it* be an external argument with a requirement that it be a surface subject. Nonetheless, none of Ruwet's arguments that directly address this point seem to involve sentences with *it*; rather, they concern expressions in various languages like *Rain falls*, which quite possibly do not display the same predicate-argument structure as *It rains*. I will thus not adopt Ruwet's conclusion, but will suppose that *ambient it* is limited to external argument positions. Indeed, as Ruwet (p. 157) notes, expressions with *ambient it* allow cognate objects. (The French (304b) translates (304a).)

- (304) a. It rained a fine little rain.
 b. Il a plu une petite pluie fine.

The possibility of cognate objects is otherwise limited to verbs that are not unaccusative (Marantz 1984).¹¹²

If this line of description is correct, then the nonoccurrence of *it* in *by*-phrases of passive sentences must be taken as something other than evidence for unaccusativity, as (305) shows.¹¹³

- (305) *A fine little rain was rained by *it*_{ambient}.

The fact in (305) is not an isolated case. Ambient *it* (in common with expletive *it*) is excluded from possessive Case positions as well, a fact that is not plausibly related to unaccusativity.

- (306) a. *Its snowing alarmed us.
 b. *Its stuffiness in that room bothered us.

Evidently, ambient *it* is allowed to occur only in a limited range of syntactic positions (nominative and accusative), independent of other facts. This is puzzling, but not an unprecedented pattern of restrictions, as can be seen in the impossibility, for example, of possessive occurrences of English reflexives (**himself's book*) or generic *one* in *by*-phrases (**Fresh corn is usually seen by one in August*).

Let us consider the nature of ambient *it* in greater detail. It is by now commonplace (see Chomsky 1981) to suggest that ambient *it* is an argument of some sort. One appeal of this suggestion is the possibility of explaining why ambient *it* behaves in a manner distinct from the expletive *it* associated with extraposed clauses. For example, ambient *it* shows up overtly in German subordinate clauses and licenses control relations in English, unlike expletive *it*.¹¹⁴

- (307) a. After PRO_i snowing, *it*_i began to rain.
 b. *After PRO_i becoming obvious that the gate was open, *it*_i seemed likely that something had happened.

In fact, ambient *it* can even function as the subject of control verbs, so long as it controls the PRO subject of a meteorological verb of the usual sort.¹¹⁵

- (308) a. It tried to rain all morning.
 b. It never managed to snow.
 c. Did it ever succeed in thundering and hailing, as they'd predicted on TV?
 d. What it's finally succeeding in doing is raining.

These examples suggest strongly that ambient *it* does not merely function as an argument, but actually has semantic value.¹¹⁶ At a first approximation, it refers to the forces in the world that are the proximate causes of weather.¹¹⁷

4.1.3.3 Emotional Weather Now let us return to SubjExp predicates. We need an analysis of predicates like *s'étonner* in which they take an argument that is (1) obligatorily external and (2) controlled by the internal Experiencer argument. An interesting possibility suggests itself: this argument is akin to ambient *it*, differing from ambient *it* only in the fact that it must be controlled.¹¹⁸

In particular, one can easily imagine that the semantics of verbs like *s'étonner* and their nonoccurring English counterparts view certain emotions as a type of "psychic weather." This view is, of course, quite in keeping with well-known metaphors and literary traditions. We are well acquainted with stormy feelings, sunny dispositions, and dark thoughts. If a poem says that "it is snowing in my soul" or "his mind is cloudy," we readily offer an interpretation, whereas a phrase like "Sue is playing chess in my soul" requires much context before we can understand it. On this view, the well-worn literary device of the pathetic fallacy, whereby nature is portrayed as having human feelings, is a simple inversion of the way UG treats human emotions.

Emotions like surprise, annoyance, and amusement are indeed like the weather in a number of respects. They are "global" (ambient), affecting one's perceptions as well as actions. They are transitory. They are somewhat unpredictable in their onset, intensity, and duration. Most important for our purposes, the proximate cause of both weather and emotions can be viewed as a force of nature, beyond conscious control of the individual. All this might lead one to expect a range of Experiencer constructions that quite literally involve ambient *it* in languages like French and English.¹¹⁹ Psychological weather differs from meteorological weather in the external world, however, in that the "natural force" that produces an emotion is internal to the individual who experiences this emotion. Furthermore, the entire phenomenon is quintessentially private: only the individual who contains the natural force that causes an emotion can experience that occurrence of that emotion. Suppose language is constructed so as to allow a language to have pronouns of "natural forces," but only if a very particular distinction is made between natural forces external to the individual and natural forces internal to the individual. If this is the case,

it would at least make sense for the ambient noun with a verb of psychic weather to be in a control relation with the Experiencer.

We are at last quite close to a possible explanation for the reflexive nature of verbs like French *s'étonner* and—by the logic already developed—for the absence of overtly occurring verbs with this meaning in English. The one missing piece is a reason why the Experiencer should function as controller rather than controllee. That is, why does English not have expressions like (309)?

(309) *Mary $\sqrt{\text{annoyed}}$ herself at Sue.

I do not have as sturdy a speculation to offer here as I have had for the preceding points, but a plausible line of attack can be imagined. One possibility might be a requirement that the lexical item used to express natural forces in meteorological and psychic weather sentences must somehow be *minimally expressive*. An appropriate notion of minimal expressivity might explain why the form used for meteorological natural forces is (in languages that use the strategies seen in English, French, and Italian) generally identical to that used for expletives: third person, neuter if available (otherwise masculine),¹²⁰ null *pro* if available (otherwise an unstressed pronoun). Sentences like (309) would violate the minimal expressivity requirement by allowing a full DP (*Mary*) to serve as the ambient natural force.

An important point needs to be made here. I have proposed that even the simplex SubjExp verbs have a kind of causal argument, one that, by hypothesis, is not due to any special causative morphology. What, then, is the semantic contribution of the causative affix that I have taken such pains to motivate? Quite simply, whereas the ambient reflexive expresses the immediate internal source of emotion, the subject of the morphological causative expresses what it has been taken to express throughout this book: elements that may stand at *any point* in the causal chain that leads to the emotion. I will call the θ -role associated with this argument *Ambient Causer* (or *A-Causer*). I have nothing profound in mind concerning the individuation of θ -roles, but simply wish to have a unique label for this argument. It may be that A-Causer is simply a Causer of a predicate that limits the argument to ambient pronouns—as a matter of s-selection. The Causer associated with *CAUS* lacks this limitation, on this view, which is the contribution of *CAUS* to the Experiencer verbs that contain it.

In any case, “active” emotions like *anger*, *surprise*, and *annoyance* will be distinguished from purely “evaluative” emotions, such as those asso-

ciated with *like*, *hate*, and *unaccusative appeal*. The verbs that speak of active emotions, unlike those that denote purely evaluative emotions, require an A-Causer argument. With the evaluative emotions, A-Causer does not necessarily occur; thus, verbs like *like* can exist in English.

4.1.3.4 Adjectival Passives In this chapter, I have proposed that verbs like English $\sqrt{\text{annoy}}$ and French $\sqrt{\text{étonner}}$ assign the role A-Causer, in addition to Experiencer and Target or Subject Matter. Furthermore, the A-Causer role is borne by a reflexive clitic as a consequence of a requirement that the Experiencer control the A-Causer role. However, the logic behind this consequence has only been explored for active forms of these verbs. Now consider the passive. Why can't the external argument be expressed as a *by*-phrase in a passive construction, as in (310)?

- (310) a. *Bill was $\sqrt{\text{annoyed}}$ by himself.
 b. *Marie a été $\sqrt{\text{étonné}}$ par soi-même.

We already have answers to this question—in fact, we perhaps have too many answers. First, if the external argument is like ambient *it*, it is excluded from the *by*-phrase as a consequence of the property seen in (305). Second, if the external argument is like other reflexives, it is restricted from the *by*-phrase by the stipulation in (301). To be sure, these are answers, but not explanations, since the relevant properties of ambient *it* and reflexives have simply been stipulated. Nonetheless, the impossibility of (310), in the context of the proposed analysis, fits into a picture that we know to be real, and that is reassuring.

More to the point, the proposed analysis, even tempered by the stipulations that rule out (310), explains a paradigm that has eluded us so far. Suppose some morphological process creates a new word out of $\sqrt{\text{annoy}}$ or $\sqrt{\text{étonner}}$ that lacks the A-Causer role entirely. Adjective formation from passive participles appears to have this property as an option. (Recall the argument in section 3.4.8.2 (example (262)) that adjectival passives are formed by the addition of a null adjectival suffix to the verbal passive.) Thus, *The door was closed* may have an interpretation in which no agent ever closed the door: perhaps it was manufactured closed and never opened. Likewise, *The island is secluded* may be quite naturally understood such that no particular agent secluded the island. Now consider $\sqrt{\text{étonner}}$ and $\sqrt{\text{annoy}}$. If adjectives are formed from the passives of these verbs, the internal Experiencer argument will be turned into an external argument (at least optionally), and the A-Causer argument may

be entirely unexpressed. If the control requirement disappears with the controlled argument, there will be no bar to the occurrence of these verbs—without any reflexive clitic.¹²¹ This, of course, is the case, as already observed during the extended discussion of adjectival passives in section 2.3.

Consider the following examples:

- (311) a. Bill was annoyed at John.
 b. Sue was quite appalled at the sight.
 c. Bill was very concerned about the price of beans.
 d. Harry was depressed about the election results.
 e. Mary was disappointed in Sue.
 f. John was disgusted with the book.
 g. Tom was pleased with his success.

That these passive forms are adjectives can be seen from the fact that listed adjectives with identical meaning block the creation of the forms under discussion.

- (312) a. *Bill was angered at Sue.
 b. Bill was angry at Sue.
 (313) a. *Bill was embittered about Sue.
 b. Bill was bitter about Sue.
 (314) a. *John was sickened about the attack.
 b. John was sick about the attack.
 (315) a. *Sue was cheered about the news.
 b. Sue was cheerful about the news.

That these are passives from the *noncausative* forms can be seen in a number of ways, partly discussed in section 2.3. The first piece of evidence, of course, is the availability of the Target and Subject Matter arguments in (311). The assumption that these are noncausatives immediately predicts the possibility of these arguments, though some might hold that this begs the question (not answered until chapter 6) of the source of the T/SM restriction.

Also important, however, are the earlier observations (in section 2.3.4) concerning the essentially stative nature of adjectival passives, where the difference between an adjectival passive with a Target or Subject Matter argument and a passive of the causative form is clear. These adjectival passives cannot do things that passives from the causative verbs can do,

for instance, occur overtly without adjectivization. The relevant paradigm is repeated from (80)–(83).¹²²

- (316) a. Sue was continually being scared by sudden noises.
b. *Sue was continually being scared of sudden noises.
- (317) a. Bill was often being enraged by totally innocent remarks.
b. *Bill was often being enraged at totally innocent remarks.
- (318) a. Sue was continually being annoyed by mysterious sounds from the cellar.
b. *Sue was continually being annoyed with mysterious sounds from the cellar.
- (319) a. John was always being deeply impressed by things that left the rest of us cold.
b. *John was always being deeply impressed with things that left the rest of us cold.

A few points should be noted. We must be careful that we do not falsely allow adjectival passives to be formed from the unaccusative psychological predicates discussed in sections 2.5.2 and 3.2.

- (320) a. The play didn't appeal to Mary.
b. *Mary wasn't appealed to by the play.
- (321) a. This mattered to John.
b. *John was mattered to by this.
- (322) a. The same idea occurred to Mary.
b. *Mary was occurred to by the same idea.
- (323) a. Smith's name escaped us for some reason.
b. *We were escaped by Smith's name for some reason.
- (324) a. The correct generalization eluded Pāṇini.
b. *Pāṇini was eluded by the correct generalization.

Of course, this follows quite straightforwardly if none of these verbs takes an A-Causer argument. They must be real unaccusatives, as I have assumed throughout this book. Nonetheless, there may be a deeper fact to explain. In English, French, and Italian, verbs whose Experiencer is dative are apparently always “real unaccusatives,” that is, verbs without an A-Causer argument. Thus, they fail to produce any sort of passive. Conceivably there is some semantic incompatibility between A-Causer and dative Experiencers, but I have no analysis to offer here.

Moving beyond psychological predicates, the current discussion gives some insight into a puzzle left over from section 2.3. While offering a critique of Belletti and Rizzi's (1988) claim that passives of ObjExp verbs are adjectival, I noted that unaccusative verbs do not necessarily form adjectival passives any better than they form verbal passives, as shown in (49a–p), repeated here.

- (325) a. *an (already) occurred event
 b. *(recently) left travelers
 c. *(newly) come packages
 d. *(recently) grown interest
 e. *a (recently) surfaced problem
 f. *(recently) descended balloon
 g. *(recently) peeled skin
 h. *(often) stunk paint
 i. *a (recently) succeeded writer
 j. *a (recently) died celebrity
 k. *a (frequently) paused machine
 l. *(well-)slept children [?]
 m. *(brightly) glistened paint [?]
 n. *a (visibly) trembled orator [?]
 o. *the (regularly) twinkled star [?]
 p. *the (already) stumbled horse [?]

Still, in a number of cases, adjectival passives can be formed from what appear to be unaccusative verbs, as I noted in (48), repeated here.

- (326) a. elapsed time
 b. departed travelers
 c. newly arrived packages
 d. newly appeared book
 e. capsized boat
 f. a fallen leaf
 g. collapsed lung
 h. blistered paint
 i. a failed writer
 j. a deceased celebrity
 k. a stalled machine
 l. well-rested children
 m. a risen Christ

- n. a stuck window
- o. drifted snow
- p. a lapsed Catholic

We are no closer than before to *explaining* which verb falls in which class. However, we now have the ability to understand why these two classes should exist. Consider a sentence that describes time elapsing, travelers departing, or boats capsizing. Each of these events might be described by a verb with one argument (the Patient or Theme to which the event denoted by the verb happened). Such a sentence would most likely be unaccusative, and would be incapable of accepting passive morphology, since it would lack an external argument. On the other hand, these same events might be described by a sentence containing a verb with *two* arguments: a Patient or Theme argument and a controlled A-Causer. In the description of reality captured by this second type of sentence, something intrinsic to time causes it to elapse; some force intrinsic to the travelers provokes their departure; and some property of the boat causes it to capsize.

The semantic difference between these two descriptions of elapsing, departing, and capsizing is negligible, but the syntactic difference is significant. In a language like French or Russian, this second type of sentence would be transitive, containing both an external and an internal argument, and would realize the external argument as a reflexive clitic. Verbs of this sort could accept passive morphology, but for the fact that ambient reflexives cannot be placed in a *by*-phrase. Adjectivization could eliminate the *by*-phrase from this passive, however, yielding legitimate adjectival passives like those in (326). This type of analysis might be behind certain instances of reflexive clitics in these languages, such as the clitics that show up in alternations like these:

- (327) a. Pierre a fermé la porte.
 Pierre closed the door
 b. La porte s'est fermée.
 the door refl-is closed
 'The door closed.'
 c. La porte est fermée. [adjectival reading]
 the door is closed
- (328) a. Marie a arrêté l'auto.
 Marie stopped the car

- b. L'auto s'est arrêté.
'The car stopped.'
- c. L'auto est arrêté.

In a language like English, verbs like those in (327b) and (328b) (or forms of *elapse* with an A-Causer) cannot show up at all, because of the absence of the appropriate clitic. Any such verb would be unusable. Therefore, at best, such sentences must have doublets without the A-Causer argument, such as the actually occurring verbs *elapse*, *depart*, *arrive*. Crucially, however, adjectival passives from such verbs *are* usable: the A-Causer argument is reassigned to a *by*-phrase by the passivization, and eliminated by the adjectivization. This, I suggest, is the explanation for the existence of examples like (326a–p). It is as if alongside *elapse*, English had a verb *s'elapse*; and alongside *capsize*, *se capsize*.

- (329) a. capsize, (Theme)
- b. *se capsize*, (A-Causer, THEME)

What is important to the logic behind the analysis is that the meaning of psychological predicates like *s'étonner* or *s'amuser* or 'be annoyed' *must* be conceptualized in this way (at least in the general case), whereas the meaning associated with *s'arrêter* *may* be conceptualized in this way. Thus, English lacks verbs of the first type, but has no problem with a pure, nonreflexive unaccusative use of *stop*.¹²³

In addition, the current discussion opens up new possibilities for the analysis of inherent reflexive verbs like French *s'évanouir*, where we might propose an analysis like the one sketched for SubjExp predicates. That is, a verb like *s'évanouir* might be reflexive because fainting and vanishing can be conceptualized as events caused by a natural force internal to the fainter or vanisher. On the other hand, it is not clear that this is the correct analysis, given the impossibility of adjectival passivization of *s'évanouir*.

- (330) *Marie est [_A évanouie].

This caution is important. Although I have committed myself to a Marantz-Kayne-style analysis of reflexive clitics (at least for French and similar languages), in which they are external arguments, I have not committed myself to an analysis in which such clitics *only* function as ambient pronouns. For example, this is clearly an incorrect suggestion for straightforward cases like (277) (*Marie se voit*),¹²⁴ and probably incorrect for a variety of other cases, including Russian *stremit'-sja* 'strive', simply on semantic grounds. I have argued that if a language needs a pronoun that

can function as an external argument yet take as a local antecedent one of its verb's internal arguments, this language must employ a reflexive clitic. I have not argued that reflexive clitics are used *only* in response to this particular set of needs.

4.1.4 Eliminating the External Argument

4.1.4.1 Adjectival Passives In the previous section, I relied heavily on the observation that adjectivization can eliminate the external argument of a passive verb, in order to explain the syntax of adjectival passives like *annoyed* in (311a). Adjectivization, of course, does not always eliminate this argument. Adjectival passives like those in (331) are acceptable.

- (331) a. The ruins of Troy were unseen by anyone until Schliemann
began his excavations.
b. The restaurant remained closed by the police for a long time.
c. Sue was much misunderstood by her peers.

On the other hand, adjectivization of passives from verbs like $\sqrt{\text{annoy}}$ *must* suppress the external argument, or else they will violate one of the conditions discussed in the previous section (the filter (301) or the similar restriction on ambient pronouns).

4.1.4.2 Nominalizations Much the same can be said about nominalizations. There is no question about the ability of nominalizations of verbs to occur without overt expression of their external arguments.

- (332) a. the annihilation of the planet
b. the realization that the world is round
c. the murder of Kennedy

If these nominalizations may be assigned structures in which the external argument of the root verb is simply missing, then we immediately explain the availability of nominalizations from verbs like $\sqrt{\text{annoy}}$. Once again, the root verb could only show up if its external argument were a reflexive clitic, unavailable in English. If nominalization may suppress this external argument, then nominalized forms of root verbs like $\sqrt{\text{annoy}}$ should be able to occur in English. This is exactly right, as I have argued at length.

As with adjectival passives, I will not propose that nominalization *must* suppress the external argument. I merely suggest that this *may* happen.

Transparently, it does not happen when the external argument is present and assigned possessive Case, or when the external argument is part of a *by*-phrase.

- (333) a. their annihilation of the planet
 b. Homer's realization that the world had ended
 c. Oswald's murder of Kennedy
- (334) a. the annihilation of the planet by the Borg
 b. the realization that the world had ended by Homer
 c. the murder of Kennedy by Oswald

As before, however, suppression of the external argument is the only option when the external argument is otherwise unrealizable. This is not a consequence limited to English. French, Italian, and Russian must also suppress the external argument with verbs like $\sqrt{\text{annoy}}$ and $s'amuser$, given the unavailability of clitic subjects of NP in all these languages (as discussed in section 4.1.2).

In many cases, it is difficult to distinguish an eliminated external argument from a *by*-phrase that is phonologically null. Yet it is important to make this distinction. Verbs like $\sqrt{\text{annoy}}$ do not show verbal passives (see (316)–(319)), yet verbal passives may occur without overt *by*-phrases. Clearly, verbal passives must not be empowered to eliminate the external argument. Perhaps the relevant distinction lies in the fact that nominalizing and adjectivizing morphologies make a semantic contribution to a word, whereas passive morphology does not.

(335) *Suppression of external argument*

Only affixation of a semantically contentful morpheme to a verb with an external argument α allows α to be unexpressed ("suppressed") in syntactic structure.

For now, we must add to this an appropriate theory of null elements that allows a *by*-phrase to be represented syntactically by something like *pro* in verbal passives. This same option can be allowed with adjectival passives and nominalizations, so long as (335) is also possible. I also presume that suppression of an external argument is not possible when the semantics of an affix crucially make reference to that external argument, as is the case, for example, with nominalizations in *-er* and adjectivizations in *-able*. The former affix produces nouns that refer to the external argument of the root verb; the latter semantically involves a modal that takes the external argument as one of its arguments. Thus, neither of these

affixes can attach to verbal roots like $\sqrt{\text{annoy}}$ (though, as we saw in section 3.4.6, these affixes may attach to the causativization of $\sqrt{\text{annoy}}$).¹²⁵

4.1.4.3 Causatives Finally, (335) will allow causativization of $\sqrt{\text{annoy}}$, *s'amuser*, and predicates of this class in the various languages we have touched on. Affixation of *CAUS* to these predicates will permit the suppression of the external argument role that would otherwise have to be filled by a reflexive clitic controlled by the Experiencer. For English, the logic is the same as it has been throughout this section: since the language lacks reflexive clitics, roots like $\sqrt{\text{annoy}}$ surface only when affixes permit the suppression of the external argument role. For French and other languages that do have reflexive clitics and therefore do allow verbs like *s'amuser*, there is an additional question. Why *must* the reflexive clitic disappear under causativization? That is, alongside (336a), why do we not find (336b)?

- (336) a. Le bruit étonne-*CAUS* Marie.
 b. *Le bruit s'étonne-*CAUS* Marie.

The answer is clear. The root of the causative verb is not, as I suggested earlier, merely marked [+reflexive]. Rather, the requirement that the reflexive clitic disappear grew out of a requirement that the A-Causer argument be controlled by the Experiencer, and an assumption that this relation requires the A-Causer to be c-commanded by the Experiencer. Clearly, this control requirement is not met in a configuration like (336b), in which the reflexive clitic is c-commanded by the Causer argument added by the causative morpheme.

Suppose now that *all* cases of obligatory reflexivity have an analysis in terms of control (though not in terms of A-Causer, as I cautioned above). If we further suppose, as I have maintained throughout, that reflexive clitics are always (in the languages we are considering) external arguments and occur only in inflected sentences (never inside NP/DP or AP), then (335) allows us to explain, and not merely stipulate, the disappearance of obligatory reflexive clitics under various sorts of morphology. We can thus dispense with (272) as an independent statement in the grammar.

4.2 Conclusions

At the outset of this journey, I made a proposal that simplified the mapping problem for psychological predicates, but raised an entirely new

problem: the T/SM restriction. In the sections that followed, I have laid the groundwork for explaining this restriction. I argued at length that semantically causative psychological predicates are morphologically complex, and I displayed some data that related the T/SM restriction to other sorts of morphological complexity. I argued for the morphological complexity on the basis of nominalization patterns. In this chapter, I backed up the analysis by providing reasons for the nonoccurrence of the roots that underlie my treatment of semantically causative psychological predicates. I began by suggesting that these roots surface in certain other languages as obligatorily reflexive predicates. I then provided a reason, perhaps plausible, for the obligatoriness of the reflexive. In particular, I argued that this reflexive is a cousin to ambient *it*, sharing ambient *it*'s restriction to external argument and differing from ambient *it* mainly in a control requirement that ended up explaining why the reflexive must be a clitic. From this analysis and independently visible properties of derivational morphology, I was able to explain why the clitic seemingly disappears under adjectivization, nominalization, and causativization.

In the next chapter, I will back up the analysis of ObjExp psychological predicates in another way, with a detailed analysis of the double object alternation. The explanatory status of the next chapter will hinge on making the same use of nominalization patterns as I have made in discussing psychological predicates. Most important, however, the discussion in this chapter will invoke ideas and principles that will lead into the explanation of the T/SM effect.

Chapter 5

Double Object Constructions and the Zero Morpheme *G*

This chapter deals hardly at all with psychological predicates or *CAUS*. Nonetheless, it will help the narrative in several ways. First, it will provide new support for using Myers's Generalization as a diagnostic of zero morphology. Double object constructions like *give Sue a book* provide evidence from command relations for null structure and (as we will see) behave just as constructions with zero morphemes are expected to behave. More important, this chapter will lead us to develop and clarify a simple theory of θ -roles and selection. This theory will lead in the next chapter to a new proposal concerning the organization of VP. In turn, this new proposal will be the key element in explaining the T/SM effect. Furthermore, the analysis developed in this chapter will itself supply an empirical backdrop which will be essential to that explanation.

5.1 The Null Element *G*

5.1.1 Case in the Double Object Construction

I start with the familiar alternation in (337).

- (337) a. Bill gave a book to Sue. [*to*-object structure]
 b. Bill gave Sue a book. [double object structure]

Let us put aside for the moment the obvious U(T)AH problem posed by this alternation (discussed, for example, by Larson (1988)). Even without this worry, the alternation ((337b) in particular) poses a number of closely related questions concerning the internal structure of the VP in these examples. One question concerns the source of Case for the two DPs in (337b). Another concerns distribution contrasts between the double object variant and the *to*-object variant. Still another concerns the internal hierarchical structure of the VP in the two alternants.

Let us begin with the first two questions. Suppose assignment of objective Case to two DPs by a single occurrence of V is impossible. Such an assumption would instantly lead us to posit some unpronounced element in double object structures that is responsible for Case on one of the two objects. I will call this unpronounced element *G*. The other object, all things being equal, would be Case-marked by V in a normal fashion. Which object would receive which sort of Case marking? If, as has often been proposed, there is an adjacency requirement on objective Case, the first object would have to be the one Case-marked by V, and the second object would be Case-marked by the null element.

This seems to be correct; the first object behaves like a DP Case-marked in normal fashion by V. Thus, the first object must be adjacent to V and must move when V is passivized.

- (338) a. *Sue gave yesterday Bill a book.
 b. ?Sue gave Bill yesterday a book.
- (339) a. Bill_i was sent t_i a book.
 b. *A book_j was sent Bill t_j.¹²⁶

Unaccusativity displays the same pattern. There is at least one (arguably) unaccusative verb in English that appears to show the double object alternation, namely, the verb *get*.¹²⁷ As expected, it is the first object in the double object structure (the Goal) that shows up in subject position. The evidence for this treatment of *get* (in the relevant use) comes from a parallelism with the behavior of verbs like *send*, seen in (340) and (341). In the variant with *to*, the Goal may be of any sort, but in the variant in which the DPs are switched, only an animate *Recipient* is possible.

- (340) a. Mary sent the book to Sue.
 b. Mary sent Sue the book.
 c. Mary sent the book to France.
 d. # Mary sent France the book.
- (341) a. The book got to Sue.
 b. Sue got the book.
 c. The book got to France.
 d. # France got the book.

If this is a requirement on the first of two objects in a double object structure, then (341b) and (341d) fall into the fold so long as the surface subject with *get* is an underlying object. This is also unsurprising from the point of view of the U(T)AH.

- (342) a. The book_i got t_i to Sue.
 b. Sue_i got t_i the book.

Evidence for this analysis comes, as usual, from the impossibility of passivizing *get*.

- (343) a. *Sue was got to by the book.
 b. *The book was got by Sue.

The fact that it is the first object that moves in examples like (342) supports the idea that Case on the first object is unexceptional, whereas Case on the second object needs an explanation in terms of some null element.¹²⁸

I conclude, therefore, that the second object of a double object construction like (337b) is a DP introduced by a null Case-assigning element *G*, much as the second object of a *to*-construction like (337a) is introduced by an overt Case-assigning element *to*.

- (344) Bill gave Sue [*G* a book].

5.1.2 C-Command Asymmetries

The existence of *G* in (344) is also supported by the binding asymmetries that distinguish the first and second objects in (344). Suppose that a relation like c-command or m-command (without any special reference to linear order) governs relations like anaphor binding, negative polarity licensing, and the relation between quantifiers and pronominal variables. The obligatory presence of *G* in a double object structure predicts that, if either object in the construction can bind the other, it will be the first that can bind the second, and not the other way around.

Barss and Lasnik (1986) have shown that such an asymmetry is indeed found. If the common assumption is correct that binding depends on command and not on linear order, then Barss and Lasnik's evidence directly supports the existence of *G*.¹²⁹ As the discussion proceeds, I will argue in a number of places for the correctness of this common assumption, thus improving on the somewhat inconclusive character of much recent debate concerning these matters' (Larson 1988, 1990, Jackendoff 1990). For now, however, we must take this matter on faith. Relevant data are illustrated in (345)–(347).

- (345) a. I showed John himself in the mirror.
 b. *I showed himself John in the mirror.

- (346) a. I showed no one anything.
 b. *I showed anyone nothing.

- (347) a. I denied every worker_i his_i paycheck.
 b. *I denied its_i owner every paycheck_i.

Attributing this asymmetry to the structure introduced by *G* is quite natural and mirrors an obvious account for similar asymmetries with *to*, as observed by Larson (1988:338).¹³⁰

- (348) a. I showed John to himself in the mirror.
 b. *I showed himself to John in the mirror.

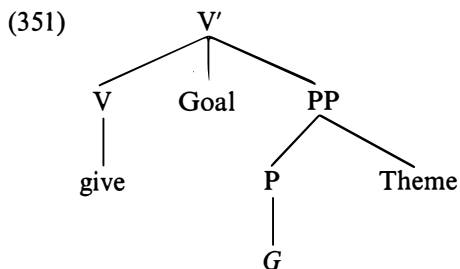
- (349) a. I showed nothing to anyone.
 b. *I showed anything to no one.

- (350) a. I gave every paycheck_i to its_i owner.
 b. ??I gave his_i paycheck to every worker_i.

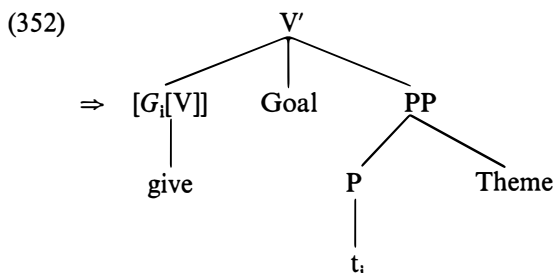
Even so, these observations do not uniquely determine the structure of double object constructions; they are also compatible with a number of different characterizations of *G*. In the next few sections, I will begin accumulating evidence that *G* is a null, affixal preposition. This will lead to a new theory concerning the structures in which *G* finds itself. Subsequent discussion will justify some of the ideas that the reader will be asked to take on faith in the next few pages.

5.1.3 *G* as an Affix

Converging evidence for the existence of *G* and for its affixal nature comes from a direction that is now familiar: Myers's Generalization. Suppose for now that *G* is a preposition, assigning Case to the Theme in double object structures just as *to* assigns Case to the Goal in *give a book to Sue*. (I return to this matter shortly.) Though I will present no argument in this chapter for *G* as a preposition (rather than, say, a null verb), I will make this assumption with the theory of the next chapter in mind. Suppose for now that *G* occurs as a preposition in a fairly traditional V' (without any binary-branching restriction) like (351).



I will now argue that *G* must move from its base position and affix (in the cases we have examined) to the governing verb. Thus, (351) is really only an underlying configuration, which affixation transforms as in (352).



This sort of affixation is forced as a consequence of a lexical property of *G*: it is marked [+affix]. Following Baker (1988), this property has the consequence that *G* must be adjoined to some nonaffixal category by S-Structure. Here, I assume, *G* must raise and adjoin to the higher verb—an instance of the sort of syntactically produced derivational morphology that Baker motivates in his investigations. The affixal status of *G* may in fact be the consequence of a general principle like (353) (see Abney 1987:152).

(353) Zero morphemes are affixes.

There is no comparable affixation of *to*, which is patently marked [–affix]. The basic data supporting this difference between *G* and *to* come from Kayne (1984a:156). Kayne observes that nominalizations related to the form with *to* are possible, whereas nominalizations directly related to the double object construction are not.

- (354) a. *Sue's gift of Mary (of) a book
 b. *John's assignment of Mary (of) a hard sonata
 c. *Bill's rental of Sue (of) an apartment
 d. *Tom's repayment of Bill (of) the money
 e. *Bill's advancement of John (of) a loan
 f. *our allocation of the society (of) funds
 g. *Mary's allotment of the political parties (of) resources
 h. *Sue's presentation of Mary (of) a medal

- (355) a. Sue's gift of a book to Mary
 b. John's assignment of a hard sonata to Mary
 c. Bill's rental of an apartment to Sue

- d. Tom's repayment of the money to Bill
- e. Bill's advancement of a loan to John
- f. our allocation of funds to the society
- g. Mary's allotment of resources to the political parties
- h. Sue's presentation of a medal to Mary

Given the structures in (351)–(352), the unacceptability of the cases in (354) strongly recalls Myers's Generalization.¹³¹ If a double object configuration necessarily involves an affixal preposition *G*, we expect affixation of *G* to inhibit further derivation. Furthermore, if the configurations with *to* do not involve zero affixation to *V*, we expect no problems with derivational morphology.

The contrasts in (354)–(355) support the analyses in this study in several ways. First, they provide further evidence of the empirical scope of Myers's Generalization. Second, they provide further evidence for the existence of zero-derived verbs. Third, the convergence of the proposed explanation for (354)–(355) in terms of zero morphology with the c-command asymmetries adduced by Barss and Lasnik and by Larson provides evidence that zero derivation may result from syntactic movement in a biclausal structure. This will be crucial to the later account of the T/SM restriction on causative ObjExp verbs.¹³²

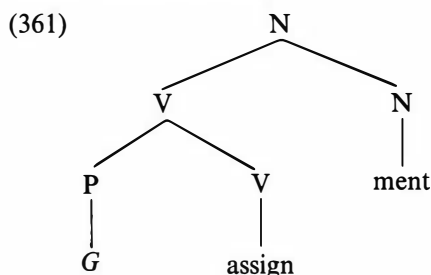
The existence of a null preposition in the double object structure gives us a grip on certain well-known phonological constraints on the construction.¹³³ For most speakers, the availability of the double object construction appears to be conditioned by the length (Oehrle 1976) or lexical class (Stowell 1981) of the main verb. Roughly speaking, main verbs that (by their phonology or morphological freedom of occurrence) seem to be Latinate are hostile to the double object construction. In contrast, the construction with *to* appears to have no such restriction.

- (356) a. Mary gave some canned food to Oxfam.
- b. Mary gave Oxfam some canned food.
- c. Mary donated some canned food to Oxfam.
- d. *Mary donated Oxfam some canned food.
- (357) a. Bill sent his regards to Sue.
- b. Bill sent Sue his regards.
- c. Bill conveyed his regards to Sue.
- d. *Bill conveyed Sue his regards.

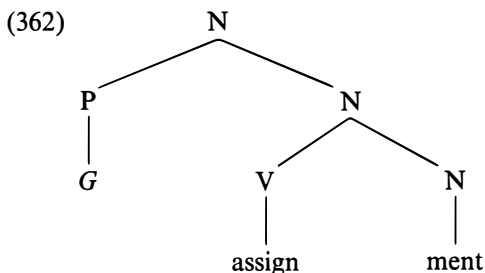
- (358) a. Mary showed her findings to the committee.
 b. Mary showed the committee her findings.
 c. Mary displayed her findings to the committee.
 d. *Mary displayed the committee her findings.
- (359) a. Tom told the story to Ben.
 b. Tom told Ben the story.
 c. Tom recounted the story to Ben.
 d. *Tom recounted Ben the story.
- (360) a. Sue wired some money to Bill.
 b. Sue wired Bill some money.
 c. Sue telegraphed some money to Bill.
 d. *Sue telegraphed Bill some money.

If *G* is an affix, the facts presented here are not at all unexpected, even if they are not explained either. Only the double object structures involve affixation to the main verb. The structures with *to* do not. Morphophonological restrictions on affixation are familiar, ranging from the effects discussed in the earlier treatment of Fabb 1988 to syllable-counting restrictions like those on the English comparative (*happier*/**curiouiser*). Thus, the contrast between the presence of morphophonological restrictions on the double object configuration and their absence with *to* reduces to the same factor that allows nominalizations from the latter structures but not with the former.

Certain questions are raised by this treatment of *G*. I have ruled out nominalizations of double object structures in the following fashion: double object structures involve an empty preposition *G*, which must be affixed to a governing word like *give* or *assign*. Myers's Generalization covers the impossibility of nominalizations like **Sue's gift of Mary of a book* or **John's assignment of Mary of a hard sonata* if the nominalizations in question involve a nominalizing affix attached *outside* the constituent [*assign G*], as in (361).



But why couldn't *G* itself attach to an already nominalized verb, as in (362), yielding no violation?



There is an easy—but rather uninteresting—answer to this question. We can claim that *G* is morphologically restricted so that it attaches only to V, in other words, that it is a deverbal affix. This will prevent it from affixing to nouns. On the other hand, this stipulation opens the possibility that some variant of English might allow *G* to affix to N, thereby showing a different (and perhaps unattested) pattern of facts. Furthermore, although morphemes do have subcategorization properties, it would be nice if we could *explain* why no double object nominalizations exist, rather than pinning the phenomenon on an accidental property of *G*. A more principled explanation might lie in a *Morphological Mirroring Principle*, which incorporates the central observation of Baker's (1985) similarly named Mirror Principle (earlier proposed by Muysken (1981)).

(363) *The Morphological Mirroring Principle*

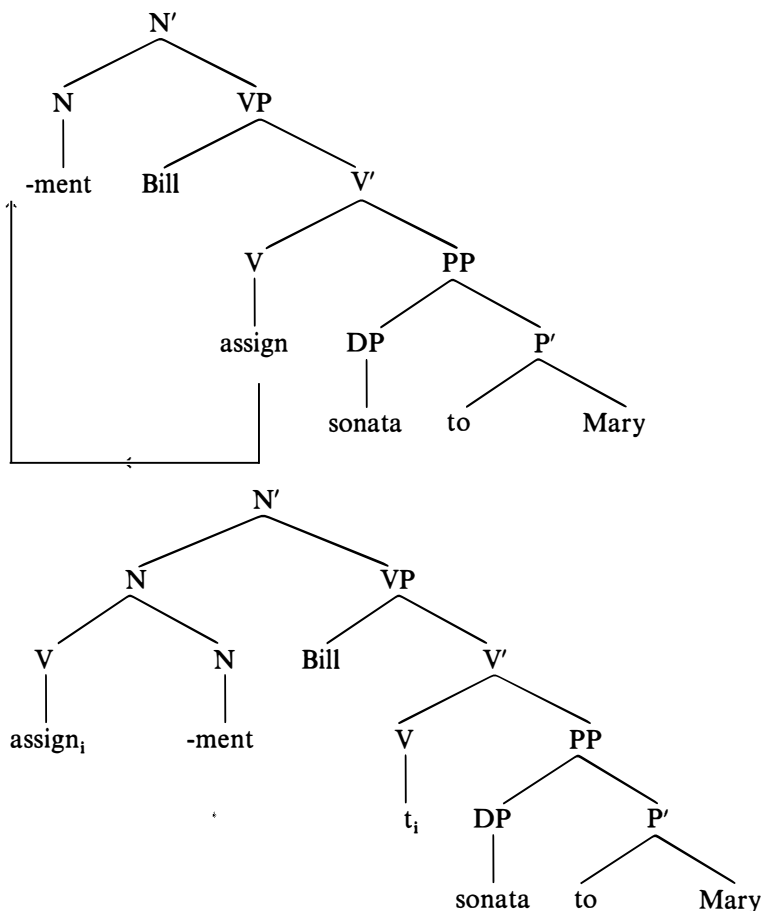
The hierarchical arrangement of heads within a word must reflect the semantic and syntactic complementation relations that hold between those heads.

Deverbal nominalizers embed constituents consisting of a verb and all realized arguments of that verb. This yields familiar glosses for *the destruction of the city* like 'the process of [destroying the city]'. Since complements to a verb are the closest units to that verb at the sentence level, heads of complements must be closer to the verb than other heads word-internally, by (363).

One might base an alternative account on the proposals of Baker (1988), who derives much of his Mirror Principle from his syntactic view of affixation. In the present context, this would involve a theory in which process readings of nominalizations involve the embedding by N of a clausal constituent, whose predicate is affixed to N by syntactic move-

ment. Thus, *Bill's assignment of a sonata to Mary* would involve something like the structure and derivation sketched in (364).

(364)



If head raising can proceed only in a cyclic fashion (perhaps for the reasons discussed by Baker (1988:362–370)), the effects of the Morphological Mirroring Principle result. Now consider a comparable structure involving *G* rather than *to*. In such a structure, *G* will first have to raise to *assign*, and the two will then raise to *-ment*, yielding the desired interaction with Myers's Generalization.

The hypothesis that process nominalizations involve syntactic embedding is not new, even in the recent literature. Lebeaux (1984) suggested syntactic embedding of much this sort for LF. (Lebeaux's hypothesis was adopted in Pesetsky 1985.) More recently, Murasugi (1990) (whose

unpublished work inspired ideas here) has explored an elaborated version of the head-raising hypothesis.

This sort of analysis (particularly one that posits D-Structure embedding, rather than LF embedding) faces certain difficulties, including some that were noted by Chomsky (1972). For one thing, it must be explained why nominalizations do not show such VP modifiers as adverbs. One possibility would be to position VP-adverbs external to VP, as constituents of the next higher category. This would place VP-adverbs as sisters of projections of Agr_O or of μ (Pesetsky 1989) in sentences, and as sisters of N-projections in nominalizations, with *-ly* inserted only in the former environment. The analysis suggested here does not, however, face any intrinsic difficulties with drifted or nonprocess readings of nominalizations, since these would involve D-Structure adjunction and not adjunction produced by syntactic movement.

One other difficulty with this construction comes from the observations concerning *-er* and *-able* in section 3.4.6. There I noted that these suffixes, unlike *-ion* and other nominalizers, are capable of attaching to zero-derived forms. If that is the end of the story, we incorrectly expect these affixes to attach in constructions with *G*.

- (365) a. *a $\text{giv-}G_i\text{-er}$ of John t_i a book
 b. *John is $\text{giv-}G_i\text{-able}$ t_i a book.

I leave this as an unsolved problem.¹³⁴

5.2 *G* and *to*

The analysis thus far suggests that *G* is a preposition that introduces Theme arguments in double object structures, much as *to* is a preposition that introduces Goal arguments in their *to*-counterparts. In the examples seen so far, when *G* introduces Theme, the Goal is a simple DP. When *to* introduces Goal, the Theme is a simple DP. Let us examine a bit more carefully what these concepts entail.

We will begin by developing a clearer view of the correspondence between the selectional and syntactic properties of verbs like *give* in these two types of analysis. Once we have done this, we can be confident of the reasonableness of the structures we have discussed.

The key questions here, as everywhere in this book, are

1. What are the lexical properties of specific predicates?
2. What principles determine the syntactic consequences of a predicate's lexical properties?

In a standard account, *give* expresses a relation among three arguments: an external Agent (mapped onto Spec,VP at D-Structure) and internal Theme and Goal (dominated by V' at D-Structure). In principle, there are two ways to view these relations: either as something assigned by *give* to various expressions (the view implicit in Chomsky's (1981) notion of θ -marking and θ -assignment)¹³⁵ or as a property borne by an expression and selected by *give*. The latter relation can be called *θ -selection*. Probably the two views can be made to describe the same set of situations; thus, they are not distinct theories but merely different points of view. Nonetheless, assuming θ -selection allows us to express certain relations more perspicuously than the (more common) view that assumes θ -assignment. This is clearest for the syntax of internal arguments contained inside PPs, such as the objects of *G* and *to*.

Standard accounts single out sisters to a predicate π as the crucial positions for the syntax of internal arguments. In the simplest cases (e.g., the Goal in (351) and the Theme in (352)), the arguments are expressed by DPs that are themselves sisters to V. The PPs indicate that there is another possibility as well. Clearly, the objects of the prepositions *G* and *to*, not the PPs themselves, function as the Theme in (351) and the Goal in (352). If Sue gave the book to Mary, *Mary* is the Goal, not *to Mary*.

A theory with θ -selection can express this quite simply. We need to allow internal θ -selection by π to be satisfied either (I) by a sister of π or (II) by an object of a preposition in a PP that is the sister of π . Let us call (I) *direct θ -selection* and (II) *mediated θ -selection*. Direct and mediated θ -selection are the only two ways in which θ -selectional properties can be satisfied.

What sort of preposition can participate in mediated θ -selection for a role R? I will make one of the simplest possible proposals about this.

(366) *Mediated θ -selection*

A preposition P can participate in mediated θ -selection for a θ -role R if P itself θ -selects R.

This assumption has straightforward consequences. *To Bill* or *G book* may be a sister to *give* precisely because *to* itself selects Goal and *G* itself selects Theme. In (337b), θ -selection of Goal by *give* is satisfied by simple sisterhood to *give* of a nominal bearing the role Goal. θ -selection of Theme by *give* is satisfied by sisterhood to *give* of a PP headed by a preposition *G*, where *G* itself selects Theme. In (337a), the situation is reversed: Theme is satisfied by simple sisterhood, and Goal by virtue of *to* itself selecting Theme. Here lies the tactical virtue of the notion

“ θ -selection”: we are not tempted to worry about a nominal argument being assigned a θ -role twice in these structures, nor do we need devices that transmit θ -roles or pass them around as in a relay race. With the concept “ θ -selection” and the sisterhood requirement as sketched here, one nominal bears one θ -role, which happens to be selected both by a nearby predicate and by a nonpredicative preposition. The issues raised in this paragraph are probably not substantive, but do constitute a simple presentation of the θ -theory that we need.

This proposal carries with it a presupposition that will be important in the next chapter: there is nothing wrong with selecting the same θ -role in different ways within a clause, so long as one participant is paired with one and only one role in a given event. Crucially, the notion “same θ -role” refers to tokens, not merely types. That is, the DP referring to the *same* Goal participant in a given event may be θ -selected by more than one element that selects the role Goal. A DP may not be θ -selected by elements in distinct clauses that θ -select distinct tokens of Goal.

We can lay out the principles that have emerged from this discussion fairly simply. First, we define *externally selected* and *internally selected* positions as they apply to standard phrase structures. Then we use these definitions to state structural conditions on the satisfaction of θ -selectional properties.

(367) *Selected positions (provisional definitions)*

- a. π *externally selects* α iff α is Spec, $\pi^{\max}$.
- b. π *internally selects* α iff α is the sister of π .

(368) *θ -selection*

- a. If π θ -selects an external θ -role R as a lexical requirement, then this requirement is satisfied if an argument bearing R occupies a position externally selected by π .
- b. If π θ -selects an internal θ -role R as a lexical requirement, then this requirement is satisfied if either
 - i. an argument bearing R occupies a position internally selected by π (*direct θ -selection*), or
 - ii. a selector of R heads a position internally selected by π (*mediated θ -selection*).

The distinction between (368a) and (368b) incorporates the observation that instances of θ -selection for external arguments are apparently never instances of mediated θ -selection. I will not have an explanation for this fact here, but it certainly appears to hold in English.

In certain cases, the semantics of matrix verb and θ -selecting preposition may set sufficient conditions for the acceptability of a particular preposition. This, I will suggest, is the case with verbs like *give*. In other cases, a preposition *P* might count as a θ -selector of role *R* selected by predicate π only because a special property of π allows *P* to select *R*. This is the case when a predicate takes an “idiosyncratic” preposition, such as *on* in *depend on*. I will call this special property *l-selection* (to suggest “lexical-selection”).

(369) *L-selection*

If a predicate π θ -selects *R* and *l*-selects α , its θ -selection for *R* is satisfied only by an argument that is also θ -selected by α .

Essentially, *l-selection* is a special case of idiomaticity. Under *l-selection*, as in an idiom, a predicate receives an interpretation only if in construction with a designated element, as a consequence of an unpredictable lexical specification. *L-selection* is not of crucial importance here (though it becomes important in Pesetsky, in preparation). Nonetheless, the simple view of *l-selection* that is made possible by the concept of θ -selection stands as an argument in favor of using θ -selection, rather than θ -marking, as our central concept.

The more interesting cases for linguistic theory (in its present state) are cases of mediated θ -selection that do *not* involve *l-selection*, since here one can see generalizations over ranges of facts. In particular, the concept helps shed light on the nature of the so-called dative alternation, because it makes clear that a mediating preposition must have semantic properties compatible in a certain respect with the verb whose θ -selection it mediates. The dative alternation involves differing patterns of direct and mediated θ -selection. Interestingly, it also involves a semantic distinction: the variant without *to* is subject to certain semantic constraints that are not found in the variant with *to*. The next section deals with these constraints in detail, picking up the main thread of discussion in its concluding paragraphs.

5.3 The Semantics of Prepositions and Mediated θ -Selection

The double object construction is found with verbs of possession. That is, what I have so far called simply “Goal” is in fact a Goal-Possessor or “Recipient.” Lying behind this name can be any theory that views θ -roles as molecular rather than atomic. Thus, a Recipient has features in

common with a Goal that is not a Possessor, while differing from it in the crucial Possessor property. Besides the basic restriction to Recipients, however, there are other semantic constraints on double object constructions. Many of these take on a new light if we accept the existence of *G* and the contrast between direct θ -selection and θ -selection mediated by *G* in this construction.

Important work on this topic has been done by Pinker (1989) (also see Gropen et al. 1989). This work has identified a number of factors that influence the acceptability of this construction. Pinker notes that "verbs can take the double-object argument structure only if they involve causation of a change of possession" (1989:102). However, this requirement is a necessary but not sufficient condition for allowing the double object configuration. Pinker calls such conditions *broad-range*.

Let us examine how Pinker's system works. For Pinker, some verbs, like *give*, denote causation of change of possession as an aspect of their lexical meaning. These verbs accept the double object structure straight-away because of the broad-range change-of-possession requirement. Other verbs, Pinker suggests, do not denote causation of change of possession as a part of their lexical meaning, but may undergo a *semantic rule* that acts directly on argument structure and adds the notion "change of possession" to their semantics. Among these verbs are certain verbs of causation of motion, such as *throw*, *fling*, and *kick*. These verbs have in their lexical meaning the notions "cause Y to go to Z," as well as a specification of the manner of causation and manner of travel. For these verbs to accept the double object structure, they must undergo a rule that replaces 'cause Y to go to Z' with something like 'cause Z to have Y'. Once this semantic rule applies, the double object structure is acceptable, provided, of course, that morphophonological requirements are satisfied (to which we return below).

Interestingly, not all verbs that might undergo this semantic rule can in fact undergo it. Consider once more the ability of verbs of causation of motion to occur in the double object configuration. Only some of these verbs in fact allow this configuration. Pinker interprets this fact as a restriction on the semantic rule that creates verbs of possession from verbs of motion. He notes that the verbs of motion that allow the double object structure are those that denote "instantaneous causation of ballistic motion." Verbs that denote "continuous causation of accompanied motion in some manner" (i.e., motion that requires "continuous imparting of

force”) do not undergo this rule. This is an example of what Pinker calls a *narrow-range* condition.

Verbs of “instantaneous causation of ballistic motion” are those that pick out a scenario in which the motion is initiated by an external causer or agent, but the continuation of motion results from natural forces such as inertia or gravity. Verbs like *throw*, *fling*, *kick*, and *flick* are thus verbs of ballistic motion, and they allow the double object structure when used to denote transfer of possession. By contrast, verbs like *pull*, *push*, *drag*, and *shlep* pick out scenarios in which an external agent is involved at all stages of the motion and therefore disallow the double object structure.

- (370) a. Mary threw John the book.
- b. Mary threw the book to John.
- c. Mary flung Sue the package.
- d. Mary flung the package to Sue.
- e. Mary kicked John the ball.
- f. Mary kicked the ball to John.
- g. John flicked her the coin.
- h. John flicked the coin to her.
- (371) a. Mary pulled the trunk to Sue.
- b. *Mary pulled Sue the trunk.
- c. Mary pushed the boulder to John.
- d. *Mary pushed John the boulder.
- e. John dragged the sack to Bill.
- f. *John dragged Bill the sack.
- g. Mary shlepped the box to John.
- h. *Mary shlepped John the box.

Pinker argues that contrasts like these arise as a consequence of the course of language acquisition. As noted most clearly by Baker (1979), constructions like the double object alternation pose a classic problem for acquisition theory. To a certain degree, these alternations appear rule-governed, yet they show gaps in productivity of the sort just illustrated. It appears that negative evidence (i.e., the absence of certain forms) is not usable data for the child acquiring language. Thus, the absence of forms like (371b,d,f,h) cannot teach the child that these examples are deviant. On the other hand, the assumption that the child learns double object structures only on the basis of positive evidence like (370b,d,f,h) cannot explain the limited productivity we do find. As noted by Marantz (1984),

if we invent a new verb *shin* whose meaning resembles the meaning of *kick* except that the action is performed with one's shin, we have no problem allowing *Sue shinned Bill the marbles*.

Pinker answers these questions in an elegant fashion. He argues for what he calls *criterion-governed productivity*, supplemented by a strong theory of UG.¹³⁶ Criterion-governed productivity (henceforth *CGP*) allows for productivity within relatively fine-grained semantic domains, for example, verbs of ballistic motion. Positive evidence for the double object configuration from one verb in a semantic domain *D* will license the double object configuration for the other verbs in that domain. On the other hand, a verb outside *D* (e.g., a verb in *D'*) will not allow the double object configuration unless positive evidence licenses it in *D'*, in which case it will be productive in *D'*.

The broad-range rule that transforms one argument structure into another is given by UG, perhaps parameterized among languages. On the other hand, the list of semantic classes that allow the broad-range rule requires positive evidence of the sort just described. This aspect of the acquisition process results in narrow-range rules, that is, instances of the argument structure alteration rule licensed in the various semantic classes of verbs that allow the double object configuration. Thus, once children learn that a verb of ballistic motion permits the double object structure, they will allow the structure with all relevant verbs of this class. *Fling* will not differ from *hurl*, nor will slang verbs like *chuck*, rare verbs like *lob*, or nonce words like *shin* differ from any other verbs of this class. On the other hand, without similar evidence for motion verbs of continuous imparting of force, children will not make such a move in this class.

The theory that I am proposing does not contain any correlate to Pinker's semantic rule. Instead, one alternant involves mediated θ -selection for Theme by *G*, and the other involves mediated θ -selection for Goal (Recipient) by *to*. On this view, the distinctions in (370)–(371) are not facts about an argument structure alternation, but facts about the distribution of the prepositions *G* and *to*. For the case under discussion, the facts adduced so far can be stated as follows:

- (372) a. With verbs of motion, the use of Theme mediated by *G* with directly selected Goal is limited to descriptions of ballistic motion and may not extend to verbs of continuous imparting of force.
 b. By contrast, the use of Goal mediated by *to* with directly selected Theme is not sensitive to this distinction.

In other words, what for the CGP theory (at least as developed by Pinker) is a fact about the acquisition of a rule that alters argument structure is for the proposed theory acquisition of the semantics of θ -roles directly selected and selected by *G* and *to*. This difference takes the phenomena studied by Pinker out of the arena of rule acquisition and places them squarely in the arena of lexical semantics. Is this move correct?

If it is, we might hope that the semantic distinctions that classify the double object structure also restrict other configurations. In fact, there are structures with phonologically overt Goal-mediating prepositions other than *to* that are sensitive to exactly the factors seen in (370)–(371). The sharpest data from English involve the preposition *at* used to express approximation to a goal.

- (373) a. Mary threw the book at John.
- b. Mary flung the package at Sue.
- c. Mary kicked the ball at John.
- d. John flicked the coin at her.
- (374) a. *Mary pulled the trunk at Sue.
- b. *Mary pushed the boulder at John.
- c. *John dragged the sack at Bill.
- d. *Mary shlepped the box at John.

Structures with *in* and *into* also appear to differ in the way structures with *G* and *to* differ, as do structures with *on* and *onto*. In all the examples below, the sentences with *in* and *on* have a pure locative, nondirectional reading (like examples with *G*). Only in (375), however, do these prepositions also have a directional reading (like examples with *to*). For example, the book in (375a) may begin outside the room and end up inside the room. In (376), by contrast, only the pure locative reading is possible for *in* and *on*. Thus, the trunk in the *on the mat* version of (376a), for example, must be on the mat during the entire event of pulling.¹³⁷

- (375) a. Mary threw the book in the room/into the room.
- b. Mary tossed the garbage on the floor/onto the floor.
- c. Mary hurled the boulder in the canyon/into the canyon.
- d. Mary kicked the ball on the field/onto the field.
- e. Mary fired a bullet in the building/into the building.
- f. Mary flung the saddle on the horse/onto the horse.
- g. Mary pushed the car in the garage/into the garage.

- (376) a. Mary pulled the trunk on the mat/onto the mat.
 b. Mary pushed the heavy box in its proper place/into its proper place.
 c. Sue dragged the sack in the office/into the office.
 d. Mary shlepped the box in the office/into the office.
 e. Mary lifted the box on the platform/onto the platform.
 f. Mary gently lowered the saddle on the horse/onto the horse.
 g. Mary drove the car in the garage/into the garage.
 h. Mary lowered the balloon on the runway/onto the runway.

In this domain at least, we can reproduce the notion of criterion-governed productivity as a fact about the semantics of prepositions. We may suppose, with the CGP theory, that positive evidence is needed to warrant the use of a preposition in a particular semantic environment, and we may take from the CGP theory the observation that the relevant semantic categories here are fine-grained. The use of *to* in (370)–(371) will warrant the use of *to* with any verb that belongs to the same classes as the verbs in (370)–(371). The use of *G*-Theme and a directly selected Goal in (370) but not (371), however, will warrant the use of *G* and a directly selected Goal with any verb that belongs to the same class as the verbs in (370), but this license will not extend to the verbs in (371) (for lack of positive evidence). The advantage of my proposed account over Pinker's version of the CGP theory lies precisely in the precedent we find for these distinctions among configurations that involve overt prepositions. Pinker posits these notions as properties of hitherto unknown rule types. This is not an unreasonable move, but it is weak, because it amounts to a description (a statement unlinked to other known facts), whereas the account proposed here amounts to the beginnings of an explanation (a description that cuts across different arrays of facts).

In fact, it is possible to be fairly specific about where the semantic distinctions lie. We cannot be dealing with a contrast between direct θ -selection of Theme and Theme mediated by *G*, because the constructions with *at*, *in*, and *on* all show the limitation to ballistic motion without showing *G*. Instead, it looks as though we are dealing with a special semantic property of θ -selection for Goal mediated by *to*, *into*, and *onto* and other sorts of θ -selection for Goal (direct θ -selection, selection mediated by *at*, *in*, *on*, etc.). Thus, the ability of a predicate of accompanied movement to take a Goal in a given structure depends on the nature of θ -selection for that Goal in the structure. Now it has not, of course, escaped

attention that the Goal-selecting prepositions compatible with continuous imparting of force all contain the morpheme *to*. Suppose we view *into* and *onto* as consisting of *in* and *on* selecting a PP headed by *to*. We may simplify our statement of the facts from (372) as follows:

- (377) A Goal arrived at because of continuous imparting of force must be θ -selected by *to*.¹³⁸

The observation encapsulated by (377) also contains a possible answer to the U(T)AH problem posed by the dative alternation. If there is a distinction between the semantic relation borne to a verb by an object of *to* and a directly selected Goal, then this distinction might be relevant to the principles that arrange arguments. If we merely assume the UAH as a guiding hypothesis, then our work is done: there is systematicity to the linking of arguments to syntactic positions. In particular, we are not necessarily committed to a derivational relation between the two alternants (contra Larson (1988)). On the other hand, if we assume Baker's UTAH, with its more rigid (hence more interesting) requirement that identical semantic relations map onto identical syntactic relations, more complex questions arise. This is the case because the semantics of *to*-objects seem to be a superset of the semantics of directly selected Goals. I return to these questions in section 6.3.2.

Of course, the full semantic picture is more complex than the discussion so far has indicated. Distinctions beyond the straightforward contrasts in (370)–(371) appear to be made. For the sake of completeness, let us examine these distinctions further. A list of verb classes that might be expected to allow the double object structure is given by Gropen et al. (1989:243–45). I reproduce the list in (378)–(379), with some minor changes in the sample verbs. The classification is theirs.

(378) *Allow the double object structure*

- a. inherently signify act of giving: *give, pass, hand, sell, pay, trade, lend, loan, serve, feed*
- b. instantaneous causation of ballistic motion: *throw, toss, flip, slap, kick, fling, shoot, blast*
- c. sending: *send, mail, ship* [cf. previous class]
- d. causation of accompanied motion in a deictically specified direction: *bring* (i.e., 'cause to go here'), *take* (i.e., 'cause to go away from here')

- e. future having (commitments that a person will have something at some later point): *offer, promise, bequeath, leave, refer, forward, allocate, guarantee, allot, assign, allow, advance, award, reserve, grant*
 - f. communicated message: *tell, show, ask, teach, pose, write, spin, read, quote, cite*
 - g. instrument of communication: *radio, e-mail, telegraph, wire, telephone, netmail, fax, ...*
 - h. creation: *bake, make, build, cook, sew, knit, toss* (when a salad results), *fix* (when dinner results), *pour* (when a drink results)
- (379) *Disallow the double object structure*¹³⁹
- a. continuous causation of accompanied motion in some manner: *pull, shlep, lift, lower, haul*
 - b. communication of propositions: *say, assert, question, claim, ...*
 - c. manner of speaking: *shout, scream, murmur, whisper, shriek, yodel, yell, ...*
 - d. fulfilling (X gives something to Y that Y deserves, needs, or is worthy of): *present, provide, supply, entrust, ...* [these take *with*]

Fortunately, matters may be somewhat less complex than they first appear. For verbs of “communication of propositions,” the problem (for both the *G* theory and the CGP theory) may reduce to the problem for verbs of motion. For verbs of “manner of speaking,” the problem may reduce to the problem for verbs of “communication of propositions.” Once the picture is simplified in this fashion, we will be in a position to reexamine the CGP theory.

5.3.1 “Communicated Message” versus “Communication of Propositions”

Consider the contrast between verbs of “communicated message” (378f) like *tell, show, ask, teach, pose, write, spin, read, quote*, and *cite*, which allow the double object structure, and verbs of “communication of propositions” (379b) like *say, assert*, and *claim*, which do not.¹⁴⁰ Let us assume that Gropen et al.’s (1989) descriptions of these two classes are essentially accurate. I will hazard a guess that this contrast is in fact a special case of the more general distinction between verbs of “instantaneous causation of ballistic motion” and “continuous causation of accompanied motion.”

The verbs of “communicated message” involve an initial act by the communicator that results in a message being received by a communicatee, but other aspects of the transmission of the message are out of the hands of the communicator. This is why these verbs in one way or another characterize the message and do not name the proposition transmitted by the communicator. Thus, if I truthfully report that Sue *told* someone that ϕ , *showed* someone that ϕ , or *taught* someone that ϕ , she might not literally have said “ ϕ .” Instead, she uttered some other set of propositions, which set off a chain of reasoning in the hearer that yielded the message ϕ . Indeed, in the case of *teach* and *show* verbal utterance is not necessary. In a slightly different instantiation of “ballistic motion,” if I *write* someone a letter (similarly for *fax*, *e-mail*, and other examples discussed by Gropen et al.), once the letter is sent I am no longer accompanying it on its journey; likewise if I *write* someone that ϕ .¹⁴¹

The verbs of “communication of propositions,” on the other hand, involve a communicative act that is supervised (or accompanied) by the communicator, so that the proposition understood by the communicatee is precisely the proposition uttered by the communicator. If I report that Sue *said* that ϕ , or *claimed* that ϕ , she must have said “ ϕ ” or some close paraphrase. Thus, with *say*, but not with *tell*, there is a sense in which Sue’s actions are responsible for every step in the communication of ϕ to a hearer. These verbs all do name verbal acts.

5.3.2 Manner of Speaking

There are at least two plausible explanations for the impossibility of the double object configuration with verbs of “manner of speaking” like *shout*, *scream*, *murmur*. Thus, if anything, we are faced with too many explanations, rather than too few.

First, when they take two objects, these verbs are clearly verbs of communication of propositions, rather than verbs of communication of message. If Mary shouts an answer to Bill, the content of her shouting must be the content of the answer, not some other proposition from which Bill can infer the answer. *Shout* thus resembles *say*, not *tell* or *show*. In fact, unlike *say*, these verbs appear to be incompatible with any sort of non-literal message.

- (380) a. Sue told Bill, in effect, that she was hungry.
 b. ?#Sue said to Bill, in effect, that she was hungry.
 c. #Sue shouted to Bill, in effect, that she was hungry.

In addition, if the account involving *G* is correct, there may be a morphological reason for the impossibility of the double object structure. In Pesetsky, in preparation, I will argue that the phonologically null complementizer found with English finite clauses is an affix that must raise to a higher nonaffixal head, typically the governing verb. As Zwicky (1971) notes (and as further discussed by Stowell (1981)), such phonologically null complementizers are marginal or impossible with verbs of manner of speaking.

(381) Sue shouted ?*(that) she was hungry.

Verbs of manner of speaking all seem to have homophonous nouns (e.g., *a shout*, *a scream*). If the verbs are derived from the nouns by a null affix, then the attachment of a null complementizer or the affixation of *G* might be a violation of the sort discussed earlier. I will leave this possibility undeveloped until the later treatment of null complementizers.

5.3.3 Location versus Possessor: Verbs of Fulfilling

Now consider the verbs of *fulfilling* described in (379d). These invite a more detailed discussion and point to subtle but important differences between the CGP theory and the zero morpheme theory of the double object alternation.

Pinker's account of these restrictions is uniform, relying on the distinction between broad-range and narrow-range conditions, coupled with CGP. This account has certain virtues. In particular, the logic of the theory provides an elegant escape from the problems noted by Baker (1979). On the other hand, the theory provides no real explanation for any of the contrasts observed. We can easily imagine possible dialects of English in which the contrasts go in the opposite direction, or in any direction that invokes the semantic distinctions posited in the CGP theory.

Of course, if the zero morpheme alternative is correct, this is not in every respect a failing; there *must* be aspects of these contrasts that have no explanation. For example, it might be that we will no more explain why *to* is indifferent to the distinctions in (370)–(371) than we can explain why violins, violas, violoncellos, and basses merit distinct names, as do oboes, English horns, and bassoons, whereas single-reed orchestral woodwinds of all sizes are lumped together as “clarinets.” On the other hand, in the discussion so far, we were able to situate the distinction in domain that can be independently investigated—the acquisition of sound-meaning pairs for lexical items. This possibility was not available

to Pinker's version of the CGP theory, giving the zero morpheme account a (provisional) advantage. Much the same advantage appears with verbs of fulfilling as well.

As noted above, verbs of fulfilling do not show the same sort of double object configuration found in (378), but they do show an apparent alternation between a structure with *to* and a structure with *with*.

- (382) a. Sue presented a medal to Mary.
 b. *Sue presented Mary a medal.
 c. Sue presented Mary with a medal.
- (383) a. Col. North provided money to the Contras.
 b. *Col. North provided the Contras money.
 c. Col. North provided the Contras with money.
- (384) a. The kolkhoz supplied grain to the villagers.
 b. *The kolkhoz supplied the villagers grain.
 c. The kolkhoz supplied the villagers with grain.
- (385) a. The book credits the discovery of radioactivity to Curie.
 b. *The book credits Curie the discovery of radioactivity.
 c. The book credits Curie with the discovery of radioactivity.
- (386) a. The thieves entrusted the loot to Bill.
 b. *The thieves entrusted Bill the loot.
 c. The thieves entrusted Bill with the loot.

It would obviously be interesting to relate the possibility of the structures with *with* to the impossibility of the structures with *G* alone.

In fact, although the Theme here is introduced by *with*, there seems to be a phonologically null element present as well. As noted by Kayne (1984a:149), the forms with *with* in these alternations pattern with double object configurations in failing to nominalize. As with *give* and *send*, the nominalizations with *to* are significantly more acceptable, wherever an appropriate noun exists.

- (387) a. Sue's presentation of a medal to Mary
 b. *Sue's presentation of Mary with a medal
- (388) a. ?Col. North's provision of money to the Contras
 b. *Col. North's provision of the Contras with money
- (389) a. the kolkhoz's supply of grain to the villagers
 b. *the kolkhoz's supply of the villagers with grain

These data suggest that *with* in these constructions is itself selected by an empty preposition G_{with} , which yields the failure of nominalization.

(390) Sue presented Mary [_{PP}[_P G_{with}] with a medal]

With here is selected by G_{with} rather than by the main verb.

Obviously, we would like to attribute the impossibility of the simple double object configuration with G to the possibility of the construction with G_{with} . This, in turn, should be related to the coherence of the semantic class identified by Pinker and his colleagues: “X gives something to Y that Y deserves, needs, or is worthy of.”

Suppose the semantics of G_{with} are similar to the semantics of G , but with the added requirement that the Theme selected by G_{with} satisfy a need of the recipient or otherwise function as a reward for the recipient. (This might be called *Satisfying Theme*.) We can assign to G the complementary semantics (*Neutral Theme*; i.e., the Theme is *not* supposed to satisfy a need or constitute a reward). Consequently, G_{with} , but not G , will be possible with predicates involving need or reward, and G , but not G_{with} , will be possible with those that do not necessarily involve need or reward.

Some clarification is necessary. Why can *give* not be used with G_{with} ?

(391) *John gave Mary G_{with} with funds.

Obviously, if the funds do not satisfy a need of some sort, (391) will be contextually inappropriate, given the semantics of G_{with} . On the other hand, acceptable sentences like *John gave Mary the funds* (with G rather than G_{with}) may be used to describe situations in which a need is satisfied or a reward given. Why isn't (391) acceptable in such a context?

This problem has been discussed under a different guise by Pinker (1989) and others in connection with another construction involving *with*. In fact, these constructions also probably involve a null preposition similar or identical to G_{with} , although in some instances, they show an alternant with a preposition other than *to*. This construction is sometimes dubbed the *locative alternation* (Fillmore 1968, S. Anderson 1971, 1977), although many of the relevant examples involve a goal rather than a location.

- (392) a. John smeared paint on the wall.
 b. John smeared the wall with paint.
 c. *John's smear of paint on the wall
 d. *John's smear of the wall with paint

As is the case with *to*, the version that uses another preposition yields a more acceptable nominalization than does the version that uses *with*. However, relevant nominalizations are hard to find. Perhaps because the majority of the relevant verbs (the so-called *spray/load* class (Levin 1989:14)) are monosyllabic and Anglo-Saxon, few of these verbs seem to produce an argument-taking nominal in the first place. Nonetheless, the following examples are relevant and, I think, illustrative:

- (393) a. The doctor injected a 2% saline solution into Mike/Mike's veins.
 b. The doctor injected Mike with a 2% saline solution.
 c. the doctor's injection of a 2% saline solution into Mike/Mike's veins
 d. *the doctor's injection of Mike with a 2% saline solution
- (394) a. The NSF lavished money on the project.
 b. The NSF lavished the project with money.
 c. the NSF's lavishment of money on the project
 d. *the NSF's lavishment of the project with money
- (395) a. The song will instill confidence into Mary and John.
 b. The song will instill Mary and John with confidence.
 c. the song's instillment of confidence into Mary and John¹⁴²
 d. *the song's instillment of Mary and John with confidence
- (396) a. Cuomo bestowed a million-dollar gift on Jones.
 b. ?Cuomo bestowed Jones with a million-dollar gift.
 c. Cuomo's bestowal of a million-dollar gift on Jones
 d. *Cuomo's bestowal of Jones with a million-dollar gift

The problems with nominalizations in the (d) examples above suggest the presence of a zero morpheme. Let us assume, for simplicity's sake, that *with* in these structures is selected by G_{with} —the same zero morpheme found with verbs like *present* and *furnish*. G_{with} here is not a semantic clone of the overt prepositions. In particular, Rappaport and Levin (1985) and Pinker argue that the predicates that take G_{with} (on my analysis) differ from their counterparts with overt prepositions in specifying a change of state in the non-Locative argument.

First, as Pinker points out, something like the distinction between “instantaneous causation of ballistic motion” and “continuous causation of

accompanied motion” is once again relevant to the availability of G_{with} . Here, gravity counts as continuous causation.

- (397) a. She dribbled paint onto the floor.
 b. *She dribbled the floor with paint.

Furthermore, G_{with} occurs exclusively with predicates that require change of state *as part of their lexical meaning*. Within his framework of rules that operate on argument structure, Pinker states the generalization as

a rule that takes a verb containing in its semantic structure the core “X causes Y to move into/onto Z” and converts it into a new verb whose semantic structure contains the core “X causes Z to change state by means of moving Y into/onto it.” ... What is special about an alternating verb is that it specifies the motion of an object or substance (and generally its manner of motion), making it eligible for the *into/onto* construction, and that this kind of motion predictably causes an effect on the surface that receives the substance. For example, when a liquid is *sprayed*, it is sent in a mist or fine droplets. However, as a result of causing such movement, a surface to which it moves predictably has an even coat of deposited liquid adhering to it. This predictability is what is crucial. (1989:79)

Pinker illustrates this point by contrasting *spray*, which allows *with*, with *pour*, which does not.

- (398) a. Sue sprayed water on the wall.
 b. Sue poured water in the glass.
 c. Sue sprayed the wall with water.
 d. *Sue poured the glass with water.

Spray specifies the end state of the wall in (398a), but *pour* says nothing of the sort in (398b).

We can see these factors at work when we look at predicates that take *with* in their alternative diatheses with other prepositions (Pinker 1989: 126–27). These prepositions (*into*, *onto*, *to*, and others) are acceptable with verbs that specify the direction of movement and final location of the Theme toward the Goal, often in terms of the kind of force used to cause the movement.¹⁴³

- (399) *Direction of movement, final location*
- a. He smeared grease on the axle. [against surface of something]
 - b. He heaped bricks on the stool. [against gravity]
 - c. She splashed water on the dog. [against gravity]

- d. The farmer scattered seeds onto the field. [with gravity]
- e. She dribbled paint onto the floor. [with gravity]
- f. He coiled the chain around the pole. [around something]
- g. He spat tobacco juice onto the table. [out from something]
- h. He attached the flyer to the wall. [via something]

(400) *Final geometry only*

- a. *They inundated water onto the field.
- b. *I clogged a cloth into the sink.
- c. *They studded metal stars onto the coat.

In the literature since Anderson 1971 another observation has been related to these data: an alleged distinction in the “holism” of the objects of locative alternation verbs. Thus, in (401a), we are likely to understand that all the hay got loaded, but we do not know whether the wagon is full. Likewise, in (401b), we are likely to understand that the wagon is full but do not know whether all the hay got loaded. Similarly in (402a–b). On hearing (402a), we naturally understand that all the grease was applied to the axle, but we do not know how much of the axle is covered. On hearing (402b), we might understand the axle to be entirely covered with grease, with some grease left in its container.

- (401) a. Sue loaded the hay into the wagon.
- b. Sue loaded the wagon with hay.
- (402) a. Bill smeared the grease onto the axle.
- b. Bill smeared the axle with grease.

The “holism” of the object might be related to the notion of “change of state” discussed above, in that the DP that can occur in object position is the one whose final state (e.g., “completely loaded” or “completely smeared”) is specified by the verb. I will not explore this idea here.¹⁴⁴

Let us now return to the occurrences of *G_{with}* and *with* that alternate with *to*. Recall that verbs in this class describe the giving of a reward or the satisfaction of a need. This characterization looks very much like the change-of-state property seen with *G_{with}* in the locative alternation. Indeed, the class of English verbs that select a Recipient and Goal and also specify a change of state in the Recipient certainly include verbs of reward or satisfaction of need. Verbs of punishment might also fall under this rubric.

- (403) a. Sue afflicted us with her vacation slides.
- b. Bill punished the class with a 10-mile run.

If this is correct, then we should have a unified analysis that explains the complementary distribution of G and G_{with} , their alternation with to , and the distribution of the locative inversion construction. Throughout, the distribution of the zero morphemes appears to be semantically more restricted than the distribution of the overt prepositions, and throughout, in Pinker's words, "predictability is what is crucial." We are interested in what can be inferred about an event from the lexical semantics of the governing predicate.

There is no mechanism in Pinker's system that explains *why* "predictability is what is crucial." Why, for example, should (398d) not be acceptable when used to describe a situation in which the glass does change state (e.g., by filling up as a consequence of pouring)?

The theory that posits the null preposition G_{with} can offer an answer. On this theory, acceptable use of (398d) would entail altering the lexical meaning of *pour* so as to allow G_{with} . Assuming that such alteration is generally impossible (except in restricted jargons) explains the deviance of the example. In fact, Gropen et al. (1991) show from experimental data that children allow structures like (398d) precisely when they have guessed wrongly at the "adult" semantics for the verb, interpreting it roughly as 'pour full'. Evidently, children recover from nonconforming semantics of this sort, a recovery that depends on the assumption that meanings cannot be freely altered. On Pinker's theory, by contrast, each "alternation" like that between *in* and *with* is an alteration of the lexical properties of a predicate. Thus, some additional caveat like "predictability is what is crucial" must be added to the discussion, and distinctions among possible alterations of lexical properties encoded in the theory. The theory that posits G does not make this sort of distinction among alterations of lexical meaning, because the work done by Pinker's lexical rule is simply done by use of one or another preposition. If a verb accepts both to and G_{with} or both to and G , this is simply because its meaning (and phonology) is compatible with both choices, not because of a rule that alters its lexical behavior.

Let us add another category of data to this discussion: verbs of deprivation. These generally pattern with G_{with} verbs, but they take a privative *of* rather than *with*. In fact, these constructions too seem to involve a zero preposition, which I will call G_{of} .

- (404) a. John robbed Sue of her money.
 b. *John's robbery of Sue of her money

- (405) a. Sue deprived John of his house.
 b. *Sue's deprivation of John of his house
- (406) a. The college divested itself of its Latvian stocks.
 b. *the college's divestment of itself of its Latvian stocks
- (407) John robbed [_{PP} Sue [_P [_P G_{of}] of her money]].

Verbs of change of state by loss also take G_{of} , where the first object is not a Source-Possessor, but simply a Source. The relevance of change of state can be seen in the contrast between *clear* and *wipe* (Levin and Rappaport Hovav 1991). Only with *clear* is a change of state specified.

- (408) a. Doug cleared dishes from the table.
 b. Doug cleared the table of dishes.
- (409) a. Doug wiped the fingerprints from the counter.
 b. *Doug wiped the counter of fingerprints.

The evidence for G_{of} is of the now familiar sort.

- (410) a. John cleared the dishes from the table.
 b. John's clearance of the dishes from the table
 c. John cleared the table of dishes.
 d. *John's clearance of the table of dishes
- (411) a. Max purged the Communists from the union.
 b. Max's Purge of the Communists from the union
 c. Max purged the union of Communists.
 d. *Max's Purge of the union of Communists

If our partitioning of the facts is correct, there is one dimension along which complementation with G_{with} and G_{of} differs from complementation with G . G_{with} and G_{of} differ from G in their indifference to whether the non-Theme is a Possessor. G_{with} and G , on the other hand, differ from G_{of} in compatibility with verbs that select Goal versus verbs that select Source. Once again, the choice of preposition can be predicted on the basis of the properties of the governing predicate.

More generally, we can fit the distribution of G_{with} , G_{of} , and G into a theoretical niche that independently exists: the relation between semantics and selection. This, in turn, is what we expect from a theory in which a preposition must select role R in order to mediate a main predicate's selection for role R.

I argue (Pesetsky 1982; in preparation) that syntactic selection by a predicate (except for *l*-selection) is predictable from semantic properties of that predicate. The notions needed in that discussion include weak concepts like semantic compatibility, but also include the stronger notion of *lexical semantics* and the derivative notions of *argument structure* and *semantic selection*. For example, a verb that describes an event involving questions may take a question as an argument and semantically select this question (e.g., *ask*), but a verb may describe an event with a question as a participant yet fail to take this question as an argument and fail to select this question (e.g., *interrogate*).

- (412) a. Sue asked (Bill) where Mary had gone.
 b. *Sue interrogated Bill (*where Mary had gone).

Similarly, the verb *put* describes motion of a Theme toward a Goal (see note 137). In any such event, there must also be a Source, yet this Source may not be expressed as an argument of *put*, since Source is not part of the argument structure of *put* and is not selected by it.

- (413) a. Bob put the book on the table.
 b. *Bob put the book from the shelf on the table.

By contrast, *carry* does select an (optional) Source.

- (414) a. Mary carried the book to the table.
 b. Mary carried the book from the shelf to the table.

These facts surely have characterizations in terms of more sophisticated concepts involving paths and motion (as in work by Gruber, Jackendoff, and others, which I will not survey here), but the general point is clear. Mere compatibility between the events described by a sentence with some predicate and some argument type does not allow this predicate to select that particular argument. Part of what is learned about a predicate is the array of arguments that it permits. In these cases at least, this information may be learned by positive evidence.¹⁴⁵

So it is with zero prepositions. The child learns which predicates that take a Theme and Goal characterize a change of state. This learning involves two factors.

1. The meaning of the predicate must be learned, and it must be observed that a Theme is part of the predicate's argument structure. If the Theme changes state, that must be observable somehow from the use of the predicate.

2. It must be known somehow that G_{with} is selected (in the general case) only by change-of-state predicates, that *G* is found only with non-change-of-state predicates, and that overt *to*, *on*, and so on, are indifferent to the change-of-state/non-change-of-state distinction.

No theory can fail to grapple with the first of these factors. Obviously, the present work will not yield an account of the acquisition of lexical meaning, but much current research suggests an interplay between contextual and syntactic observation as a key to this process.

The second of these factors occupies the same logical position in the present discussion as CGP does in a theory with argument-structure-changing rules. The key to Pinker's theory was conservativity across narrow semantic classes in the applicability of these rules. The version of my account that is logically the closest to Pinker's would replace his notion of conservativity in rule acquisition with conservativity with respect to the distribution of prepositions like *G*, G_{with} , *to*, and *on*. In this theory, the forms with G_{with} would be limited to change-of-state predicates because these are the predicates with which this preposition is found. Forms with *to* would not be limited to change-of-state predicates, because they are found with a variety of predicate classes.

These two approaches both raise certain questions and problems, whose answers can only come with more empirical work. Nonetheless, some speculations about their solution can be advanced.

First, if these approaches are correct, why is there not more divergence among speakers concerning the distribution of the various alternants? If the only pressure not to use G_{with} (or G_{of}) with non-change-of-state predicates is the lack of exposure to expressions like **Sue gave Mary with a present*, and if there is a range of narrow semantic classes that can accept G_{with} only if evidence is supplied, then one would probably expect systematic variation among speakers of a sort that seems not to be found.

This question is closely related to another. Are the semantic conditions on alternations like the dative and locative similar across languages? Certainly, a variety of languages display these alternations. For the locative alternation, it has been suggested by Rappaport and Levin (1985:36), cited by Pinker (1989:97), that "when a language manifests the alternation, the verbs that participate in the alternation fall into the same broad semantic class as the English locative alternation verbs." If this is correct, then we can abandon theories that require positive evidence to license an alternant for a given category of predicate in favor of a theory driven more by UG.

Finally, the hypothesis that the forms that take *with* involve a zero morpheme G_{with} raises a puzzle that is closely linked to the previous two questions. If the child's task is simply to discover the distribution of various types of complements, and the child is presented with VPs like *smear the axle with grease* and *present Mary with a prize*, why does the child not simply assume that a meaning for *with* is being discovered? Why does the child instead assume that *with* is selected by an inaudible element? And why is this element so similar in its semantics to the other inaudible element posited for *give Mary a book*?

This last puzzle suggests strongly that we will end up with a more UG-driven theory than those considered so far. Since there is no phonetic evidence for zero morphemes, the reasons for their existence in linguistic representations must be sought in abstract properties of the linguistic system and in properties of language acquisition. What scattered and fragmentary evidence I have concerning languages other than English makes it plausible to suggest that the set of zero noninflectional morphemes is fixed. Although not all available zero morphemes are necessarily found in any given language, zero morphology is otherwise a truly universal portion of vocabulary. If this is so, then the syntax and semantics of G and G_{with} may be completely given by UG. The child's task is simply to discover whether the language being acquired has these morphemes, and to discover which predicates have semantics compatible with one or another of these morphemes. However, I have no positive evidence concerning this process, and I must leave it as a topic for future investigation.

To summarize: If a child knows the semantics of these zero prepositions and the semantics of predicates, nothing needs to be learned about which predicates take which prepositions. The significant difference between the zero morpheme theory and Pinker's version of CGP lies simply in the fact that the child's conservativity lies now, not in the acquisition and application of a semantic rule, but in the acquisition of the semantics of predicates. Use of a verb of accompanied causation of motion will not be extended to ballistic environments without positive evidence, and a verb whose usage warrants a semantics involving change of state will not be extended to cases that do not involve change of state without positive evidence. Productivity on the part of the child will lie, not in extending a semantic rule to verbs similar to the one for which the rule has already been posited, but in matching preposition semantics to verbal semantics and in the free usage of semantically appropriate verb-preposition pairs.

Clearly, the key question is whether the acquisition of lexical semantics follows the course necessary to make this scenario work. This too awaits further research.

Some loose ends regarding the zero morpheme approach are dealt with in appendix A and appendix B at the end of this chapter.

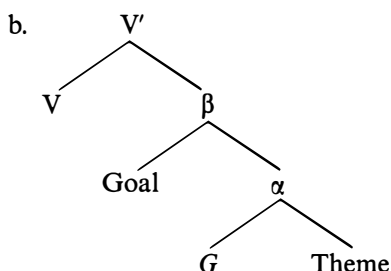
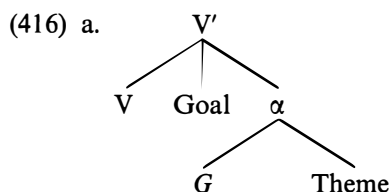
5.4 Structures for *G*

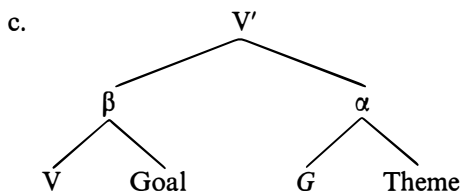
In the discussion so far, I have motivated the existence of *G* (and its congeners) and have situated *G* as part of a general view of θ -selection and lexical learning for prepositions. During this exposition, I have conservatively assumed a traditional structure for the V' in which *G* finds itself. In fact, as I show in this section, there is some evidence against such traditional structures. I also consider some alternatives to my identification of *G* as a preposition. The topic is of special importance because my resolution of the debate over the structure of constructions with *G* will be crucial, both to my solution to the T/SM problem and to the evidence that supports this solution. Thus, it is important to keep track of the trees presented in this chapter so that we may see the forest more clearly in the chapters to come.

The proposal I will develop in this section and the next chapter continues a line of research initiated by Kayne (1984a:chap. 6). Kayne proposes a prohibition on government by α in structures with multiple branching like (415) (also see Larson 1988:380).

(415) [_P α X Y ...], where α , X, Y are immediate constituents of β

If (415) were impossible, then (416a) could not be a structure for double object constructions, with the Goal and the constituent containing *G* and Theme both immediate constituents of V' . This would leave (416b) and (416c) as possibilities.





(416c) is not compatible with the binding phenomena of (345)–(347) if c-command is the relation important to these phenomena (nor would any corresponding structure with *to* be compatible with (348)–(350)—this was Kayne's (1984a:chap. 6) proposal for *to*-datives). If m-command is the relevant relation, and if α is a maximal projection, then (416c) remains a plausible structure.

In fact, one empirical consideration, introduced into recent discussion by Larson (1988), picks (416b) over (416a) and (416c). Only (416b) allows examples like (417a–b) to be instances of constituent coordination.

- (417) a. Sue gave [Bill *G* the book] and [Mary *G* the record].
 b. Bill sent neither [John *G* the letter] nor [Mary *G* the postcard].

Of course, movement cannot treat the two objects as a factor,

- (418) a. *[Bill *G* a book] (though) she gave ...
 b. *[Who *G* his paycheck] should we send?

but at least coordination does pick them out as a syntactically relevant unit, a fact expressed by the constituency in (416b). The problem with movement will be taken up in chapter 7.

5.5 The Nature of *G*

If we trust in the weak considerations that have led us to (416b), we need to raise anew the question of the identity and nature of *G*.

I have assumed so far that *G* is a preposition whose role in the sentence is similar to that played by other prepositions. On the other hand, if a constituency like (416b) is correct, we might question this assumption. An obvious alternative is to view *G* as a *predicate*. There are two sub-possibilities: *G* might be a predicate (or trace thereof) that heads a small clause embedded by the main verb of *V'*, or else *G* might itself be a trace of that main verb. The former I will call the *small clause* theory. The latter is consistent with *VP shell* theories of the sort pioneered by Larson (1988).

5.5.1 Small Clause Theories

Consider first the *small clause* theory. Versions of this proposal have been advanced by Guéron (1986), Hoekstra (1988), Mulder (1992, esp. pp. 64–65), and others, following general suggestions by Kayne (1984a). This theory is seductive because of the following observation: many predicates that participate in the double object construction have a natural semantic decomposition into a causative predicate embedding an expression of possession. It is obviously attractive to associate *G* with the possessional expression.¹⁴⁶

- (419) a. Bill gave [Sue *G* a book].
 b. Bill *MAKE* [Sue *HAVE* a book]

In these structures, if we accept small clause theories, *G* is not a mediating preposition, but a predicate that takes the Goal and Theme as arguments.¹⁴⁷ *MAKE* simply embeds the possessional expression in a manner paralleled by overt periphrastic causative structures.¹⁴⁸ If we accept this proposal, identifying *G* with the possessional component of (419), we still need to ask whether *G* is null because the possessional component is lexically null, or whether it is null because it is the trace of the verbs pronounced *give*, *send*. This is the same as asking whether *give*, *send*, and so on, are pronunciations of the causative component or of the possessional component. Only the latter view seems plausible, since the choice of pronounced form dictates the obligatoriness and optionality of the Theme and Goal arguments. Double object verbs that select Theme and Goal differ greatly in the obligatoriness of Theme and Goal arguments. *Give* requires both a Theme and a Goal to be present, except when *give* means something like *donate*, when Goal may be omitted. Thus, (420) is most naturally used only as a statement about a charitable contribution.

- (420) I gave \$20.

Assign also takes an optional Goal.

- (421) a. The teacher assigned the students the homework.
 b. The teacher assigned the homework to the students.
 c. The teacher assigned the homework.
 d. *The teacher assigned the students.

Give and *assign* differ from *teach*, which takes an optional Theme and an optional Goal (with both omitted only in certain uses), and from *write*, which has a similar diathesis.

- (422) a. Bill taught the students French.
 b. Bill taught French to the students.
 c. Bill taught the students.
 d. Bill taught French.
 e. Bill taught (at Harvard).

- (423) a. Sue wrote Mary a letter.
 b. Sue wrote a letter to Mary.
 c. Sue wrote Mary.
 d. Sue wrote a letter.
 e. Sue wrote (last week).

Feed differs from all of these, in taking an optional Theme but a (perhaps weakly) obligatory Goal. Some argument must be present.

- (424) a. John fed the cow some hay.
 b. John fed some hay to the cow.
 c. John fed the cow.
 d. ??John fed some hay.
 e. *John fed (last week).

If the forms pronounced *give*, *send*, and so on, were simply manifestations of the causative predicate, we would not expect them to control the distribution of arguments in the embedded clause. We do not generally find that sort of phenomenon in clausal embedding. Thus, the only viable version of the proposal that sees *G* as an independent predicate is one that sees it as the trace of *give*, *send*, and so on, where these forms are possessional predicates that become causative by movement to a higher null causative expression.¹⁴⁹ This is in fact quite explicitly the proposal outlined by Mulder (1992).

- (425) Bill *MAKE* + gave_i [Sue t_i a book]

This type of analysis rests crucially on two factors: first, the attractiveness of the parallel between (419a) and (419b); second, the responsiveness of this analysis to the binding and coordination facts presented in the preceding sections. There are several problems with the small clause approach. These problems seriously reduce the attractiveness of the parallel in (419) and ultimately suggest that this parallel—though compelling—is an illusion born of examining a subset of the relevant facts. The first problem concerns the nominalization patterns I have already discussed. The second concerns domains for pronoun and anaphor binding. This

problem was noted by Larson (1988) and concerns the extension of Barss and Lasnik's (1986) asymmetries to syntactic domains where there is no enticing semantic decomposition to justify structure.

5.5.1.1 Nominalization Problems The alternant with *to* behaves for co-ordination purposes like the alternant with *G*.

- (426) a. Sue gave [the book to Bill] and [the record to Mary].
 b. Bill sent neither [the letter to John] nor [the postcard to Mary].

What would the small clause theory say about these cases? If *give* and *send* are embedded predicates in the alternant with *G*, we must presume that they have the same status with *to*. That is, the *to*-alternant should also involve a *G* that functions as the trace of *give* or *send* raised to *MAKE*. (Since our discussion of mediated θ -selection largely concerned *to*, its results could be transposed to this framework.) This leaves us without any obvious explanation for the contrasting behavior of the two alternants in nominalizations, which I attributed to the absence of *G* in the *to*-alternant. If we abandon the idea that *G* is a trace of *give* or *send* and make it a predicate by itself, then we can recapture this contrast by assigning *to* the same status; but then we run afoul of the selection problem discussed above.

5.5.1.2 Complete Functional Complex Problems If *G* is a predicate that selects Goal and Theme, these three elements collectively form what Chomsky (1986b) calls a *complete functional complex (CFC)*. Since this functional complex is not associated with any obvious verbal inflection, it is an example of a small clause. Consequently, on theories like Chomsky's, a phrase like *Sue G a book* in (419a) should behave like an opaque domain for anaphor binding. As (427) illustrates, small clauses generally are opaque for anaphor binding.

- (427) a. *Sue considered Bill angry at herself.
 b. *The boys_i made the girls fond of each other_i.

The same is true of complements to causative predicates.

- (428) a. *Sue_i made the rocks land on herself_i.
 b. *The boys_i made the girls think about each other_i.

Crucially, the same is *not* true of the constituent formed by *G*, Goal, and Theme.¹⁵⁰

- (429) a. Sue_i showed Bill *G* herself_i.
 b. The boys_i gave the girls *G* pictures of each other_i.

This consequence is not unique to Chomsky's (1986b) system. In the theory of Reinhart and Reuland (1993), for example, an argument reflexive is said to "reflexive-mark" its predicate. (I assume that a similar sort of marking would be required for argument reciprocals.) This has the consequence that this predicate must be semantically reflexive: two of its arguments must be in a binding relation. Since that is not true of the arguments of *land* in (428a), for example, the deviance of that sentence is predicted. By parity of reasoning, if *G* is a possessional predicate, it should be reflexive-marked by *herself* in (429a) and run afoul of the requirement that *G* be a reflexive predicate. Since this conclusion is false, the small clause analysis of the constituent with *G* is problematic for a theory like Reinhart and Reuland's as well.

To be sure, some convention or other could be developed to sidestep this problem. Any theory that posits a special enough relation between *G* (or its antecedent) and the causative predicate could also posit a loosening of binding theory resulting from this special relation. The point remains, however, that this result does not follow by itself from the system. Furthermore, we would also leave unexplained the continued absence of CFC effects in the *to*-alternant, which invites a small clause analysis for all the same reasons the *G*-alternant does (as we just saw).¹⁵¹

- (430) a. Sue_i showed Bill to herself_i.
 b. The boys_i sent the letters to each other_i's relatives.

5.5.1.3 Rightward Command Problems The third problem for the small clause theory is raised by important observations made by Larson (1988). Without addressing small clause theories directly, Larson nonetheless notes that the command asymmetries and coordination possibilities that motivate small-clause-like constituency for the sequence *Theme G Goal* motivate very similar constituencies for sequences whose elements cannot be related by any plausible small clause semantics. (431) shows this for coordination. I cannot think of a plausible interpretation of *a record on Thursday* in (431a) in which some predicate links *a record* with *on Thursday*. In (431b), linking the second object of a double object construction in a constituent with one of the adjuncts actually undermines the small clause that can otherwise be formed by the first two objects.¹⁵²

Coordination

- (431) a. Mary bought [a book on Friday] and [a record on Thursday].
 b. John gave Bill [money in Boston] and [supplies in New York].
 c. Sue will speak to Mary [about linguistics on Friday] and [about philosophy on Thursday].
 d. Kremer will perform this concerto [in Rome on Tuesday], [in Lund on Wednesday], and [in Somerville on Thursday].
 e. Starker will play [Bach in Rome on Tuesday], [Kodaly in Lund on Wednesday], and [Simpson in Somerville on Thursday].

A range of phenomena, including (not limited to) anaphor binding, seem to slice sequences off from the right edge of V' . If we continue to adhere to a command-based system, then this tells us something about the constituency of V' : "The farther right you go, the lower you are," or "Rightward is downward."

Anaphor binding(432) $L \rightarrow R$

- a. Sue spoke to these people_i about each other_i's friends in Bill's house.
 b. John spoke to Mary about these people_i in each other_i's houses on Tuesday.
 c. Sue plays concerts in these countries_i on each other_i's national holidays.
 d. Mary danced with these people_i in each other_i's hometowns.

(433) $R \rightarrow L$

- a. *Sue spoke to each other_i's friends about these people_i.
 b. *Sue spoke to Mary about each other_i's flaws in these houses_i.
 c. *Mary danced in each other_i's cities with these mayors_i.

Pronouns as bound variables(434) $L \rightarrow R$

- a. Sue spoke to each employee_i about his_i paycheck.
 b. Sue spoke to Mary about each employee_i in his_i house.
 c. Gidon Kremer performed in every Baltic republic_i on its_i independence day.
 d. Mary danced with no employee_i in his_i hometown.

(435) $R \rightarrow L$

- a. ??Sue spoke to his_i friends about each employee_i.
- b. ??Sue spoke to Mary about its_i flaws in each house_i.
- c. ??Mary danced in his_i hometown with no employee_i.

Negative polarity

(436) $L \rightarrow R$

- a. Sue spoke to no linguist about any conference.
- b. John spoke to Mary about no linguist in any conference room.
- c. Mary danced in no city with any prominent citizen.
- d. Mary danced with no linguist at any NELS party.

(437) $R \rightarrow L$

- a. *Sue spoke to any linguist about no conference.
- b. *Sue spoke to Bill about any problem in no conference room.
- c. *Mary danced in any city with no prominent citizen.

Here we can begin to cautiously support our insistence on a command relation for binding, rather than a relation that mentions precedence. The constituency that a purely command-based account demands for (432)–(437) is also the sort of constituency that the coordination data motivate, if we stick to the simplest theory of that phenomenon—one that singles out constituents. This constituency includes at least the structure in (438) (I omit many details that distinguish among analyses).

(438) $[V [\text{object}_1 [\dots \text{object}_2 [\text{adjunct}_1 [\text{adjunct}_2 \dots \text{adjunct}_n]]]]]$

The behavior of coordination converges with the behavior of binding in a way that we can express with theory if we stick to the simplest theories of both phenomena, in which constituent structure is the key. This reasoning is due to Larson, though he does not state the argument quite as starkly.

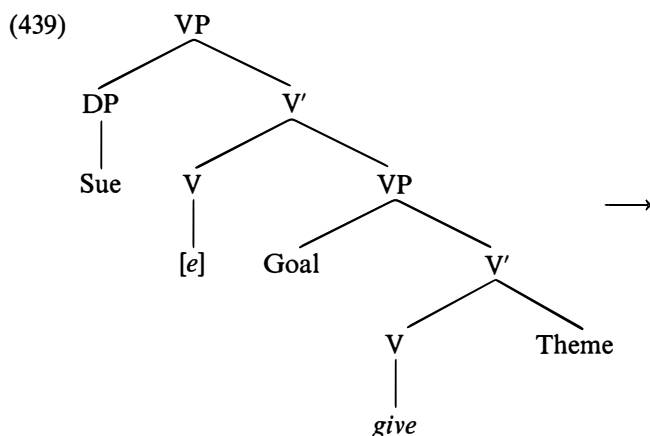
Small clause theories of double object structures thus reach an impasse. The central argument for these theories is the convergence of constituency-based accounts of coordination and binding with a plausible semantic decomposition for predicates that participate in double object constructions. In more complex cases, however, the same reasoning about coordination and binding leads to structures that have *no* basis in seman-

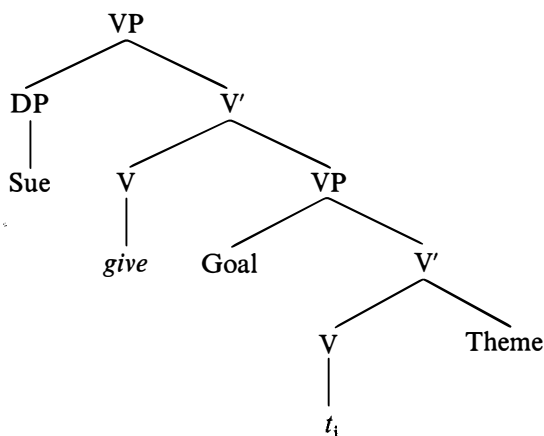
tic decomposition. In the case of (431b), the structures that must be posited actually undermine the decomposition. To summarize: The sequence *DP G DP* in a double object construction does not behave for binding theory like a clause whose predicate is *G*. Furthermore, the constituency that looks “small-clausal” at first glance turns out to be part of a more complex general structure for *V'* that undermines the beguiling similarity between double object constituency and clausality. Thus, we should turn away from theories that posit *G* as an independent predicate. The nature of double object constructions can ultimately be determined only in the context of a general account of the internal structure of *V'*.

5.5.2 VP Shell Theories

The CFC problem and the rightward command problem arise from viewing *G* as a verbal predicate (or its trace) distinct from the main predicate of *V'* in the context of a standard theory of the syntactic representation of argument structure. One approach that avoids these difficulties is Larson’s (1988) proposal for *VP shells*, according to which *G* is in fact a trace of the main predicate of (the highest) *V'* itself. My presentation of this proposal will ignore some complexities of Larson’s analyses and will more closely resemble the simpler version of Larson’s system developed by Aoun and Li (1989).

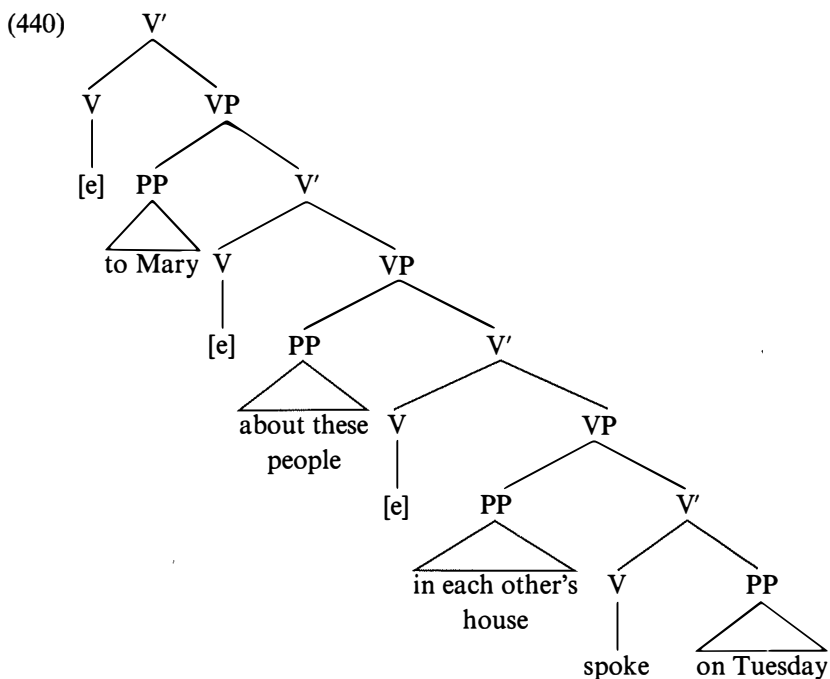
VP shell theories involve derivations like (439), where the lower VP is an instance of a VP shell.





This analysis hinges upon two features of θ -assignment. First, θ -assignment by a predicate like *give* is restricted to the specifier and complement of the position occupied by *give*. Second, θ -assignment is not “complete” at D-Structure, as it is in the classical model of *LGB*. Only after the verb *give* raises to the higher V-position does it assign its Agent role to *Sue* in the specifier of the higher V-position. Since complete θ -assignment is a definitional property of D-Structure in the *LGB* model, Larson’s analysis entails abandoning that model. (See Chomsky 1993 for discussion.) Of course, this observation is not a criticism. The existence of D-Structure as a “pure representation of GF- θ [the mapping between θ -roles and grammatical functions]” has been a contentious point throughout the recent history of generative syntax. Indeed, the existence of this level has played a relatively minor role in actual analyses. Instead, it has mainly served a technical function as an anchoring point for derivations mapping to S-Structure, PF, and LF.

The abandonment of this assumption allows more radical structures in which *none* of the main verb’s lexical properties are satisfied at D-Structure. This is how Larson explains the “rightward is downward” correlation discussed above, as summarized in (438). Consider (432b) (*John spoke to Mary about these people_i in each other_i’s houses on Tuesday*). This, for Larson, may have an initial structure like (440), where the subsequent derivation moves the verb *speak* through the various empty V-slots.



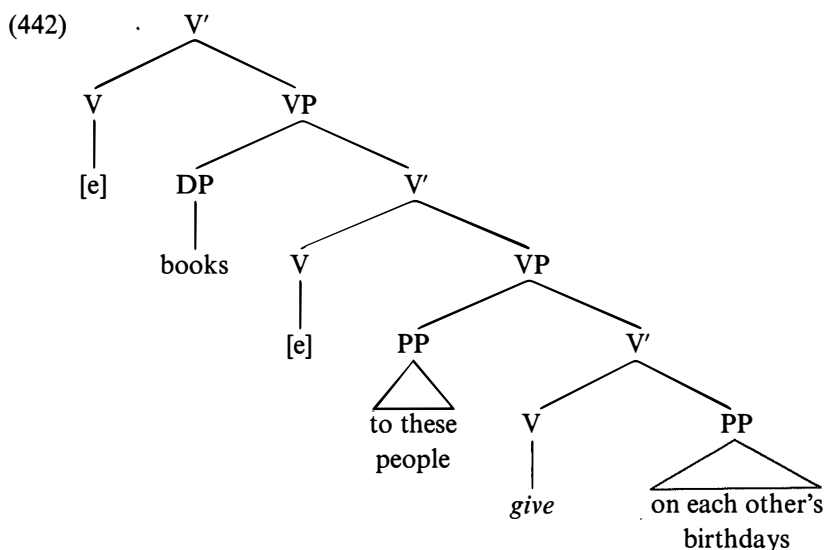
This structure is, in fact, forced if we maintain (as Larson does) the following principles:

1. An adjunct semantically related to *V* must be sister to a projection of *V*.
2. Adjuncts are lower than arguments in a thematic-relational hierarchy of some sort.
3. Syntactic branching is binary.

If we ignore the branching structure introduced by *PP* in our calculations of *c-command*, this sort of structure exactly captures the previously noted rightward-downward correlation for binding. (Though as it happens, problems that arise from ignoring *PP*s in command calculations will furnish the most important argument against *VP* shell theories in the next chapter, not to mention the solution to the *T/SM* problem.) Crucially, nothing clausal is implied by trees like (439) that group Theme and Goal together in a constituent. No particular relation between Theme and Goal is entailed either. Any semantic relation between these two elements results from the internal structure of the main verb's semantics, but syntactic constituency goes its own way.

At this point, however, we can already observe one explanatory gap in Larson's proposal: the apparent absence of a natural account for the contrast between double object and *to*-constructions when they are nominalized. On Larson's account, both structures involve his equivalent of *G*—the trace of the main verb—separating the first from the second object in each construction. This is necessary if only because the object of *to* can bind into adjuncts. The exact shape of the relevant trees depends on the details of the analysis assumed, but the general form of the structure for examples like (441) must be (442). Note the V-slot between the first and second objects.¹⁵³

(441) Sue gave books to these people_i on each other_i's birthdays.



5.5.3 Then What Is Correct?

Delaying until the next chapter a fuller critique of Larson's proposal, we can still begin to see the attractiveness of alternatives that do not identify *G* with the trace of a main predicate (or with a null main predicate), but rather treat it on a par with prepositions like *to*. It becomes natural to allow analyses without *G* for alternants with *to*, and to posit a relation between *G* and the Theme, but no relation between *G* and the Goal. No CFC problems arise, and nominalization patterns are accounted for.

On the other hand, we have not made this analysis consistent with the coordination and binding data that motivated the discussion of small clause and VP shell theories in the first place. As things stand, we are faced with a VP shell theory that seems to handle command and coordination phenomena (with caveats to be introduced shortly), but lacks an account of nominalization patterns. Opposed to this is a prepositional theory of *G* that has not so far been adapted to handle command and coordination phenomena, but does account rather naturally for nominalization patterns. In the next chapter, I develop the prepositional theory in a manner that incorporates some of the structural properties of VP shells and yields, at last, the answer to our main riddle: the T/SM restriction.

In this chapter, I have presented an analysis of many of the major features of the English double object construction, along with an analysis of the similar locative alternation. This analysis used the tools developed in previous chapters to posit certain zero prepositions, whose properties are in many ways similar to the properties of *CAUS*. One key difference lay in the ability to argue that the empty preposition *G* is affixed by a rule of syntax. This argument was based on the c-command asymmetries identified by Barss and Lasnik (1986), which motivate a preaffixation syntactic structure in which *G* heads a PP. The analysis presented in this chapter thus supports a treatment of derivational morphology like Baker's (1988). It has a further, more technical consequence, which is important to the rest of the story. The discussion in this chapter shows that Myers's Generalization acts like a filter on morphological structures built in the syntax, on top of whatever other role it may play in excluding illegitimate words. In the next chapter, I will argue that *CAUS* has a syntactic analysis very similar to that of *G* and its kin. The results of this chapter will then ensure that we do not lose the interaction between *CAUS* and Myers's Generalization that was developed in the previous chapter.

In this chapter, I have dealt only cursorily with one obvious question concerning double object configurations: the U(T)AH problem posed by the alternation with *to*. Furthermore, I have not dealt with one traditional trial by fire for analyses of the double object configuration: its behavior in English particle constructions.

The next two chapters will provide just such analyses, once I have developed the theory of argument structure and constituency that explains the T/SM effect. In fact, results concerning this interaction will be the extra dividend we earn once we finally settle the thematic hierarchy paradox from the previous section.

Appendix A: Future Having versus Choosing

Verbs of “future having” appear to allow a double object construction like those we have examined so far.¹⁵⁴

- (443) a. Mary promised John a nice raise.
 b. Sue bequeathed us her record collection.
 c. Bill guaranteed John a pension.
 d. The commandant permitted Dostoevsky one last cigarette.
 e. The doctor prescribed Bill some Valium.

These verbs have in common with *G* verbs the string *V DP DP*, and also seem to involve a zero morpheme.

- (444) a. *Bill's guarantee of John of a pension
 b. *Mary's promise of John a nice raise

One might argue that these verbs involve *G tout court*, since their meanings involve a Theme and a Recipient, albeit within the scope of some sort of intensional operator that blocks inferences from *Bill guaranteed John a pension to John got a pension*. The only interesting bar to this analysis is the observation (Pinker 1989:119, Gropen et al. 1989:247ff.) that this class does not appear to observe the morphophonological constraint otherwise characteristic of constructions we have analyzed as containing *G*. Verbs of future having with the stress pattern of *permit*, *prescribe*, and *guarantee* allow the double object structure, whereas verbs like *convey* and *recount* do not. In this respect, the zero morpheme here is more like *G_{with}*. *G_{with}* also fails to obey this constraint, as (382)–(386) and (393)–(396) (or examples like *deprive* for *G_{of}*) illustrate.¹⁵⁵

Obviously, we might posit a separate morpheme *G_{fur}* of future having and postulate that this morpheme, like *G_{with}*, does not obey the relevant conditions. Alternatively, we might seek a more principled analysis. For example, we might propose that these structures contain the same morpheme *G* found with verbs like *give* and *show*, but posit another empty morpheme as well. This extra morpheme would reflect the irreality of the possession relation.

- (445) Bill promise John_i [$\emptyset$ [_{*t*_i} *G* a pension]]

Suppose *G* raises to $\emptyset$ and thereby does not violate the morphophonological condition barring the attachment of *G* to Latinate morphemes. Then *G* + $\emptyset$ would raise to the higher verb, producing the data in (444). We would need to require *G* to cooccur with $\emptyset$ while still allowing *to*, since *to* does not interfere with nominalizations.

- (446) a. Bill's guarantee of a pension to John
 b. Mary's promise of a nice raise to John

I will leave these questions open, since I have no evidence that might shed further light on them. However, the type of complex raising structure suggested here will return as a crucial part of the next chapter.

Appendix B: G_{with} and Idiosyncrasy

A leading idea of Pinker's work, which he inherits from earlier research by Levin and Rappaport, among others, is the *predictability* of alternations like the double object alternation. In my treatment of these alternations, I have transferred this notion of predictability from the domain of "alternations" to the domain of preposition selection and preposition semantics. However, this domain exhibits occasional idiosyncrasy, related to the concept of l-selection introduced earlier. It would not be surprising to discover that verbs may select *G* or G_{with} as a matter of l-selection—the same type of selection that dictates *on* with *depend* and *at* or *to* with *look*. This appears to be the case with verbs like *envy* and *begrudge* for G_{to} and *serve* for G_{with} . None of these verbs allows an alternant with another preposition.

- (447) a. Sue envied/begrudged Mary her intelligence.
 b. *Sue envied/begrudged her intelligence to Mary.
 c. *Sue's envy/grudge of Mary of her intelligence
- (448) a. Bill served John with a summons.
 b. *Bill served a summons to John.
 c. *Bill's service of John with a summons

These cases should be distinguished from another domain in which overt prepositions appear worse than double object configurations: causative uses of verbs like *give* (Oehrle 1976).

- (449) a. The wind gave Sue an idea.
 b. ?*The wind gave an idea to Sue.
- (450) a. The war brought John only grief.
 b. ?*The war brought only grief to John.

I will return to examples like (449)–(450) in the next chapter, where they will provide crucial evidence for the account I will give for the T/SM effect with Experiencer verbs.

Chapter 6

The Target/Subject Matter Restriction Solved: Cascades

By giving a coherent and plausible description of the bound roots posited in the morphological analysis of verbs like English *annoy*, chapter 4 provided a foundation on which to build an account of the T/SM restriction. The previous chapter made this foundation more solid, by showing a wider domain in which zero affixation interacts with nominalization. It also included a preliminary discussion of concepts and structures that will be crucial in accounting for the T/SM restriction.

In chapter 4, I suggested that the bound root is a form that, for semantic reasons, involves a controlled, external A-Causer argument that, for syntactic reasons, can only be realized as a reflexive clitic. Application of nominalizing or causativizing morphology to such a root removes the external argument and, with it, the need for the reflexive clitic. Adjectival passivization has the same consequence. Thus, in English, these predicates show up nominalized, causativized, and as adjectival passives; they show up in their pristine form only in languages that have the necessary clitic. Nominalization and adjectival passivization do not invoke any T/SM restriction, but causativization does. In this chapter, I will propose an account for the T/SM restriction with ObjExp causative verbs. This account will explain the other previously examined cases of this restriction as well as a variety of new phenomena.

In and of itself, the T/SM restriction is a rather small fact. Nonetheless, it is one of those small facts that (if I am correct) allow us to discover important properties of language. In particular, before we are done, our investigation of the T/SM restriction will lead to a complete revision of current conceptions of phrasal organization. Remarkably, this revision arises not merely from the T/SM restriction, but from the solution to the problems brought up at the end of the previous chapter. We begin where we have just left off, and then rejoin the T/SM restriction.

6.1 From VP Shells to Cascades

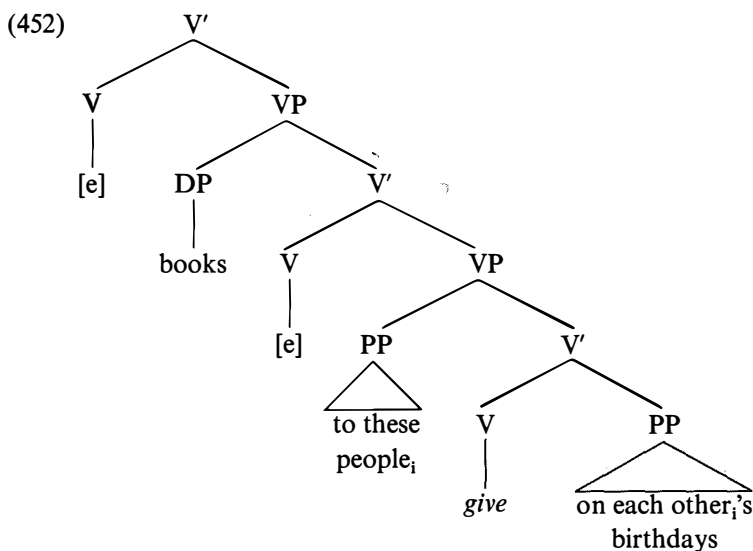
6.1.1 A Novel Structure: Binding Evidence

The last chapter ended with the observation that something was right about the VP shell hypothesis, insofar as it imposes a general equivalence between “rightward” and “downward” among dependents of V. The evidence that “rightward” roughly entails “downward” came from a convergence between binding asymmetries and coordination patterns. At the same time, the VP shell hypothesis did not seem to offer a natural account of the nominalization contrast between double object and *to*-object structures. In fact, the VP shell hypothesis seems to fall short in a much more serious fashion.

Consider again the “rightward-downward” link as it manifests itself in examples like the ones in (432)–(437) and (441), some of which are reproduced here.

- (451) a. Sue spoke **to** these people_i about each other_i’s friends in Bill’s house.
 b. John spoke to Mary **about** these people_i in each other_i’s houses on Tuesday.
 c. Sue gave books **to** these people_i on each other_i’s birthdays.

If we take these examples (and their cousins in (432)–(437)) as evidence for c-command, then we have to worry about the structure introduced by the prepositions boldfaced in (451). Consider the VP shell structure for (451c) in (442), reproduced here.



The PP *to these people* c-commands *each other* in (452), but the antecedent of *each other*—the simple DP *these people*—does not. This is a familiar problem from the literature on binding theory. Even in a traditionally organized V', certain PPs must be disregarded when computing command relations for binding. These problems were noted (without any solution) by Reinhart (1981:631–32, 1983:175–76) and by Jackendoff (1990: 430ff.), but they do not seem to have received any systematic or convincing treatment in the literature. A VP shell organization eliminates the embarrassment that PPs must not count for command of α when α 's licenser precedes α , but not when it follows. Nonetheless, the fundamental problem raised by PPs remains.

PPs are a problem even if one revives aspects of the earliest command-based accounts of binding (e.g., Langacker 1969), which included a role for linear precedence as well as command. (Jackendoff (1990) has recommended reviving this approach, as part of a critique of Larson 1988). In this approach, the data we have examined would follow from a precedence condition, if PP is consistently disregarded in the computation of command. For example, we might define a relation called *EBPP-command* ("everything-but-PP-command") as a component of the notion "binding" (cf. Lasnik's (1976) notion of *Kommand*).

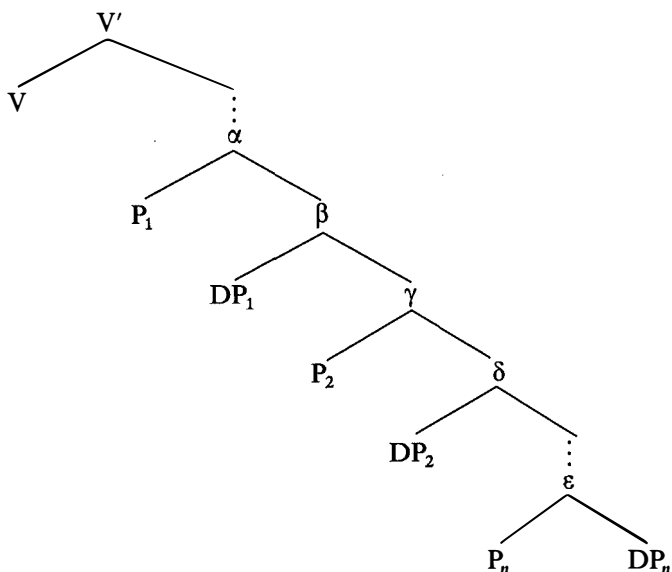
- (453) a. α EBPP-commands β iff the first non-PP that dominates α also dominates β .
 b. α binds β iff α EBPP-commands β , α precedes β , and α and β are coindexed.

Such a theory would account for the interaction of prepositions with binding phenomena. Nonetheless, (453) would not explain this interaction. (453) cannot tell us *why* PPs fail to count for command, precisely because this fact is stipulated.¹⁵⁶

There is an alternative approach, however, which I will pursue in this chapter. Unlike its predecessors, this approach is not just a way around a difficulty for c-command; instead, it solves in one stroke an array of problems—including, ultimately, the T/SM restriction. This alternative retains the simplest notion of c-command—one that does *not* distinguish among category types. The behavior of prepositions for binding phenomena is taken, not as evidence for a failure of command theories, but as an argument for novel forms of constituency. In particular, the present problem suggests that the structure of a VP whose terminal elements are shown in (454) resembles (455).

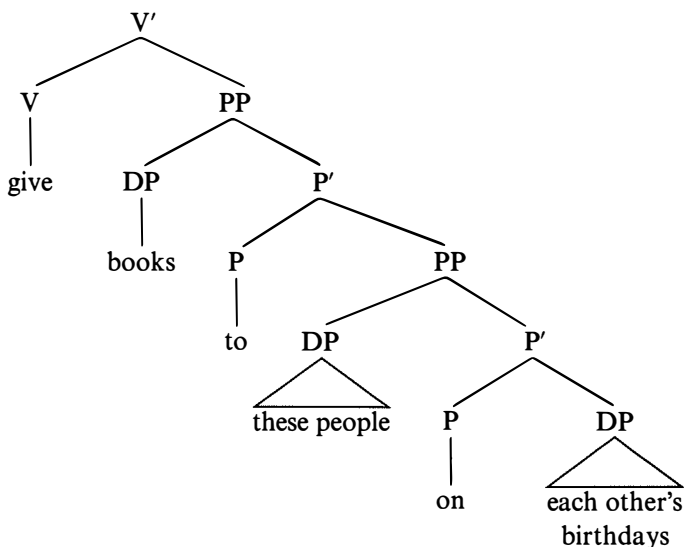
- (454) V ... P₁ DP₁ P₂ DP₂ ... P_n DP_n ...

(455) ...



In (455), DP_1 asymmetrically c-commands all the material to its right, as do DP_2 and DP_n . This is just what is required if we are to solve our binding problems in the simplest possible way. In (455), the sequences $DP_1 P_2 DP_2 \dots P_3 DP_n \dots$ and $DP_2 \dots P_3 DP_n \dots$ are constituents. For (451c), a structure of this sort might look like (456).

(456)



To summarize: We have asked whether the aberrant behavior of prepositions argues for a new sort of structure—(455)–(456)—or for an alternative to simple c-command. I will now begin to argue in favor of the new sort of structure. Structures like (456) I will call *Cascade structures*. These structures differ in appearance from the familiar type. I will show, however, that the differences of principle are small. These differences boil down to two important points. First, the branching in these structures is binary. Second, and most novel, the relation “internal argument of α ” (“*internally selected by α* ”) does not correspond exclusively to the syntactic relation “sister of α ”; rather, it may also correspond to the relation “specifier of the sister of α .” Thus, the Theme argument of *give* in (456) occupies the specifier position of *give*’s sister, the PP headed by *to* (which satisfies the Goal requirement of *give*). Likewise, the object of *to* occupies the specifier position of the PP headed by *on*. The object of *on*, however, occupies a rather traditional place as the sister to *on*. This sort of structure is as law-governed as any (as we will see), but clearly differs from the traditional alternatives. Nonetheless, it is what we need if we are to have a simple account of the binding phenomena we have been examining.

Is this the right move? As always in discussions like these, the strength of the overall argument for Cascade structures rests on the quantity and quality of the evidence that converges on the solution. We begin with coordination phenomena that furnish just such converging evidence for Cascade structures.

6.1.2 Coordination Evidence

Coordinations like (457) have already provided general evidence for a view of command and structure in which “rightward” generally entails “downward.” With Larson (1988), I have tried to maintain the simplest view of coordination, according to which it involves only constituents. If we resolve the preposition problem for command by positing structures with novel constituents, then these constituents should be able to serve as conjuncts in coordination. This prediction is true. Coordination of the novel constituents of (455) can be observed in (458).¹⁵⁷

- (457) a. Sue will speak to Mary [about linguistics on Friday] and [about philosophy on Thursday].
 b. Kremer will perform this concerto [in Rome on Tuesday], [in Lund on Wednesday], and [in Somerville on Thursday].
 c. Sue gave books [to these people on Friday] and [to those people on Saturday].

- (458) a. Sue will speak to Mary about [linguistics on Friday] and [philosophy on Thursday].
 b. Kremer will perform this concerto in [Rome on Tuesday], [Lund on Wednesday], and [Somerville on Thursday].
 c. Sue gave books to [these people on Friday] and [those people on Saturday].

This sort of conjunction should not be possible at all if we assume a VP shell structure (and continue to adhere to Larson's conjecture that only constituents participate in conjunction). In a Cascade structure, however, the coordinations in (458) involve constituents every bit as real as those in (457). Furthermore, the constituency that allows us to maintain the simplest theory of coordination is the same constituency that allows us to maintain the simplest notion of binding. This sort of convergence constitutes a strong first argument for Cascade structures. Either phenomenon by itself admits a variety of accounts, but these classes of accounts intersect (as far as I can see) at one point: Cascade structures. In this light, it is quite encouraging and surprising that this point of intersection also allows us to maintain the simplest (constituent-based) theories of both binding and coordination.¹⁵⁸

6.1.3 Obligatory Cascades?

The command and coordination phenomena of the previous section are both consistent with Cascade structures as a configurational option, but do not tell us whether Cascade configurations are "obligatory"—that is, whether there are alternatives to the constituencies seen in the Cascade trees we have examined. For example, *must* DP_1 c-command DP_2 and DP_n , in (454), and *must* DP_2 c-command DP_n or are there alternatives? This remains an open question because structural properties whose only role is to *license* certain elements or relations can only be detected when those elements or relations are present. By contrast, structural properties that "*delicense*" (exclude) certain elements or relations can provide a test for this type of obligatoriness. A configuration that excludes some element or relation σ is the only available configuration if σ is in fact excluded.

In fact, there is evidence that the command relations incorporated in Cascade structures are obligatory. The evidence comes from the condition that restricts c-command of names by pronouns, grouped in *LGB* with the binding principles for pronouns and anaphors as Principle C of the binding theory.¹⁵⁹ This condition has just the desired property; it "*delicenses*" certain relations of coreference.

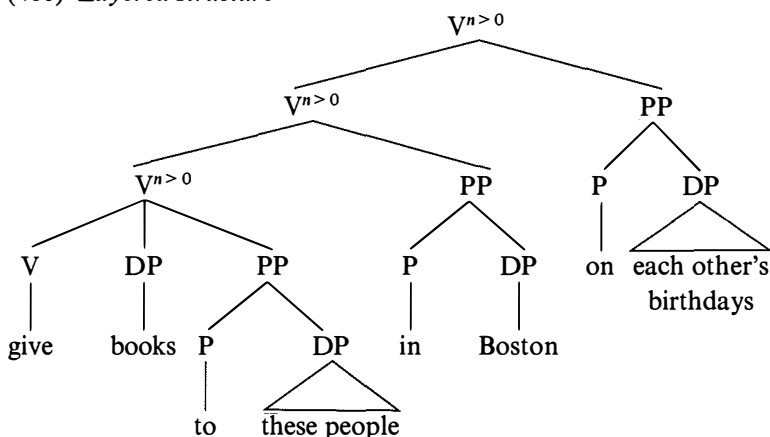
- (459) a. *Sue spoke to him_i about Bill_i's mother.
 b. *Mary danced in it_i with the owner of the hall_i.
 c. *Mary played quartets with them_i at [John and Sue]_i's party.
 d. *John threw the ball to him_i on Friday during John_i's speech.

If the computations that enter into coreference were allowed to ignore Cascade configurations, the coreference relations in (459) would be possible, contrary to fact. This means that Cascade structures *must* be formed, as part of the computational mechanism that deals with syntax.

6.1.4 Principles for Cascade Structures

What principles of grammar could yield structures like (455)–(456)? Let us begin by considering the principles that underlie traditional (non-Cascade) structures. I will call the traditional organization of phrases like VP a *Layered structure*, since selected arguments in these structures occupy layers distinct from other elements. Consider, for example, a Layered structure for a VP like the one in (451c), to which I have added a locative phrase.¹⁶⁰

(460) *Layered structure*



A Layered structure, like a Cascade structure, is generated by projecting certain semantic and lexical properties onto a tree that obeys X-bar theory. However, Layered structures do not have the “downward” orientation of Cascade structures. Indeed, “rightward is upward” is a more appropriate slogan for a Layered structure than “rightward is downward.” Furthermore, Layered structures are not restricted to binary branching as Cascade structures are.

How can the differences between these modes of configuration be characterized? The discussion of θ -selection in the last chapter has already introduced some useful notions. In a Layered-structure representation, if *give* θ -selects an argument, it must be a sister to *give* when direct θ -selection holds. Otherwise, the phrase headed by a preposition that mediates θ -selection must be a sister to *give*. These notions were spelled out in (367) and (368), repeated here. (The subscript *L* stands for *Layered*.)

(461) *Selected positions (Layered)*

- a. π externally selects_L α iff α is Spec, $\pi^{\max}$.
- b. π internally selects_L α iff α is the sister of π .

(462) *θ -selection*

- a. If π θ -selects an external θ -role *R* as a lexical requirement, then this requirement is satisfied if an argument bearing *R* occupies a position externally selected by π .
- b. If π θ -selects an internal θ -role *R* as a lexical requirement, then this requirement is satisfied if either
 - i. an argument bearing *R* occupies a position internally selected by π (*direct θ -selection*), or
 - ii. a selector of *R* heads a position internally selected by π (*mediated θ -selection*).

Now consider the relation that holds between a main predicate and its *arguments* in the broadest possible sense of that term. For example, consider the relationship that holds between *give* and *books*, *give* and *these people*, *give* and *Boston*, *give* and *each other's birthdays* in (460) (the "Theme relation," the "locative relation," the "temporal relation," etc.). Let us call the categories that bear these relations to the main predicate *semantically related* to that predicate. Semantic relatedness has a structural correlate in traditional, Layered structures. This correlate is spelled out in (463).

(463) *Semantic relatedness (Layered structures)*

A category α is *semantically related*_L to a predicate π iff α or a θ -selector of α is a sister to a projection of π .

Viewed from one perspective, (463) enforces a sisterhood requirement on semantic relatedness: elements semantically related to π must be sisters or daughters of sisters to projections of π . Consequently, (463)

also has the effect of imposing an m-command condition on semantic relatedness.

Let us consider what alternative to (461) would produce Cascade structures. Cascade structures differ from Layered structures because (461b) and (463) are different for Cascade structures—the structural correlates of semantic relatedness and internal selection. The changes are minor and technical but the consequences are far-reaching. I have already informally characterized the necessary change to the notion of selected positions. Here is a restatement:

(464) *Selected positions (Cascade structures); version 1 of 3*

- a. π *externally selects* α iff α is Spec, $\pi^{\max}$.
- b. π *internally selects* α iff
 - i. α is the sister of π , or
 - ii. α is the specifier of the sister of π .

I will revise this statement shortly. For the moment, let us turn to the other important notion: semantic relatedness. In a Cascade structure, semantic relatedness must involve *c-command* rather than m-command. This is part of the import of the binding evidence in (459).

We might suppose that any head c-commanded by V counts as semantically related to that V, but such a statement would allow too many instances of semantic relatedness. For example, we do not want to allow the object of a preposition inside a nominal direct object of V to bear a semantic relation to V. *Science* in (465) bears no semantic relation to the verb *interrupt*.

(465) interrupt [_{DP} a discussion [_{PP} about science]]

Nor do we want to allow categories in embedded CPs to bear a semantic relation to elements outside those clauses. Thus, we must set things up so that DP and CP block semantic relatedness to categories outside DP and CP. Given our treatment of “mediated θ -selection” by prepositions like *to* with *give* or *on* with *depend*, DPs and CPs are special in another way: they are the only categories that count as bearing θ -roles. Accordingly, I will call them *argument categories*.

(466) Argument categories: DP, CP

It is now possible to give the syntactic correlates in Cascade structures of the relation “semantically related,” which holds of argument categories in VP.

(467) *Semantic relatedness (Cascade structures)*

- a. An argument category α is *internally semantically related*_C to a main predicate π iff
 - i. α is c-commanded by π , and
 - ii. there is no *argument category* Ω such that π c-commands Ω and Ω dominates α .
- b. An argument category α is *externally semantically related*_C to a predicate π iff α occupies Spec, π .
- c. An argument category α is *semantically related*_C to a predicate π iff α is internally or externally semantically related to π .

I will leave the roster of “semantic relations” unspecified here, but it includes internal θ -roles as a special case.

(468) If π θ -selects α as an internal argument, α is semantically related to π .

(468) and the condition that blocks semantic relatedness across DP and CP conspire to prevent θ -selection across a DP or CP boundary, which is in general the right result.¹⁶¹

Let us assume further that (469) presents facts, with the content of the three dots to be determined empirically (though not in this book).

- (469) a. If an argument category bears the relation Causer, Means, Location, Time, . . . , or Instrument to some predicate, it occupies a position semantically related to that predicate.
- b. If an argument category occupies a position semantically related to a predicate, it bears some semantic relation to that predicate.

Finally, it will be useful to define *Cascade*.

(470) The *Cascade* of a predicate π comprises

- a. the set of categories semantically related to π , and
- b. the categories that θ -select them.

I will also speak of π as the *head* of its Cascade. The notion “head of a Cascade” provides the concept needed to explain the behavior of zero morphemes below.

Let us now fill in some miscellaneous missing details and minor consequences of the system.

6.1.5 Loose Ends and Consequences

6.1.5.1 Raising to Object and Semantic Relatedness In general, the structural correlates of semantic and lexical notions are just as well defined for Cascade structures as they are for traditional, Layered structures. In one respect, however, Cascade syntax displays an ambiguity of function not found in some versions of Layered structures. Nothing in the discussion so far reserves the position of either a verb's sister or the specifier of a verb's sister for a θ -selected argument of that verb. This contrasts with the situation in *LGB*, where the sister position to α is reserved for elements θ -marked by α . Nonetheless, semantic relatedness leaps into the fray here, even though this is a broader concept than θ -selection. An argument in a clausal complement to V is, by hypothesis, not semantically related to V. On the other hand, a DP that has raised from a CP complement to a position in which a θ -selected object of V can occur will perforce be semantically related to V, yielding a contradiction. Thus, Cascade structures, like the system in *LGB*, rule out "raising to object" of the classic type. Even though I am allowing object position (the sister to V) to be a projection of a nonargument, it must still bear a semantic relation to V.¹⁶²

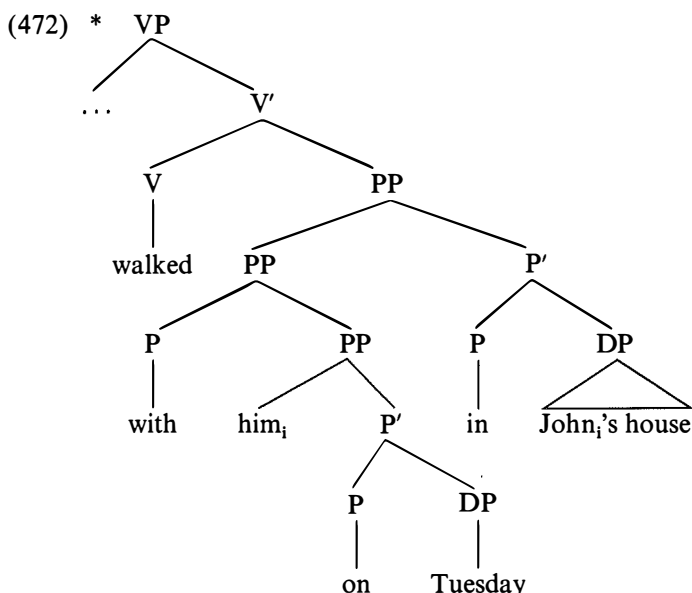
6.1.5.2 Semantic Relatedness outside VP Nothing in the current proposals eliminates the possibility (mentioned briefly above) that some PPs in a clause might be semantically related, not to V, but to some higher element in IP or CP. This would explain judgments like the following (from Reinhart 1983:60):¹⁶³

- (471) a. We sent him_i to West Point in order to please Ben_i's mother.
 b. We'll just have to fire him_i whether McIntosh_i likes it or not.
 c. Rosa won't like him_i anymore, with Ben_i's mother hanging around all the time.

I leave open the question of whether modifiers of higher categories are arranged in a Cascade-like fashion or not.

6.1.5.3 No PP in Specifier Position Let us return to the data from Principle C considered in section 6.1.3 above. I noted that these data motivate a certain "obligatoriness" for Cascade structures. Yet the principles for Cascade structures given in this section allow configurations that will bypass Principle C for these examples. These configurations display a PP

in a specifier position, mediating θ -selection by a higher predicate. Such is the case in (472), which incorrectly admits coreference between *him* and *John*.



I will deal with this problem with a stipulation concerning Cascade structures. (473) seems to be what is needed, and in essence restates a condition proposed by Kayne (1984b).

(473) *Specifier stipulation*

If α occupies a specifier position in the Cascade of a predicate, α is an argument category (i.e., CP or DP).

In its present state, (473) is nothing to be proud of. On the other hand, if we adopt (473), we also can eliminate the stipulated asymmetry between the way a predicate can satisfy θ -selection for an external θ -role and the ways a predicate can satisfy θ -selection for an internal θ -role. In (368) (repeated in (462)), I allowed θ -selection for an internal θ -role to be satisfied by either direct or mediated θ -assignment, but required θ -selection for an external θ -role to be satisfied only by direct θ -assignment. Since (368) allows θ -selection for an external θ -role to be satisfied only by material in an externally selected position, and since (464) identifies only the specifier as an externally selected position, and since (473) bars PP from specifier position at D-Structure, mediated θ -selection of an external argu-

ment is impossible, even without stipulating this in (368). Accordingly, (368) may be simplified.

(474) *θ-selection (revised)*

Let τ range over (internal, external). If π θ -selects a τ θ -role R as a lexical requirement, then this requirement is satisfied if either

- a. an argument bearing R occupies a position τ -ly selected by π (*direct θ -selection*), or
- b. a θ -selector of R heads a position τ -ly selected by π (*mediated θ -selection*).

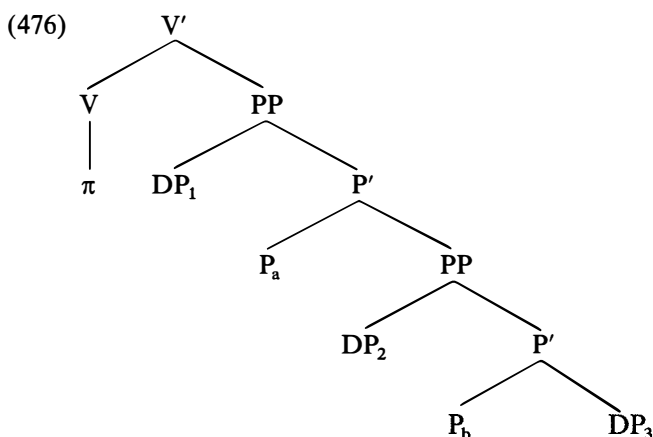
6.1.5.4 No More than Two Obligatory Internal Arguments The restriction that limits Cascade structures to binary branching is crucial to the results I have presented (assuming, as I have, that Cascade structures conform to X-bar theory). Without this restriction, for example, we might lose our account of the Principle C effect in (459). This would happen if *to him* and *about Bill's mother* could be structured as in (475), with a ternary-branching V'.

(475) [_{V'} spoke [_{PP} to him_i] [_{PP} about Bill_i's mother]]

Similar problems would follow if (459c) allowed a ternary-branching higher projection of V, which immediately dominated constituents [*Mary played quartets*]_i, [*with them*]_i, and at [*John and Sue*]_i's party. The restriction to binary branching is what forces θ -selected arguments of α to be "tucked into" the specifier of α 's sister, yielding the binding results we have examined.

If we were to impose a similar restriction on traditional, Layered structures, which do not allow this sort of "tucking in," we would, contrary to fact, see that each predicate was permitted a maximum of one internally θ -selected argument. Allowing selection to look into the specifier of the selector's sister as well as at the sister itself raises the maximum to two. But this in fact seems to be correct.¹⁶⁴ We cannot actually observe this fact directly, since the concept of θ -selection is theory-internal. However, we can observe it in action by noting the absence of verbs that take more than two *obligatory* arguments. It is important to be careful about the logic of this observation: not all θ -selected arguments are obligatory, but only among obligatory arguments is it possible to detect the maximum number of arguments that a verb can have.

There is one loophole in our discussion of Cascade structures, which I will now close. If a preposition could mediate θ -selection of more than one θ -role, then a predicate π could satisfy θ -selection for three roles in the structure (476).



Suppose π selects the θ -roles borne by DP_1 , DP_2 , and DP_3 . Now suppose P_a θ -selects the roles borne by DP_2 and DP_3 , and P_b θ -selects the role borne by DP_3 . π 's selectional properties can now be satisfied, even though π selects three roles.

- DP_1 satisfies the first selectional property of π , since it occupies the specifier of π 's sister.
- DP_2 satisfies one of the selectional properties of P_a , since it occupies the specifier of P_a 's sister. DP_3 satisfies the second selectional property of P_a , since the head of the sister to P_a θ -selects the role borne by DP_3 .
- Since P_a can successfully select the roles borne by DP_2 and DP_3 , and since P_a is the head of the sister to π , DP_2 and DP_3 can also satisfy the selectional properties of π . Thus, all three instances of θ -selection by P_a can be satisfied in this structure.

In fact, this case simply does not arise, because of the following observation:

(477) Prepositions internally θ -select a maximum of one θ -role.

This observation is true even if the edifice of Cascade Syntax that we are building should crumble. As far as I can tell, (477) is an accurate low-level, theory-independent observation. There is no prepositional counterpart to *hand* or *put*, requiring two expressions to satisfy lexical properties. More

generally, prepositions seem to take a maximum of one internal argument, obligatory or not. I do not have an explanation for this.¹⁶⁵ Nonetheless, since (477) *is* a fact, the result reported in this section stands.¹⁶⁶

Finally, we must note that the distinction between “argument” and “nonargument,” or between “ θ -selected category” and “non- θ -selected category,” is hard to discern when the phrase under discussion is not obligatory in its structural environment. For example, English speakers have the intuition that *bet* and *wager* allow three internal arguments.

- (478) a. Sue bet [John] [\$500] [that the world would end on Tuesday].
 b. Bill wagered [me] [a day's pay] [that the world would end on Wednesday].

Likewise (as Stan Dubinsky (personal communication) has pointed out), a similar intuition attends the various impedimenta of buying and selling, as well as the itinerary phrases with verbs of motion.

- (479) a. Mary sold [her car] [to John] [for \$500].
 b. John traveled [from Boston] [through Kansas City] [over the mountains] [to Los Angeles].

Still, in none of these cases do we find more than two obligatory arguments. If there is some other distinction to be drawn (as revealed perhaps by extraction from weak islands), it is something looser than θ -selection (but perhaps more restrictive than semantic relatedness; see section 7.1.5 for related discussion).

6.1.5.5 A Minimality Condition on θ -Selection In this section, I will present empirical motivation for a slight extension in the domain of internal selection for Cascade structures. As it turns out, this extension is also a conceptual improvement over my previous proposal (in that it has conceptual links to other syntactic proposals).

The verbs in (480), with the diatheses given, require two internal arguments. This is not a problem for Cascade structures. The first argument occupies the specifier position of the PP headed by the preposition that mediates θ -selection for the second argument.

- (480) a. Fred handed *(some toys) *(to the little children).
 b. Bill cleared *(the crumbs) *(off the tables).
 c. The unscrupulous prosecutor plied *(his witnesses) *(with drinks).
 d. Mary put *(the books) *(on shelves).

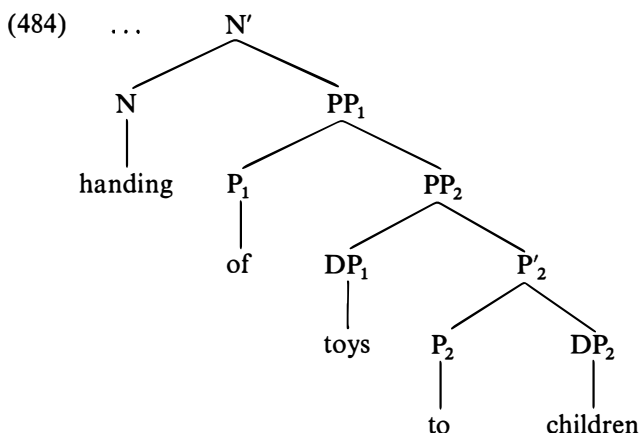
New consider nominal counterparts to these structures in which the first argument must be introduced by *of*. The best examples are gerunds in *-ing*. Not surprisingly, the two arguments remain obligatory.

- (481) a. the handing *(of toys) *(to little children)
 b. the clearing *(of crumbs) *(off tables)
 c. ?the plying *(of witnesses) *(with drinks)

In the structures of (481), the first DP c-commands the second, just as in V's with two PPs (e.g., *speak to these people about each other's friends* in (451a)).

- (482) a. The handing of the toys to each other's creators is an old tradition at the annual Old Toymakers' Dinner.
 b. As long as you clear some crumbs off some tables, you can keep your job, but the clearing of no crumbs off any tables is grounds for dismissal.
 c. The plying of each witness with his favorite drink was a favorite tactic of Hamilton Burger.
- (483) a. the handing of [toys to children] and [weapons to adults]
 b. the clearing of [crumbs off tables] and [dirt off chairs]
 c. the plying of [witnesses with wine] and [judges with whiskey]

These data motivate Cascade structures like (484).



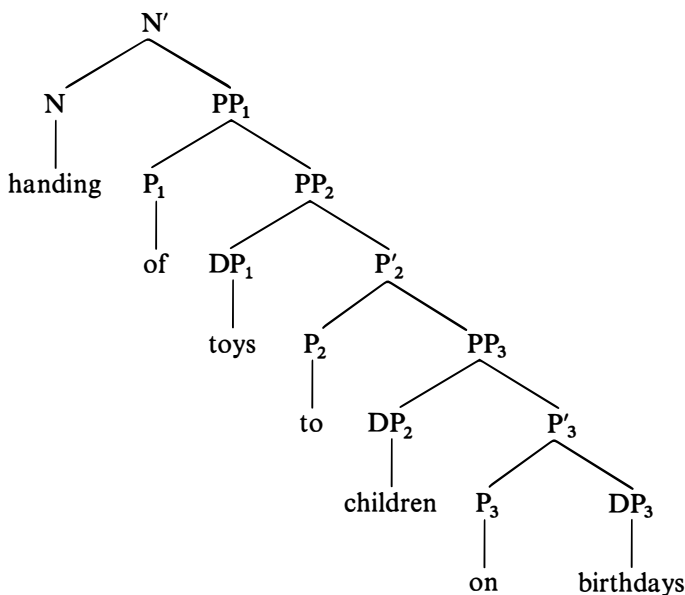
No problem arises with θ -selection of DP_1 , as long as we assume that *of* is a fairly all-purpose θ -selector (much like *G*, as we will see). Among its many virtues, *of* θ -selects Theme, which is one of the two roles that *hand-*

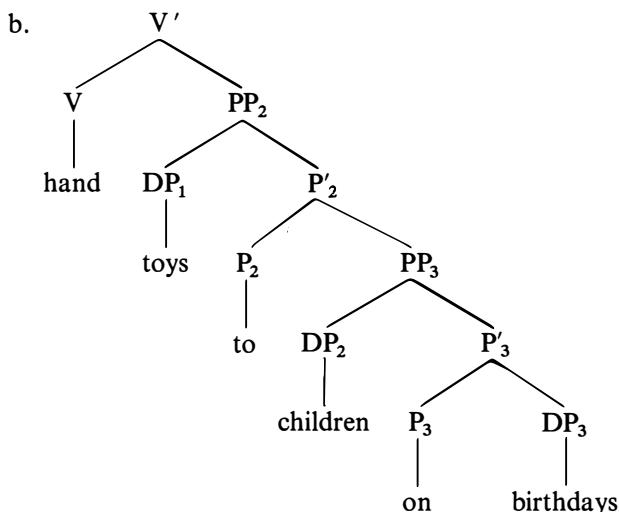
ing requires. DP_1 thus bears Theme and satisfies that requirement of *handing* via mediated θ -selection.

The problem arises with the second argument. PP_2 is neither the sister of N nor the specifier of the sister of N. The same, of course, is true of DP_2 . Consequently, there is no position in which θ -selection for the second argument can be satisfied under the definition in (464). Fortunately, however, there is an extremely appealing reformulation of the notion “internally selects” for Cascade structures that allows the sister, the specifier of the sister, and PP_2 in (484) to count as internally selected positions, and nothing else. I owe this reformulation to suggestions by Ken Wexler and Akira Watanabe (personal communication). I will lead up to this reformulation gradually.

Let us add a non- θ -selected expression to (484) and compare the result with its verbal counterpart. I have numbered the nodes uniformly, hence the absence of PP_1 in (485b).¹⁶⁷

(485) a.





In each structure, we want DP_1 and PP_2 to be accessible to θ -selection by *hand*, and no other nodes. DP_1 in each case is the nearest specifier c-commanded by *hand*. This seems to be the key. We can replace (464) with (486).

(486) *Selected positions (Cascade structures); version 2 of 3*

- a. π externally selects α iff α is Spec, $\pi^{\max}$.
- b. π internally selects α iff
 - i. π c-commands α , and
 - ii. there is no specifier σ such that π c-commands σ and σ c-commands α .

In other words, internal θ -selection by α is blocked by intervening specifiers. When the sister to α has a specifier, internal θ -selection can access the sister to α and its specifier, just as I have claimed up to this point. Because mediated θ -selection is possible, internal θ -selection can be satisfied even in cases where the actual argument α lies on the far side of the specifier barrier, so long as α is selected by a category whose maximal projection lies on the near side of that barrier.

The novelty in (485a) is the fact that the sister to *handing* lacks a specifier. In that case, internal θ -selection can continue to look down the tree until it finds a specifier. No node c-commanded by that specifier will be accessible to internal θ -selection, but mediated θ -selection works as always. DP_2 serves as an obligatory argument of *handing* because it is

θ -selected by P_2 , whose maximal projection is *not* c-commanded by the nearest specifier to *handing*.¹⁶⁸

The naturalness of this revised concept of “internally selected position” becomes apparent when we put this discussion together with the observation (section 6.1.5.3) that only argument categories may occupy specifier positions. In addition, we should note that the binary-branching, X-bar geometry of Cascade structures entails that all (nonmodifier) maximal projections that have a nonnull c-command domain are specifiers. (A specifier must be an argument and an intervening argument must be a specifier.) Consequently, we can replace *specifier* in (486) with *argument category*.

(487) *Selected positions (Cascade structures); version 3 of 3*

- a. π externally selects α iff α is $\text{Spec}, \pi^{\max}$.
- b. π internally selects α iff
 - i. π c-commands α , and
 - ii. there is no *argument category* σ such that π c-commands σ and σ c-commands α .

The role of σ in (487) now looks exactly like the role typically played by intervening categories in minimality phenomena (Rizzi 1990). Internal selection is a relation between predicate categories and argument categories. It is quite in keeping with the most recent discoveries about syntax that this relation can only hold between a predicate and the *nearest* argument category. In informal terms, a predicate that internally θ -selects an argument looks down the tree for the nearest argument, satisfying itself along the way, if it must, by categories whose head takes the relevant argument, but looking no farther than the nearest argument. Clause (bii) of (487) can thus be thought of as a *minimality* condition on the notion “internally selects” (Rizzi 1990, Chomsky 1993).¹⁶⁹

This change in the notion of internal selection does not disrupt the result that limits the number of internal θ -selected arguments to two. DP_3 , for example, could not be a θ -selected argument in (485b) because DP_1 and DP_2 intervene between the main predicate and DP_3 , as well as between the main verb and PP_3 . If PP_2 or PP_3 lacked a specifier, DP_3 would be accessible to internal selection by the main predicate—but then the V' would not have three internal arguments.

6.1.6 The Next Step

So far, this chapter has been devoted to motivating Cascade structures in V' and N' . A variety of arguments were presented for these structures, and their nature has been clarified. These arguments and clarifications served an essential purpose—fleshing out and supporting the reality of these rather novel structures. Now let us return to the discussion at the beginning of the chapter, where the basic motivation for Cascade structures was spelled out. There, two sorts of observations were important: the constituents of Cascade structures were shown to function as conjuncts in coordination, and the novel c-command relations among maximal projections in Cascade structures were shown to be exactly what is needed to explain the distribution of binding phenomena.

Another observation should go hand in hand with these. In addition to containing novel c-command relations among maximal projections, Cascade structures contain novel c-command relations among heads. In the next section, I will show that these novel c-command relations are also real. Not the least instantiation of this observation will be the explanation, at long last, for the T/SM restriction.

6.2 Cascade Interactions among Prepositions

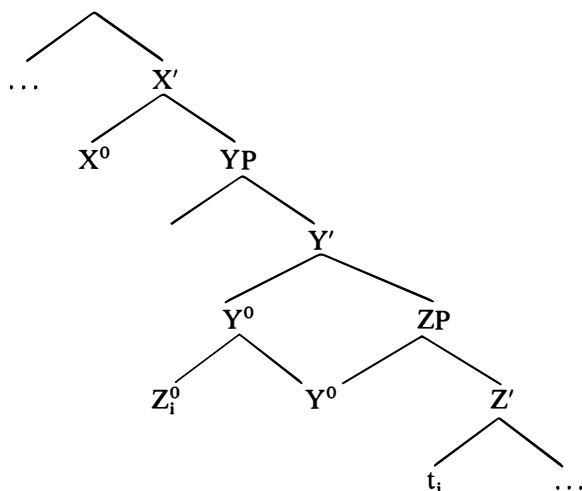
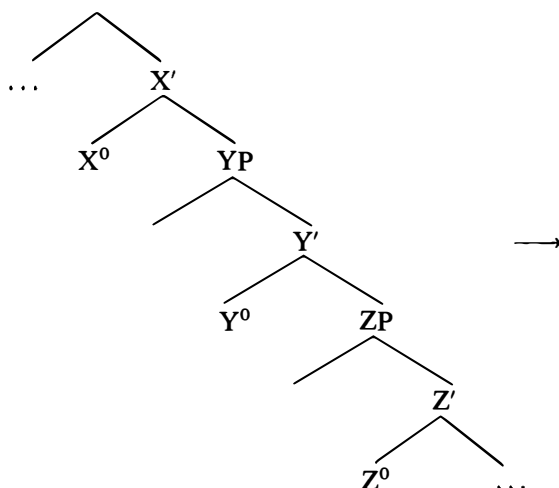
Current syntactic theory provides exactly one tool by which to judge command relations among syntactic heads. This is Travis's (1984:131) *Head Movement Constraint*, extended to derivational morphology by Baker (1988).

(488) *Head Movement Constraint (HMC)*

An X^0 may only move into the Y^0 that properly governs it.

I will not devote much attention here to Chomsky's (1986a) and Baker's suggestions that the HMC can be derived from the Empty Category Principle (ECP), nor to later proposals (Rizzi 1990, Chomsky 1993) that attribute the HMC to a general notion of minimality. This line of research seems correct, but is at present not important to the discussion. What is important here is the *effect* of the HMC. For our purposes, the HMC amounts to the observation that movement of an X^0 cannot skip over an intervening head, where the simplest notion of intervention, as always, involves c-command. Thus, in order for Z^0 to move to X^0 in (489), Z^0 must first adjoin to Y^0 , forming a new category headed by Y^0 . Only then can $Z^0 + Y^0$ move to X^0 .

(489)



The correctness of this scenario will be crucial in the sections to come. Most important will be the idea that the constituent Z^0 – Y^0 on the right side of (489) is headed by Y^0 . Consequently, it is the lexical properties of Y^0 that determine whether this constituent can move to X^0 . If Y^0 is nonaffixal, then even adjunction of Z^0 to Y^0 won't get Z^0 to X^0 .

In the literature on these issues, the HMC typically rears its head in the inflectional system, where V must often raise past various categories on its way to Infl (Agr_S) or Comp. If, for example, we assume that the

constituency relevant to the HMC is provided by Layered structures like (460), there should be no detectable HMC interactions among the various prepositions. If, on the other hand, the HMC uses the command relations of Cascade structures, we might expect to discover novel interactions among the prepositions. For example, in a structure like (485a), P_1 should block any attempt by P_2 to move to V, unless P_2 can adjoin to P_1 , and the two of them move together to V.

In general, we look for an interaction between the HMC and Cascade constituency wherever the presence of an otherwise acceptable preposition creates a problem. This is exactly the path we will follow in explaining the T/SM effect. The T/SM argument, as it surfaces in the nominalizations and adjectival passives that we have examined, is typically introduced by a preposition. The nominalizations in (490), reproduced from (199), have adjectival passive counterparts (*agitated*, etc.). The relevant prepositions are boldfaced.

- (490) a. Bill's continual agitation **about** the exam was silly.
 b. Mary's constant annoyance **about/at/with** us got on our nerves.
 c. Despite Sue's frequent amusement **at** the goings-on, she was, deep down, quite upset.
 d. John's constant embarrassment **about** his looks was unnecessary.
 e. Notwithstanding our continual surprise **at** the news from the East ...

In section 3.3, I argued that the T/SM restriction does not arise from any semantic problem. This suggested a syntactic explanation, but no obvious possibilities were available at that point. In the discussion that followed, I laid the groundwork for such an explanation by developing the proposal that causative psychological predicates contain a phonologically null causative morpheme. The syntactic origins of this causative morpheme in the context of Cascade structures will give us our explanation. The best path to this explanation, however, runs through an empirical territory quite close to the T/SM effect, involving a different set of verbs. These are the double object verbs discussed in the previous chapter.

6.2.1 The T/SM Restriction Explained

6.2.1.1 Oehrle's Observation The phenomenon in question seems to have been first noted by Oehrle (1976). Crucially, this phenomenon also

involves causative semantics. Oehrle notes that examples like (491) have three types of readings.

(491) Nixon gave Mailer a book.

On two of these readings, Nixon is asserted to have performed an action. These readings differ in whether the book changes ownership or not, a difference that is not of interest here. A third reading, however, is of great interest. Oehrle paraphrases this reading as (492).

(492) “Mailer wrote a book which he wouldn’t have been able to write if it hadn’t been for Nixon.”

The key fact, noted by Oehrle, is the absence of the “third reading” in the alternant with *to*.

(493) Nixon gave a book to Mailer.

Oehrle identifies this third reading as causative. This seems to be correct. For example, if we replace *Nixon* with a DP that cannot be the Agent of an action, but can only be a Causer, the alternant with *to* lacks any non-deviant reading.¹⁷⁰

(494) a. The war years gave Mailer his first big success.
b. *The war years gave his first big success to Mailer.

(495) a. Interviewing Nixon gave Mailer a book.
b. *Interviewing Nixon gave a book to Mailer.

(496) a. The space program had some technological consequences. For one thing, it gave the world Teflon.
b. *... it gave Teflon to the world.

This phenomenon extends to certain other double object verbs. (497a–d) are Oehrle’s examples (p. 71) (he uses “#” instead of “*” as the relevant diacritic). The other examples are my own. (501) shows the effect with a verb that takes *for* rather than *to*.

(497) a. Katya taught me Russian.
b. Katya taught Russian to me.
c. Lipson’s textbook taught me Russian.
d. *Lipson’s textbook taught Russian to me.

(498) a. Bill told Sue the answer.
b. Bill told the answer to Sue.

- c. Talking to Bill for just a few seconds would have told Sue the answer.
 - d. *Talking to Bill for just a few seconds would have told the answer to Sue.
- (499)
- a. Bill showed Sue the answer.
 - b. Bill showed the answer to Sue.
 - c. Talking to Bill for just a few seconds would have shown Sue the answer.
 - d. *Talking to Bill for just a few seconds would have shown the answer to Sue.
- (500)
- a. Valente guaranteed Scorsese the prize money.
 - b. Valente guaranteed the prize money to Scorsese.
 - c. The absence of competition guaranteed Scorsese the prize money.
 - d. *The absence of competition guaranteed the prize money to Scorsese.
- (501)
- a. Sue rented John the apartment.
 - b. Sue rented the apartment to John.
 - c. ?A smile and the offer of a check rented John the apartment.
 - d. *A smile and the offer of a check rented the apartment to John.
- (502)
- a. Bill got Mary the prize.
 - b. Bill got the prize for Mary.
 - c. Hard work got Mary the prize.
 - d. *Hard work got the prize for Mary.

This paradigm is not found with all verbs that show a double object structure alternating with *to*. For example, *hand* does not seem to produce the relevant effect, at least in the idiomatic usage of (503), nor does *bring* in (504).¹⁷¹

- (503)
- a. Sue handed Bill a golden opportunity.
 - b. Sue handed a golden opportunity to Bill.
 - c. The conversation with Sue handed Bill a golden opportunity.
 - d. The conversation with Sue handed a golden opportunity to Bill.
- (504)
- a. Bill brought Sue a modicum of happiness.
 - b. Bill brought a modicum of happiness to Sue.

- c. Talking to Bill for just a few seconds would have brought Sue a modicum of happiness.
- d. Talking to Bill for just a few seconds would have brought a modicum of happiness to Sue.

Furthermore, there is variation among speakers concerning which verbs show Oehrle's effect. This is not, however, a reason for despair or irritation over unstable and murky judgments; and in fact I will have an explanation for this variation. What is important is simply the existence of the effect.

6.2.1.2 Evidence from Nominalizations Recall the analysis of the double object alternation in chapter 5. There, we saw that the double object variant without the overt preposition involves a zero preposition *G*. This zero preposition must affix itself to the governing verb (as the impossibility of nominalization showed). By contrast, as is evident, the overt prepositions *to* and *for* are nonaffixal. I now suggest that this distinction is essential to the explanation of Oehrle's contrasts.

Our starting point will be the correctness of Oehrle's observation that the forms that resist overt *to* (and perhaps *for*) describe causes of events—not activities by agents. Suppose these causative uses involve affixation of the zero morpheme *CAUS* to an otherwise agentive verb. (In other words, suppose the agentive use is primary.) This morphological process will eliminate the Agent role, just as A-Causer was eliminated by affixation of *CAUS* in chapter 4, forming a causative verb from an agentive verb. The resulting verb—for example, *CAUS-give*—will thus not be glossed 'cause to give' but rather 'cause to be given' or, perhaps, 'cause to get'.

Evidence of a familiar sort from nominalizations supports an analysis in which causative use of verbs like *give* involves *CAUS*. The evidence is independent of Oehrle's observation, since it does not depend on the presence of *to*.

- (505) a. Smith's gift of a new building
 - b. *our efforts' gift of a new building¹⁷²
- (506) a. Valente's guarantee of the money
 - b. *the absence of competition's guarantee of the money
- (507) a. Sue's rental of the apartment
 - b. *the smile's rental of the apartment

- (508) a. Bill's procurement of the prize
 b. *hard work's procurement of the prize

Suppose now that *CAUS*, like *G*, starts life at D-Structure as a syntactically independent head. If this is the case, we must ask about its structural position before affixation. The most familiar type of proposal would identify *CAUS* as a verb that embeds the caused proposition. If this were correct for Oehrle's cases, there would be no obvious source of an interaction between *CAUS* and prepositional arguments of the embedded verb.

I will take a different road. In particular, let us consider the following hypotheses:

1. *CAUS* is a clause-internal preposition like those in (509).
 (509) a. Sue yelled *out* of frustration.
 b. Mary objected to the show *because of* Bill's remarks.
 c. Mary jumped *for* joy.
 d. John died *of* consumption.

In particular, *CAUS* does not introduce a selected argument, but rather has the status of *because* or the *wh*-phrase *why* (i.e., what has often been called an "adjunct"). That is why, for example, neither the causative use of *give* nor the presence of overt causative prepositions is obligatory.

2. *CAUS* is [+affix], and, like *G*, must be moved to the main verb.

- (510) *CAUS* must be part of the main verb by PF.

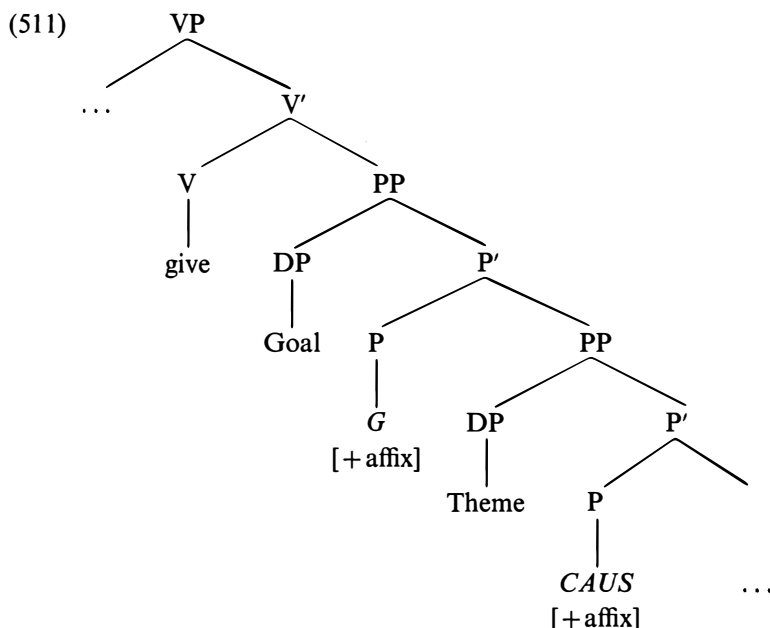
The general idea of a causative prepositional affix is not original here, have been first developed by Walinska de Hackbeil (1986) for the causative *en-* in *enlarge*, *embitter*, *endear*, and so on. Here I extend this idea to the zero causative morpheme.

3. *The properties of this affixation are given by Cascade structures.* As a consequence, since *CAUS* is not part of the θ -selectional domain of the main predicate, it is situated *lower* in the Cascade of V than θ -selected arguments Theme and Goal. For example, when θ -selection for Theme is mediated by *G*, a structure that also includes *CAUS* will look roughly like (511). Consequently, movement of *CAUS* to V requires, as an intermediate step, adjunction to each preposition that intervenes between it and V. If one such preposition is nonaffixal, *CAUS* cannot move to V. (This point was made in the discussion following (489).)

If these hypotheses are correct, then affixation of *CAUS* to a verb like *give* requires prior affixation to *to* or *G*. Of these two prepositions,

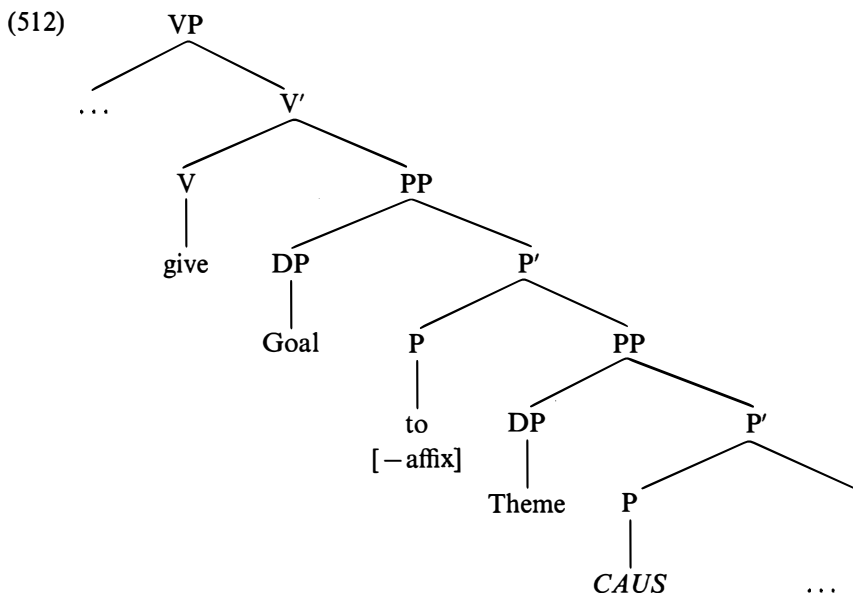
however, only *G* can affix to *V*, since only *G* is an affix. Oehrle's effects are explained by these hypotheses. Let us see how this works.

Consider the structure for the alternant with *G* when *CAUS* is also present, namely, (511). (Remember that the agentive version of this example lacks *CAUS*.) For the moment, I omit the material represented by the three dots.



In (511), *CAUS* must raise to *give*, by hypothesis. Movement of *CAUS* to *give* in one step is blocked by the HMC. Adjunction of *CAUS* to *G* is possible, however, since *CAUS* is [+affix]. This forms a new category headed by *G*. Since *G* itself is [+affix], the category *CAUS* + *G* headed by *G* can itself function as an affix and raise to *give*, as desired.

Now compare (511) with the alternant (512) that contains *to* rather than *G*.

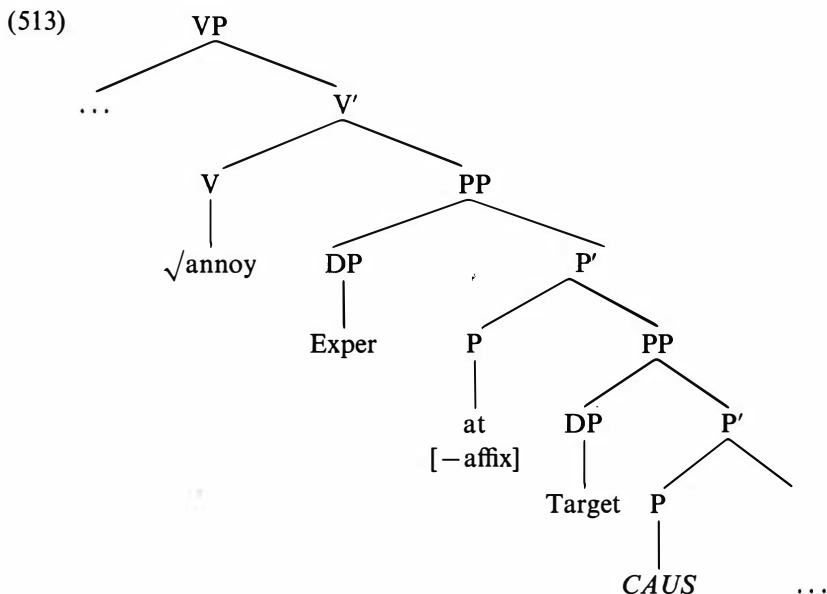


Here, as in (511), *CAUS* must raise to *give*. Here, as in (511), *CAUS* can only do this by first adjoining to the intervening preposition. But here, unlike in (511), there is a problem. *CAUS* can adjoin to *to*, but the resulting category headed by *to* is headed by a nonaffix. Therefore, this new category cannot raise to V. Consequently, *CAUS* has no acceptable way of moving to *give*, and the structure does not have a legitimate outcome. This explains Oehrle's observations.

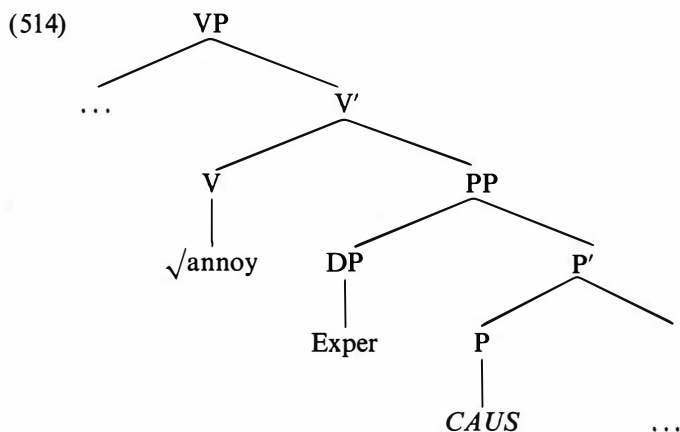
6.2.1.3 The T/SM Restriction Explained We can now explain the T/SM effect with causative ObjExp verbs in exactly the same fashion. Let us assume that here too, *CAUS* starts life as an independent preposition, lower than selected arguments in the Cascade of the main verb (e.g., $\sqrt{\text{annoy}}$). If *CAUS* is once again an “adjunct” preposition, akin to the *wh*-word *why*, then its location low in the Cascade continues to be unsurprising: *CAUS* is not θ -selected to be the head of its Cascade. This is why, in a language like French or Russian, in which $\sqrt{\text{annoy}}$ may surface as a reflexive verb, there is no requirement that *CAUS* occur at all.¹⁷³

CAUS here, as with *give*, must adjoin to the head of its Cascade (e.g., a root like $\sqrt{\text{annoy}}$). This movement once again is blocked by any nonaffixal prepositions that intervene in the Cascade representation. The prepositions that introduce T/SM arguments are nonaffixal and intervene in ex-

actly this way. *CAUS* in (513) cannot raise to $\sqrt{\text{annoy}}$ without adjoining first to *at*, but the phrase headed by *at* that is formed by this adjunction cannot itself proceed on to $\sqrt{\text{annoy}}$, because it is headed by a nonaffix.



By contrast, consider a VP with $\sqrt{\text{annoy}}$ and *CAUS* in which the T/SM argument is missing. This configuration, (514), should yield a good result and does.



Because this structure (unlike (513)) contains no T/SM argument, there is no problem raising *CAUS* to $\sqrt{\text{annoy}}$. Nothing relevant intervenes.

The structure in (514) is unproblematic in a Cascade Syntax. In (514), the Experiencer θ -selected by $\sqrt{\text{annoy}}$ (in a usage that does not include a Target argument) is the specifier of the sister to $\sqrt{\text{annoy}}$. The actual sister to $\sqrt{\text{annoy}}$ is a projection of *CAUS*, a preposition that is unrelated to the θ -selectional properties of $\sqrt{\text{annoy}}$. This is fine, so long as we include nothing in our theory that reserves the sister position to a verb for an argument of that verb. Indeed, we have included no such provision. A nonargument may head the sister to a category X if X does not have two arguments of its own. If X does have two arguments of its own, like $\sqrt{\text{annoy}}$ when it does take a Target, then the nonarguments must be elsewhere.

6.2.1.4 The Position of Non- θ -Selected Arguments There is no incompatibility between the structures posited by Cascade theory to explain command facts and those posited to explain T/SM effects and Oehrle's observations. Consider examples like (515)–(516).

- (515) a. Interviewing Nixon gave nobody a book at any time.
 b. *Interviewing Nixon gave a book to nobody at any time.
- (516) a. Lipson's textbook taught [John and Mary]_i Russian during each other_i's sabbaticals.
 b. *Lipson's textbook taught Russian to [John and Mary]_i during each other_i's sabbaticals.

The unacceptability of the (b) examples is quite clearly that accorded to sentences displaying Oehrle's effects, rather than sentences in which negative polarity licensing or bound anaphora fails. (The judgments do not change their character when the negative polarity items and the reciprocals are replaced by ordinary DPs.) We have already explained these effects as a consequence of *CAUS* attempting to move to the main verb from a head position c-commanded by *to*, where *to* is an intervening nonaffixal head that blocks this movement.

Now, why are the (a) examples acceptable? Clearly, *CAUS* must be able to start in a position higher than the prepositions *at* and *during*, since these, like *to*, are nonaffixal. In fact, no principle prevents such an initial structure since *at* and *during* are unselected. Although (517a) would disallow raising of *CAUS*, (517b) would allow it (since *G* is affixal). No independent principle chooses (517a) over (517b).

- (517) a. *... *give* [_{PP} Goal *G* [_{PP} Theme [_{P'} [_{P₀} *at*] [_{PP} any time [_{P'} *CAUS* Causer]]]]]
 b. ... *give* [_{PP} Goal *G* [_{PP} Theme [_{P'} [_{P₀} *CAUS*] [_{PP} Causer [_{P'} *at* any time]]]]]

Both structures are acceptable because neither *CAUS* nor *at* (*during*) is selected by *give* (nor is either selected by the other).

6.2.2 Causers and Backward Binding

Now let us fill in the three dots following *CAUS* in (513) and (514). The key to the analysis of the T/SM constraint and Oehrle's effects has been the claim that *CAUS* starts lower than prepositions that mediate internal θ -selection. That is why these prepositions can block the affixation of *CAUS* to the head of its Cascade. This observation immediately recalls Belletti and Rizzi's (1988) analysis of "backward binding" with Experiencer predicates. Just as the preposition *CAUS* starts out lower than prepositions that mediate internal θ -selection, so—for Belletti and Rizzi—the argument that I have called Causer (their "Theme") starts out lower than (other) internal arguments. This initial "lowness" accounts for the fact that anaphors inside the Causer argument can be bound by the Experiencer.

- (518) [*Pictures of each other_i*] annoyed the kids_i.

Crucially, just this sort of backward binding is found with causative uses of verbs like *give*.¹⁷⁴

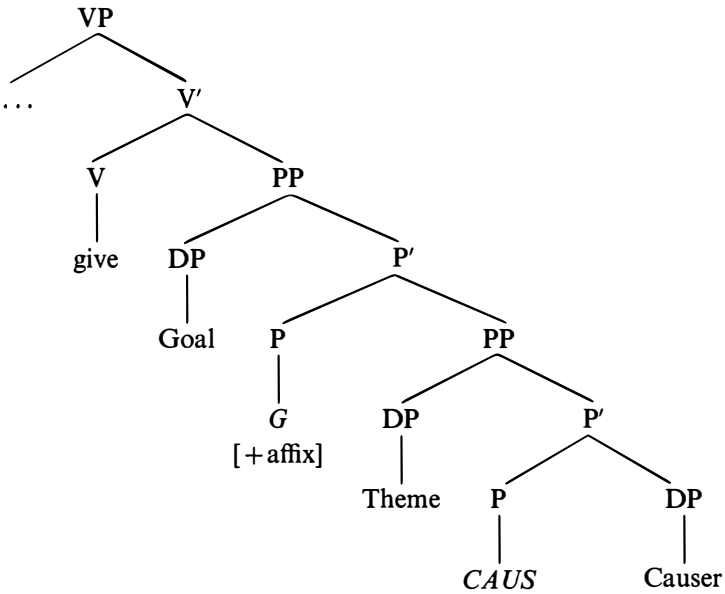
- (519) a. Each other's remarks gave John and Mary a book *t*.
 b. Those books about himself taught Bill the meaning of caution *t*.

These predicates thus behave like Experiencer predicates with respect to both Myers's Generalization and backward binding.

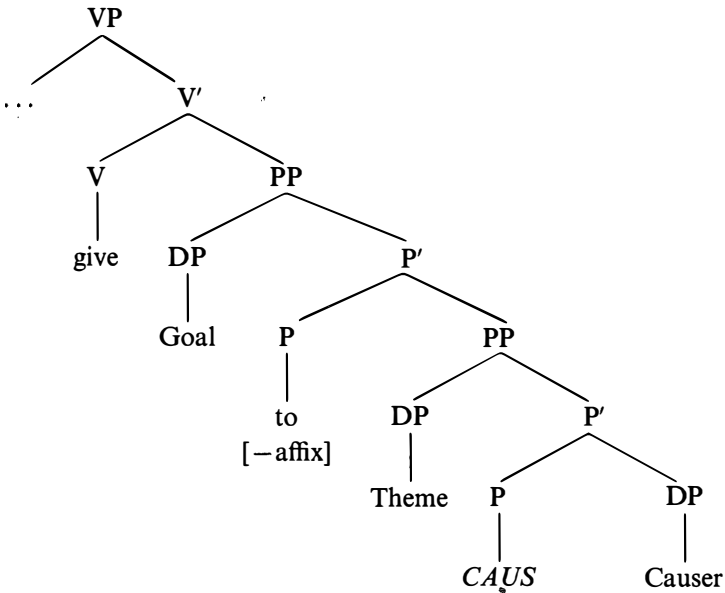
As already noted, affixation of *CAUS* by movement from a VP-internal position makes sense if *CAUS* is a preposition like those in (509). This means that the underlying *object* of *CAUS* is the Causer. Evidence for the underlying position of *CAUS* thus converges strikingly with evidence for the underlying position of Causer. The convergence is striking because the arguments are from quite different empirical domains.

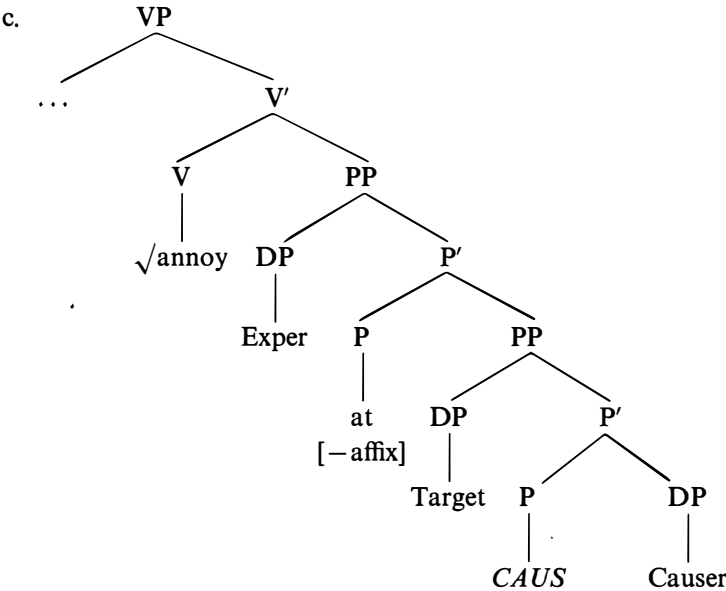
The types of structures we now assume are those in (520).¹⁷⁵

(520) a.



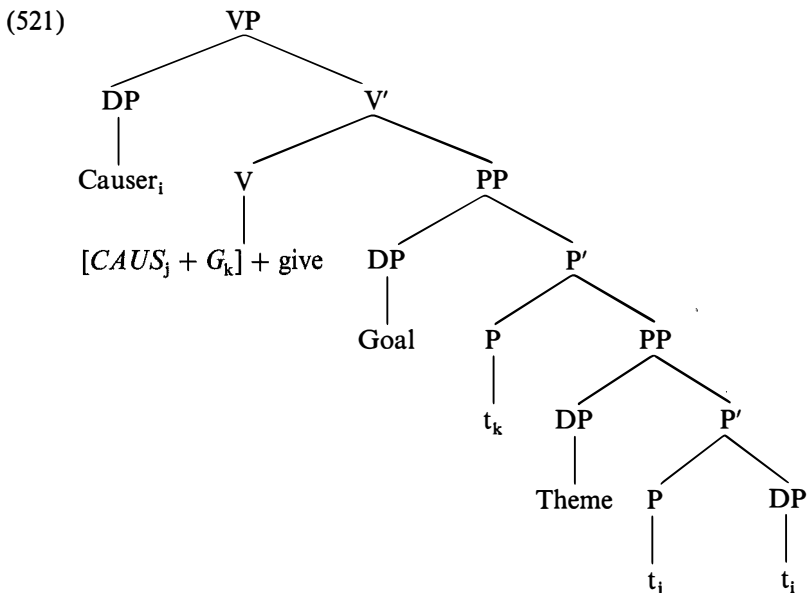
b.





Recall that only (520a) yields a legitimate surface output, since only in (520a) can *CAUS* raise to the highest V (adjoining to [+affix] *G* in the process).

Let us assume that *CAUS* does not license Case on its object (unlike the overt causative prepositions). The Causer argument, particularly if it is nominal, must move to a Case position. The subject position is now available for this purpose, as shown in (521).



Movement of the Causer raises questions about the thematic status of the subject position in (521) (drawn as VP-internal, though little hinges on this). The verb *give* without *CAUS* assigns the role Agent to its external argument. Likewise, as discussed in chapter 4, $\sqrt{\text{annoy}}$ without *CAUS* assigns the role A-Causer to its external argument. I argued (also in chapter 4) that the presence of affixal *CAUS* on a verb causes that verb to lose its ability to assign its external argument role. The relevant principle was (335), reproduced here.

(522) *Suppression of external argument*

Only affixation of a semantically contentful morpheme to a verb with an external argument α allows α to be unexpressed (“suppressed”) in syntactic structure.

Thus, even if *give* without *CAUS* has an external Agent argument role to assign, it does not have this role to assign after *CAUS* affixes to *give*.

But now an important question arises: Where in the derivation does *CAUS* first attach to *give*? We might think that this question was already answered when we posited syntactic movement of *CAUS* to *give*. But a theory in which *CAUS* attaches to the head of its Cascade by movement coexists somewhat uneasily with the assumption that *CAUS* deprives *give* of its external Agent argument. Taken together, these two assumptions

entail that there is some stage in the derivation at which the external θ -role of *give* and $\sqrt{\text{annoy}}$ is not suppressed, then a point at which *CAUS* attaches, followed immediately by a stage at which the external θ -role is gone. This sort of derivation, for example, is incompatible with the Projection Principle familiar from *LGB*, which imposes uniformity on the mapping between structure and lexical properties throughout the derivation. For example, Chomsky's formulation contains the following clause, where L_i and L_j are levels of representation chosen from among D-Structure, S-Structure, and LF:

- (523) If α selects [= " θ -marks"] β in γ at L_i , then α selects β in γ at L_j .
(1981:38)

Even if one accepts Chomsky's (1993) more recently proposed retreat from the Projection Principle, it is not clear whether the relation between a given position and the θ -role associated with it should be allowed to vary during the course of a derivation. Certainly, there is no independent evidence for such a possibility—for example, no way in which causative *give* reveals the hidden presence of an Agent or *annoy* that of an A-Causer. Quite the contrary: it was the elimination of A-Causer in English that permitted roots that select this role to surface as parts of pronounceable words. Although I cannot narrow the range of possibilities to just one, there is an appealing response to these questions that helps solve the remaining problems we have with *CAUS*. This response itself stems from a proposal incorporated in Chomsky's (1993) minimalist program. It involves accepting adjunction of *CAUS* to V as a syntactic process, without accepting this as the process that attaches *CAUS* to V in the first place.

The literature on V-movement and inflectional morphology before Chomsky 1993 wrestled with a paradox. Following earlier traditions, this literature assumed that syntactic movement is responsible for the attachment of inflectional morphology to the main verb of the sentence (in the Romance and Germanic languages, at least). English, however, presented a problem. The main verb in English appears not to raise to high inflectional positions by S-Structure. For example, it cannot appear to the left of VP-initial adverbs.

- (524) *Mary speaks_i barely [_{VP} t_i French].

At the same time, the relation between the inflected verb and negation suggests that the English verb, like the verb in French, must move to some inflectional position after all. This is because negation is incompatible

with the inflected main verb, a phenomenon easily explained as an instance of the HMC if the English verb must raise over it.

- (525) a. *Sue not speaks French. →
 b. *Sue speak_i not [t_i French].

Now (525) seems to show that the English verb must move to some sort of Infl, whereas (524) shows that it does not move by S-Structure. This is not yet a problem, since movement of V to Infl in English can simply be forced to occur in the mapping from S-Structure to LF (after Spell-Out in the system of Chomsky 1993). The paradox arises because the English verb form displays affixal inflection.

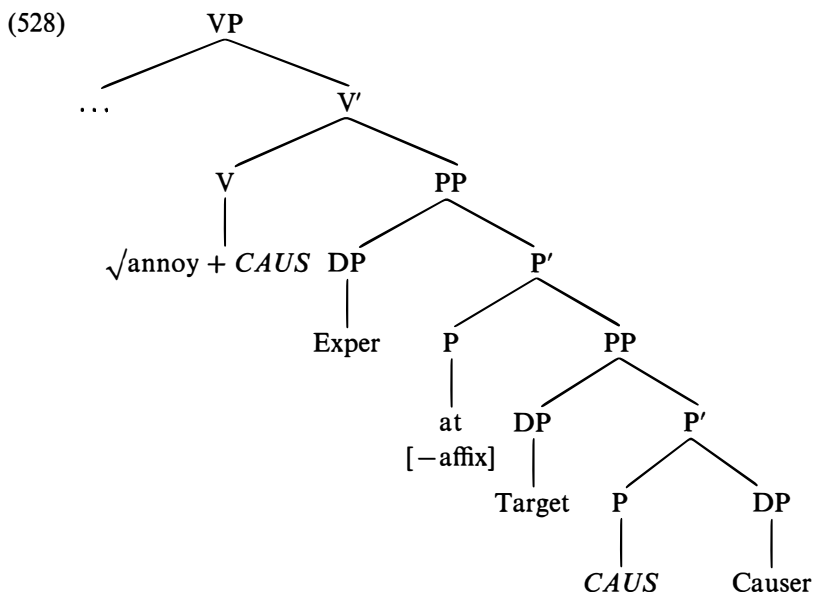
- (526) Mary speaks French.

If the verb moves to get a morpheme attached to it, and this movement happens only at LF, then PF should not see the morpheme already attached. In other words, there should be no *-s* on *speaks* in (526). Chomsky (1993) resolves this paradox by separating movement of V to an inflectional position from the process that attaches inflectional affixes to that V. In his conception, an inflected verb emerges from the lexicon with all its morphology already attached. The function of movement to inflectional positions is to legitimate or “check” the features of the morphology. If these features are *strong*, the movement must take place somewhere in the mapping to PF, since strong features are illegitimate at that level. If the features are *weak*, the movement must take place in the mapping to LF. The inflectional features of English are weak. Thus, *speaks* in (525) and (526) must move to some inflectional node (Chomsky’s Agr_s) by LF in order to legitimate the presence at LF of the morpheme that realizes that node’s features. A separate principle called *Procrastinate* ensures that the movement will not happen any earlier than the branching-off point between PF and LF (S-Structure, in earlier theories). Chomsky suggests that inflectional features disappear both from the inflectional node and from the verb once the two are joined by syntactic movement. This saves the derivation from “crashing” at LF, where (unchecked, unlegitimated) inflectional features like those in (527) are impermissible.

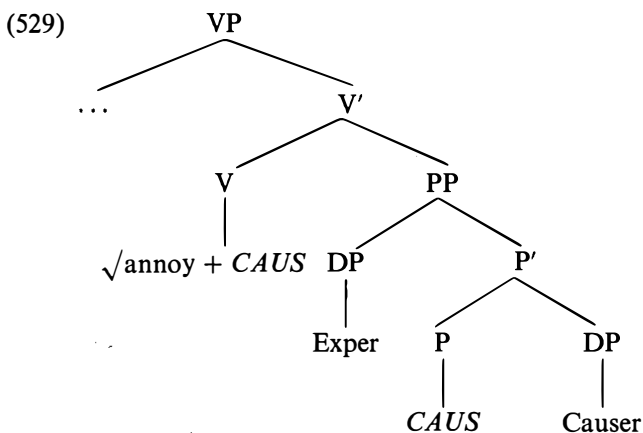
- (527) Mary [Agr_s 3sg] speak-[/s/, 3sg] French

Suppose the same sort of analysis is applied to forms with *CAUS*. If we suppose that *CAUS* is affixed to V in the lexicon, then the suppression of the verb’s external role is a property that holds from the point at

which the relevant verb is introduced into structure. The questions debated above disappear, just as they do for English V-movement. We assume that *CAUS* bears certain features that must disappear (or, alternatively, be “discharged”) by some relevant level—PF or LF. Since the affixal status of intervening elements seems to matter, we may assume that these movements take place by PF (what used to be called “S-Structure movement”), where affixal status matters. In Chomsky’s terminology, the features of *CAUS* are said to be *strong*.¹⁷⁶ (528) illustrates the appropriate revision of the structures proposed here for Experiencer predicates under the Minimalist assumption that head movement may have a checking function, rather than an affixation function. The nonaffixal preposition *at* creates the now-familiar problem.



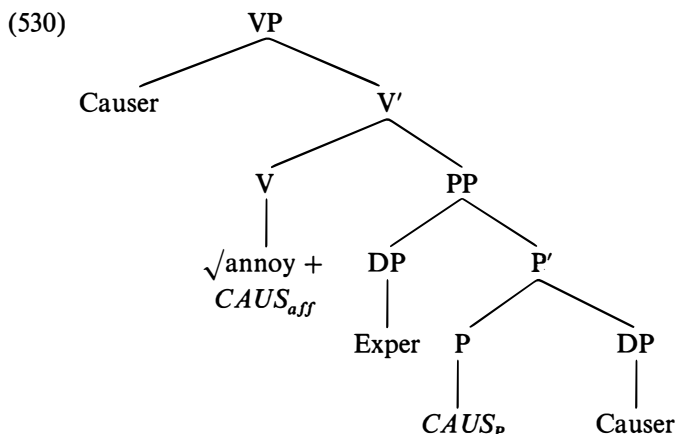
If *at* and its object are missing from (528), as in (529), the lower occurrence of *CAUS* can raise to the higher, and all is well.



Thus, if we adopt Chomsky's checking theory concerning the link between affixation and movement, the problems connected with θ -role suppression in the syntax disappear.¹⁷⁷ Furthermore, a new possibility is raised—one that solves another fundamental problem that arose early in the discussion of Experiencer predicates.

As the first step in considering this possibility, let us call the prepositional *CAUS* "*CAUS_p*" and the *CAUS* that starts out affixed to the verb "*CAUS_{aff}*." In the analysis so far, *CAUS_p* and *CAUS_{aff}* are similar enough that each discharges the features of the other. I have also argued that *CAUS_p* θ -selects Causer. Let us now entertain the possibility that *CAUS_{aff}* also θ -selects Causer. In other words, the two occurrences of *CAUS* are similar down to their θ -selectional properties. On the other hand, we know that causative prepositions take Causer as an internal argument, whereas causative verbs seem to take Causer as an external argument. I will not explain this difference between *CAUS_{aff}* and *CAUS_p*, but let us take it as a fact. We therefore might not be surprised if *CAUS_{aff}* externally selects its Causer argument, whereas *CAUS_p* (by analogy with the overt causative prepositions) internally selects its Causer argument. In other words, the similarity between the two does not extend to the type of θ -selection, perhaps a syntactic consequence of the distinction between verbal affix and freestanding preposition.

In other words, positing two occurrences of *CAUS* as a way out of syntactic θ -role suppression, we now expect two occurrences of Causer, as in (530).



The higher occurrence is the external argument of the main verb, and the lower is the internal argument of *CAUS_p*. I proposed that movement takes place between these two positions. The proposal was important because a “low” initial attachment site for Causer went hand in hand with the low initial site of *CAUS* that motivated my treatment of T/SM effects and Oehrle’s Generalization. The evidence came from backward binding, which was treated in the spirit of Belletti and Rizzi’s analysis—as a consequence of A-movement.

This whole proposal was made at a point in the discussion where the higher argument position was identified as athematic. Suppose now that this higher position is actually thematic, but is associated with *exactly the same* θ -role as the higher position. Does anything change? That depends on whether movement to a θ -position is explicitly barred by some principle of grammar, or whether the only thing barred is movement among positions that bear distinct θ -roles. In fact, a ban on movement among positions that bear distinct θ -roles makes sense if, for example, chains are the objects of θ -theory and θ -roles are uniquely assigned. On the other hand, a ban on movement among positions whose semantics are indistinguishable would have to be an “add-on”—a purely syntactic constraint with no external motivation. Many such constraints are known, but there is no evidence for this one. Therefore, I assume that movement of Causer from object of *CAUS_p* to subject of *V + CAUS_{aff}* is in fact *movement from one θ -position into another θ -position*—a case unforeseen in earlier discussions of the θ -Criterion, but one there is no reason to forbid. We have two identical Causers, occupying two semantically identical positions. We simply permit movement to take place from one to the other.¹⁷⁸

- (531) Movement may not proceed from a position in which θ -role R_1 is θ -selected into a position in which a distinct θ -role R_2 is selected. (Movement from a position in which R_1 is selected into a position in which R_1 is also selected is not prohibited.)¹⁷⁹

In fact, however, our conclusions are stronger and more dramatic. Our very first discussions of Experiencer predicates yielded a paradox (section 2.5.1). Backward binding with Experiencer predicates seemed to be admirably explained by the hypothesis that the argument in subject position of these predicates moved from some lower position. Yet, Belletti and Rizzi's arguments notwithstanding, there seemed to be no evidence supporting nonthematic status for the subject position, and some evidence pointing the other way. For example, the ability of causative Experiencer predicates to license accusative Case seemed at odds with the unaccusative analysis. (531) dissolves the paradox. These predicates take an external argument *and* allow movement into that external argument position—for a principled reason.

Additionally, the availability of passive forms for Experiencer predicates seemed in conflict with the unaccusative analysis, as in the following examples based on (78)–(79):

- (532) a. In those days, Bill was often being frightened by one thing or another when I would come home from work.
 b. In those days, Bill was again and again being annoyed by Mary's behavior.

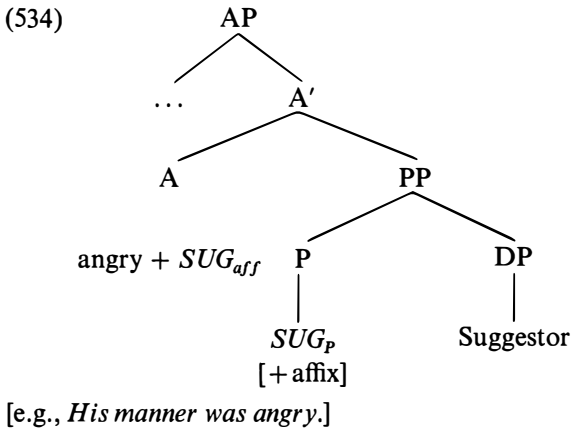
We can also add examples with causative *give* and *teach*.

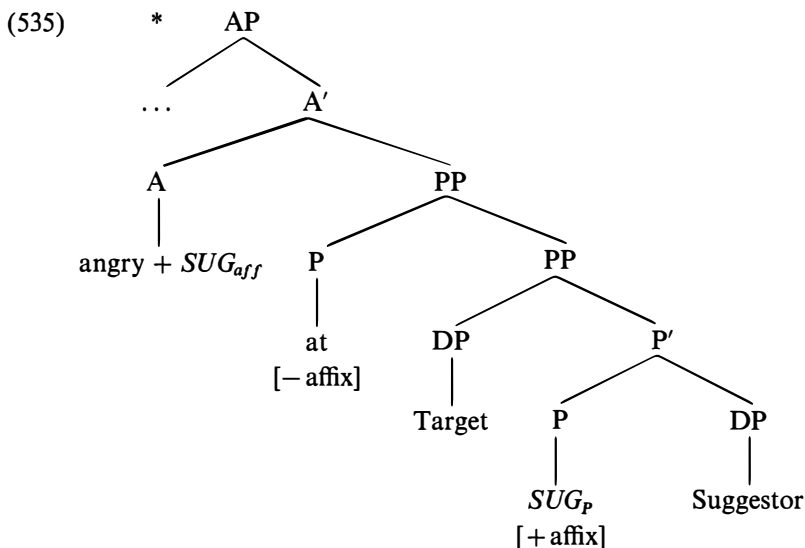
- (533) a. Mailer was being given his finest ideas by a series of interviews with Nixon when Nixon suddenly stopped talking to him.
 b. Sue was taught Russian by Lipson's textbook.

In the framework of ideas developed here, these examples involve movement from the object of *CAUS_P* to the object of *by*—once again an instance of movement between two positions bearing identical θ -roles in the same clause.¹⁸⁰ The “dethematization” of the Spec, VP position is no different with these predicates than with any other. Crucially, nothing violates the generalization that bars verbal passives with predicates that lack external arguments.

6.2.3 *SUG*

A loose end in the discussion so far is the analysis of the T/SM restriction with *SUG*. Constructions with *SUG* were introduced as a demonstration that the T/SM effect is independent of the ObjExp construction. An analysis exactly parallel to that developed for *CAUS_P* can account for the T/SM restriction with *SUG* as well. Suppose that *SUG* arises as an independent preposition inside AP, and suppose that this preposition must raise to A in structures analogous to those in which *CAUS_P* must raise to V. I will call the object of this preposition the *Suggestor*. I assume, as with *CAUS*, that affixation of *SUG* to A dethematizes the subject of A and assigns the Suggestor role to the subject position of AP. Just as with *CAUS*, this occurrence of *SUG*—call it *SUG_{aff}*—must be licensed under adjunction by a freestanding occurrence of the same morpheme—call it *SUG_P*. In an entirely familiar manner, the raising of *SUG_P* to *SUG_{aff}* is blocked in Cascade structures by intervening prepositions, such as a preposition that introduces a Target or Subject Matter argument. Thus, *SUG_P* can raise to A in (534), but not in (535).





[e.g., **His manner was angry at Mary.*]

The only troubling note concerning *SUG* is the apparent absence of any overt preposition whose semantics mirror those of *SUG*. The best I can do is point to a phrase like *judging from*.

(536) John was angry, judging from his manner.

This may or may not be a problem. It needs to be shown, for example, whether this gap is remedied in other languages, an investigation I have not undertaken.

6.2.4 Summary

The various pieces of the discussion fit together quite tightly. To summarize: We saw early in the chapter that Cascade structures receive independent support from coordination, which reveals constituency, and from binding phenomena, which reveal command relations among XPs. The T/SM restriction followed immediately from the command relations among X^0 s in the very same structures. The only assumption needed was to posit *CAUS_p*. The problem with *CAUS_p* was its association with the dethematicizing effect. This problem disappeared once we adopted a checking theory of morphological features. This theory immediately resolved the one remaining problem from the opening chapters: the apparent conflict between the A-movement suggested by backward binding

with Experiencer predicates and the evidence against an unaccusative analysis for these predicates.

The outline of the solutions to the original problems is now clear. As usual, however, gaps remain. In this case, there are two: an outstandingly large one that will engender a separate chapter concerning XP-movement, in which I argue that Cascade structures and Layered structures coexist in the grammar, and another (to which I turn next) that concerns the lack of correlation between the distribution of backward binding and the distribution of T/SM and T/SM-like effects.

6.3 More on *CAUS*

The previous sections accounted for the T/SM effect and its congeners by positing adjunction of *CAUS_p* to a V containing *CAUS_{aff}*. Since both species of *CAUS* are phonologically null, and head movement is an abstract process, we must ask how children acquiring language (as opposed to linguists) come by this analysis. After all, if there were no *CAUS* or no movement of *CAUS*, there would be no T/SM restriction.

In section 3.4.7, I presented two reasons why a language learner would posit the existence of a null causative morpheme. First, if the language learner has reason to know that a semantically causative verb fails to externally θ -select what would otherwise be its external argument, morphology must be posited that can account for the external argument's absence. One such external argument is A-Causer, with verbs that involve psychological states like annoyance. UG requires such a verb to θ -select A-Causer. When A-Causer is missing, therefore, the finger of the Language Acquisition Device points to morphology. For verbs like *give*, UG somehow must dictate that Agent is the external argument of choice. This will lead the language learner to the presence of *CAUS* when a homophonous verb *give* selects a Causer rather than an Agent. If there is any subtlety to the learner's decision concerning whether Agent or Causer is primary in the diathesis of a predicate, we expect some differences among speakers. As noted earlier, this seems to be the case. For example, although most or all speakers detect Oehrle's effects with *give*, such effects are not universally detected with *show*.¹⁸¹

- (537) a. Talking to Sue showed Bill many things:
 b. %Talking to Sue showed many things to Bill.

We might conclude, then, that the use of *show* does not fully determine whether the verb's semantics are such that it can have a Causer subject without an underlying Agent subject role being suppressed by *CAUS* or whether the use with an Agent subject is primary. What factors might come into play in a speaker's decision is a question worth a great deal of study.

Second, verbs like *break* and *grow* that alternate between an inchoative sense and a causative sense (with no change in phonological form) also seem to involve *CAUS*. The evidence here came from Myers's Generalization and the impossibility of action or process nominalizations of these verbs. In section 3.4.7, I noted that the existence of an apparent alternation seems to be the crucial factor. Thus, *grow* has no causative nominalization, but *cultivate* does. In some cases, *CAUS* might be posited because A-Causer has been suppressed, but in other cases, *CAUS* simply adds an argument to a genuinely unaccusative predicate. Here, the existence of an alternation is apparently crucial.

6.3.1 Problems with Syntactic Affixation of *CAUS*

Suppose all instances of *CAUS* involve affixation in the course of the syntactic derivation—derivations like those discussed in the preceding chapter. We naturally expect that θ -selected arguments mediated by overt prepositions should be impossible whenever *CAUS* is present. But this is false for the class of inchoative verbs that includes *break* and *grow*, if Goal and other telic arguments are taken to be θ -selected arguments. (It is also false for verbs of "accompanied motion" like *walk* and *gallop* in their causative uses.) If this assumption is correct, then some instances of *CAUS* are not attached by syntactic head movement.

- (538) a. John broke the cookie into little pieces.
- b. Sue grew the seeds into beautiful plants.
- c. Sue flew the plane to Los Angeles.
- d. Bill rested the coffee cup on the table.
- e. Tom dropped the book onto the floor.
- f. Harry rolled the bowling ball down the lane.
- (539) a. John walked the dog to his grandmother's house and back.
- b. Sue galloped the horse through the woods.
- c. The wind ran my car into a brick wall.
- d. Mary jumped the horse over the hurdle.

What distinguishes the causatives with *CAUS* in (538)–(539) from those that disallow prepositions mediating θ -selection? There are too few “data points” to permit a firm conclusion. Nonetheless, I can venture something more than a guess. When the T/SM effect shows up with ObjExp verbs, and when Oehrle’s effects show up with causative uses of *give*, the causative morphology suppresses an external θ -role that the verbal root would otherwise require. With ObjExp verbs, it is A-Causer; with *give*, it is Agent. By contrast, causative morphology in (538)–(539) adds a θ -role to a clause, without suppressing any θ -roles that would otherwise be assigned. I suggest that this contrast is crucial to the difference between *CAUS* moved to V in the syntactic derivation and *CAUS* affixed to V from D-Structure on.

(540) An affix that suppresses a θ -role has strong features.¹⁸²

The same factor will dictate that *SUG* bears strong features in the cases we have examined, since *SUG* suppresses the Experiencer role otherwise assigned by an adjective like *angry*.

On the other hand, (540) does not force *CAUS_{aff}*, when attached to unaccusative uses of *break* or *grow*, to bear strong features. This is because *CAUS_{aff}* in this usage suppresses no θ -role when added to these predicates. Likewise, the deadjectival use of the causative prefix *en-* discussed by Walinska de Hackbeil (1986), mentioned in section 6.2.1.2, does not bear strong features (and hence does not require syntactic affixation), since no θ -role is suppressed. Note that no HMC problems are created by *to* in (541b) or by *towards* in (541c), in contrast to the T/SM effect with *CAUS*.¹⁸³

- (541) a. We enlarged the scope of the investigation.
 b. For the things you do endear you to me. (John Lennon and Paul McCartney, *The Beatles* [“White Album”])
 c. The riots embittered us forever towards the police.

The same line of attack may be relevant with cases closer to the semantic class of emotion predicates. (These are the “counterexamples” to the T/SM restriction that were revealed by the dictionary search discussed in section 3.3, mentioned in note 56.) Consider, for example, verbs of inspiring and discouraging, exemplified in (542).

- (542) arouse, discourage, dispose, encourage, excite, galvanize, goad, incline, inspire, provoke, shame, stimulate, turn (against)

These verbs allow an argument that is introduced by an overt preposition (as documented in (543)), yet they may well contain *CAUS_{aff}* or some other null morpheme, given the impossibility of nominalization documented in (544).

- (543) a. Sue's remarks aroused us to action.
 b. The rain discouraged us from our tasks.
 c. These results inclined us toward the more difficult course.
 d. Sue's remarks inspired them to action.
 e. It provoked him to rage.
 f. Mary shamed us into going to the movies.
 g. It stimulated us to greater exertions.
 h. His remarks turned Mary against him.
- (544) a. ?*the remarks' arousal of John and Mary to action
 b. ?*the rain's discouragement of us from our tasks
 c. *its inclination of us toward the more difficult course
 d. *its inspiration of John and Mary to action
 e. *its provocation of Sue to rage
 f. *its stimulation of us to greater exertions

Suppose we accept the idea that *arouse* is bimorphemic, composed of $\sqrt{\text{arouse}}$ and *CAUS_{aff}*. If the underlying root is not required to take an A-Causer external argument, then perhaps no suppression of an external argument is necessary in the derived causative form. If that is the case, the analysis will proceed as it does in (538), and nothing like a T/SM effect is expected.

The same story may be told about other verb classes that might look like counterexamples to the T/SM restriction (depending on one's expectations). Verbs of accustoming and alienation are one such class.

- (545) a. The orientation lectures acclimatized us to our new surroundings.
 b. ?The passage of time accustomed the Berliners to their wall.
 c. His remarks alienated the voters from the party.
 d. These actions estranged Bill from his profession.
 e. These measures will habituate the workers to loud noise.
- (546) a. ??the lectures' successful acclimatization of Sue to her new surroundings
 b. ??its successful alienation of the voters from the party

- c. ??the passage of time's successful estrangement of Bill from his profession
- d. ??the measures' successful habituation of the workers to loud noise

These proposals are ad hoc, of course, in that we lack an independent theory that tells us when A-Causer is required by lexical semantics. Consequently, I advance them cautiously, more as a suggestion for research than a firm result. Nonetheless, it is striking that these counterexamples, like the phenomenon itself, seem to cluster around semantic classes of verbs. This in itself suggests an explanation that is rooted in lexical semantics; my proposal is the beginning of such a suggestion.¹⁸⁴

We thus have three types of semantically causative verbs: (1) those that contain strong *CAUS* (causative *give*, causative *annoy*), (2) those that contain weak *CAUS* (causative *break* and *grow*), and (3) those that do not contain *CAUS*. Furthermore, although this book has given no principled account of which verbs fall in which class, there is nonetheless order in the distribution of these classes, and therefore material for future investigation.

Let us consider further the semantically causative verbs that do not contain *CAUS*. Does it follow from the fact that a verb has a Causer subject without the need for *CAUS* itself (*CAUS_{aff}* with or without *CAUS_p*) that the verb may not cooccur with *CAUS_p*? I see no reason to reach this conclusion. For example, alongside (547) we find structures like (548).

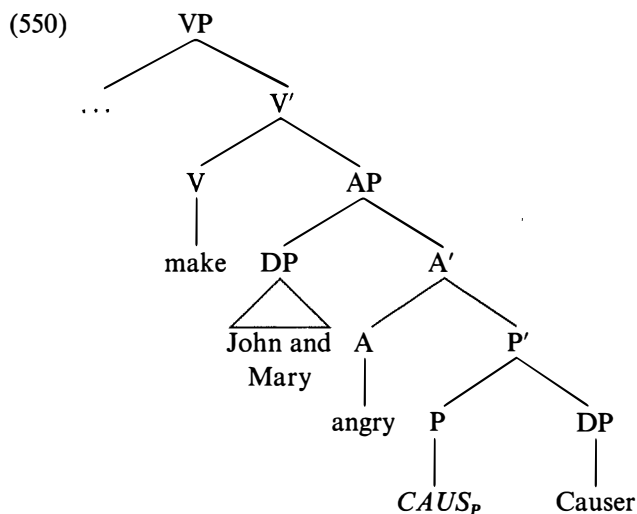
- (547) a. *Causer* made John and Mary angry.
- b. *Causer* forced John and Mary to confront their problems.
- c. *Causer* used to make Sue blush.
- d. *Causer* harmed John and Mary.
- (548) a. *Causer* made John and Mary angry *CAUS_p Causer*.
- b. *Causer* forced John and Mary to confront their problems *CAUS_p Causer*.
- c. *Causer* used to make Sue blush *CAUS_p Causer*.
- d. *Causer* harmed John and Mary *CAUS_p Causer*.

Naturally, *CAUS_p* should fail to license Case on the Causer, in keeping with the properties it has had throughout this discussion. This requires the sort of movement of the lower Causer to the position of the higher Causer that we have also seen throughout. This, of course, sets the scene for

backward binding in these constructions. Indeed, as we saw in section 2.5.1, backward binding is found here. The following examples are excerpted from (124)–(126) in that section:

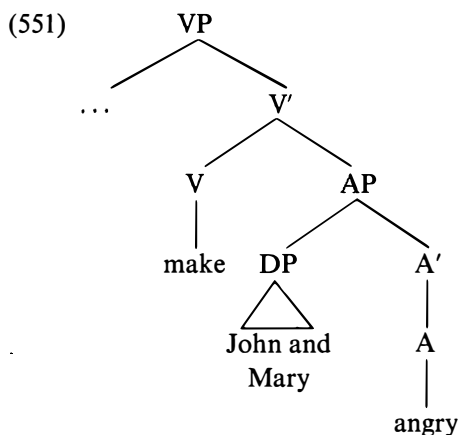
- (549) a. Each other's remarks made John and Mary angry.
 b. Each other's criticisms forced John and Mary to confront their problems.
 c. Pictures of herself used to make Sue blush.
 d. ?Each other's criticisms harmed John and Mary.

We now have a problem. To license this sort of backward binding, the relevant structure must be inside the complement to *make*. Otherwise, material in the embedded clause cannot license the anaphor.¹⁸⁵



If (550) is the correct structure, we expect nonaffixal *angry* to block movement of *CAUS_p* to *make*, assuming that this sort of movement must take place.¹⁸⁶ In fact, however, there is no reason why it should take place. Let us continue to suppose that *CAUS_p* has strong features, as it must if it is to move in the other constructions that we have examined. Note, however, that the presence of *CAUS_p* in these sentences adds nothing to the semantics, nor does the presence of an extra Causer. Furthermore—and this is unique to the present case—the occurrence of *CAUS_p* in these structures also adds nothing to the syntax, by hypothesis, since *CAUS_p* does not have to be strong so as to discharge features on *CAUS_{aff}*.

(since the latter is missing). Therefore, there is in fact no need for *CAUS_P* and the lower occurrence of Causer to exist at any particular stage of the derivation. Suppose therefore that *CAUS* and its object can be freely deleted, since the semantics of both are represented elsewhere in the structure. The strong features of *CAUS_P* disappear with the preposition, and (most important) we correctly expect that the intervention of *made* in (549a) and (549d) will not create any problem.¹⁸⁷



In fact, omitting differences that arise from SOV order (to be discussed in section 7.3), a stronger form of the same analysis can be given for the Japanese causative Experiencer constructions also discussed in section 2.5.1. Recall that these constructions are overtly periphrastic. The standard analysis of Japanese *sase* treats it as a predicate of causation much like *make*. Much as in (550), backward binding was found ((552) repeats (131a)).

- (552) [Zibun_i-ga gan kamo sirenai koto]-ga Hiroshi_i-o
 refl-NOM cancer may-have fact-NOM Hiroshi-ACC
 nayam-ase-ta.
 worry-CAUSE-past
 'The fact that himself_i may have cancer worried Hiroshi_i.'

We observed, however, that these constructions fail a test for unaccusativity discovered by Miyagawa (1989). A floated quantifier cannot mark the spot from which the Causer has moved, according to any A-movement analysis of backward binding. I repeat (140).

- (553) a. [3-tu-no omosiroi koto]-ga Hanako-o
 3-cl-GEN interesting thing-NOM Hanako-ACC
 yorokob-ase-ta.
 happy-make-past
 b. ??[*Omosiroi koto*]-ga 3-tu Hanako-o yorokob-ase-ta.
 c. *[*Omosiroi koto*]-ga Hanako-o 3-tu yorokob-ase-ta.

In order to explain backward binding with constructions like (549), I posited the Cascade structure in (550), which no principle forbids. This raised a problem if *CAUS_p* were required to discharge its strong features: *angry* should block raising. Allowing it to delete instead made sense, and sidestepped any issue of feature discharge. If *CAUS_p* deletes, its object must delete as well (if only because of the geometry of Cascade structures). Now suppose we strengthen this conclusion slightly, to require otiose elements to delete. This follows from Chomsky's (1986b, 1993) principle of Full Interpretation, which allows elements to survive at LF only if they contribute to semantic interpretation. An otiose occurrence of *CAUS_p* and Causer makes no such contribution. It follows then that there will be no lower Causer position at LF, hence no position for a floated quantifier to modify, whenever the associated *CAUS_p* is otiose.

This analysis has obvious consequences for English cases in which the T/SM effect (and its congeners) is observed. The trace of *CAUS_p* must be allowed to delete and perhaps must be forced to delete (along with its Causer object) even when movement has taken place to *CAUS_{aff}* in order to discharge the latter's features. Because of the function of *CAUS_p* movement, deletion would have to wait until after movement. This consequence has interesting implications for the nature of other phenomena. First, whatever accounts for the HMC cannot be a constraint on LF representations, since LF will never contain a trace of *CAUS_p*. Instead, it must be a constraint on movement, requiring movement of heads to proceed by shortest steps, as in Chomsky's (1993) minimalist theory.

Interestingly, the same logic tells us that the c-command condition on binding must also have a derivational character. When we find backward binding without an expected T/SM effect, the position that licenses backward binding under c-command is missing at LF. This entails that binding relations cannot strictly be read off LF representations, the hypothesis entertained by Chomsky (1993). Chomsky allows only two sorts of constraints, those that hold at the "interface" levels LF and PF, and those that govern derivations. Whereas the HMC falls in the latter group, he

proposes that conditions on anaphora fall entirely in the former group. This cannot be correct, if the analysis suggested here is on the right track. I will leave the questions raised by this observation to be answered in later work.

6.3.2 *G*, U(T)AH, and the Dative Alternation

In section 5.3, we observed systematic differences between the sort of Goal that is directly θ -selected by a verb like *give* in *Mary gave Sue a book* and the sort of Goal that is directly θ -selected by *to* and related prepositions (summarized in (377)). This observation eliminates the possibility of movement between the object of *to* and the directly θ -selected Goal of a verb like *give*, even if *to* is subsequently deleted. Such movement creates a chain that contains two distinct θ -positions.

(554) *Mary gave Sue_i a book (to t_i).

This accords with the facts of backward binding, which provide no evidence that the Goal in a double object structure is ever lower than the Theme in Cascade Syntax.

- (555) a. Sue showed John and Mary each other's friends.
b. Sue showed each other's friends John and Mary.

- (556) a. I entrusted the adults in the room with each other's children.
b. *I entrusted each other's children with the adults in the room.

On the other hand, by analogy with our treatment of backward binding in periphrastic causatives, we might expect the possibility of movement from the object of *G* to a directly selected Theme position, with subsequent deletion of *G* and the trace it θ -selects on grounds of redundancy. We discovered no relevant differences between the Theme θ -selected by *G* and its directly θ -selected counterpart.

(557) Mary gave a book_j to Sue (*G* t_j).

In (557), deletion of *G* is motivated by the intervention of *to* between *G* and V, exactly as with *CAUS* in the preceding section. (If nothing happens to *G* it will be an illegally unaffixed zero morpheme.) In fact, backward binding here provides evidence that an occurrence of Theme in *give Theme to Goal* truly is c-commanded by the Goal, as in (557). In particular, the Goal may bind into the Theme, with roughly the acceptability of "backward binding" in general, as noted by Burzio (1986:199, 203). The (b)

examples of (558)–(559) form minimal pairs with their counterparts in (555)–(556).¹⁸⁸

- (558) a. Sue showed John and Mary to each other's friends.
 b. Sue showed each other's friends to John and Mary.
- (559) a. I entrusted the adults in the room with each other's children.
 b. I entrusted each other's children to the adults in the room.

Data from quantifier scope presented by Aoun and Li (1989) seem to go in a similar direction. According to Aoun and Li, structures with *to* allow a scope ambiguity not found in structures with *G*. In my judgment, the data are rather murky, but, if correct, they would support the analysis given here, if scope mirrors c-command throughout the derivation in a manner similar to anaphora binding.

- (560) *Unambiguous* (per Aoun and Li (1989))
 a. Sue gave every child some problem.
 b. Sue gave some child every problem.
- (561) *Ambiguous*
 a. Sue gave some problem to every child.
 b. Sue gave every problem to some child.

These data support the possibility that an occurrence of Theme in the environment of Goal may be c-commanded by Goal, even in examples with *to*. Similar analyses might be assigned to Themes with other Goal and Locative arguments, in view of data like these:

- (562) a. John threw each other's newspapers at Sue and Bill.
 b. ?Mary placed pictures of himself on top of Bill. [e.g., at a funeral]

The same sort of analysis can account for backward binding with true unaccusative Experiencer verbs like *appeal*, which occupied center stage in earlier chapters, if *G* can θ -select Target as it does Theme and Patient.

- (563) [Each other's suggestions]_i appealed to John and Mary (*G* _{*t*_i}).

We might now ask whether the occurrence of Theme below Goal, Location, and Experiencer arguments is only an option—detectable through backward binding—or whether, when Theme and these other arguments cooccur, an instance of Theme is always found in a lower slot. The latter conclusion would follow if arguments were arranged according to a thematic hierarchy like the one partially specified in (564).

(564) ... Goal/Location/Target ... > Theme ...

(564) would allow us, with Larson (1988), to return to a view of the dative alternation that strictly obeys a relatively coarse-grained U(T)AH. We would not have to rely, as we did in section 5.3, on distinctions between directly selected and mediated Goal to maintain an account of the alternation that obeys the U(T)AH. Instead, both alternants would display initial structures that satisfy (564),

- (565) a. *give* [Goal [*G* Theme]]
 b. *give* [to [Goal [*G* Theme]]]

with the PP headed by *to* moving to specifier position in (565).¹⁸⁹

In fact, the argument for this new alternative can be made stronger. In section 2.1, I mentioned an important discovery by Levin and Rappaport (1988a). They note that verbs of motion whose meaning “includes a specification of inherent direction” seem to be unaccusative across languages, whatever the agentivity of their subject. Suppose we view these verbs as incorporating a Goal argument into their lexical meaning. If this is so, then (564) will force unaccusativity on any verb in this class that θ -selects a Theme. At D-Structure, the Theme θ -selected by a verb that contains Goal will need to be c-commanded by that verb. This is not the case if the Theme is an external argument. Thus, Levin and Rappaport’s observation supports the hierarchy in (564), in turn supporting the proposed interpretation of the backward binding phenomena with *to*.

6.4 Grand Summary

This is a good point to summarize the entire discussion.

The overriding theme of this book has been the hunch that the availability and syntactic positioning of arguments is not a matter of chance (like the association of sound and meaning), but arises from laws governing the structure of lexical entries, from laws regulating the projection of these entries onto syntactic structure, and from laws governing syntactic structures themselves. Of course, as I acknowledged in the opening chapters, in any given domain this hunch might be wrong. Searching for principled explanations may sometimes be a fool’s errand. However, only by following the hunch that language is law-governed can we discover what laws there are. This I have done in the domain of semantically causative Experiencer predicates like *annoy*.

These predicates appeared to present U(T)AH problems, since they appeared to display the same θ -relations as verbs like *love*—with the opposite mapping of roles to syntactic positions. I considered the most obvious way around this problem: a simple unaccusative analysis of these predicates like the proposal of Belletti and Rizzi (1988). Backward binding phenomena made this proposal look appealing, since it jibed with the U(T)AH-friendly hypothesis that the surface subject of *annoy* is inserted in a position lower than the surface object. On the other hand, this solution was at odds with other tests for unaccusativity—in particular, the availability of verbal passivization and the unavailability of quantifier float in Japanese. We can now see, with the wisdom of hindsight, that the construction was passing tests for movement into subject position, and failing tests for the LF existence of a chain linking a θ -position to a non- θ -position.

I concluded from this discussion that the simple Unaccusative Hypothesis would not do. Instead, I proposed another solution to the U(T)AH problem—a solution based on the observable semantic differences between the subject of verbs like *annoy* and the object of verbs like *love*. This hypothesis eliminated the U(T)AH problem at one stroke, but left us without an account for backward binding and with no account for the fact that the θ -roles identified as distinct cannot cooccur within a single clause (the so-called T/SM restriction).

To solve these problems, a more sophisticated understanding of the properties of predicates like *annoy* was needed. I argued (using Myers's Generalization) that these predicates include a zero causative morpheme *CAUS*, a morpheme also found with certain other semantically causative predicates. I made this analysis plausible by identifying (in English and in languages like French) the root that this morpheme attaches to. In turn, this discussion made it clear that affixation of *CAUS* (like affixation of many other derivational morphemes) eliminates the external argument of the predicate to which it attaches.¹⁹⁰ The extended discussion of *CAUS* and its θ -suppressing property set the scene for the analysis of T/SM phenomena in this chapter.

The scene was also set, in the preceding chapter, by an extended discussion of the double object alternation in English. There, I made two distinct points that were relevant in the present chapter. First, I argued that the double object alternant as well as the alternant with *to* involves a preposition situated between the two objects. The phonologically zero preposition *G* found in the double object alternant was shown to be an

affix on the main verb, by the same reasoning that led to this conclusion for *CAUS*. Second, I began examining the constituency of VP by considering the debate in the literature over the constituency of double object constructions.

In this chapter, the threads came together. The discussion of constituency zeroed in on a new proposal: Cascade structures. Evidence from different domains converged on these structures. First, they yield the right chunks for coordination. Second, they yield the right command relations for binding phenomena. Next, I pulled in the thread involving the morpheme *CAUS*. If *CAUS* is a preposition in a Cascade structure that moves to the main verb, then the T/SM effect is a case of the Head Movement Constraint. Supporting evidence for this came from the thread introduced in the previous chapter. If *CAUS* moves as described, it should be compatible in the double object alternation with the zero affixal morpheme *G* and incompatible with the nonaffixal morpheme *to*. This turned out to be the case.

Finally, I made sense of the θ -deleting action of *CAUS* (though I did not explain this property) by situating affixation of *CAUS* in the context of a checking theory of derivational morphology based on Chomsky's (1993) proposals. This entailed the presence of two Causer positions in many of the structures under consideration, including those involving Experiencer predicates. This in turn explained why the subject of these predicates behaves like an external argument, while behaving in other respects like a target for movement. In other words, it reconciled the evidence for and the evidence against an unaccusative analysis for predicates like *annoy* (and causative *give*).

In short, Cascade structures, a checking theory for derivational morphology, and the existence of zero causative morphology conspire to explain the central puzzle of this book, while solving numerous puzzles of morphology, lexical semantics, binding theory, and coordination. There is consequently good reason to stick with these proposals as we fashion a solution to the most pressing problems that they face.

The plotline that began with puzzles concerning Experiencer verbs has run its course. However, since this book is a technical report and not a mystery novel, our task is not done just because we have nabbed the guilty party. Before we can accept the structures that make the proposed solution possible, we must understand the theoretical model that is implicated in acceptance of Cascade structures. That will be the topic of the next chapter.

Chapter 7

Cascade Syntax and Layered Syntax

Since the mid-1960s, it has been taken as a truism that only constituents may be affected by a transformational operation. This truism, unlike many others, had an empirical basis. First, many well-motivated transformational operations of earlier transformational grammar indeed affected only constituents. Furthermore, analyses whose transformational operations affected nonconstituents (e.g., Question Inversion and Negative Insertion in Chomsky 1957) were improved in a number of ways when the operation was reformulated so as to pick out a constituent (e.g., Aux). Consequently, the constraint that transformational operations affect only constituents was built into the formalism for rule systems in later transformational grammar (see, e.g., Chomsky 1977:chap. 4). In more modern principles-and-parameters approaches, this assumption survives intact: the variable α in Move α ranges over constituents; chains are sequences of positions, where a position is a constituent; and it is argued that structural constraints on binding relations also respect constituency. The constituency that emerges from this literature, of course, is the constituency that I have called Layered structures.

The conclusions reached in the previous chapter throw a monkey wrench into this tradition, which consequently must be reexamined. Actually, the monkey wrench was always there, waiting to be thrown—precisely because Layered constituency is wrong for coordination, for binding relations, and for head movement. At worst, then, the previous chapter has called attention to a series of real problems with traditional constituency. At best, it has argued for the relevance of Cascade structures to these problematic phenomena.

On the other hand, movement suggests quite different conclusions. If the structures provided by Cascade theory were to provide constituents for Move α or Form Chain (Chomsky 1993), we would be able to move

constituents that in fact cannot be moved, as in (566), and unable to move sequences that can in fact undergo movement, as in (567).

- (566) a. *[To John about himself] Mary spoke ____.
 b. *I wonder [to whom about physics] Mary spoke ____.
 c. *[In the garden on Tuesday] Bill said [he gave the book to Sue ____].
 d. *Sue asked me [in which garden on Tuesday] Bill said [he gave the book to Sue ____].
- (567) a. [To the adults] Sue showed the kids ____ on each other's birthdays.
 b. [To none of the officials] did Sue send her money ____ on any of these days.
 c. [On which table] did Tom put the book ____ during its construction?
 d. [To the children] were given books ____ on each other's birthdays.

The examples in (567) are particularly puzzling if constituency for non-head movement is defined over a Cascade Syntax. In each example of (567), there is direct evidence for the sort of structure predicted by the Cascade theory, since the binding and licensing relations involving reciprocals, negative polarity items, and pronouns as bound variables depend on the existence of Cascade theory structures before movement (or under some reconstruction convention). Nonetheless, the constituency motivated by these various binding and licensing phenomena is not the constituency that determines what undergoes movement. Instead, what undergoes movement are the constituents of Layered structures.

7.1 Layered Syntax and Movement

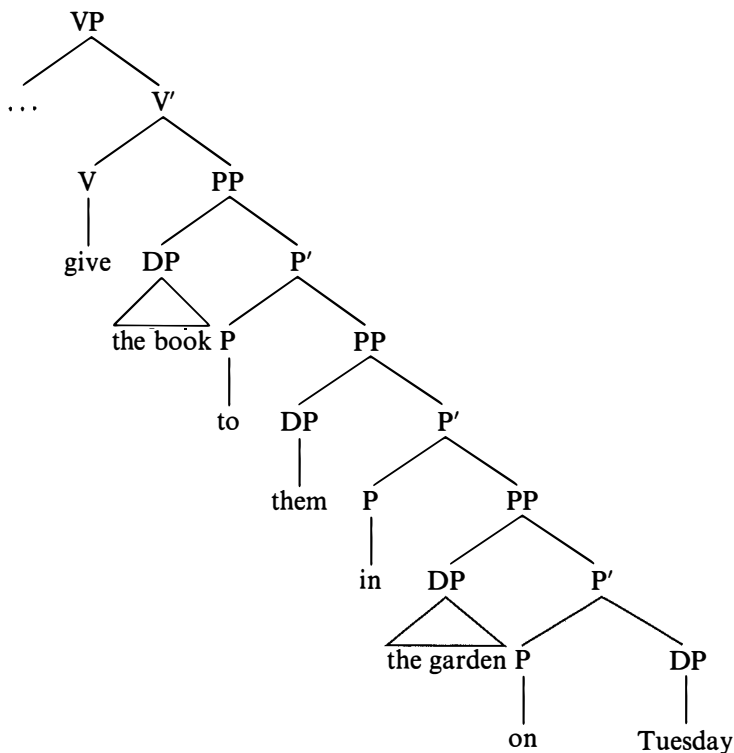
7.1.1 Conflicting Evidence?

We might beat a hasty retreat at this point, in company with most writers on this question, who assume (quite tacitly) that where constituency evidenced by movement is at odds with constituency evidenced by other phenomena, the observations based on movement have a prior claim on our structural affections. Auxiliary assumptions (e.g., assigning a special status to certain instances of P or PP) are then made to explain data that point to structures other than those motivated by movement. In this chap-

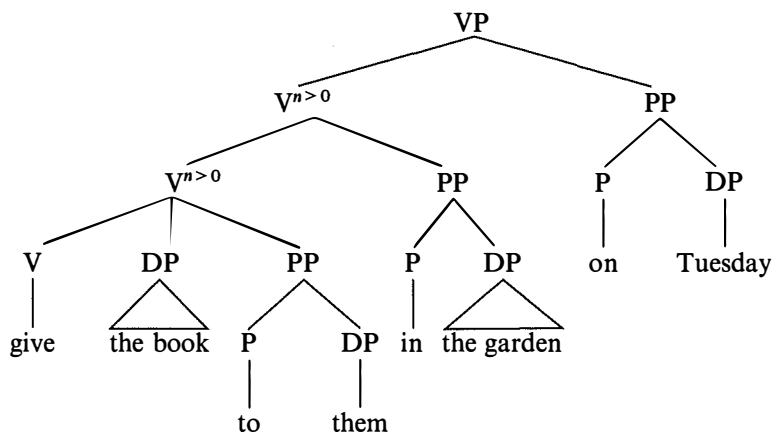
ter, by contrast, I would like to confront the issue squarely. If factorization for movement suggests a constituency distinct from that motivated by other phenomena, then I will take this as evidence for two distinct structural analyses of predicate-headed phrases like VP.

The principles of Cascade Syntax yield VPs like (568) for verbs like *give* with semantically related locative and temporal expressions.¹⁹¹

(568) *Cascade Syntax*



If we stick to the idea that the structural description of a movement process always involves groups of categories that form a structural unit (i.e., constituents), then we cannot escape the conclusion that phrasal movement reveals a very different organization of the same VP. Not surprisingly, phrasal movement motivates a Layered structure like (460), repeated here, in which the two arguments of *give* are its sisters, in which PPs have their traditional structure, and in which nonargument PPs are adjoined to projections of V higher than the minimal V'.

(569) *Layered structure*

The evidence for this organization of the VP is given in (570)–(571). The examples of fronting in (570) require appropriate context (e.g., for (570b) something like *John said he would give the book to them in the garden . . .*).

- (570) a. . . and give the book to them in the garden on Tuesday he did.
 b. . . and give the book to them in the garden he did on each other's birthdays.
 c. . . and give the book to them he did in the garden on each other's birthdays.
 d. ?* . . and give the book he did to them on Tuesday in the garden on each other's birthdays.
 e. * . . and give he did the book to them on Tuesday in the garden on each other's birthdays.
- (571) a. The book he gave to them in the garden on each other's birthdays.
 b. To them he gave the book in the garden on each other's birthdays.
 c. In the garden he gave the book to them on each other's birthdays.
 d. On each other's birthdays he gave the book to them in the garden.

In other words, I will suggest that Cascade Syntax and Layered Syntax are not competing proposals about the hierarchical organization of sentences, but represent aspects of this organization that are both relevant to syntactic phenomena.

7.1.2 Summary of Definitions and Principles

Before I spell out how this system works, it will be useful to collect the principles discussed so far. The sorts of conditions that map θ -relations to syntax— θ -selection, l-selection, and the principle of θ -Position Uniqueness—are exactly the same for Layered Syntax as for Cascade Syntax. The differences lie in the details—in particular, the nature of the syntactic positions relevant to θ -theory and semantic relatedness, as well as the branching structure of the trees. Henceforth I will use subscript C for terms as they apply to Cascade structures, and subscript L when Layered Syntax definitions are intended.

Common to Cascade and Layered Syntax

(572) *θ -selection* [originally (474)]

- Let τ range over (internal, external). If π θ -selects a τ θ -role R as a lexical requirement, then this requirement is satisfied if either
- an argument bearing R occupies a position τ -ly selected by π (*direct θ -selection*), or
 - a θ -selector of R heads a position τ -ly selected by π (*mediated θ -selection*).

(573) *Miscellaneous statements*

- If π θ -selects α as an internal argument, α is semantically related to π . [originally (468)]
- Argument categories: DP, CP [originally (466)]
- If an argument category bears the relation Causer, Means, Location, Time, . . . , or Instrument to some predicate, it occupies a position semantically related to that predicate. [originally (469a)]
- If an argument category occupies a position semantically related to a predicate, it bears some semantic relation to that predicate. [originally (469b)]

Cascade Syntax Only

(574) *Branching_C* is binary.

(575) *Selected_C positions* [originally (487)]

- π *externally selects* α iff α is Spec, $\pi^{\max}$.
- π *internally selects* α iff
 - π c-commands α , and
 - there is no *argument category* σ such that π c-commands σ and σ c-commands α .

(576) *Semantic relatedness_C* [originally (467)]

- a. An argument category α is *internally semantically related_C* to a predicate π iff
 - i. α is c-commanded by π , and
 - ii. there is no argument category Ω such that π dominates Ω and Ω dominates α .
- b. An argument category α is *externally semantically related_C* to a predicate π iff α occupies Spec, π .
- c. An argument category α is *semantically related_C* to a predicate π iff α is internally or externally semantically related to π .

Layered Syntax only(577) *Branching_L* is *n*-ary (i.e., not exclusively binary).(578) *Selected_L positions* [originally (461)]

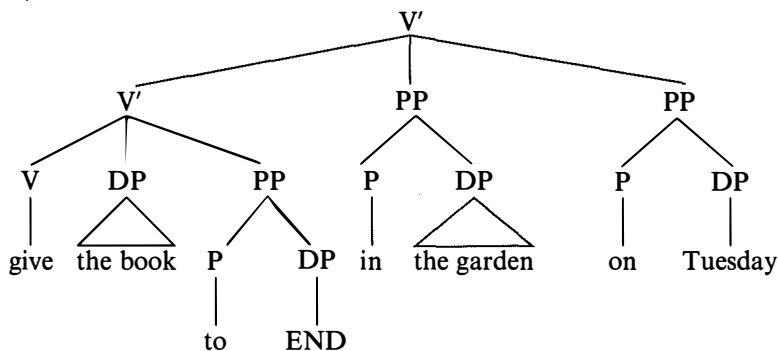
- a. π *externally selects_L* α iff α is Spec $\pi^{\max}$.
- b. π *internally selects_L* α iff α is the sister of π .

(579) *Semantic relatedness_L* [originally (463)]

A category α is *semantically related_L* to a predicate π iff α or a θ -selector of α is a sister to a projection of π .

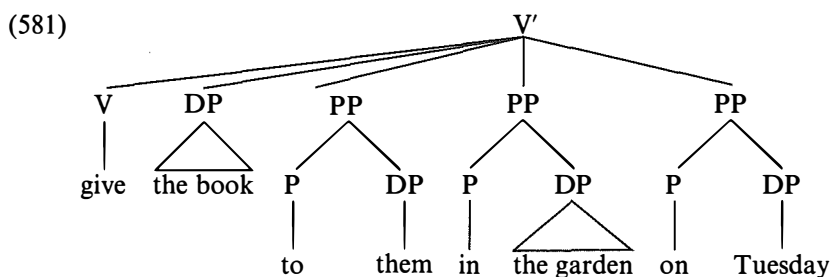
There is a certain amount of freedom in the Layered Syntax concepts, which is worth noting, though it appears to be innocuous. The requirements in (577)–(579) allow PPs whose objects are not θ -selected by V to asymmetrically c-command elements that are θ -selected by V. The requirements do not force any asymmetric c-command relations among “nonargument” PPs. For example, a locative and a temporal PP might be sisters in the Layered structures produced by (577)–(579), as in (580).

(580)



These options do not pose any empirical problem, since the evidence from (570)–(571) is not incompatible with a theory that allows this sort of flat structure, so long as it also allows the more articulated structure of (569). To cut down the number of structures we need to consider, I will generally discuss only structures like (569).

Similarly, the requirements in (577)–(579) do not actually exclude *non-argument* PPs from the lowest V'. They merely force *argument* PPs into the lowest V'. Thus, although a structure like (581) does not support movement of verb and θ -selected arguments leaving non- θ -selected material behind, we so far have no empirical reason to exclude this structure as an alternative to (569) or (580).



In this case, we will shortly see some reason to allow structures like (581) alongside “more layered” structures like (569) and (580).

7.1.3 The Organization of the Grammar

There are a number of ideas one might explore in determining the relation between Cascade and Layered representations. For example, one might decide that Layered representations are formed, perhaps by transformation, from Cascade representations at the interface between syntax and semantics (i.e., LF). (Erich Groat (1992) has pursued this possibility in unpublished work.) This would accord with the fact that the “rightward is upward” character of Layered structures corresponds to interpretation. For example, compare (582a) with (582b).

- (582) a. John plays quartets in foreign countries on weekends.
 b. John plays quartets on weekends in foreign countries.

If there is any difference in the scope of modification in these examples, the PP to the right has wider scope (Andrews 1982a, 1983). Thus, (582a) restricts quartet playing in foreign countries to weekends. On the other hand, (582b) most naturally restricts weekend quartet playing to foreign

countries. These preferences can be read off the asymmetric c-command relations of Layered structures, but not Cascade structures. On the other hand, although these data might argue that Layered structures are relevant to interpretation, they do not tell us whether Cascade structures are irrelevant at LF.

In this section, I will assume that they are not. In particular, I will propose that the phrasal organizations of Layered and Cascade Syntax are both relevant throughout the syntactic derivation. Layered and Cascade structures are parallel organizations of each step of the derivation. To each Cascade structure corresponds a Layered structure, and to each Layered structure corresponds a Cascade structure. This is not too strange. Cascade and Layered structures are nothing more than distinct structures assigned to individual arrays of functional heads, θ -selecting heads, and argument categories, where these structures are the result of mappings from the relations among these elements to structures. The following correspondence principles relate the two structures:

(583) *Cascade-Layered correspondence principles*

- a. α occupies a position semantically related_C to β in a Cascade structure iff α occupies a position semantically related_L to β in the corresponding Layered structure.
- b. α occupies a position externally selected_C by β in a Cascade structure iff α occupies a position externally selected_L by β in the corresponding Layered structure.
- c. α occupies a position internally selected_C by β in a Cascade structure iff α occupies a position internally selected_L by β in the corresponding Layered structure.
- d. α occupies a position that bears a structural relation *R* (complement, specifier, ...) to a functional category *F'* in a Cascade structure iff α occupies a position that bears a structural relation *R* (complement, specifier, ...) to *F'* in the corresponding Layered structure.
- e. Terminal node α *precedes* β in a Cascade structure iff α precedes β in the corresponding Layered structure.¹⁹²

Let us now return to movement. For reasons that I will make clear later, I adopt a "copy theory of movement" (Chomsky 1993). I assume that movement is a procedure that takes a constituent (the *Source*) and copies this constituent to some other position (the *Target*). To the Source is added an instruction that the terminal elements are not pronounced

(Chomsky 1993); this is what I informally indicate as the “trace” of movement. As Chomsky notes, the presence of a copy, rather than a null element, makes “reconstruction” effects unsurprising, at least for $\bar{A}$ -movement. Source positions can count for binding theory, even if binding theory applies after movement, because the contents of the Source position are present even after movement. A-movement must have the property that the Source position need not feed binding theory. For Chomsky, this fact is captured by supposing that A-movement does not leave a full copy. I will assume that a full copy is left by A-movement, but that some other factor allows this copy to escape binding theory.

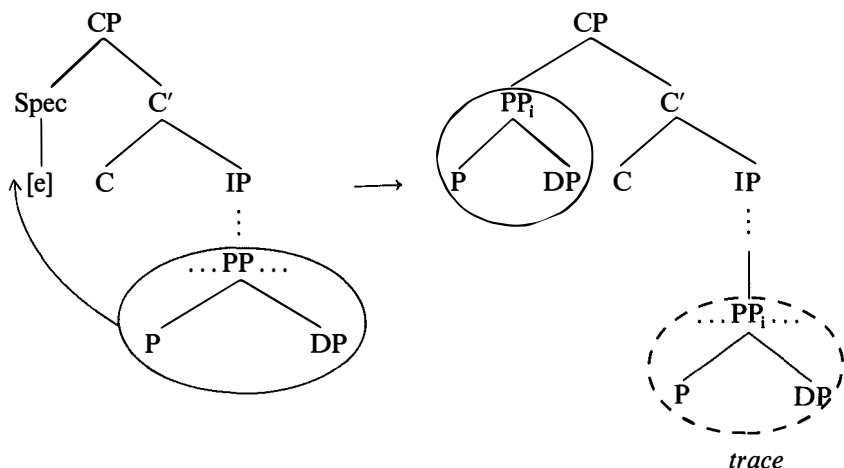
If the source is a nonhead (e.g., a maximal projection), movement must relate two successive structures in Layered Syntax. This time, it is Cascade Syntax that “tags along.” Of course, the contents of both the Source and the Target in nonhead movement will have correspondents in Cascade Syntax. Here, however, we must make an important observation. The principles of Cascade and Layered structures (alongside X-bar theory and whatever principles govern functional categories) are nothing more than specifications of the ways in which heads and bigger constituents organize themselves into still bigger constituents. Consequently, there is no reason that the organization of the copy produced by movement should have to match the organization of the original.

I will use the term *L-constituent* to designate a group of elements that forms a constituent in Layered Syntax but not in Cascade Syntax. The term *C-constituent* will be used for the opposite case, and the term *LC-constituent* will be used for a group of elements that forms a constituent in both structures. Since argument categories stop Cascades, DP and CP will always be LC-constituents. The maximal projections of predicates (NP, VP, and AP) will be LC-constituents as well. This leaves PPs as the only maximal projection C-constituents we are likely to find, with the exception of some exotic possibilities we will consider briefly below.

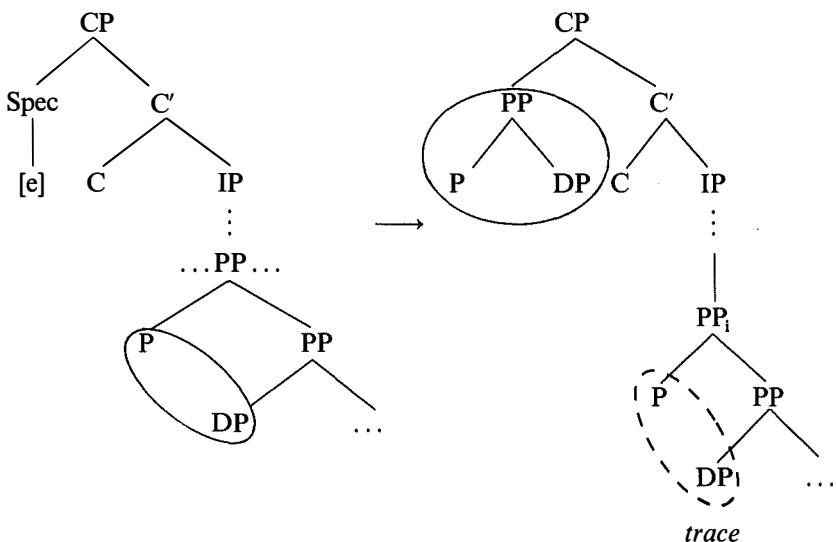
Suppose an L-constituent PP moves to Spec,CP. This means that a copy of the PP is substituted for the Spec,CP, with an instruction not to pronounce the original. How will this look in Cascade Syntax? The Source will be a P with its θ -marked argument in the specifier position of its sister, both before and after movement. The only difference in the Source will be the instruction to PF that leaves the Source unpronounced (i.e., a trace). The Target position must satisfy (583d). In other words, since Spec,CP is an LC-constituent, the contents of Spec,CP must be the same in Cascade and Layered structures. This is no problem in the present

case, since the Cascade correspondent to the L-constituent PP organizes itself as a Cascade PP once it is copied out of its original position. The trees in (584) demonstrate how this works.

(584) *Layered structures*

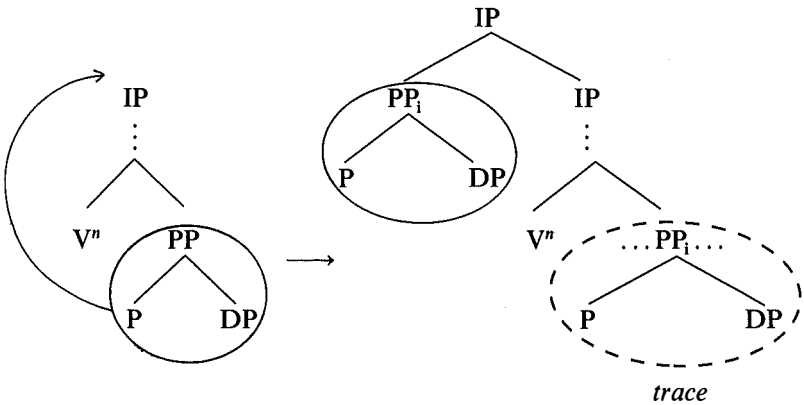


Cascade structures

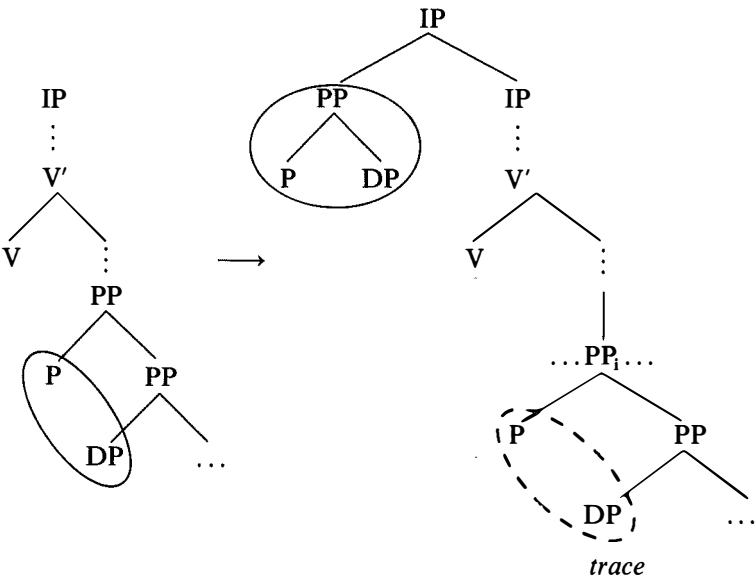


Adjunction of an L-constituent PP to some maximal projection (e.g., IP) will yield a similar reorganization, as shown in (585).

(585) *Layered structures*

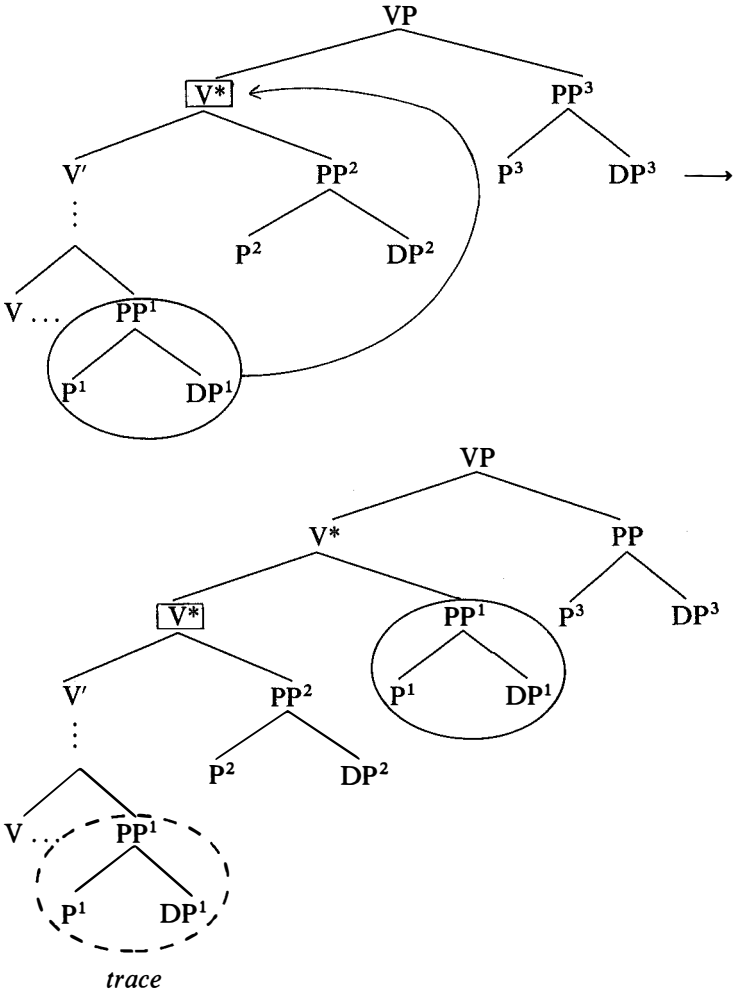


Cascade structures

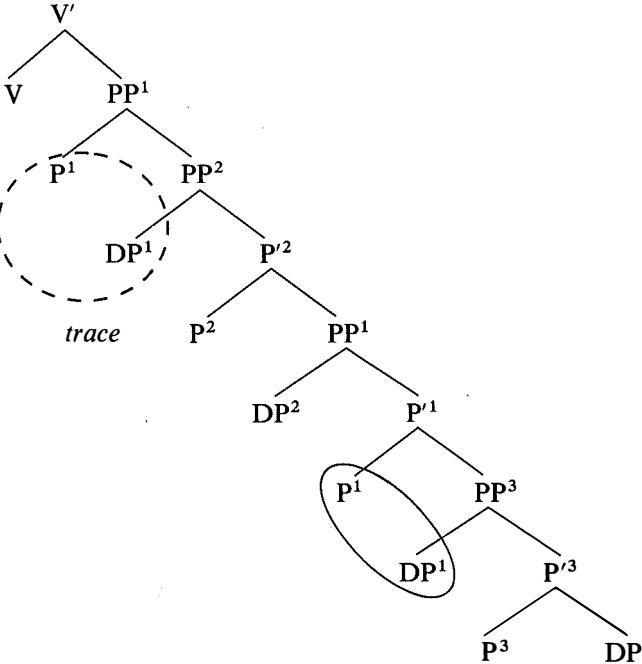
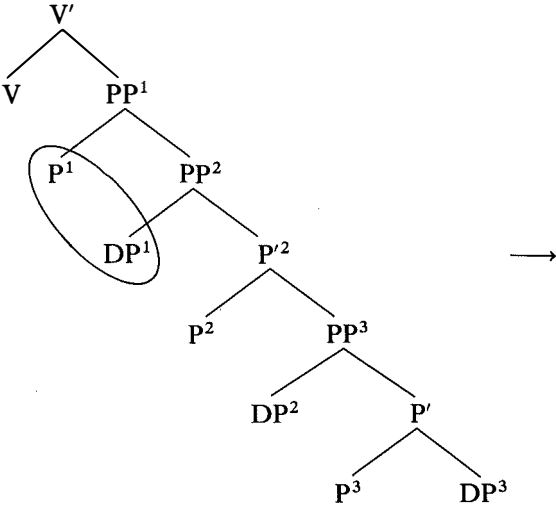


On the other hand, this sort of reorganization does not occur if the Target position is not an LC-constituent. For example, if a PP could raise in Layered Syntax from an object position to a higher but still semantically related position, it might turn out that neither Source nor Target is a constituent in Cascade Syntax. In (586), PP¹ adjoins to the boxed V-projection V*, where neither Source nor Target is a constituent in Cascade Syntax.

(586) *Layered structures*



Cascade structures



If the Source of movement is a head, the operation could just as well relate two successive structures in Cascade Syntax as it could relate successive structures in Layered Syntax. Obviously, heads are LC-constituents

par excellence. Crucially, however, the HMC must examine Cascade Syntax, in order to derive the T/SM effect and its congeners. Further issues arise if movement of *CAUS* is movement of a non- θ -selected preposition, since under those circumstances it will be movement from a non-L-marked PP in Layered Syntax. I return to this issue later. For now, let us make the simplifying assumption in (587).

(587) Head movement always maps a Cascade structure onto a Cascade structure.

Thus, the key point to remember for now is that movement of everything except heads is a Layered Syntax operation, which analyzes and copies Layered Syntax structures. With respect to movement, Cascade Syntax merely follows along, building its structures at each step according to the correspondence principles in (583). Henceforth, I will call this theory the *Dual System*.

7.1.4 Movement and C-Command

The Dual System is strongly supported by cases in which phrasal movement interacts with phenomena that involve c-command. In this section, I will discuss only simple examples. Consider, for example, the contrast among the topicalizations in (588) and (589).¹⁹³

- (588) a. About each other's vacation plans, Sue said Mary had spoken to the kids ____.
- b. *To the kids, each other's friends said Mary had spoken ____ about vacation plans.
- (589) a. *To him_i, Sue said Mary would never speak ____ about John_i.
- b. To him_i, John_i said Mary would never speak ____ about Sue.

As is known, the binding properties of constructions with $\bar{A}$ -movement involve evaluation at the Source position, where Principle C must rule out certain interpretations and Principle A may allow certain interpretations. On the copying view of movement, this means nothing more than the assumption that copied, unpronounced material enters into binding relations if locally $\bar{A}$ -bound (here, I follow Chomsky (1993)). These binding relations, of course, are Cascade phenomena. The Cascade theory then explains the (a) examples of (588)–(589) by organizing the VP such that the object of *to* in (588a) and (589a) is in fact the specifier of the following PP before movement. After movement, the relevant P and DP positions are not pronounced, collectively constituting the trace of movement, but are still organized as predicted by the Cascade theory.

(588b) is much more interesting. Let us assume, as we have throughout, that PP is as important as any other projection to the calculation of command relations. If so, then (588b) is excluded for a simple reason: there is no point at which *the kids* c-commands *each other*. The PP containing *the kids* blocks c-command by *the kids* of the reciprocal. (*The kids* occupies an A-position (object of *to*), so that cannot be a factor.)

Alan Munn (personal communication) raises the possibility that some appropriate formulation of the Weak Crossover Condition might furnish an alternative account of (588b). To exclude this possibility, (589b) is relevant. In (589b), the same command considerations are at stake, for the reason that *him* does not c-command *John*. But here Weak Crossover might be expected to cause the example to degrade (if we hold that Weak Crossover has that effect on (588b)). Instead, the structure of the configuration saves coreference. This leads us to conclude that Weak Crossover is not relevant in either pair of examples.

We also cannot argue for these cases that there is something special about the binding properties of categories that are contained as proper subparts of $\bar{A}$ -positions. In (590), *herself* is close enough to *Sue* to satisfy Principle A of the binding theory only after *wh*-movement takes place, and does so even though it is contained inside an $\bar{A}$ -position.

(590) Sue_i asked [which pictures of herself_i]_j John said that Bill had bought t_j.

Likewise, VP-fronting constructions like ... *and speak to them_i about each other_i's vacation plans* are unproblematic.

In the context of the Dual System, the examples in this section furnish an argument that the branching structure induced by PP counts for command relations. Indirectly, this is yet another argument for Cascade Syntax and against any theory that disregards PP in command calculations. Simultaneously, the examples supply an argument for reorganization of nonconstituents copied by Move α as a constituent at the Target site.

7.1.5 Layered Syntax and Adjunct Islands

So far, I have assumed that the relevance of Layered Syntax to the theory of movement begins and ends with the determination of the Source and Target of movement—in particular, nonhead movement. Is Layered Syntax responsible for other things as well?

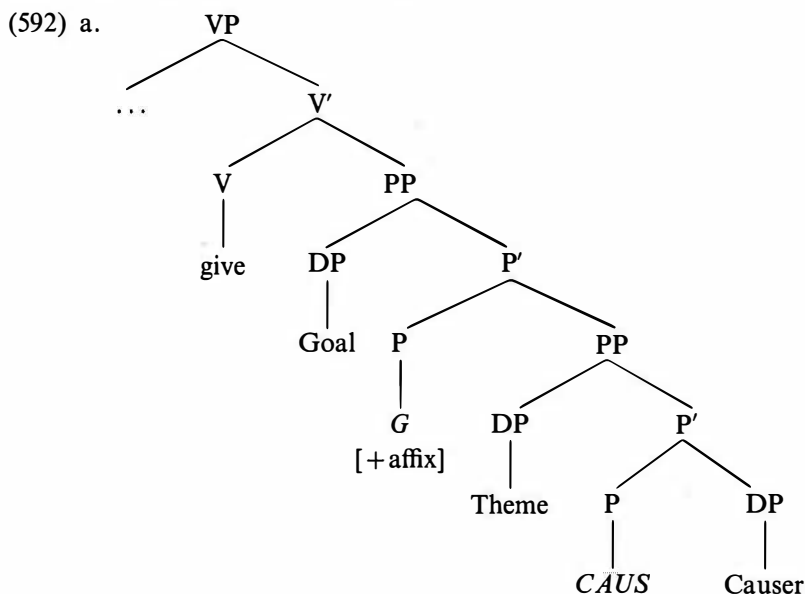
Conceivably, the theory should assign to Layered Syntax responsibility for the familiar island constraints on XP-movement. If this speculation is

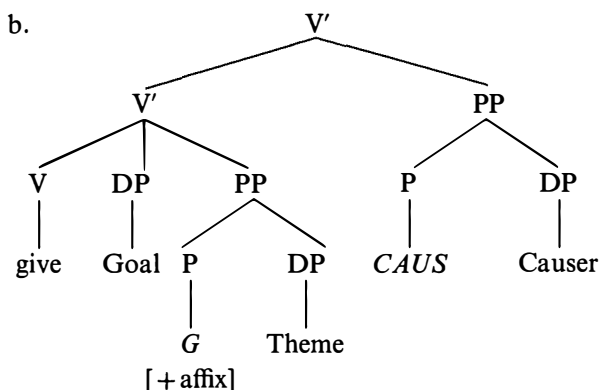
correct, it should come as no surprise to syntacticians. The articulations of Layered Syntax (in its traditional guise as the *only* syntactic structure) were mapped out by researchers largely in response to the constraints on XP-movement. Notions like “I-marking” (Chomsky 1986a) and research topics like “PP-extraction” are naturally pursued only with structures that include traditional PPs and the traditional distribution of complements and specifiers. Only in Layered Syntax, for example, is there a structural distinction between adjuncts and arguments in VP. This sort of distinction has often been held to regulate contrasts like the following:¹⁹⁴

- (591) a. a picture which_i Bill looked at t_i
 b. ?a concert which_i Bill slept during t_i
 c. ?the compartment that_i Bill slept inside t_i on the train

If a structural distinction needs to be made in order to account for these phenomena, it is Layered Syntax that makes it.¹⁹⁵

An obvious problem for any theory that excludes extraction from adjuncts is the existence of A-movement from the object position of *CAUS_P*, if, as I argued, *CAUS_P* heads a phrase whose syntax is that of locatives, temporals, comitatives, and other “adjunct” expressions (see note 175). In Layered Syntax, for example, a Cascade structure like (520a), repeated here as (592a), may look like (592b).





In (592b), movement of the Causer to subject position would constitute movement out of an adjoined PP.

One should wonder, however, if movement from PPs headed by prepositions like *CAUS* is really generally forbidden. Worries arise most readily for $\bar{A}$ -movement. First, *wh*-movement from causative PPs with overt heads seems fully acceptable.

- (593) a. What_i did Beethoven die of t_i?
 b. The kind of pain you wince from t_i is not the kind of pain you die from t_i.

Furthermore, *wh*-movement from a variety of locative, temporal, and comitative PPs seems substantially better than (591b).

- (594) a. a concert which_i Bill slept [at t_i]
 b. a lute concert that you don't have to take caffeine pills [before t_i]
 c. a mandolin concert that you don't have to wear a hearing aid [to t_i]
 d. a town that_i Sue has given lectures [in t_i]
 e. the room that John threw his books on the bed [in t_i]
 f. the person that John put the books on the shelves [with t_i]

What distinguishes (591b–c) from (594a–f) is obscure, but it may have something to do with the semantic complexity of the preposition. *During* requires examination of both boundaries of an event. Likewise, *inside* requires examination of more spatial boundaries than *in*. (Compare a person standing up in a box, who is *in* the box, but not *inside* the box, with the same person fully contained by the box, who is both *in* and *inside* the

box.) In any case, it seems unlikely that $\bar{A}$ -extraction of the object of a non- θ -selected preposition should be excluded by a theory of islands.

Turning to A-movement, which is our real concern, we can make similar points, though the overall shape of the phenomenon is less clear here. Certain occurrences of locative prepositions can apparently be stranded under passive.

- (595) a. This bed_i was slept [in t_i] by Bertrand Russell.
 b. A stage this splendid_i should be acted [on t_i] only by the finest actors.

I know of no comparable examples with temporal, causative, or comitative prepositions.

- (596) a. *1987_i was acted [in t_i] by many previously unemployed actors.
 b. *Abdominal pain_i has been winced [from t_i] by many brave soldiers.
 c. *Sue_i was walked [with t_i].

On the other hand, both temporal and locative DPs, as well as “exchange” phrases, appear in subject position in a construction discussed by Perlmutter and Postal (1984).

- (597) a. 1984 saw the opening of a new bridge.
 b. The 19th century witnessed the growth of capitalism.
 c. This room has seen many fine parties.
 d. This room sleeps five people.
 e. The table sits twelve.
 f. \$10 can buy you a used copy of *Zero Syntax*.

A case can be made that these constructions involve extraction of a DP from a PP headed by a null preposition. First, the inability of these constructions to passivize suggests that the surface subject is not the initial subject, as Perlmutter and Postal noted. These are unaccusative constructions.

- (598) a. *The opening of a new bridge was seen by 1984.
 b. *The growth of capitalism was witnessed by the 19th century.
 c. *Many fine parties have been seen by this room.
 d. *Five people are slept by this room.
 e. *Twelve are sat by the table.
 f. *A used copy of *Zero Syntax* can be bought by \$10.

Second, the preverbal subjects of these constructions behave like DPs and not like fronted PPs. For example, although it is well known that the PP

fronted under locative inversion blocks inversion of the inflected verb, as can be seen in (599), that is not the case with the subjects in (597), as can be seen in (600).

- (599) a. Under the bridge sat John.
 b. ?Is it the case that under the bridge really sat John?
 c. *Did under the bridge really sit John?
- (600) a. Did 1984 really see the opening of a new bridge?
 b. Has this room seen many fine parties?
 c. Does this room sleep five people?
 d. Can \$10 buy you a used copy of *Zero Syntax*?

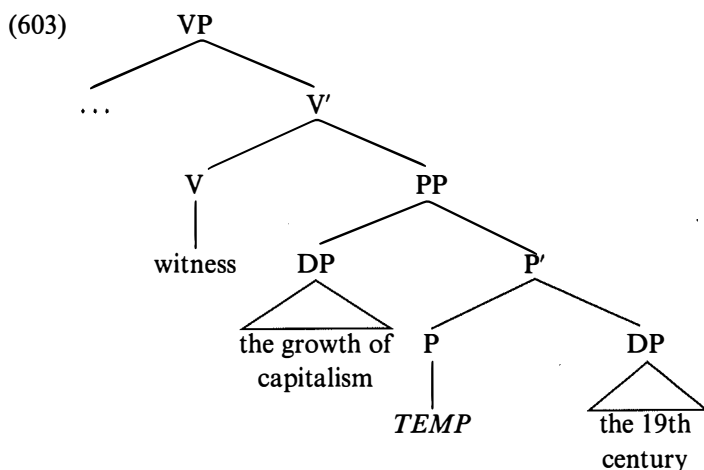
Furthermore, the null preposition may incorporate in the head of its Cascade, like *CAUS* and *G*. In particular, this preposition shows behavior that is familiar from *CAUS*. We can observe the absence of nominalizations (where fair tests can be constructed),

- (601) a. *the 19th century's witnessing of the growth of capitalism
 b. *\$10's purchase of a used copy of *Zero Syntax*

and, possibly, effects akin to Oehrle's Generalization.

- (602) a. \$10 can buy that linguist a used copy of *Zero Syntax*.
 b. ??\$10 can buy a used copy of *Zero Syntax* for that linguist.

I thus suggest an analysis like that proposed for *CAUS* in the Cascade Syntax—for example, involving null prepositions like *TEMP* in (603).¹⁹⁶



A fuller representation of (603) would display two occurrences of *TEMP* (*TEMP_{aff}* and *TEMP_P*), with *TEMP_{aff}* dethematizing the main verb *see* much as *CAUS_{aff}* dethematizes verbs to which it is attached. The main difference between the null prepositions in (598) and *CAUS* lies in the fact that the constructions of (598) are truly unaccusative; the null preposition that is affixed to the verbs here does not add an external θ -role.

These observations are important because locative, temporal, comitative, and “exchange” phrases are non- θ -selected arguments par excellence in the contexts we have been looking at. By showing that they are not islands for extraction, we have reason to worry less about extraction of Causer from PPs headed by *CAUS_P* in the Layered Syntax. Thus, even if Layered Syntax is the type of representation over which island constraints are stated, we must rethink any theory that makes movement from adjunct PPs impossible or deviant.¹⁹⁷

The hypothesis that Layered Syntax configuration dictates island phenomena is attractive, in part, because island effects do single out certain adjuncts, as well as subjects, from which nothing can be extracted.

(604) *a book which_i [the cover of t_i] is torn

Adjuncts and subjects have in common the fact that they are not sisters to a lexical head. This is the core of Huang’s (1982) Condition on Extraction Domain, as well as Kayne’s (1984a:chap. 3) proposal for ECP extensions. The same idea informs Chomsky’s (1986a) notion of l-marking. If these approaches are correct, then there is in turn an attractive approach to our problems within the Dual System. In section 7.1.2, I noted that the (simplest) definition of semantic relatedness for Layered structures admits a particular structural ambiguity in the positioning of non- θ -selected PPs inside VP. In addition to occupying canonical adjunct positions, these PPs may optionally be sisters to V^0 , as illustrated in (581). This was the Layered Syntax analogue of the earlier observation that Cascade Syntax does not reserve θ -selected positions for arguments that are in fact θ -selected. It merely must be the case that if a category is θ -selected, it occupies the appropriate sort of position.

Suppose now that the option illustrated in (581) is restricted (by principles unknown) to phrases like those extracted from in (594). That is, certain adjunct PPs may optionally occupy the sort of position otherwise reserved for arguments. If this is correct, we expect that extraction from these phrases should be possible, even if l-marking or a similar notion

blocks extraction from nonargument PPs in other positions. Of course, an alternative explanation might start by throwing away the supposition that a structural distinction of this sort regulates extraction, but there are actually empirical considerations that support the structural approach. Recall that (581) is only an option. Adjunct phrases (unlike θ -selected phrases) may also sit outside V' in the Layered Syntax. The existence of this possibility is illustrated by the patterns of V-projection fronting seen in (570). We could as well have used ellipsis as a test for Layered structure constituency. As is known, V-projection fronting and V-projection ellipsis seem to single out the same constituents. I repeat (570) as (605) and add examples of ellipsis in (606).

- (605) a. ... and give the book to them in the garden on Tuesday he did.
 b. ... and give the book to them in the garden he did on each other's birthdays.
 c. ... and give the book to them he did in the garden on each other's birthdays.
 d. ?*... and give the book he did to them on Tuesday in the garden on each other's birthdays.
 e. *... and give he did the book to them on Tuesday in the garden on each other's birthdays.
- (606) a. John [gave the book to them in the garden on Tuesday], and Mary did too.
 b. John [gave the book to them in the garden] on Tuesday, and Mary did on Wednesday.
 c. John [gave the book to them] in the garden on Tuesday, and Mary did in the concert hall on Wednesday.
 d. *John [gave the book] to his friends in the garden on Tuesday, and Mary did to her colleagues in the concert hall on Wednesday.
 e. *John [gave] the book to his friends in the garden on Tuesday, and Mary did the pamphlet to her colleagues in the concert hall on Wednesday.

If extraction from a locative or temporal PP requires that PP to be a sister of V' in Layered Syntax, then such extraction should be incompatible with a fronting or ellipsis structure that requires the PP to be a non-sister of V'. Relevant examples with fronting might run afoul of the fact that fronting itself creates an island,

- (607) *I know that Bill gave the book to them on Tuesday. Tell me what day_i [give the book to them] he did on t_i.

but no such considerations arise for parallel facts with ellipsis.

- (608) a. *I know that Bill [gave the book to them]_j on Tuesday. Tell me what day Mary did _____j [on t_i].
 b. *I know that Sue [gave the book to them on Tuesday]_j in the big garden. Tell me which garden_i Mary did _____j [in t_i].

This provides an argument in favor of the view that extraction from adjuncts depends on situating these adjuncts in the sort of position where θ -selected arguments are found. In turn, this suggests a reason why extraction from a PP headed by *CAUS_{aff}* is not deviant, though we still lack an understanding of the restrictions that limit situating adjuncts in Layered Syntax θ -selected position and the further restrictions that govern A-movement out of these adjuncts.¹⁹⁸

Thus, we can cautiously suggest that the distribution of labor between the two types of representations is as in (609).

- (609) a. *Layered Syntax*: XP-movement, island conditions on XP-movement, XP-ellipsis, interpretation of modification relations
 b. *Cascade Syntax*: everything else

(609) is only a guideline for research, not a fully tested hypothesis. I will adopt (609) as a working assumption, but we should bear in mind its provisional and rough nature as we proceed.

7.1.6 Conclusions

The Dual System is certainly not the simplest imaginable theory of grammar. Obviously, a theory with one structural representation is simpler than a theory with two. The claim here is not that the Dual System is a null hypothesis for syntax, but that it is a *correct* hypothesis for syntax. In the next section, I will try to show this by using the interaction between Cascade and Layered Syntax to explain many of the properties of heavy shift, among the oddest and least understood of syntactic processes.

I will then briefly consider another domain in which the interaction between the two representations is illuminating: particle structures. This discussion will also include a speculation concerning the double object alternation in English and its relation to the U(T)AH.

7.2 Heavy Shift

So far, evidence for the dual nature of the Dual System has been closely tied to the distinction between “arguments” and “adjuncts” inside *V'*. Conceivably, one might consider developing an alternative account of the possibilities for movement and ellipsis that directly reflected this distinction in some way. For example, one might develop a theory in which *V*-projection fronting were replaced with some convention that fronted semantically defined groups of some sort—not headed constituents in *X*-bar theory. It is thus of some interest to see whether the appropriate distinctions actually involve adjuncts and arguments, or whether instead they involve structural elements in a phrase marker. Furthermore, it is useful to see whether the ideas and definitions that enter into the Cascade and Layered sides of the system, when put through a workout, help us explain previously inexplicable phenomena. These are the goals of the present section, which deals with heavy shift constructions.

If a DP argument of *V* is focused or otherwise “heavy,” it may be pronounced farther to the right in *VP* than is normal (Ross 1967, 1986). This is the phenomenon generally called *heavy NP shift*, which I will anachronistically rename *heavy DP shift*. In (610), the gap marks the normal position of the shifted DP.

- (610) a. We gave ____ to John on Friday [a brand-new toy].
 b. John introduced ____ to Sue and Mary without thinking [a new teammate who was to catapult them to Olympic glory].
 c. Tom threw ____ at his brother yesterday [something dangerous].
 d. Harry put ____ on this table [the new Ming vase he'd bought].
 e. Mary offended ____ by neglecting to smile [her favorite uncle from Sweden].

7.2.1 The Clause-boundedness of Heavy Shift

This rightward displacement is upward-bounded. A shifted phrase associated with an embedded clause may not, for example, appear after material associated with a matrix clause, as the tense-adverbial interactions in (611) demonstrate.

- (611) *Sue claimed [that she will give _____i to Mary] yesterday [a big book]_j.

Although the data are murkier, it also appears that a shifted phrase may not occur after material associated with VP-external parts of a clause. We can see this in the degradation of shifted material placed to the right of two of the adverbials identified as VP-external in (471).¹⁹⁹

- (612) a. ??We'll just have to fire ____, whether McIntosh likes it or not,
[that person you hired yesterday].
b. ??Rosa won't like ____ anymore, with Ben's mother hanging
around all the time, [the things she used to like].

Let us assume, then, that heavy DP shift places a constituent at the right periphery of VP (but see note 209). The most standard possible analysis of this phenomenon assumes that this movement is rightward adjunction to VP. Following our observations so far, this adjunction must map a Layered structure onto another Layered structure, with Cascade structures tagging along.

Now we know something about positions adjoined to VP, if the proposed analysis has been on the right track so far. A position adjoined to VP in the Layered Syntax is a position for elements semantically related to V (see (573d)).²⁰⁰ If a DP is adjoined to V, that DP is semantically related to V and bears some semantic relation to V. In this way, for example, we excluded processes like raising to object—processes that raise an element of an embedded clause to a position of semantic relatedness outside that clause. The same factors should govern adjunction to VP.

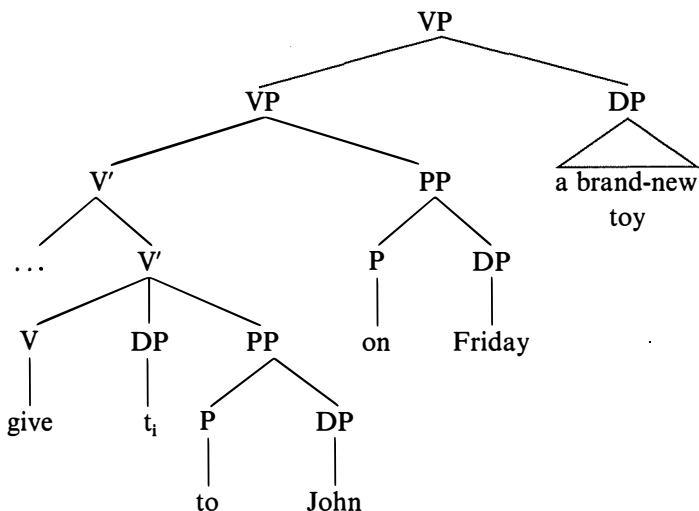
In fact, this consequence seems correct. Granted that the rightward movement that we are discussing is adjunction to VP, why is it clause-bound? If adjunction of α to VP always entails a semantic relation between α (or an argument category in α) and V, then the clause-boundedness of heavy shift is immediately explained: only an element that functions as an argument of V can adjoin to a projection of that V. That is the semantic content of “semantic relatedness.”²⁰¹

Indeed, the need for Cascade Syntax to tag along might even explain why heavy shift operates rightward, rather than leftward. In English, at least, argument takers precede all arguments except those in specifier position. Suppose this is a condition that holds of all representations. It follows that adjunction into a semantically related position must be rightward movement, or else we arrive at a contradiction. The analysis does not, of course, explain why other sorts of movement are not rightward, but at least it appears to solve one hitherto unsolved corner of the puzzle.²⁰²

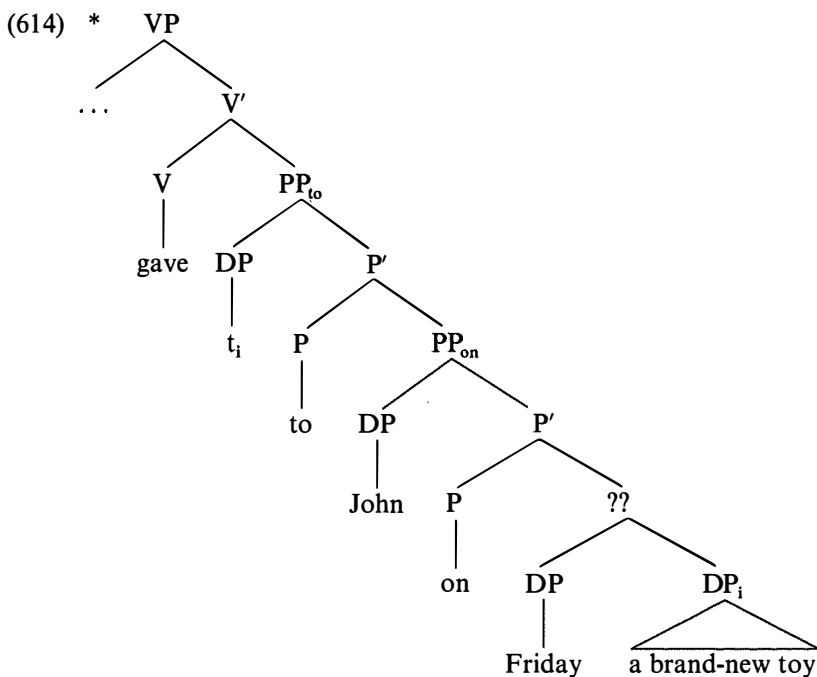
7.2.2 G and Heavy Shift

Still, if the Dual System is correct, the classic analysis of these phenomena as DP-adjunction cannot be exactly correct. Suppose a Layered Syntax DP is adjoined to VP, as in (613).

(613)



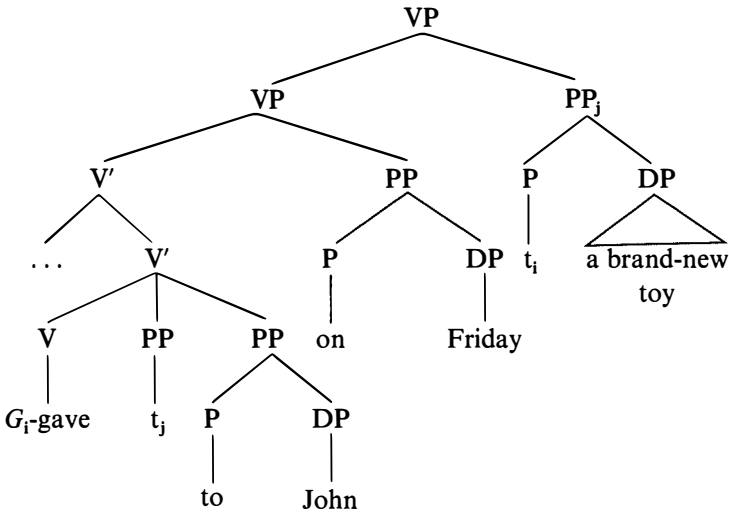
As (614) illustrates, no well-formed Cascade structure corresponds to (613). We cannot find a position in Cascade Syntax on the right periphery of VP, semantically related to *give* but not θ -selected by it, that could be occupied by *a brand-new toy* after movement.



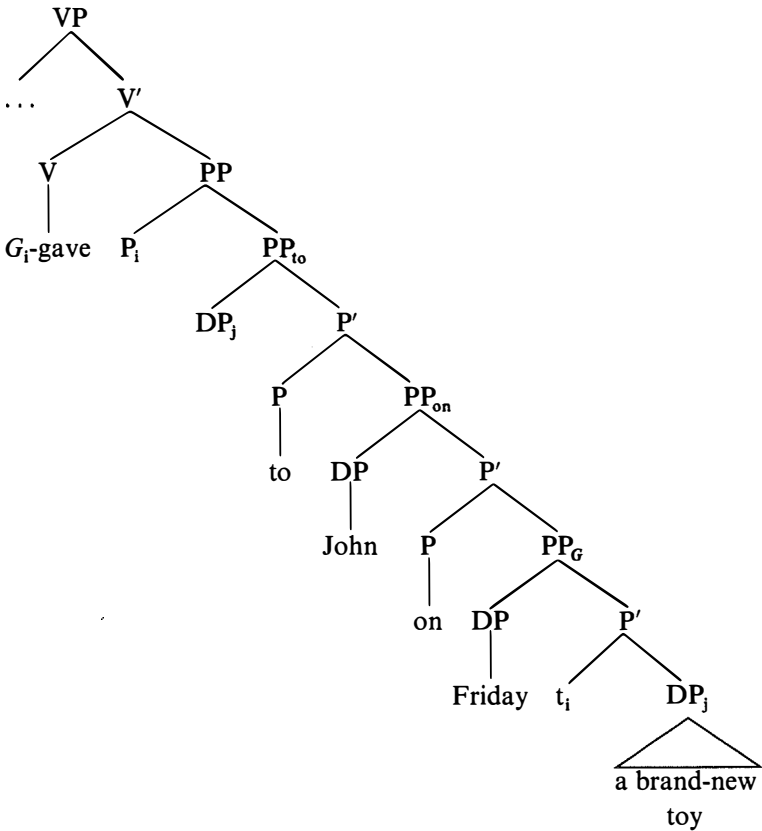
In general, the sequence ... *DP DP* ... inside VP is inadmissible in Cascade Syntax. On the other hand, there is another analysis of these structures that does allow a Cascade structure. We must analyze the shifted constituent as a PP, rather than a DP. Of course, there is no overt P in the examples we have considered, but this fact merely suggests that the PP is headed by a null preposition. Suppose the null preposition *G* is able to θ -select DPs that bear the role Patient as well as Theme (alternatively, we might posit a distinct null preposition for Patient). If this is possible, then the shifted element in our examples can be, not a DP, but a PP headed by *G*. More accurately, the shifted element is a PP headed by the *trace* of *G*, which must, as always, adjoin to V.²⁰³ (The adjunction of *G* to V must, of course, take place before heavy shift, if, as I suggested in the previous chapter, the HMC is a constraint on derivations.)

Heavy shift, illustrated in (615), is now exactly the type of movement that we already examined in (586).

(615) a. *Layered structure*



b. *Cascade structure*



Heavy PP shift²⁰⁴ is not an artifact of this approach, but exists in any case, as noted by Larson (1989) (the examples are mine).

- (616) a. John depends ____ for his livelihood [on royalties from his book].
 b. Sue sent copies of her book ____ on Tuesday [to the prize committee].
 c. Bill spoke ____ about linguistics [to several speech pathologists].
 d. Sue put her books ____ during the concert [in the coatroom].

There is a certain difference between overt PP shift as in (616) and apparent heavy DP shift as in (610). One might see this difference as a challenge to my analysis of apparent heavy DP shift as PP shift, but I do not think the threat is serious.

With apparent heavy DP shift, the displaced constituent is strongly required to be phonologically “heavy.” As discussed by Rochemont (1978) and Stowell (1981), the “heaviness” requirement on the nominals that appear in heavy DP shift constructions is a complicated matter. Focus appears to be at least as important as, if not more important than, mere phonological weight. The nature of the requirement is murky, but the requirement itself is sharp.

With rightward PP shift, however, the facts are more subtle. Concerning examples like those in (617), Larson (1989) reports an intuition that (617a) is the “unmarked” order (also see Larson 1990:608). He finds the opposite order “most natural only when the outer complements are stressed or ‘heavy,’” in (617c). I agree with this observation. To me, the contrast is clearer with *depend* in (618).

- (617) a. Max talked to Mary about Bill.
 b. Max talked about Bill to Mary.
 c. Max talked about Bill to all of the other witnesses.
- (618) a. John depends on royalties for his livelihood.
 b. John depends ____ for his livelihood [on royalties].
 c. John depends ____ for his livelihood [on royalties from his many books].

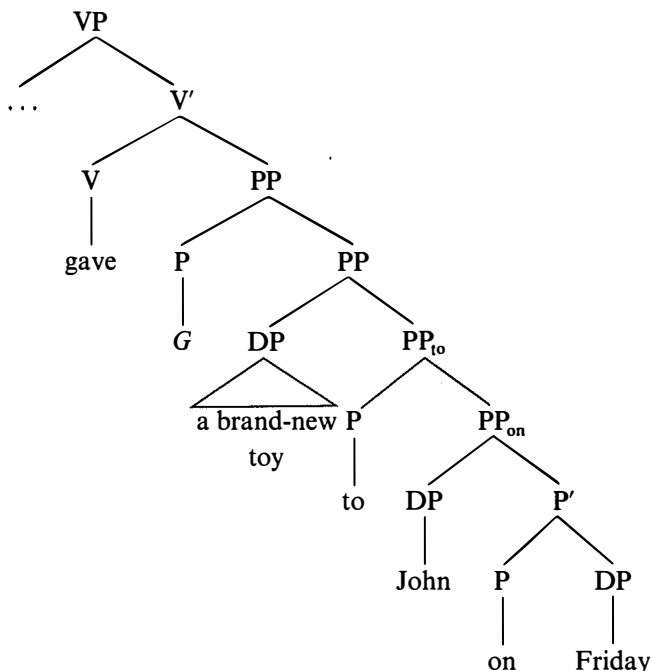
The contrast becomes clearer still when the object of the relevant preposition is a phonologically weak pronominal.

- (619) a. Max talked to 'er about 'em.
 b. ??Max talked ____ about 'er [to 'em].
 c. Max talked ____ about 'er [to those of 'em who were present].
- (620) a. John depends on 'er for 'em.
 b. ??John depends ____ for 'em [on 'er].
 c. John depends ____ for 'em [on some of 'er].

Obviously, we do not find the sharp requirement of focus or heaviness with overt prepositions that we observe with apparent heavy DP shift. This might lead us to question the claim that apparent heavy DP shift is simply shift of a PP headed by a null preposition. Still, there is some heaviness effect in every case of rightward shift. I suggest that the heaviness requirement shared by all types of shift is a consequence of the factors that motivate movement in the first place. Chomsky (1991, 1993) proposes that movement occurs only when motivated by the need to satisfy lexical requirements of particular words and morphemes (e.g., the “strong features” discussed in the previous chapter). Suppose that focused and other heavy phrases carry features that impose requirements satisfied by movement to a VP-peripheral position. Then movement to such a position will be limited to focused and other heavy phrases. I will leave open the question of why this factor appears to be stronger in some cases than in others and why the movement is not obligatory.

Finally, a reminder concerning the structural conditions on θ -selection is perhaps timely. Unless *G* is somehow inserted in the course of the derivation, a Cascade Syntax structure underlying (615b) must be (621) (itself presumably derived from structures like those posited in section 6.3.2).

(621)



Note that the Theme *a brand-new toy* does not occupy the specifier of the sister of *give*. It is, however, the closest argument category to *give*, and as such can satisfy internal θ -selection by *give* for Theme under the notion of θ -selection that includes a minimality clause (i.e., (575b)). In this respect, *G* is occupying the sort of position that *of* occupies in a nominal.

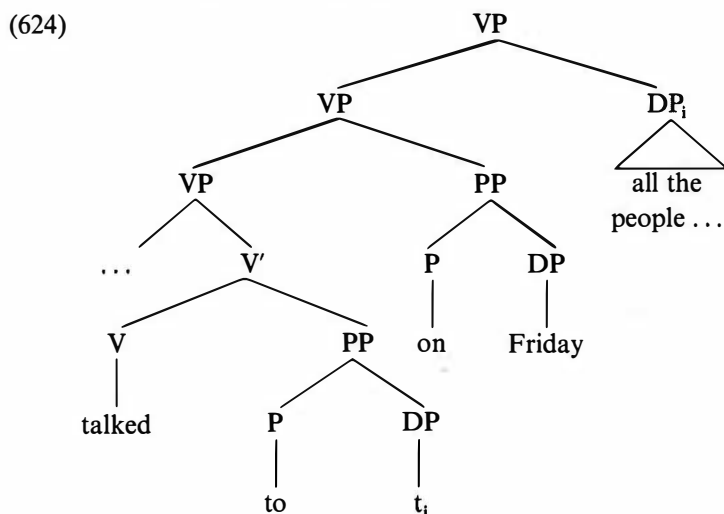
7.2.3 The Absence of Preposition Stranding

Let us turn to more direct evidence for the proposed analysis of heavy shift. In particular, let us consider arguments for the analysis of apparent heavy DP shift as heavy PP shift (with *G*). Previous investigators have noted the general impossibility of preposition stranding with heavy shift (Ross 1967, Bresnan 1976, Stowell 1981). In English, extraction operations like *wh*-movement can extract the θ -selected argument of a prepositional; however, even in English, heavy shift cannot strand a preposition.

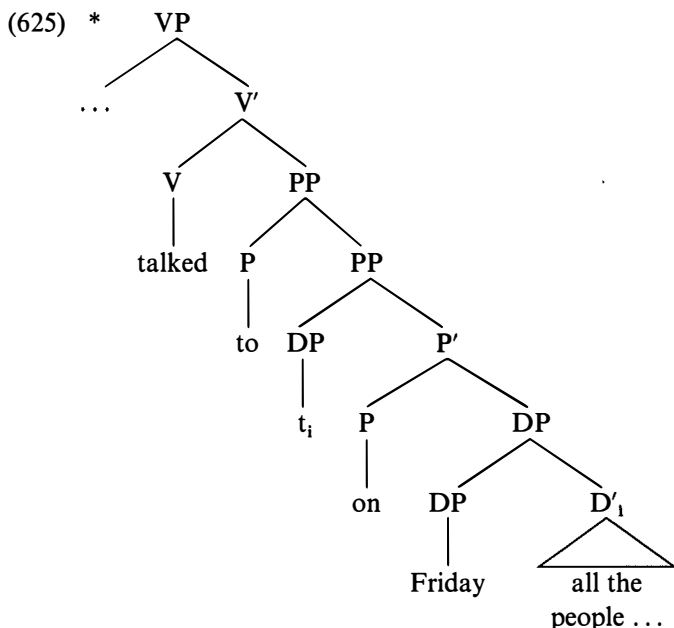
- (622) a. Who_i did John talk to t_i on Friday?
 b. What_i did Leona think about t_i in her jail cell?
 c. This is a county which_i you can buy firecrackers in t_i before major holidays.

- (623) a. *John talked to t_i on Friday [all the people he should have spoken to on Thursday] $_i$.
 b. *Leona thought about t_i in her jail cell [all the unmade beds in her hotel] $_i$.
 c. *You can buy firecrackers in t_i before major holidays [all the southern counties of New Hampshire] $_i$.

This property is straightforwardly explained by the present analysis. DP shift should be impossible, whether the DP is an object of a verb or an object of a preposition. Of course, as with extraction of a DP object, there should be nothing illegitimate about the Layered Syntax extraction of the θ -selected DP argument of a preposition, as in (624).



If the present analysis is right, Cascade Syntax properties of derived representations can indirectly filter Layered Syntax operations. When we try to construct a Cascade Syntax counterpart to (624), there is no legitimate position in which to attach the argument category DP such that it is semantically related to *talked* but not θ -selected by it, preserving linear order. As before, the problem is finding a maximal projection that can have *Friday* in its specifier position and in which *all the people* is semantically related to *talked*. One possibility is (625).



In this structure, *Friday* occupies the specifier position of the heavy-shifted DP. However, *Friday* cannot be semantically related to *talked*, since it is contained inside an argument category, in violation of (469).

The key property of the Dual System is the requirement that each step in the derivation have a well-formed structure both in Cascade Syntax and in a Layered Syntax. Even though movement here maps one Layered Syntax representation into another, Cascade representations *must* be able to tag along. The impossibility of a DP not θ -selected by P, semantically related to but not internally selected by V, accounts for the impossibility of preposition stranding under heavy shift.

7.2.4 The Uses of *G*

This analysis entails a wider use of *G* than the discussion has so far revealed. I proposed first that *G* θ -selects Theme, and I have restricted this notion implicitly to the cases to which the label was first applied by Gruber: an object in motion or at rest with respect to some other Goal, Source, or Location, plus metaphorical extensions of this image to semantic fields like vision and conversation. If I am correct about heavy shift, either *G* itself or some distinct null preposition must also be allowed to

θ -select Patient, Target, Subject Matter, and a variety of other arguments for which an appropriate thematic label other than Theme might be developed, such as the objects of *offend*, *recognize*, *buy*, and *use* in (641).

- (626) a. Sue kicked ____ several times [*G* the open door].
 b. Mary destroyed ____ on Tuesday [*G* the bookcase she had constructed so cleverly].
 c. Bill loved ____ with all his heart [*G* the kind of syntax they did when Grandma was a girl].
 d. John resembles ____ at times [*G* a poor lost soul].

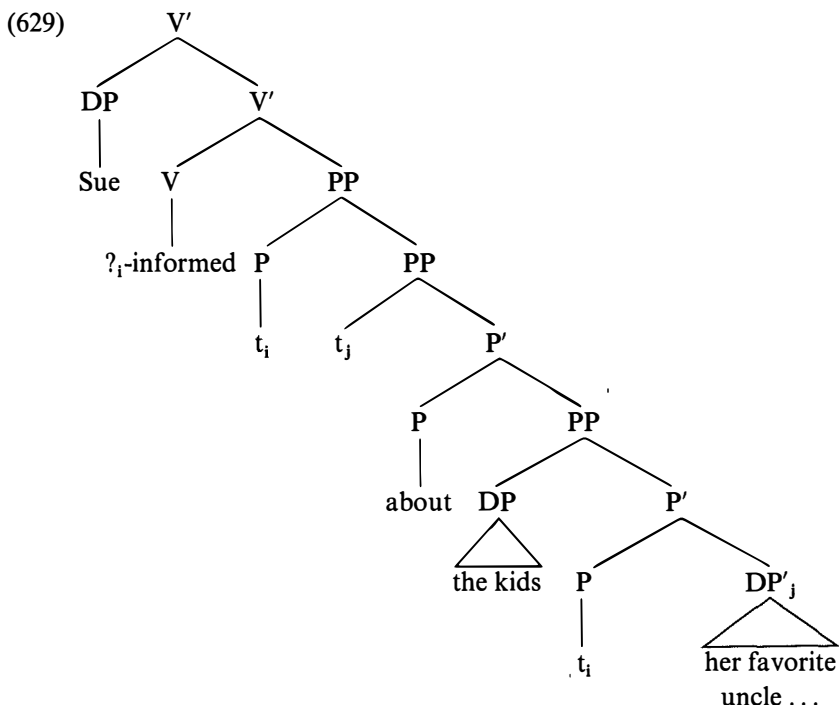
This conclusion exposes both a weakness and a strength in the current theory. The weakness is the absence of an explanation for the nonoccurrence of double object causative psychological predicates, for example, (627a) with the D-Structure representation in (627b). Understand the invented verb *noy* to have the meaning of *annoy*; the point here is that no real verb behaves like *noy* (see note 175).

- (627) a. The book noyed Mary this kind of syntax.
 'The book made Mary annoyed at this kind of syntax.'
 b. ... *CAUS_{aff}*-√noyed Mary *G* this kind of syntax *CAUS_P* the book

Clearly, there must be some sort of incompatibility between the use of Target mediated by *G* in (626c) and the presence of causative morphology in (627a)—an incompatibility not found when *CAUS* raises through *G* in the examples I took from Oehrle.

On the other hand, the present account of heavy shift leads us to expect that the phenomenon will be "thematically governed." This explanation seems to be correct, and is a strength of the account. If an argument bears a θ -role that cannot be selected by a null preposition, it will not occur in shifted position. This appears to be the case with Goal. The incompatibility of heavy shift with a Goal unmediated by *to* is well known for the simple double object construction, but quite crucially extends to other cases of unmediated Goals.

- (628) a. *Mary gave ____ a book [every student who didn't have one].
 b. *Sue informed ____ about the kids [her favorite uncle from Cleveland].



Some other simplex objects also cannot be shifted—again, I suggest, because of the lack of an appropriate preposition.

- (630) a. ??John helped with the homework all the students who came to him.
 b. ??Harry confused with Artur the once-famous composer Anton Rubinstein.

On the other hand, we do not expect these limitations to be genuinely “thematic” at heart. We are not, after all, dealing with a thematic condition on a syntactic rule, but with a syntactic condition that interacts with the repertoire of null prepositions available to the lexicon. Thus, for example, although a bare DP Goal cannot be heavy-shifted, and although there is no null preposition that θ -selects Goal, there is no reason why a PP headed by a lexical Goal-selecting preposition like *to* or *in* should not be able to undergo heavy shift. Indeed, this is fully possible.

- (631) a. Sue gave the book ____ on Friday [to John].
 b. Bill would speak ____ about politics [to everyone who he could stop in the hallway].
 c. Mary will put the computer ____ on Friday [in the room downstairs].

Support for this view comes from another line of investigation as well. *G* might also have a variant that absorbs the external θ -role and fails to assign Case, just like *CAUS* (*CAUS_p* and *CAUS_{aff}*). Let us entertain the possibility that a non-Case-assigning form of *G* plays exactly this role in forming English middles. Movement of the object of *G* to subject position can proceed as it does with the object of *CAUS*. This leads us to expect not to find nominalizations of middles, for the same reason *G-V* does not normally nominalize. This appears correct.

- (632) a. *the bureaucrats' easy bribery
 b. *cities' easy destruction
 c. *Capone's easy rehabilitation
 d. *the book's easy translation
 e. *the account's easy transference
 f. *Charlemagne's easy coronation
 g. *the play's easy performance
 h. *the book's easy translation

The unacceptability of these forms follows straightforwardly if they represent attempts to nominalize a verb whose head is a zero affix, in violation of Myers's Generalization.

- (633) a. [[bribe_v] *G* _v] ry _N]
 b. [[destroy_v] *G* _v] tion _N]

Recall that there were two exceptions to Myers's Generalization: *-able* and *-er*. In the explanation of Myers's Generalization in terms of Fabb's categories in section 3.4.8.2, these were the two affixes that could attach freely to previously suffixed forms (group D). All things being equal, if the null affix theory of middles is correct, we expect that *-er* and *-able* should contrast with the nominalizers in (632) and should attach to middle verbs. As it happens, this seems to be a real possibility for *-er*. Levin and Rappaport (1988b) have observed that *-er* can attach to "cooking verbs" and have argued that these are instances of affixation to a middle.

- (634) baker (= potato), broiler, fryer, roaster, steamer (= clam), ...

They note,

The nominals in [(634)] do not receive the interpretation that would be expected if they were derived directly from the transitive or unaccusative uses of the related verb. . . . The actual interpretation of [the nominal *broiler*] is closer to the interpretation that the related verb receives in the middle construction, *This chicken broils well*. . . . This account of the middle construction could also explain why non-event *-er* nominals based on the middle use of the base verb appear to be rather freely coined. We have found newly coined nominals of this type in recipe sections of magazines: *dunker* (= type of doughnut), *sipper* (= a drink), and *dipper* (= vegetable or fruit to be used with dips). Consider also the following example suggested to us by D. Pesetsky: *This noun makes a good incorporator*. . . .

Further evidence for claiming that some *-er* nominals are derived from the middle use of the base verbs comes from the fact that such nominals show the Affectedness constraint also exhibited by the middle construction. Middles are only derived from verbs with +affected objects, and *-er* nominals that appear to be related to the middle use of verbs are only derivable from verbs with +affected objects. Thus it is impossible to attribute an internal interpretation to *seer*, a nominal based on the verb *see*, which does not have an +affected object and does not have a middle form (**This planet sees well*). (pp. 1078–79)

Unfortunately, there is an inconsistency between my use of this observation and Levin and Rappaport's claims about the nature of *-er*. Levin and Rappaport defend the generalization noted by Fabb (1984) and Burzio (1986) that limits *-er* to verbs with an external argument. This generalization correctly predicts that prototypical unaccusative verbs of existence and appearance generally do not allow *-er* affixation (examples from Levin and Rappaport 1988b).

- (635) *disappearer, *appearer, *dier, *laster, *ender, *transpirer,
 *waner, *exister, *happener, *occurer, *emanator, *collapser,
 *wilter

If middles involve dethematization of the external argument, then either Levin and Rappaport's examples are not instances of *-er* affixation to middles, or else *-er* is not limited to verbs with external arguments. I will not attempt to unravel this puzzle here. I will simply note that nominalized middles display the effects of Myers's Generalization, including the exception (*-er*) that proves the rule. This supports the analysis in terms of *G*.

Preposition stranding is impossible, just as with heavy shift. If the movement to subject position were to strand a preposition, it would strand it either in the configuration [*G*[*P* . . .]] (where *P* itself θ -selects Theme or Patient), which would involve movement from a Case-marked

position, or in the configuration $[P[G \dots]]$, in which case G would have been unable to raise to V over the preposition.²⁰⁵

(636) *Sue_i talks to t_i easily.

Note now that the thematic restrictions on heavy shift also hold of middles.

- (637) a. *These kids give ____ books easily.
 b. *Sue informs ____ about disasters easily.
 c. ??John helps ____ easily.
 d. *Anton Rubinstein confuses ____ with Artur easily.

This is just as we expect if medial G has the properties we are attributing to G in general.

There is a problem with this proposal, however, for which I do not have an interesting solution. Among the motivations for positing the existence of G , I listed in section 5.1.3 the possibility of viewing morphophonological constraints on the double object construction as constraints on the affixation of G . In this way, for example, I explained the impossibility of (638a). On the other hand, just this affixation of G is now posited when heavy shift of a Theme applies, and no morphophonological restriction is found, as can be seen in (638b).

- (638) a. *Mary G_i -donated Oxfam t_i some canned food.
 b. Mary G_i -donated ____ to Oxfam t_i some canned food.

Either the conclusion in section 5.1.3 was completely wrong, or some other fact intervenes in (638a). One might think that the morphophonological restriction, like many of the semantic restrictions on the double object construction, is linked to direct θ -selection of Goal rather than to affixation of G , but the acceptable unshifted counterparts to (628) suggest that this is not correct. Instead, both factors appear to be relevant: affixation of G is blocked by morphophonological factors when the host to which G attaches directly θ -selects Goal.

7.2.5 Layered Syntax Raising Is Cascade Syntax Lowering

The whole discussion of heavy shift is irrelevant, of course, if the Layered Syntax and Cascade Syntax analysis of this phenomenon is not correct in other respects. In this section, I will turn to evidence for the general approach—evidence that will support the Dual System—as well as for the specific hypothesis that Cascade Syntax tags along behind Layered Syntax when XP-movement applies.

When a PP undergoes heavy shift, there is an important difference between the derivational steps in Layered Syntax and the corresponding steps in Cascade Syntax. In Layered Syntax, the operation involves raising, since the adjoined position c-commands the source, but the consequences for Cascade Syntax look just like lowering. This can easily be seen by inspecting the trees in (615), and might look like a disturbing consequence.

In fact, however, this consequence is among the most interesting *results* of the whole approach. First, let us see how heavy shift behaves like raising.

The most straightforward test involves the constituencies for V-projection fronting and for V-projection ellipsis. The most natural examples involve V-projection ellipsis, where the V-projection layer that excludes the shifted phrase can be omitted, just as predicted by a Layered structure like (615a).

- (639) a. Bill has given ____ to John brand-new toys more often than Sue has [_{VP} ...] brand-new items of clothing.
 b. Melchior sang ____ during Saturday's concert the role of Siegfried better than Flagstad did [_{VP} ...] the equally demanding role of Brünnhilde.
 c. Sue will no doubt offend ____ in the near future her favorite uncle from Cleveland. Mary certainly will [_{VP} ...] her big-spending cousins from Tucson.

Examples with V-projection fronting provide the same sort of evidence, though context, pauses, and intonation are crucial in guaranteeing intelligibility.²⁰⁶

- (640) a. I said Bill would give ____ to John [all his brand-new clothes], and [give ____ to John]_i he has [_{VP} t_i] every item of clothing he bought during his trip to Italy.
 b. I had a hunch that Melchior would sing ____ during Saturday's concert [the role of Siegfried], and [sing ____]_i he did [_{VP} t_i] that terribly demanding part, without any difficulty.
 c. I thought Sue would offend ____ in the near future [her favorite uncle from Cleveland], and [offend ____]_i she did [_{VP} t_i] that pompous bastard.

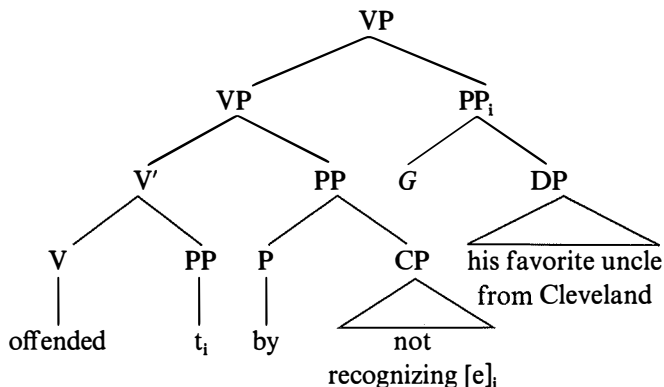
Suppose, furthermore, that the c-command requirement on the antecedents of traces is a Layered Syntax requirement. If this requirement is a

general property of bound gaps, then we are not surprised to see the requirement holding of parasitic gaps, particularly if (with Kayne (1984a), and not Chomsky (1986a)) we view the LF relation between the parasitic gap and its ultimate binder as identical to the relation between a trace and its binder.²⁰⁷

We are then not surprised that heavy-shifted phrases license parasitic gaps inside VP, as noted by Engdahl (1983a).

- (641) a. John offended [by not recognizing ____] his favorite uncle from Cleveland.
 b. Sue bought [in order to use ____ when she was bored] a new computer with fancy speech recognition hardware and a port for an espresso machine.

(642) *Layered structure*



Chomsky (1982) demonstrates that licensing of parasitic gaps is a property of binders in $\bar{A}$ -positions. Let us assume that the relation of a parasitic gap to its binder is a movement relation, and thus is sensitive to Layered structures.²⁰⁸ Although a PP adjoined to a nonzero projection of V occupies a semantically related position in Layered structure, this position is also an $\bar{A}$ -position in the typology relevant for licensing variables (since it is not a position in which θ -selection or Case checking can take place).²⁰⁹

Now to the key point. Despite all this evidence for heavy shift as raising, binding phenomena with heavy shift constructions behave as if the construction involved *lowering*. Some of the relevant facts were observed by Larson (1989), but the following examples are my own. I have not constructed minimal pairs for presentation here, wishing instead

to present a variety of examples, but careful examination of minimally differing examples does not change the picture. The (a), (b), (c), and (d) sentences that follow exemplify

- a. Principle C
- b. anaphor binding
- c. pronouns as bound variables
- d. negative polarity licensing

(643) *Interactions between θ -selected object and shifted phrase (with G)*

- a. *We gave ____ to him_i on Friday [John_i's brand-new toy].
- b. We gave ____ to them_i at the interviews [copies of reports on each other_i].
- c. Bill heard ____ from each committee member_i on Friday [a report on his_i activities].
- d. Tom threw ____ to none of these people on Tuesday [any set of keys that had "Do Not Copy" stamped on them].

(644) *Interactions between non- θ -selected argument and shifted phrase (with G)*

- a. *We gave ____ to him at [that time]_i [a copy of the report we'd finished on [Friday]_i].
- b. I sent ____ to Sue for the kids_i [copies of each other_i's report cards].
- c. Bill heard ____ from Sue on [each day]_i [a message about what to do [then]_i].
- d. Tom threw ____ to Mary on no occasion [any set of keys that had "Do Not Copy" stamped on them].

(645) *Interactions between θ -selected object and shifted phrase (overt P)*

- a. *Mary talked ____ about him_i on Friday [to Bill_i's mother].
- b. John talked ____ about the men_i on Tuesday [with each other_i's supervisors].
- c. Rosa spoke ____ about each defendant_i [to his_i lawyer].
- d. Bill depended ____ for none of these items [on any charity from his friends].

(646) *Interactions between non- θ -selected object and shifted phrase (overt P)*

- a. *Mary talked ____ about Bill at [that time]_i [to people who'd seen him on [Friday]_i].
- b. [impossible to construct]

- c. Bill relied ____ in each major city_i [on its public transportation system].
- d. Sue looked ____ at no time [at any unrelated data].

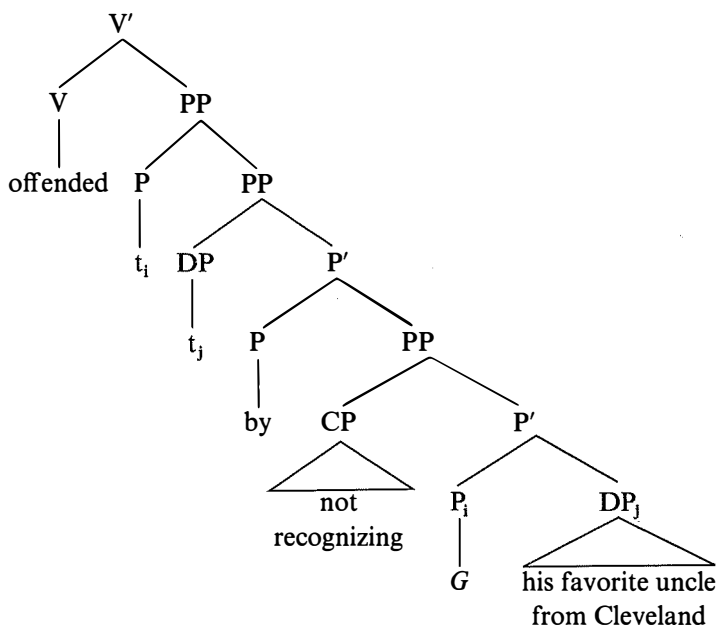
Other instances of binding that should be excluded in the Dual System are also deviant. The examples in (648) are drawn from Jackendoff 1990: 431–34, with the Weak Crossover deviance in (648a) weakened in accordance with my judgment.

- (647) a. ??Mary invited ____ to her home on his_i birthday [each boy in the class]_j. [Weak Crossover]
 b. *The social worker described ____ to themselves_i [the members of the gang]_j. [Principle C, with anaphor antecedent]
 c. *John put ____ on any table [none of these documents]. [negative polarity]
- (648) a. ??I talked ____ about her_i mother [to every girl]_i.
 b. *I talked ____ about themselves [to John and Bill].
 c. *I talked ____ about any of the boys [to none of the girls].

The Dual System requires that the c-command requirement on traces of XP-movement can be satisfied in the Layered Syntax, even though the c-command requirement on binding relations appears to be a Cascade Syntax requirement. Except for this fact, the status of heavy shift as a Layered Syntax raising operation is unremarkable, and the apparent status of heavy shift as a Cascade Syntax lowering operation is explained as a straightforward instance of the Cascade Syntax derivation “tagging along” with Layered Syntax.

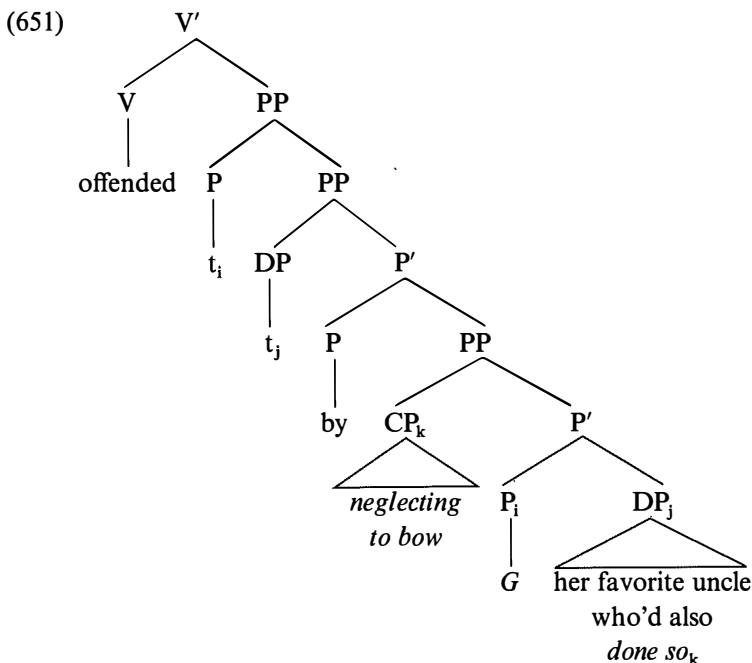
Raising and lowering properties of heavy shift can be observed in the same examples. For example, as (649) illustrates, parasitic gap constructions like those in (641) still behave binding-theoretically as though the shifted phrase is c-commanded by the adjunct.

(649)



We can see this via Principle C effects that seem to hold of the relation between the pronominal *do so* construction and its VP antecedent.

- (650) a. Sue offended ____ by [neglecting to bow]_i [her favorite uncle who'd also [done so]_i].
 b. *Sue offended by also [doing so]_i her favorite uncle who'd [neglected to bow]_i.
 c. By also [doing so]_i, Sue offended her favorite uncle who'd [neglected to bow]_i.



Not surprisingly, the parasitic gap, like the real gap, must be the sort of gap that can be a PP headed by *G*. This is another way in which the Layered and Cascade structures interact. I assume that, whatever the analysis of parasitic gaps, the various gaps must match in category.

- (652) a. ??Sue offended ____ [by not talking to ____] her favorite uncle from Cleveland. [cf. (623a)]
 b. *Sue offended ____ [by informing ____ about the kids] her favorite uncle from Cleveland. [cf. (628b)]

We thus resolve what otherwise would be a paradox. For the purposes of V-projection fronting, V-projection ellipsis, and the c-command condition on traces, shifted elements are higher than the rest of the VP. For the purposes of binding theory, shifted elements are lower than the rest of the VP. This is a remarkable property of the system. It is in no sense an “add-on.” The theory would be more complex if adjunction to VP did not establish a new configuration of semantic relatedness. Instead, the properties of heavy shift arise from the very guts of the system.

There is one complication in the data, which actually supports the general analysis. The object of a shifted PP (including the object of *G*) cannot

with any ease be an anaphor bound by the object of a PP to its left. This phenomenon has been discussed by Reinhart (1983), Jackendoff (1990), and Larson (1990), among others.

- (653) a. John talked to the men about each other.
 b. *John talked about the men to each other.
 (Jackendoff 1990:430ff., Larson 1990:608)

- (654) a. I spoke with Rosa_i about herself_i.
 b. *I spoke about Rosa_i with herself_i.
 (Reinhart 1983:177)

One might be tempted to attribute these facts, with Larson (1989), to a structure in which the first DP fails to c-command the second. Such a structure, of course, would be incompatible with the facts of (643)–(646) —specifically, with (643b), (644b), and (645b). In particular, although the object of the first preposition may not bind an anaphor that *is* the object of the second preposition, it may bind an anaphor contained *within* the object of the second preposition, as those examples show.²¹⁰

On the present analysis, we are left with only (653)–(654) to explain. In fact, a straightforward account is available if we remember that $\bar{A}$ -movement typically has no effect on the action of Principle C. When a constituent undergoes $\bar{A}$ -movement, its subconstituents are evaluated for Principle C in the original position from which the constituent was moved. Thus, for example, with the caveats explored by Lebeaux (1988) (cf. Chomsky 1993), *wh*-movement does not eliminate Principle C violations.

- (655) a. *He_i said Sue had purchased this picture of Tom_i.
 b. *[Which picture of Tom_i]_j did he_i say Sue had purchased t_j?

On the view of movement posited by Chomsky (1993) and adopted here, this is a result of viewing $\bar{A}$ -movement as copying. The structure of (655b) actually includes a silent clone of *which picture of Tom* at LF, which triggers Principle C effects. Now it is true that the Dual System situates the $\bar{A}$ property of heavy shift in Layered Syntax, but binding relations for Principle C are evaluated in a Cascade syntax. If we need to decide whether the trace of movement is or is not a copy that is relevant for binding theory, we must ask, in this system, whether movement targeted an A-position or an $\bar{A}$ -position. Heavy shift targets an $\bar{A}$ -position; therefore, its trace is a copy relevant for binding theory.

If this is correct, then more accurate representations of (653b) and (654b) would be (655a) and (656b), where italics mark the trace.

- (656) a. *John talked [*to each other*] about the men [to each other].
 b. *I spoke [*with herself*] about Rosa [with herself].

For Principle C, the trace versions of *to each other* and *with herself* in (653b) and (654b) will behave just as pronouns do in familiar unshifted sentences like (657a–b).

- (657) a. *John talked to them_i about the men_i.
 b. *I spoke with her_i about Rosa_i.

When the anaphor is not the object of the preposition, but a subpart of that object, as in (645b), there is no Principle C violation, even when we examine fuller representations, (658a). The structure in (658a) no more violates Principle C than (658b) does.

- (658) a. John talked [*with each other's supervisors*] about the men_i on Tuesday [with each other_i's supervisors].
 b. John talked with her_i supervisor about Mary_i.

This theory also lets us understand why replacing the anaphor with a pronoun in (653b) and (654b) does not improve matters.

- (659) a. *John talked about the men_i to them_i.
 b. *I spoke about Rosa_i with her_i.

If c-command were the sole blocker of (653b) and (654b), then the same lack of c-command could be expected to allow pronouns to replace the anaphors. On the other hand, if Principle C in the position before movement is at stake, then these examples are predicted to be no better, which is the case.²¹¹

The cases of quantifier binding in the (c) examples of (643)–(646) raise no new problems, since the bound variable is a subconstituent of the object of the second preposition. Thus, the “reconstruction property” of heavy shift has none of the consequences found with Principle C violations. On the other hand, the copied structure might be relevant to whatever factors influence Weak Crossover. If this is the case, we can account for Larson’s judgments of deviance in examples like (645c) by positing that the original position of *to his lawyer* to the left of *about each defendant* is relevant to crossover.

- (660) Rosa spoke [*to his_i lawyer*] about each defendant_i [to his_i lawyer].

Finally, negative polarity licensing does not involve binding at all, at least in the traditional sense. Consequently, no problems should arise in the (d) examples of (643)–(646). Still, should these examples be judged deviant by some speakers (as they are by Larson), one might argue that the unlicensed negative polarity item in the original position of the shifted phrase is a source of some trouble.

7.2.6 Heavy CP Shift

Since CP, like DP, is an argument category, the matrix VP of examples like (661a) must contain a null preposition between the first and second objects, as indicated. Otherwise, either *Bill* would have to occupy Spec,CP, which would sever its semantic relation to *tell*, or else some other illegal structure would have to be assigned. I will assume that the preposition here is *G*, by analogy with (661b).

- (661) a. I will [_{VP} tell Bill *G* [_{CP} that the world is round]].
 b. I will [_{VP} tell Bill *G* [_{DP} a story]].

That *G* may take CP as an object can also be seen in the availability of heavy CP shift.

- (662) a. I will tell Bill ____ on Friday [_{CP} that the world is round].
 b. Sue told Mary ____ on Friday, after telling Sue ____ on Tuesday, [_{CP} that the world is round].

There is one difficulty with this proposal, which I will not solve. Allowing CP to be an object of *G* flies in the face of the usual prohibition against overt prepositions with CP objects in English.

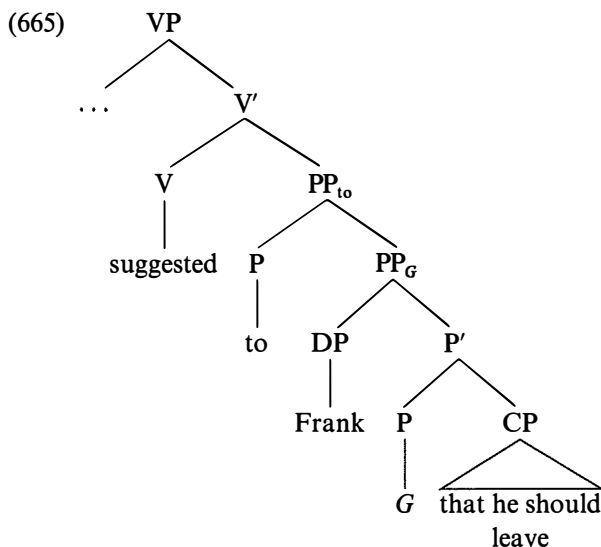
- (663) a. *Sue talked about that the world will end.
 b. *Bill is content with that the world will end.

I will leave this problem unresolved, since there is, to my knowledge, no known explanation for the constraint displayed in (663).²¹²

As it happens, the phenomenon of heavy CP shift provides another demonstration of the explanatory power of several aspects of the Dual System. Consider another type of CP complementation involving two θ -selected arguments.

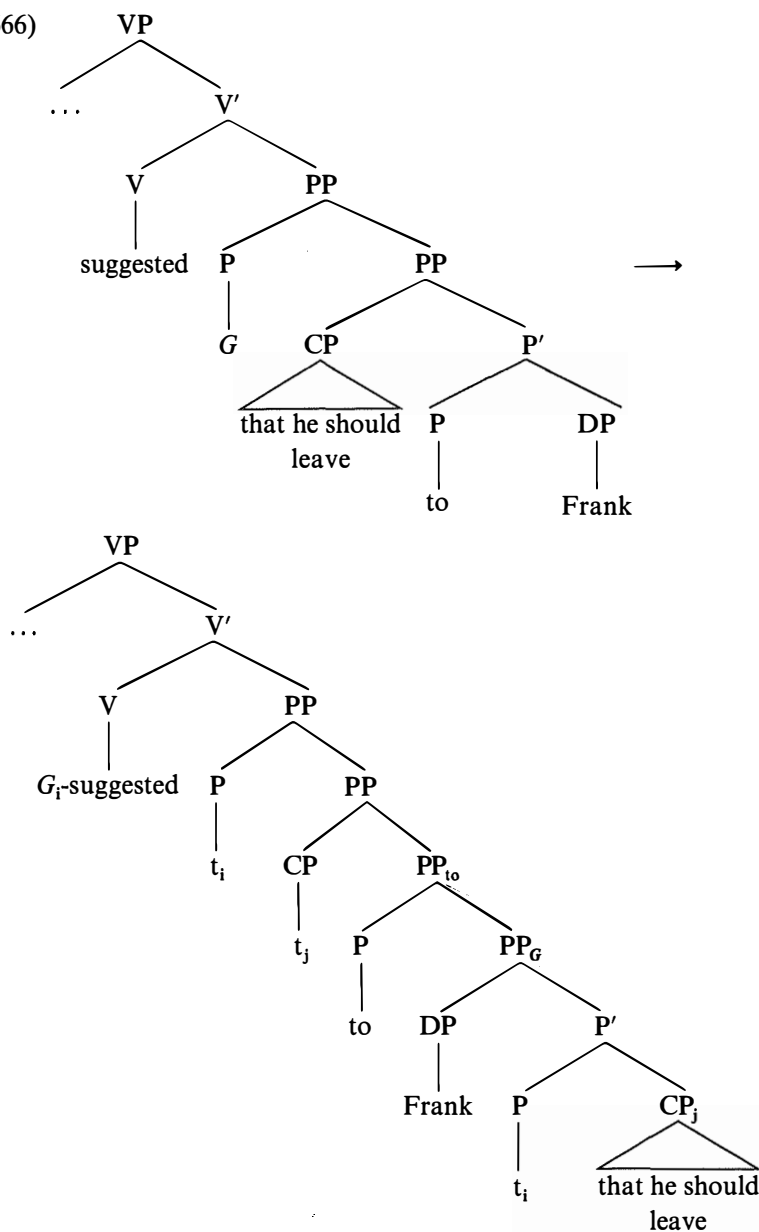
- (664) a. Mary suggested to Frank that he should leave.
 b. Sue protested to the judge that Joyce's rights had been violated.
 c. Melvin confessed to the priest that he had sinned.
 d. Anthony signaled to the gorilla that dinner time was near.

Once again, the CP complement must be θ -selected by a null preposition, if only for the same reasons that motivated the null preposition in (661a). However, our hypotheses tell us something more. Suppose the structure of (664a) were (665) *tout court*.



This structure poses no problems for θ -selection. However, putting our discussion of the T/SM effect and Oehrle's effects together with our discussion of the affixal property of *G* in chapter 5, the structure poses another problem. Since *to* is nonaffixal, *G* should be unable to raise to *suggested* over *to*. Therefore, (664a) must actually be an instance of heavy CP shift, with *G* and its object originating in a PP structurally adjacent to *suggested*. I give the relevant structures in the Cascade derivation in (666).

(666)



Interestingly, this is also the analysis (minus the Cascade structures) first suggested by Stowell (1981), who proposed that CP in these constructions is a first object of the verb, forced to undergo heavy shift by his Case Resistance Principle.²¹³ In fact, this conclusion, inexorably forced by the Dual System, also seems to be correct. The argument comes from an observation attributed by Postal (1986:225) to Alexander and Kunz (1964:26) and Higgins (1973).

Consider passive sentences that involve preposition stranding. These are often called *pseudopassives*. The preposition in these constructions must be affixed to V by a morphological process whose exact nature is unclear. What is clear, however, is that this process is not only tree-dependent, but also dependent on adjacency between V and P.

- (667) a. *Sue_i was given a book to t_i.
 b. *The book_i was looked recently at t_i.

(668) *Sue_i was given-[to_j] books t_j t_i.

The trace of heavy PP shift counts as a blocker of adjacency.

- (669) a. *Sue_i was given ____ to t_i [G all the books we had acquired at the library].
 b. *Sue_i was heard ____ from t_i [G great things].
 c. *Money_i was depended ____ for t_i [G on Sue].

Crucially, constructions like (664) behave just like heavy shift constructions (examples from Postal 1986).

- (670) a. *Frank was suggested to that he leave.
 b. *The doctor was said to by Marvin that his liver was inflamed.
 c. *The judge was protested to that Joyce's rights had been violated.
 d. *The investigator was confessed to that Tom was an extraterrestrial.

When the CP argument is optional, as it is with *confess* and *signal*, the availability of pseudopassive correlates with the absence of the CP.

- (671) a. The priest was confessed to by Melvin (*that he had sinned).
 b. The gorilla was signaled to by Anthony (*that dinner time was near).

Thus, the system forces on VPs with a nominal and clausal argument a unique analysis, and independent evidence confirms that this analysis is correct.

7.2.7 The Verb-Particle Construction and *G*

Finally, we need to back up the claim that heavy shift involves a null preposition.²¹⁴ In doing so, fortunately, we can tie up some loose ends (and create some new ones). If *G* is not limited to the double object construction, but can introduce garden-variety DPs that bear the appropriate θ -role, we might expect *G* to act as a *deus ex machina* for difficulties with Case assignment. In one instance—the Cascade of nominalizations—we must prevent *G* from acting in this way. In fact, we have already discovered that Myers's Generalization takes care of this problem. Furthermore, for one-argument nouns and adjectives like *father* and *proud*, even if there is no nominal or adjectival affix that produces an effect of Myers's Generalization, there may be a general requirement that arguments be introduced with *of* (an instance of *l*-selection), as discussed by Chomsky (1986b). In another case, however—the verb-particle construction—*G* plays just the *ex machina* role it should.

Our discussion of heavy shift included the observation that the θ -role of Goal is never θ -selected by a phonologically zero preposition in English. This observation explained why Goal arguments never display “apparent heavy DP shift”—that is, heavy PP shift with a null preposition (see (628)–(630)). English particle constructions provide another use for this observation. In this book, I have adopted a traditional view of Case, in which verbs are directly responsible for licensing objective Case, without the intermediacy of a specifier-head relation with a functional category. With this tradition, I also assume the existence of an *adjacency condition* on objective Case of the sort developed by Chomsky (1980), Stowell (1981), and others: the verb and the element whose objective Case it licenses must be string-adjacent. V-movement and heavy shift may render this adjacency opaque (as may head-last word order, discussed below), but I know of no compelling arguments that lead us to reject this requirement, once traces of V and peculiarities of OV order are taken into account.

In work in progress, Bobaljik (1994) argues that adjacency is quite generally the environment for discharge of features (in Chomsky's (1993) sense). (It is quite possible to uncouple Chomsky's theory of checking from his specific assumptions about the structural conditions on checking.) For example, V may move to Infl to discharge inflectional features, or else, Bobaljik suggests, it may discharge features under adjacency with Infl of a certain sort. If, for example, these features are “strong,” then adjacency between V and Infl (or their traces) must hold at PF. It is for

this reason that the interposition of a DP between V and Infl requires overt V-raising in SVO Germanic languages. Adverbial material, however, is ignored. Such is the case under so-called object shift, as discussed by Holmberg (1988). In Danish, for example, movement of the pronoun *den* between Infl and V (to the left of adverbial *ikke* 'not') is possible only when V is otherwise able to move to Infl, as is the case when V moves through Infl to Comp in (674). When V remains in VP, as it does in the embedded clause of (672) and as it does when participial in (673), object shift of *den* 'it' is impossible.

(672) a. Det var godt at Peter ikke købte den.
 it was good that Peter not bought it
 'It was good that Peter didn't buy it.'

b. *Det var godt at Peter den ikke købte.
 it was good that Peter it not bought

(673) a. Hvorfor har Peter ikke købe den?
 why has Peter not bought it
 'Why hasn't Peter bought it?'

b. *Hvorfor har Peter den ikke købe?
 why has Peter it not bought

(674) Peter den ikke købte.
 Peter it not bought
 'Peter didn't buy it.'

The same factor may govern discharge of objective Case features, and even discharge of the strong features of empty morphemes such as *G*, as we will see. Let us begin with Case features.²¹⁵ Let us examine constructions in which a particle intervenes between V and the direct object. Since the particle blocks adjacency between V and the direct object, it follows that any sequence of the apparent form *verb particle DP* must actually be analyzed as in (675). These must be additional cases in which *G* marks a direct object in a construction other than the double object configuration.

(675) verb particle *G* DP

(676) a. Sue sent out *G* the message.
 b. We threw away *G* the notice.
 c. Ernie put down *G* the duckie.

If the properties of *G* in this construction jibe with those I attributed to *G* in the heavy shift construction, then the DP most immediately

following the particle should be thematically restricted. In particular, it should not be able to bear the role Goal. This seems to be correct, and it immediately explains the contrast between the following (a) and (b) examples in many idiolects (adapted from Emonds 1976:82–83):

- (677) a. The secretary sent out *G* a schedule to the stockholders.
 b. %The secretary sent out the stockholders *G* a schedule.
- (678) a. Some student paid back *G* his loan to the bank.
 b. %Some student paid back the bank *G* his loan.
- (679) a. John read off *G* the figures to Mary.
 b. %John read off Mary *G* the figures.
- (680) a. A clerk will type out *G* a permit for John.
 b. %A clerk will type out John *G* a permit.

Objective Case on the Goal argument in the (b) examples cannot be licensed by the verb without violating the adjacency condition, but is straightforwardly licensed by *G* in the (a) examples.

I assume that the particle in these constructions forms part of V. It is not, for example, detachable under coordination.

- (681) *John turned up the air conditioning and down the heat.
- (682) John [_V[_V turned] up] *G* the air conditioning.

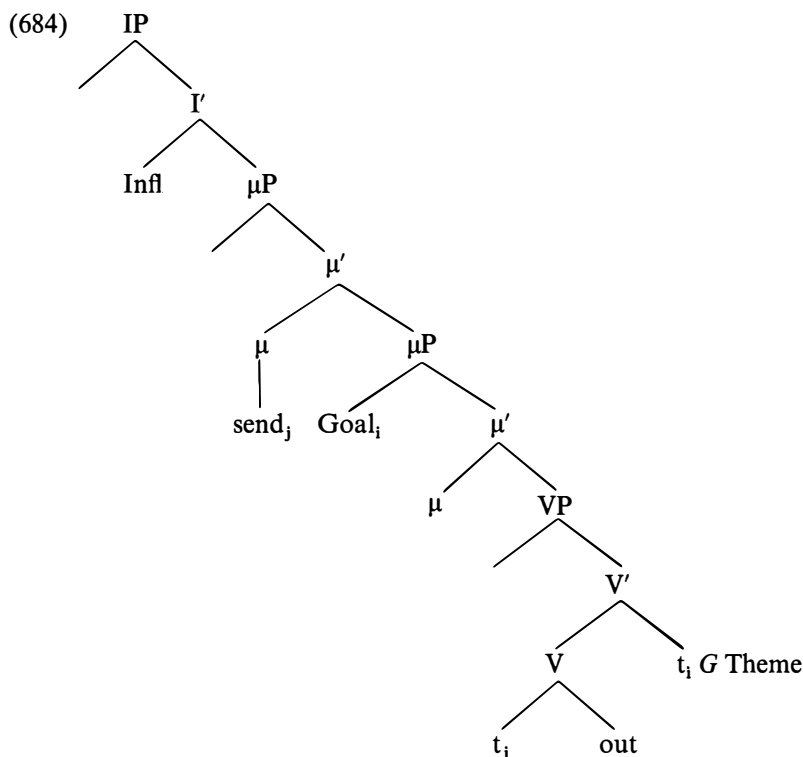
Nonetheless, it may be that adjacency to the inflection on V, rather than adjacency to a right bracket labeled V, is necessary for the discharge of Case features.²¹⁶ Idiolects that accept the (b) examples can be accounted for if these idiolects impose a weaker requirement, so that mere adjacency to a V-bracket is sufficient for Case licensing. In these idiolects, there is no need for *G*, hence no thematic restriction on the construction.

Let us return to the idiolects in which the (b) examples of (677)–(680) are unacceptable. Even in these dialects there is an alternative order of elements that saves the structure: *verb Goal particle Theme*.

- (683) a. The secretary sent the stockholders out *G* a schedule.
 b. Some student paid the bank back *G* his loan.
 c. John read Mary off *G* the figures.
 d. A clerk will type John out *G* a permit.

The best extant analysis of the ordering seen in (683) is Johnson's (1991). I will adopt the essentials of his analysis, even though I will not adopt his approach to objective Case (which eschews adjacency). Johnson

argues that the order in (683a) is derived from the ordering in (677b) by raising both the main verb and the first DP over the particle to higher positions. Following a usage from Pesetsky 1989, Johnson calls the projections that host the verb and the object μP . For the moment, let us adopt this usage, as in (684).



Movement of Goal here is motivated by reasons of Case. In a checking theory like Chomsky's (1993), this means that the Goal bears strong Case features that need to be checked. In the present version of this approach, these features are checked against comparable strong features borne by V. Given the analysis in (675), the strong features of V are optionally present, but the strong features of DP are obligatorily present. (This is the counterpart in a checking theory to the Case Filter of *LGB*.) When V bears strong Case features, as in (684), it can move if movement provides an opportunity for its features to be checked. The unique feature of this configuration is that the lower V can move out of the higher V, in the manner widely assumed for separable prefix constructions in other Germanic languages

(Bach 1962, Bierwisch 1963). Once it leaves its particle behind, adjacency can be satisfied, as (684) illustrates.²¹⁷

Of course, this sort of movement is not limited to double object structures, but quite generally derives the order *verb object particle*.

- (685) a. Sue sent_j the message_i [t_j out] t_i.
 b. We threw_j the notice_i [t_j away] t_i.
 c. Ernie put_j the duckie_i [t_j down] t_i.

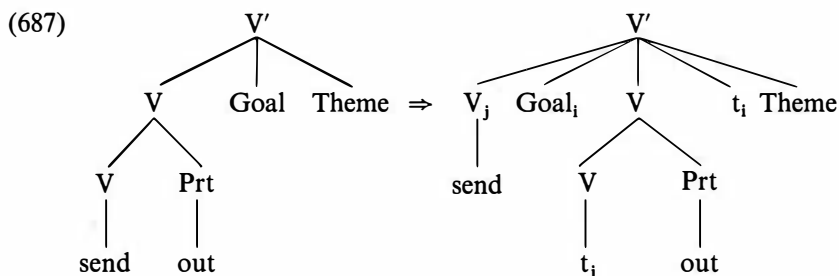
The structure produced by this analysis yields the correct constituency for coordination. For example, the lower μP may be a conjunct, as may VP.

- (686) a. John turned [μP the air conditioning up] and [μP the heat down].
 b. John turned the air conditioning [VP down on Friday] and [VP up on Saturday].

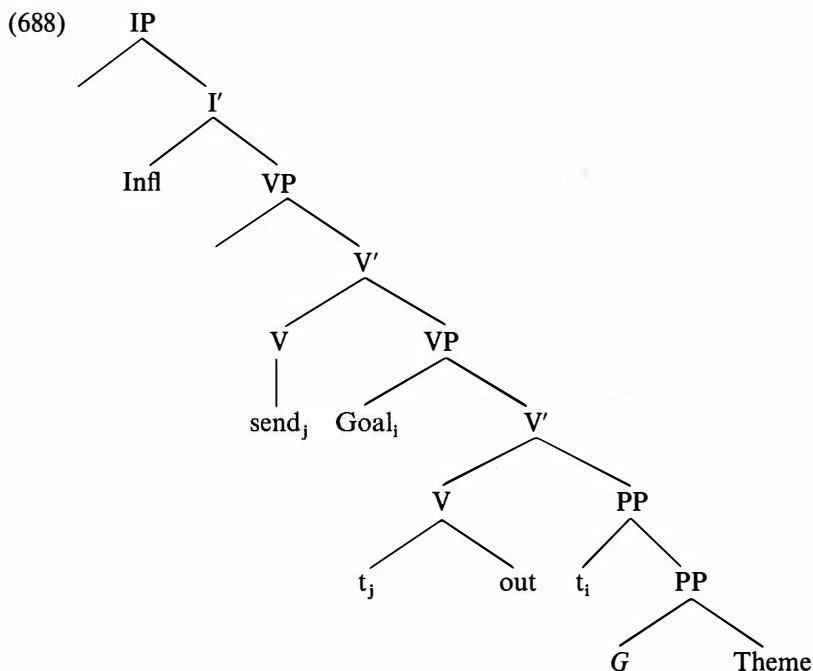
Like Cascade structures (and VP shells), the structures of Johnson's analysis provide small clause-like constituency in (686a), without requiring small clause semantics. (686b) shows that this is the correct result, since there is no sensible analysis of *down on Friday* according to which it forms a semantic unit.

Let us now ask what μP is. Johnson assumes that μP is a functional projection relevant to the assignment of Case. In particular, he analyzes μP (like Chomsky's Agr_OP) as the projection responsible for objective Case in all configurations. This may be correct, but (like Chomsky's Agr_OP) it still has the status of speculation. I would like to present a speculation of a different sort. If μP is a functional projection, then most of (684) is identical in Cascade and Layered Syntax. On the other hand, there is an approach that involves μP only in the Cascade Syntax.

In the repertoire of transformational operations, earlier work occasionally posited local permutations of constituents that share the same mother node. Loosely following Emonds (1976), I will call these operations *local transformations*.²¹⁸ With the advent of binary-branching structures of the sort discussed in chapter 5, operations of this sort no longer seemed to have a place. Suppose we analyze the movements of V and DP in the *verb object particle* structure as Layered Syntax operations of this sort, moving V and the first DP leftward for all the Case reasons just discussed—but with the movement confined to the Layered Syntax V'. (The mother of the V in (687) is the category that immediately dominates the lexical category V, namely, the verb and its particle.) The result would be (687).



When we consider the Cascade structure corresponding to (687), an interesting picture emerges. The new position of V in (687) is simply the normal head position. What follows has a perfectly reasonable mapping to Cascade structure, and organizes itself into something that resembles in part a VP shell configuration. Putting it differently, in this analysis, μP does not exist in Layered Syntax, and corresponds to VP in the Cascade Syntax.



Although there is no equivalent of the second μP in this structure, the constituency relevant to coordination is (as far as can be detected) the same in (688) as in (687).

What follows is more speculative still, yet its conclusion is striking enough that it is worth considering. The analysis of verb-particle constructions sketched above can explain—essentially without stipulation—the single most interesting exception to the T/SM restriction that we have observed in this book.

I begin with two proposals. The first concerns *G* and *CAUS_P*. I have assumed that features of *G*, perhaps related to its status as a zero morpheme, force it to move to V. If this movement is motivated by the need to discharge strong features on *G*, one may wonder if mere adjacency to V will suffice in (688), no movement of *G* to V being required. (This leaves the analysis of *G* in the double object configuration unaffected, since *G* is not adjacent to V in *V DP G DP*. *G* will need to move to V in this environment, just as in previous discussion.) This assumption presumes that *G* does not need to be adjacent to the inflectional morphemes of V in order to discharge its features. In other words, *G*, unlike objective Case (in some idiolects), is not affected by the particle. Suppose now that *CAUS_P* has the same property as *G*. It merely needs to be adjacent to a V containing *CAUS_{aff}* to discharge its features.

My second proposal concerns the trace of the moved DP in the verb-particle construction. I have generally been supposing that something like the Projection Principle forces θ -selectional requirements to be met throughout a derivation. In (688), however, the requirement that *send* have a Goal is met in two separate ways: through *send* and *Goal* in their final positions, and through their traces.²¹⁹ Suppose that this redundancy allows (or even forces) t_i to delete. (The reasoning parallels that which motivated deletion of *CAUS_P* and *Causer* in certain instances in the last chapter.) The same considerations will hold whenever this sort of local movement takes place.

Before putting these two proposals together, let us review the account of the T/SM effect from the last chapter. Let us take up once more the sort of underlying structure I have posited for T/SM violations. In a structure like (528), *CAUS_P* (the lower occurrence of *CAUS*) is free to affix to *at* in an attempt to move to the complex *CAUS_{aff}-√annoy*. The problem with this structure is the absence of any next move that can bring *CAUS_P* to a position where it can discharge its features. In previous discussion, a next move was always needed, because it was assumed that *CAUS_P* could only discharge its features via adjunction to the head of its Cascade. The movement of an *at*-headed phrase is impossible because *at* is not an affix. (Additionally, I exclude excorporation as an intermediate step in move-

ment.) If my speculations here are correct, we now have another possibility to consider. $CAUS_P$ might discharge its features under adjacency to the complex $CAUS_{aff} - \sqrt{annoy}$ after adjoining to *at*. In (528), however, *at* is not itself adjacent to this complex, so that possibility also fails.

We are now in a position to recall an observation from section 3.3. As part of a series of arguments that the T/SM restriction has a syntactic basis, I noted that semantically causative ObjExp idioms that contain particles appear able to violate the T/SM restriction (the following examples repeat (173)–(178)).

- (689) a. *The election results really irritated Sue at the media.
b. The election results really riled Sue up at the media.
- (690) a. *The check calmed Bill about the accident.
b. The check calmed Bill down about the accident.
- (691) a. *Her remarks really depressed Bill about it.
b. Her remarks really got Bill down about it.
- (692) a. *The news cheered Sue about her plight.
b. The news cheered Sue up about her plight.
- (693) a. *The lectures excited Bill about classical music.
b. The lectures turned Bill on to classical music.
- (694) a. *The article angered Bill at Mary.
b. The article really pissed Bill off at Mary.

The speculations in this section immediately suggest an account for these facts. The analysis of an example like (689b) involves leftward movement both of *riled* and of *Sue*.

- (695) The election results really riled_i Sue_j up t_i t_j at the media.

Suppose $CAUS_P$ is adjoined to *at* and the trace of Sue (t_j) is deleted, as discussed above. The complex $CAUS_P + at$ will obviously be adjacent to the trace of *riled*. If *riled* itself contains $CAUS_{aff}$, all features can be discharged, and the structure will be acceptable.

- (696) The election results really [$CAUS_{aff} + riled$]_i Sue_j up t_i
 $CAUS_P + at$ the media.

If the sorts of movement described by Johnson do not take place (i.e., if we find the order *verb particle object*), then the two bearers of $CAUS$ are not adjacent, and the T/SM effect is expected to resurface. Indeed it does,

although the *verb particle object* order is itself somewhat doubtful with these verbs.

- (697) a. ?The election results really riled up Sue (*at the media).
 b. ?The check calmed down Bill (*at the accident).
 c. ?Her remarks really got down Bill (*about it).
 d. ?The news cheered up Sue (*about her plight).
 e. ?The lectures turned on Bill (*to classical music).
 f. ?The article pissed off Bill (*at Mary).

The assumptions of this section lead in many directions.²²⁰ Confirmation of the explanations offered here awaits further exploration of Bobaljik's ideas concerning adjacency and of the mechanisms by which redundant traces can and must disappear during the derivation. That is why I have been cautious and have called the discussion speculative. If correct, however, the analysis would support the view of the T/SM restriction taken in this book, since every rule benefits from the existence of an exception that proves it.²²¹

7.3 Word Order Variation

In this section, I take up a final important loose end. The problem concerns the position of head, θ -selected arguments, and semantically related elements in languages where the order of head, complement, and specifier differs from that of English. In particular, I will consider problems that arise when V appears after VP, N after NP, or P after PP. A Dual System like the one proposed here imposes rather narrow restrictions on the kinds of analyses one can give to languages in which some or all heads do not precede their complements.

In fact, the restricted range of possibilities offered by the Dual System seems to coincide with a very restricted range of possibilities that are detectable empirically. In particular, there do not appear to be any languages whose VP is a mirror image of English—for example, where *give the book to the kids on each other's birthdays* has the word order and head-final, specifier-final structure in (698) (ignoring DP-internal constituents).

- (698) [[[[[each-other's-birthdays on p_v] the-kids pp] to p_v] the-book pp] give v']

As Haider (1993) notes, the general direction of branching (at least as revealed by patterns of c-command) in complement-head languages seems to be rightward, just as in English—not leftward, as in (698). This suggests the possibility that the specifier-initial, head-initial configuration of English Cascade structures is universal.²²² If that is the case, then apparent head-final order must be the consequence of something other than basic, underived constituent ordering. In this respect, the Dual System converges strikingly with conclusions reached by Kayne (1994) (also Zwart (1993)).

One obvious explanation for verb position in an OV language is rightward V-movement.²²³ Consider a language in which the elements that mediate θ -selection precede what they θ -select, but where V follows VP—in other words, consider a prepositional OV language. Such a language, to a rough approximation, is German, on the now-traditional analysis of Bach (1962) and Bierwisch (1963), as modified by Den Besten (1989). Consider the order of elements in the subordinate clause of (699).

- (699) daß sie jedem_i ein Paket an seine_i Privatadresse schicken
 that they everybody_i a package to his_i home address send
 werden
 will
 'that they will send everybody a package to his home address'

(699) displays the ordering *indirect object – direct object – semantically related element – verb*—exactly the ordering found in English. This ordering is perfectly compatible with the Dual System if V' is verb-initial in the syntax.

- (700) [_V schicken jedem_i G ein Paket an seine_i Privatadresse]

The surface position of *schicken* must then be the consequence of some subsequent operation. One possibility would be movement of the verb to the head of a right-headed functional category.²²⁴

- (701) [_V t_j jedem_i G ein Paket an seine_i Privatadresse] schicken_j

The claim that the underlying order of V'-internal elements in V' is that in (699) is controversial. More commonly, it is claimed that θ -selected arguments follow non- θ -selected elements in underlying structure, with the order in (699) derived by rightward scrambling of the arguments. Evidence against this view comes from the fact, again noted by Haider, that the order in NP is the same—and there is no NP-internal scrambling.

- (702) a. Briefe an den Präsidenten aus Langeweile schreiben
 letters to the president out of boredom write
 b. an den Präsidenten aus Langeweile Briefe schreiben
 c. das Schreiben von Briefen an den Präsidenten aus Langeweile
 the writing of letters to the president out of boredom
 d. das Schreiben an den Präsidenten aus Langeweile von Briefen
- (703) a. das Schießpulver im 13. Jahrhundert erfinden
 the gunpowder in the 13th century invent
 b. im 13. Jahrhundert das Schießpulver erfinden
 c. die Erfindung des Schießpulvers im 13. Jahrhundert
 the invention the-GEN gunpowder-GEN in the 13th century
 d. *die Erfindung im 13. Jahrhundert des Schießpulvers

Furthermore, reorderings of dative and accusative, even when they precede non- θ -selected elements, invoke highly specific stress patterns typical of scrambling. This is unexpected if scrambling is involved whenever θ -selected arguments precede other elements in V'. Thus, it looks as if German V'-internal constituent ordering is quite close to that of English, with V-positioning due to some other factor, perhaps head movement.

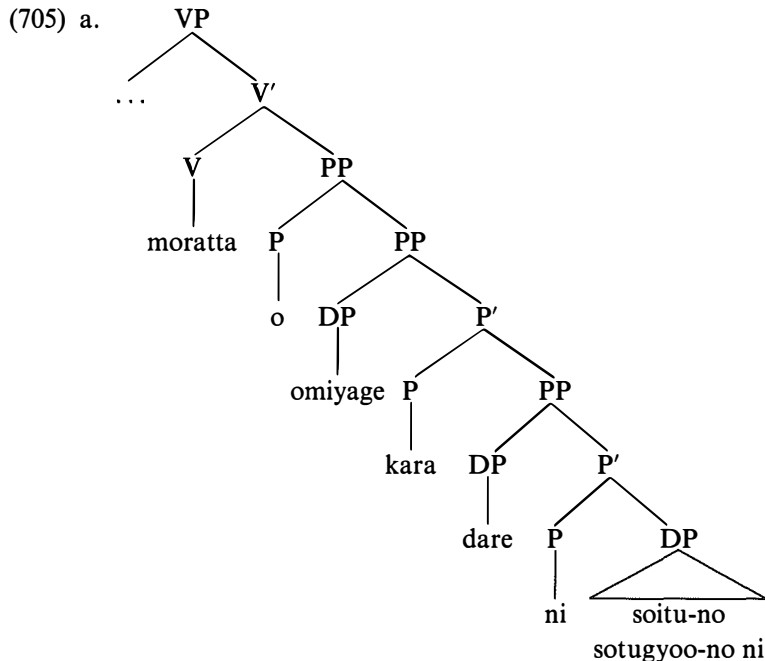
Syntactic movement is not the only available account of head positioning, however. Marantz (1988) has stressed the fact that an element X that precedes another element Y in the syntax may nonetheless be pronounced as a suffix on Y. This is possible so long as X and Y have some linguistically significant relation to each other, such as complementation or adjacency. Perhaps the head-maximal projection relation is a significant relation in this sense. If so, a verb that heads a VP may be pronounced as a suffix on that VP, despite its syntactic position internal to the VP. This proposal removes the head-initial/head-final parameter from the syntax in favor of the phonology. Perhaps this is a novel suggestion, but it is neither worse nor more complex than more traditional alternatives.

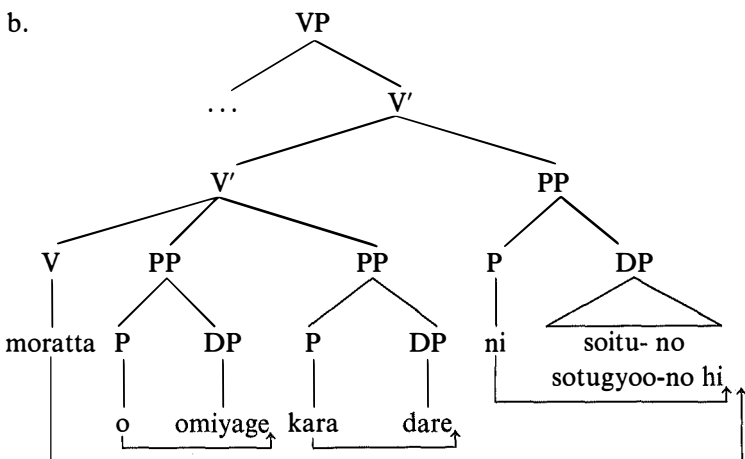
An advantage of this proposal is the immediate possibility of adapting the Dual System to "fully" head-final languages like Japanese, where postpositional structures do not adapt themselves easily to the system. As is apparent, for example, from Hoji's (1985) extended discussion of bound variable anaphora, Japanese postpositional phrases seem to display the binding properties that we have associated with Cascade structures. In Japanese, as in German, the basic structure for V' seems to be right-branching, with PPs disregarded much as in English (as far as one can tell). This can be seen, for example, in the Principle C effect of (704a) and in the availability of *soitu* 'that guy' as a variable bound by *dare* 'who' in

(704b). Both phenomena indicate that c-command “ignores” the postposition *kara* ‘from’.²²⁵

- (704) a. John-ga omiyage-o kanozyo_i-kara [Mary_i-no
John-NOM present-ACC her_i-from [Mary_i-GEN
sotugyoo-no hi]-ni moratta.
graduation day]-at received
‘*John received a present from her_i on Mary’s_i graduation day.’
- b. John-ga omiyage-o dare-kara [soitu-no
John-NOM present-ACC who_i-from [that-guy_i-GEN
sotugyoo-no hi]-ni moratta no?
graduation day]-at received Q
‘Who_i did John receive a present from on his_i graduation day?’

If we claim that Japanese word order is in all relevant respects identical to that of English, but with heads pronounced at the right bracket of their Layered structure maximal projections, the fundamental facts of Japanese word order are straightforwardly accounted for. (705a) is a Cascade structure for (704b); (705b) is the Layered structure, with pronunciation position indicated by the arrows.





It is hard to imagine facts that could pose a problem for this proposal, as long as the phenomena that depend on c-command relations behave identically in the two languages. (Coordination of the P's of (705a) can be excluded because a single P would have to suffix to two PPs in Cascade Syntax.) Thus, although the Dual System narrows the class of possible explanations for head-final order, a plausible and simple account is available—one that invites further research.

7.4 Potential Psycholinguistic Perspectives on the Dual System

This chapter and the preceding one have offered the promised account of the T/SM constraint and have grounded the account in a Dual System for phrase structure that is motivated from all sides. These results, in turn, allow us to maintain the view that the linking problems of psychological predicates should be solved by distinguishing Causer from Target and Subject Matter arguments, thus eliminating one outstanding threat to the simplest view of linking phenomena, discussed in the opening chapters of this book.

As I emphasized in section 7.1.6, the Dual System is not a “minimalist” theory of syntactic organization. If one imagines a deity designing grammar so that it would look maximally simple to the linguist, such a deity would surely not develop the Dual System. Cascade Syntax seems entirely unnecessary. Nonetheless, I have argued that Cascade Syntax, as integrated into the Dual System, is correct. Put differently, the Dual System is what one is led to by the convergence of a variety of grammatical

phenomena, whatever the “plausibility” one might assign to this theory before considering the facts that it explains. Reasons for the Dual System might therefore best be sought outside the traditional realm of grammatical investigation.

Recall the division of labor suggested in (609), repeated here.

- (705) a. *Layered Syntax*: XP-Movement, island conditions on XP-movement, XP-ellipsis, interpretation of modification relations
- b. *Cascade Syntax*: everything else

This presentation was adequate in its context, but might turn out to be misleading—suggesting as it does that Layered Syntax is somehow special, with Cascade Syntax the “real” (default) representation. One might equally well suggest that Layered Syntax plays this role, acting as the only structure that represents semantically contentful relations among items in structure. Besides representing modification relations, it might very well be the Layered Syntax constituency that assigns binders to variables, scope to scope-bearing elements, and—ultimately—arguments to argument takers. By contrast, Cascade Syntax might simply prove to be a provisional representation—a “sketch” of syntactic structure, created by the human sentence-processing system as an intermediate step on the way to Layered Syntax, which is what the semantics looks at.

As it happens, a fairly simple algorithm can provide a Cascade parse of a surface string—at least for a clearly head-initial language. Cascade structures inside categories like VP and NP are uniformly right-branching, except that argument categories form constituents on their own. Put in terms of labeled bracketings, the algorithm looks like this:

- (706) *Preliminary: identify argument categories.*
 - a. Scanning from left to right, place a left bracket in front of each word.
 - b. Enclose each argument category (and each category not semantically related to a predicate) in a pair of brackets.
 - c. Close brackets opened inside paired brackets inserted by (b) at the right edge of the pair.
 - d. Close open brackets at the right edge of the sentence.

The string in (707), for example, will undergo the algorithm as indicated in (708).

- (707) The old folks spoke to these people about us in Bill’s house.

- (708) a. [The [old [folks [spoke [to [these [people [about [us [in [Bill
[’s [house.
 b. [[The [old [folks] [spoke [to [[these [people] [about [[us] [in
[[[Bill] [’s [house].
 c. [[The [old [folks]]] [spoke [to [[these [people]]] [about [[us] [in
[[[Bill] [’s [house]].
 d. [[The [old [folks]]] [spoke [to [[these [people]]] [about [[us] [in
[[[Bill] [’s [house]]]]]]]]]]].

The process of forming a Layered structure from a Cascade structure might then be seen as a rebracketing that closes categories earlier than they are closed by (708d), for example, moving the end of the category opened by *to* to a position following the argument θ -selected by *to* (*these people*). For head-final languages, obviously, the development of Layered structure would have to deviate slightly from this simple plan (the details depending on which proposals from the previous section turn out to be the best supported by further evidence).

If Cascade structures are simply “immature” Layered structures, the sensitivity of binding phenomena and coordination to representation before rebracketing might then be taken as evidence that judgments of binding and coordinability are made by the human parser (for some reason) earlier in the language-understanding process than are other judgments. A comparable claim would need to be pursued concerning those facts about affixation of *CAUS*, *SUG*, and *G* that seemed sensitive to Cascade syntax. These speculations suggest an obvious program of psycholinguistic research, which would seek to discover whether Cascade-sensitive phenomena actually cluster together in real-time processing, for example, producing measurable effects earlier than other phenomena of sentence processing. Needless to say, investigation of these questions lies in the future. Nonetheless, I have included these speculations as an example of a not utterly implausible answer to the obvious question: Why two modes of syntactic structure?

7.5 Conclusions

The main results have already been summarized in section 6.4. This chapter and the preceding one have displayed the long-delayed account of the T/SM constraint. This chapter has grounded the account in a Dual System for phrase structure that, as I have been at pains to show, is motivated

from all sides. These results, in turn, allow us to maintain the most encouraging perspective on linking.

Superficially, there is a certain tension between the main results of this book and the results of this chapter. The body of the discussion was devoted to maintaining the simplest possible view of the lexicon, according to which the meaning of an item determines most (if not all) of its syntactic properties. At the same time, holding onto this simple view of the lexicon-syntax interface led us down a syntactic path that is not as straight as we might have suspected. Have we merely ended up with the familiar “syntactician’s trade-off,” whereby an analytic simplification of one module of grammar leads immediately to an equal and opposite complication of another module of grammar? I suspect that this has not happened here.

First, the Dual System explains phenomena that were not explained adequately by traditional theories—phenomena unrelated to the lexical questions with which we began.

Second, the Dual System is (if correct) a fundamental property of UG (or its implementation in sentence understanding), not the response of an individual speaker to specific experience. In this respect, it is a construct of a very different type than the lexical-item-specific stipulations concerning *like*, *appeal*, and *please* that we have been trying to avoid throughout this book. Nor is the Dual System a simple replacement for these stipulations; obviously, awarding a stipulation the title “property of UG” is a spurious achievement. Rather, the stipulations were avoided by a closer investigation of verbal semantics. It was the solution of a problem raised by this investigation (the T/SM constraint) that led us to the Dual System.

At this point, I will conclude the discussion of the Dual System, as I concluded the discussion of the T/SM effect at the end of the previous chapter. This investigation does not represent itself as a *Summa Syntactica*, an oracle on all syntactic questions. Instead, it is a mining expedition that has discovered one or two veins of syntactic ore. Whether it has struck gold or base metal will, of course, only be known in time to come.

1. I ignore the possible splitting of “Infl” into separate categories for agreement and tense, central to Pollock 1989. The point still stands. I also ignore the question of whether the affix occupies Infl at D-Structure (the assumption made by Emonds and Pollock, as well as in Chomsky 1991, or whether Infl merely legitimates the occurrence of an affix on V, as in the checking theory proposed in Chomsky 1993.

2. Except for certain verbs of motion; see below. There may also be parameterization, as argued by Marantz (1984) and Levin (1983). In this case, the child has a more difficult task: to determine first (somehow) whether the language being acquired is deeply ergative or deeply accusative, with acquisition in its later stages proceeding as discussed in the text. For some discussion tending in the opposite direction, see Baker 1988:427–28 and Anderson 1976.

3. For the sake of simplicity, and substituting *D-Structure* for Relational Grammar’s *initial stratum*, we might revise the UAH to make it more nearly parallel with Baker’s UTAH.

(i) *Universal Alignment Hypothesis (revised)*

There exist principles of UG that predict the structural relations between items from the thematic relations holding between them at the level of D-Structure.

4. Actual practice in Relational Grammar seems to interpret the UAH as something closer to Baker’s UTAH. I am not aware of any discussion of this matter in the literature. Whether or not my literal reading of Perlmutter and Postal’s UAH has actually been assumed by anyone, it is a reasonable fallback if the UTAH proves too strong. See the discussion of (25) below.

5. This truism is independent of issues connected with the “epistemological priority” of certain grammatical entities (e.g., *Agent*, *precedes*) over others (e.g., *subject*) (*LGB*) discussed above. The truism is also independent of whether the traffic in lexical acquisition proceeds uniformly from semantic observations to syntactic conclusions, as suggested, for example, by the theory of s-selection and c-selection developed by Grimshaw (1979, 1981), Pesetsky (1982), and Chomsky (1986b); whether syntactic observations can also help the language learner acquire semantic data, as suggested by Gleitman and Landau’s (1986) work on the acquisition of

vision vocabulary by blind children; or whether the traffic is two-way (Pinker 1989). Whether lexical acquisition goes from semantics to syntax, from syntax to semantics, or different ways at different ages, any close correspondence between semantic and syntactic properties of lexical items will help the language learner.

6. A nice pair suggested by Beth Levin (personal communication).

7. Throughout this study, I assume that arguments are assigned θ -roles, and that these θ -roles are categorized in the familiar manner, perhaps cross-classified as in Rozwadowska 1988 or Dowty 1991. This may be a distorted picture, as emphasized in much work by Jackendoff (e.g., Jackendoff 1987). Jackendoff argues for a “fine-grained theory of argument structure, where thematic roles appear as positions in a detailed conceptual representation” (p. 408). But in fact Jackendoff is calling for a “medium-grained theory” by comparison with the theory assumed here and the alternative that might distinguish verbs of quiet speech from verbs of noisy speech. It does not seem to me that Jackendoff’s discussion has any immediate consequences for the work reported here. Indeed, in Jackendoff’s theory, the linking problem does not disappear; it is merely streamlined. Even though the arrangement of arguments for any given verb may be predictable from the arrangement of arguments in the structure that gives the verb its meaning, it looks as though statements about linking are still needed internal to this structure. Thus, instead of specifying that arguments functioning as Causers must be external, it would be necessary to specify that a particular argument of an abstract predicate is external. This may be a correct move for other reasons, but it leaves the present discussions largely unaffected. I suspect that similar points can be made with respect to any proposal that purports to do without θ -roles.

8. Case Grammar (Fillmore 1968) also assumed the UTAH, in essence, and provided a means of analyzing (20)–(22) that was not unlike the Unaccusative Hypothesis (also see Chomsky 1972:170–80).

9. Beth Levin (personal communication) points out that Bresnan and Kanerva’s theory could be thought of as respecting something like the UTAH as well, if we focus not on grammatical relations, but on the initial classification of arguments.

10. If Rosen is correct, then unaccusative verbs exist, but their distribution is unpredictable. In a language like Italian, one might suppose that an arbitrary classification of verbs could be learned much as nominal gender is learned in a language like French or German. Information about nominal gender in these languages is presumably provided to the child by the definite articles and adjectives that modify these nouns. In Italian, the choice of auxiliary could provide comparable evidence, leading the child to a correct bifurcation of the set of verbs. It is far less clear what evidence would tell the child that the “verbal genders” correspond to unaccusative and unergative representations at D-Structure. Additionally, I do not know if the other languages cited by Rosen have similarly perspicuous markers of verb class.

11. I propose a partial explanation for this observation in section 6.3.2.

12. Rosen (1984:65) notes that the distinction seen in (27a–b) is found in Albanian and Dutch as well.

13. Paul Postal (conference question, 1987) has asked how an agentive *by*-phrase in a passive clause can be considered consistent with the UTAH version of this claim. In Relational Grammar, as in most transformational approaches before the mid-1970s, *by*-phrases are initial subjects, which are placed in the *by*-phrase (*en chômeage*) as part of the passive rule. This sort of analysis (in terms of DP-postposing followed by DP-preposing) in contemporary transformational theories would violate the otherwise well-motivated Projection Principle of *LGB* and is thus unacceptable.

On the other hand, if the linking rules do not require that the Agent of a predicate *P* must be a D-Structure subject of IP, but merely require the Agent to be assigned to the highest argument position associated with *P*, then Postal's question may be answered. Either we may view the VP-external passive morpheme itself as the true bearer of the Agent role, as suggested by Jaeggli (1986b) and by Baker, Johnson, and Roberts (1989) (with the *by*-phrase some sort of adjunct to the passive morpheme) or we might place the *by*-phrase itself in a VP-internal subject position, as hinted at by Fukui and Speas (1986). In this book, I take the former view, for simplicity's sake, but I consider the matter open. In general, data that motivate the concept "1-chômeur" or the interchange of subject and object (Perlmutter and Rhodes 1988) in the Relational Grammar literature might instead support an elaborated theory of VP and functional projection subjects, such that the surface subject is a Spec,IP, whereas a 1-chômeur or an object with subject-like properties occupies some lower specifier position.

14. This in essence is the conclusion of Ruwet (1972), who argues that many of the facts attributed by Postal (1971) to Psych Movement follow instead from differences in the association of thematic roles with grammatical functions. Some of Ruwet's work, but not his conclusions, anticipate the arguments presented in chapter 3 against Belletti and Rizzi's (1988) proposals.

15. The minor differences assumed for these structures would, of course, motivate a weakening. See the discussion of "equivalent" structures above.

16. This conclusion was reached independently by Hoekstra (1991) for Dutch, in work dating from 1988.

17. The examples with question marks in brackets are of somewhat dubious unaccusativity, if cross-linguistic criteria are applied (Beth Levin, personal communication). Also, since adverbs sometimes improve adjectival passives, I have included the parenthesized adverbs in (49) to give each example the maximum benefit of the doubt.

18. Postal (1986:217) claims that adjectivization of pseudopassives is itself sporadic. His starred examples seem to me to improve greatly when adverbs are added: an **(often) bumped-into wall*, *certain *(often) met-with difficulties*, and so on.

19. The situation may be different in Italian and Dutch (Hoekstra 1984:178–79), where adnominal participles may be formed from unaccusatives. These participles, of course, have active meaning.

I offer the following speculation. These constructions may not be adjectival. Instead, they may be sentential constructions with a null or deleted copula, of the sort we might postulate for adnominal progressives in English like *The man walking the dog is Bob*. If we were to analyze the English construction as involving an adjective *walking*, we would have to explain why this adjective assigns objective Case, a phenomenon otherwise unknown in English (except, perhaps, for *near* and *worth*). An analysis involving a sentential adnominal modifier, with internal structure like [PRO *copula verb + ing*], would avoid this difficulty. Conceivably, the null copula itself could be analyzed as an adjective. In any case, English cannot produce constructions of this sort in the active perfect tense, because the copula is not used as an auxiliary here. By contrast, Italian and Dutch do employ the copula with the active perfect—specifically with unaccusatives, as it happens. Thus, constructions like *the arrived man* in these languages are expected to be possible. However, they show, not an adjectivized passive participle, but an active sentential adnominal with a null copula. Catalan is like English in lacking *be* as a perfect auxiliary. On a cursory glance, Catalan appears to have only sporadic instances of the construction seen in (48)–(49) (Eulàlia Bonet, personal communication), thus patterning with English.

In any case, the English facts call into question B&R's suggestion concerning ObjExp passives, since English ObjExp predicates pose linking problems no less serious than their Italian and Dutch counterparts.

20. In principle, one might look for other cliticization possibilities with verbs like *affascinare* 'fascinate', but relevant examples seem not to exist.

21. Similar observations are suggested by unpublished work by Friederike Moltmann (1991) on German passives with *werden* 'become' (cf. *venire*) versus *sein* 'be' (cf. *essere*). Juan Uriagereka (personal communication) notes that the same analysis seems to be correct for the comparable auxiliary alternations in Spanish.

22. For Sandro Zucchi, *spaventato* is for some reason more acceptable in this context than is *terrificato*, a judgment on which most other informants concurred. Some speakers (e.g., Anna Cardinaletti (personal communication)) prefer *terrorizzato* 'terrified' as a lexical item over *terrificato*, which yields the judgment in (66b). Also, (66b–c) are improved if the phrase *questa prospettiva* is replaced by some more concrete DP, as in (66a).

23. Wasow (1977:340) notes, citing McCawley (1970), that *very much* can modify certain active verbs: for example, *John very much respects your family*. Crucially, the bare adverb *much* cannot: **John much respects your family*.

This discussion jibes with a valuable discussion of these modifiers by Borer (1990). Borer notes that *very* is generally possible with a participle only if *very much* is possible with the corresponding verb. Thus, there may be participial adjectives that disallow *very*, and the ungrammaticality of **very jumping* does not tell us that *jumping* must be a verb. Nonetheless, even though some adjectives exclude *very*, no verbs allow *very*. This is a fact that can be established independent of the present discussion.

Additionally, note that although few nonparticipial adjectives may be modified by *much*, *very* is a common modifier of adjectives. I am grateful here for comments by Guido van den Wyngaerd (personal communication).

24. Irrelevant here is the ability of some stative adjectives to be used in the progressive if accorded an agentive interpretation, as in Partee's (1977) *John is being easy to please*, or *John is just being silly*. None of the examples discussed in this section is crucially agentive.

25. The sentences in (75) do not have to meet the requirements that allow non-judgmental stative verbs like *know* to be used in the progressive (*John was knowing more and more answers as the class went on* vs. **John was often knowing one thing or another*).

26. The interpretation described by Baker (1989) is reminiscent of Tenny's (1987) notion of an argument that "measures out the event." Tenny suggests that only direct objects may measure out the event denoted by their predicate, thus predicting certain instances of unaccusativity. If Tenny is correct, then the interpretation of *depress* seen in (74a) might indeed involve an unaccusative structure, with *the situation* originating in direct object position. Tenny's restriction seems too strong, however. First, sentences like *The situation was making Mary depressed* show the same pattern and interpretation as (74), yet they cannot involve unaccusativity, for reasons discussed in connection with example (128) below. Second, other apparently transitive structures show subjects "measuring out the event": if *The children recited the poem* concerns a recitation *separatim* and not in unison, then the subject *the children* measures out the event as surely as *the apples* does in *John ate the apples*.

27. See Wasow 1977:340 for a version of this point.

28. This observation was already made by Postal (1971:43).

29. Grimshaw notes, correctly, that a *by*-phrase is also found with adjectives and adjectival passives. This fact is irrelevant here, however. All that is important is that the idiosyncratic prepositions seen in the (b) sentences of (80)–(83) force these examples to be adjectival passives.

30. Den Besten cites earlier work by Kraak and Kloster (1968) and by Hoekstra and Moortgat (1979), which I have not consulted.

31. This is Den Besten's claim, though one wonders if the facts are as nebulous in Dutch as they are in English, where many adjectival passives allow *by*-phrases.

32. I have altered Den Besten's gloss in the interests of clarity.

33. I am grateful to Luigi Burzio for invaluable help, both theoretical and factual, with this section.

34. I think that even the use of 'see' in (106d) falls under this generalization: compare (106d) to a usage of 'see' in which it means 'realize' (an inherently non-controlled state).

(i) ??pro hanno visto che due più duo faceva quattro.

saw that $2 + 2 = 4$

'Somebody saw that $2 + 2 = 4$.'

35. Much the same appears to hold of any use of a corporate name in place of an individual representing the corporate body, though not all examples are equally successful (e.g. (id)). The material in brackets in (ia–g) indicates the sort of interpretation that can be given to the subject of the sentence.

- (i) a. The mafia robbed Mrs. Johnson. [one mafioso, acting for the mob]
- b. The CIA came for Charley. [one agent]
- c. Purity Supreme accepted our check. [one cashier]
- d. ??The KGB punched me in the nose. [one agent]
 [*better*: The KGB came for Charley and punched him in the nose.]
- e. *The CIA seemed to me to be crazy. [one agent]
- f. *The phone company has been arrested by the police. [if one repairman
 was arrested]
- g. *The Mafia got a phone call yesterday. [if one mafioso received a call]

36. The gloss of *pro* as 'somebody' is misleading, if I am correct; I simply follow B&R here.

37. Note the auxiliary *essere* in these examples. Even on B&R's account, this auxiliary choice tells us that *venire* here is unaccusative. See also Burzio's (1986: 332–36) discussion of the *venire a* construction.

38. Even if the corporate third person plural requires the role of Agent, this does not constitute an illicit meddling of specific θ -roles in the syntax (cf. B&R), since Agent can be regarded as a feature relevant to selection, much like [+concrete], which similarly plays no role in syntactic rules.

39. Luigi Burzio (personal communication) observes that this example degrades noticeably if *sono* is replaced by the appropriate form of *venire* (*vengono*). If *venire* is limited to verbal passives—which I argued against above—this degradation supports B&R's proposal. However, since *venire* is also incompatible with generic tense, this alone may explain the degradation, if generic tense is a precondition for generic *pro*.

40. The gloss is B&R's. Generalization (111) correctly predicts that the corporate reading will not be available for *pro* here, a fact overlooked by B&R.

41. I owe the germ of this observation to Martin 1986. Martin's examples involve control rather than binding of reflexives and reciprocals. A strict c-command condition on control of PRO is somewhat problematic, so I have used reflexives and reciprocals throughout.

42. In Pesetsky 1987a, I wrote that in the Generative Semantics literature, the binding facts discussed above "do not seem to have been noticed, but they might easily have been captured by an extrinsic ordering of anaphora rules with Psych Movement (Lakoff's (1970) 'Flip')." At the time, I was unaware that the basic binding facts, along with an analysis of essentially this character, were first presented by Akatsuka (1969) (as reported in Akatsuka 1976:64).

43. The initial *s* of the causative morpheme *sase* is dropped after most consonants.

44. Akatsuka also gives examples of non-Experiencer verbs that display the same properties; see (123)–(126).

45. The examples and judgments are due to Shigeru Miyagawa (personal communication) and Kiyoshi Kurata (personal communication).

46. Bennis (1986:141) suggests that the passives seen in (152) are not “really passives.” He notes that the passive auxiliary can be replaced by *raken* ‘get, become’ and that a *door*-phrase is also possible in the active.

- (i) a. Ik wordt/raak daardoor geërgerd.
 I am/get that by irritated
 b. Hij ergert mij door zijn gedrag.
 he irritates me by his behavior

At best, however, (ia) might show that some alternative passive is available. Similarly, (ib) demonstrates that another source for the *door*-phrase should be considered, but Bennis provides no arguments in favor of the claim that true passives from *hebben* ObjExp verbs are impossible. Clearly, further investigation is necessary, particularly in light of the seeming synonymy of (151)–(152). Everaert (1986: 112–15, esp. fn. 21) presents additional arguments (involving inherent reflexives) for the non-unaccusativity of the *hebben* ObjExp verbs.

47. Notice, however, that B&R’s idea that inherent accusative Case assignment licenses lexical DPs is in conflict with Icelandic, where the data provided by Andrews (1982b) show that inherently Case-marked DPs must also be in a structural Case position. In this light, the objects in (153)–(154) might actually be objects of a phonologically null preposition, as suggested by René Mulder (personal communication).

48. As a reviewer notes, “Lack of a verbal passive is consistent with an unaccusative analysis, but not definitive evidence for one, since verbs might be unpassivizable for other reasons.” I grant this possibility, but, absent such other reasons, propose unaccusativity as the determining factor.

49. An alternative account of the failure of passivization in the starred Italian examples of (148)–(149) might blame it on the assignment of dative Case by V. This account would not extend to the English examples in (153)–(154) and would not account for the contrast between the somewhat “marginal” impersonal passives in (i) (from Burzio 1986:180) and the utterly impossible impersonal passives in (ii) (thanks to Alessandra Giorgi for these data).

- (i) a. (?)?Gli fu sparato addosso.
 to him was fired upon
 b. ?Gli fu detto del pericolo.
 to him was told of the danger
 c. ??Gli fu parlato a lungo.
 to him was talked at length
 (ii) a. *Gli fu piaciuto.
 to him was pleased
 b. *Gli fu mancato.
 to him was lacked
 c. *Gli fu riuscito.
 to him was succeeded

Alessandra Giorgi, in particular, finds the examples in (i) less marginal than Burzio does, with (ib) perfect, but reports a solid contrast with the examples in (ii).

50. I return in chapter 6 to the role of prepositions in c-command calculations. It will not be necessary to disregard *to* at that point.

51. In this section, I repeat the relevant examples from (30)–(36) for convenience, but maintain the original numbering.

52. Here I disagree with Zubizarreta (1988), who claims that (161b) is in fact a contradiction and who argues against my (1987b) presentation of the distinction between Causer and Subject Matter.

53. Note also:

- (i) a. John worries/is concerned about Bill driving at night.
b. Bill driving at night worries/concerns John.
- (ii) a. Mary feared/was afraid of another possible tornado.
b. ??Another possible tornado frightened Mary.

The subject Causer and the object Subject Matter roles differ in factivity. In (ib), but not (ia), for example, it is presupposed that Bill drives at night. The oddness of (iib) presumably derives from the presuppositions associated with Causer arguments.

54. I leave open whether Agents are special cases of Causers, or else whether Agents might occupy a distinct place on a hierarchy like (166).

55. Unfortunately, speakers differ on corresponding cases in Dutch. Dutch has at least four verbs that lie roughly in the semantic field of *please/appeal to*: *bevallen*, *verheugen*, *plezieren*, and *behagen*. Of these, the first takes the auxiliary *zijn* ‘be’ characteristic of unaccusatives, and the other three take *hebben* ‘have’. Some speakers, but not all, seem to report a reading for *bevallen* that allows X to *bevallen* Y even though Y does not like X. An unresolved issue among my informants is whether this situation can or must be attributed to “perversity” in Y, so that Y is presupposed to like things that he evaluates negatively. Contrariwise, some speakers report that *plezieren* does not allow the interpretation of *please* discussed in the text, though others report the opposite. For *verheugen* and *behagen* I have only one speaker’s data (Ellen-Petra Kester, personal communication). She reports that *verheugen* may translate *please* in the environment discussed in the text, but *behagen* may not. I leave this tangle of data to experts and native speakers. I am grateful to René Mulder, Guido van den Wyngaerd, and Ellen-Petra Kester for discussion.

56. The search did in fact turn up a small number of potential counterexamples, to which I return in section 6.3.1.

57. If the content of the impossible third argument is given as an afterthought (or right-dislocation), it often becomes more acceptable—even without periphrasis of the main verb. The result is certainly not incoherence. Target examples are, for some reason, better than Subject Matter examples.

- (i) a. ?The article in the *Times* angered Bill—but not at the government.
- b. ?My remarks irritated Sue—with Bill, not with me.
- c. ?The Chinese dinner satisfied Bill—not with his trip as a whole, but at least with the gastronomic portion of it.
- d. ?The distant rumbling frightened Mary—but not of another tornado.
- e. ??The television set worried John—but not about the veracity of Bill's alibi.
- f. ??The problem of lexical entries bores John—not with his life as a linguist, of course, but just with the problem of lexical entries.
- g. ??Something Bill had said bothered Mary—not about her future, precisely, but about something far more serious.

58. Alec Marantz (personal communication) notes that something like this constraint was proposed by Williams (1981:100) as part of his analysis of Japanese causatives. A requirement of this sort was also suggested by Joan Bresnan in 1978 class lectures at MIT (as Tim Stowell has reminded me), but does not seem to be proposed in any of the papers anthologized in Bresnan 1982. Other works often presuppose (179), yet the condition has rarely been formulated explicitly; see Lasnik's (1988:4) straw-man proposal that **The army destroyed the city by a missile* might be ruled out because it contains two Agent arguments, if "the intentional/unintentional difference between [*the army* and *the missile* is] not thematic but rather a question of vagueness."

59. This resembles the approach taken by Grimshaw (1991). She accepts the arguments in Pesetsky 1987a,b concerning the distinctions between Causer and other non-Experiencer arguments of ObjExp and SubjExp predicates. She argues, however, that the Causal distinction is relevant only in the "Aspectual dimension" that determines which argument is the subject. In thematic structure, which determines the identification of one argument as external ($\neq$ subject), these arguments are regarded as the same, as in B&R's view. The subject of ObjExp verbs is thus a deep subject, as I have argued here, but is not marked as external. To the extent that this theory has the consequences for passive sketched in section 2.3.4, it seems to be incorrect. To the extent that I can successfully argue that the problem raised in this section has a different solution, Grimshaw's distinctions are redundant (in this domain). Nonetheless, with certain alterations in her account of passive, Grimshaw's theory is not inconsistent with the one presented below.

60. One could of course develop a more sophisticated theory of Thematic Diversity to handle these cases. Perhaps θ -roles are decomposable into features (as suggested by Rozwadowska (1988)). Letting α and β stand for the features relevant to Causer, Target, and Subject Matter, we might assign to Causer [$+\alpha$, $+\beta$], to Target [$+\alpha$, $-\beta$], to Subject Matter [$-\alpha$, $+\beta$]. Causer now shares a feature with Target and with Subject Matter, but Target and Subject Matter do not share a feature. Thematic Diversity could be strengthened so as to exclude cooccurring arguments that share a feature (rather than all features). Such a theory would only gain plausibility in the context of a developed theory of θ -role structure that gave content to " α " and " β ."

61. The example is constructed so as to eliminate a reading in which the material following *likes* is a single DP.

62. Compare **What about Sue Mary hates ...* and **What Mary discarded about Sue was a book* for arguments that *about Sue* in (180d) is not a modifier of *what* or its trace.

63. Higgins's observations were brought to my attention by Richard Kayne (personal communication). The examples in (181) are Higgins's; those in (182) are my own.

64. Many thanks to Noriko Kawasaki for assisting me with data collection concerning this point, and to Michiko Terada, Hajime Hoji, and Yoshi Kitagawa for extensive discussion.

65. One possibility, broached by Jackendoff (1975) and Wasow (1977:359), is to derive these nominals from adjectival passives in *-en*. This possibility has been effectively argued against by Amritavalli (1980).

Zubizarreta (1987) argues that these nominals encode "Lexico-Semantic Structure" more directly than the corresponding verbs, which acquire their "ObjExp" character via the intermediacy of a level of "Lexico-Syntactic Structure." This hypothesis does not seem to capture the absence of causative force in the nominalizations, which I take to be a central fact.

66. The Causer also cannot be expressed with a *by*-phrase: **Amy's fright by the scarecrow*, **Sam's annoyance by Dave* (Dowty 1991:558). Dowty proposes that these cases are excluded by a general constraint (noted as a possibility by Rappaport (1983:131)) barring *by*-phrases with stative nominals. As part of a sustained argument against the sort of θ -theory assumed here, Dowty suggests that this statement of the generalization is preferable to proposals that bar some particular configuration of θ -roles in nominalizations. Dowty supports the claim that the relevant nominals are stative by noting an incompatibility with predicates of events.

- (i) the boredom of the class (# that happened ten minutes after the lecture started)
- (ii) the unintentional fright of the children (# that occurred when they saw the scarecrow)

On the other hand, I do not see any reason to believe that the distribution of *by*-phrases in English is sensitive to stativity (or to anything other than the factors that otherwise distinguish subjects; see Lasnik 1988). Consider, for example, the *by*-phrases that cooccur with the nominals in (iii)–(v), which also pass Dowty's test for stativity.

- (iii) the possession of narcotics by the children (# that happened last week)
- (iv) the ownership of Saabs by factory workers (# that occurred when they got the pay raise)
- (v) the knowledge of algebra by high school teachers (# that happened last year)

Furthermore, even if *by*-phrases could be excluded as Dowty suggests, it is hard to see how that exclusion would extend to subject Causers like those in (208). This criticism does not prejudice the possibility that the results of this book might be recast in terms compatible with Dowty's general view of thematic relations, as developed in the cited paper.

67. A question concerning my interpretation of (186)–(192) is posed by an observation of Eulàlia Bonet (personal communication). She notes that some of these examples improve if the possessive is replaced by a postnominal *of*-phrase, as (i) shows. The examples in (ii) show this improvement is not universal.

- (i) a. ?the anger of your remarks
- b. the optimism of her expression
- c. the irritability of his comments
- d. ?the sadness of his words
- (ii) a. *the pride of his bearing
- b. *the fearfulness of his behavior
- c. *the annoyance of her manner

If (i) shows genuine nominalizations of adjectives like those noted by Higgins, then we cannot claim that there is a general ban on such nominalizations. There is another interpretation of these examples, however. In (i)–(ii) it is plausible that the *of*-phrase is not the logical subject of the adjective inside the nominal, but a location phrase modifying the nominal itself, much like the PPs in underived nominals like *the middle of the room* or *the men of California*. These phrases too may not occur in possessive position.

- (iii) a. ??He stood in the room's middle.
- b. ??California's men voted against the proposal.

Furthermore, the acceptability of the *of*-phrases roughly correlates with the acceptability of *in*-phrases.

- (iv) a. the anger in your remarks
- b. the optimism in her expression
- c. the irritability in his comments
- d. the sadness in his words
- e. ??the pride in his bearing
- f. ??the fearfulness in his behavior
- g. ??the annoyance in her manner

I will thus assume that the cases pointed out by Bonet are not nominalizations of adjectives like those noted by Higgins, but argumentless nominalizations of the normal adjectives, with a locative modifier.

68. If *CAUS* attaches *only* to verbs, the (b) examples of (218)–(221) would be simple violations of subcategorization, hence irrelevant to Myers's Generalization.

69. I am grateful to Hagit Borer (personal communication) for discussion of these examples.

70. I am making traditional assumptions about the difference between inflectional and derivational morphology: Tense, agreement, person, number, and gender affixes are inflectional. Other morphemes are derivational. Inflectional morphemes are external to derivational morphemes.

We cannot then dispose of *-able* by categorizing this morphology as inflectional, since *-able* may be followed by the derivational morpheme *-ity*—unheard of in the English inflectional system.

Actually, passive *-ed* seems to pose similar problems, displaying an ambiguous status. I will suggest that it can be followed by a null adjectivizer (in the adjectival passive construction), which would seem to categorize it as derivational. On the other hand, it may be applied to zero-derived forms (e.g., *documented*), which would seem to characterize it as inflectional. These forms do not generally tolerate further affixation (e.g., **documentedness*), but they do tolerate adjectivization (*undocumented aliens*). This raises serious questions, which I will not answer here.

71. A terminological note: I will reserve the term *nominalization* for action and state nominals like *destruction* and will always refer to forms in *-er* by explicitly mentioning the morpheme *-er*.

72. The weeding process was not terribly careful, because of the poor tools available to me; so the total number is probably somewhat smaller than 1,545. Obviously, if there is reason to question the representativeness of the data examined here, a more painstaking study will be necessary.

73. Suffixes not on Fabb's list were also not trapped. In this connection, Donca Steriade (personal communication) has raised questions about nouns in *-ation* like *documentation* and *fragmentation*. On the plausible assumption that the innermost roots of these words are the nouns *document* and *fragment*, we must ask about their derivation. Forms in *-ation* are often built on verbs, for example, verbs in *-ize* (*civilization*, *nominalization*). If these forms are uniquely deverbal, then words like *documentation* would have to have the following analysis, violating Myers's Generalization:

(i) [[document_N] ∅_V] + ation

This conclusion is not necessary, however. Denominal instances of *-ation* exist, though perhaps they are less common than deverbal instances.

(ii) a. salivation, pollination, carbonation, orchestration, origination,
calumnation, formulation, gravitation, ...

b. *from*: saliva, pollen, carbon, orchestra, origin, calumny, formula, gravity,
...

Many of these forms, of course, are best analyzed as denominal verbs in *-ate* to which *-ion* has been added.

(iii) salivate, pollinate, carbonate, orchestrate, originate, calumniate, formulate, gravitate, ...

This analysis is less straightforward with *documentation*, since **documentate* does not exist. Here there are two possibilities. One is to posit an unanalyzed suffix *-ation* in place of *-ate* + *ion* or alongside it. We would have to allow this suffix, like

-ate, to attach to nouns as well as verbs. *Documentation* would be an instance of this unanalyzed suffix. The other possibility is to give -ation a unique analysis as -ate + ion and to allow morphology to produce *documentate as a possible but nonoccurring word (perhaps blocked by the verb *document*).

The notion “possible but nonoccurring” seems necessary in any case, as argued by Allen (1978). It would also be applicable to forms in -ation derived from verbal roots like those in (iv) (Marchand 1969:260).

- (iv) a. accusation, realization, damnation, information, restoration, ...
- b. *accusate, *realizate, *damnate, *informate, *restorate, ...

If we were to eschew possible but nonoccurring words, and if -ation were always unanalyzable, we could not explain the systematic relatedness of pairs like *celebrate/celebration*, *integrate/integration*, *separate/separation*. The roots *celeber, *separ, and *integer (in the relevant sense) do not exist as occurring words in English, yet these roots produce obviously related words with attachment of -ate and -ation. Furthermore, we would not expect the -ate portion of -ation to trigger the same allomorphy in roots in -ify as does the -ate portion of adjectives in -atory and -ative (forms from *Webster's Seventh Collegiate Dictionary*).

- (v) a. certify, classify, edify, justify, ossify, pacify, purify
- b. certification, classification, edification, justification, ossification, pacification, purification
- c. certificatory, classificatory, edificatory, justificatory, ossificatory, pacificatory, purificatory
- d. justificative, significative

The allomorphy supports the independence of the -ate portion of these suffixes, despite the fact that verbs like *certificate, *classify do not occur independently.

If examples of V + ate + ion, exist, even when X + ate does not occur independently, then we may expect that N + ate + ion can also exist without the possibility of N + ate. Such would be the case with *documentation* under the analysis in (vi).

- (vi) [[[document_N] ate_V] ion_N]

74. I have not investigated whether zero agentive noun formation also generally behaves exceptionally with respect to Myers's Generalization, but *bore* ‘someone who bores’ (see Kiparsky 1982:7) is one clear example of this. Blocking here rules out *borer.

75. Though not the lexicalized examples like *irritable*, *puzzler*. Note also that *worrier* (cf. (203)) is only derived from the SubjExp verb *worry* (‘someone who worries’), whereas *puzzler* is only derived from the causative ObjExp verb *puzzle* (cf. (204)).

76. This observation is due in part to Chomsky (1972).

77. A reviewer finds (235d) better than the other (d) examples. The present theory does not provide a place for nondiscrete judgments of this sort, since it rules the (d) examples out with a single constraint.

78. This observation is independent of Smith's (1972) claim that Anglo-Saxon nominalizing morphemes can never be affixed to causative predicates. (229)–(234) are then irrelevant, since there is another reason to rule them out, but examples like (235) and (236) still make the point. To explain Smith's observation, one might pursue the tantalizing (and anti-etymological) possibility that *CAUS* for some reason is considered part of the Latinate vocabulary of English.

79. This hypothesis is especially reminiscent of Lakoff's (1970) proposals, as discussed above. It also has points of similarity with Walinska de Hackbeil's (1986) analysis of a variety of English causative constructions (which precedes my own). She also assumes a zero morpheme with causative force, though the analyses she proposes do not coincide with the analyses in this book.

80. Other factors may block certain nominalizations. For example, *injure* is semantically causative and lacks an inchoative homonym (**Bill injured yesterday*), but *injury* appears to lack any nonresult reading whatsoever (**the injury (of John) lasted an hour*).

81. Beth Levin (personal communication) notes that this proposal entails a certain sort of unlearning. The language learner must revise the lexical entry for a causative verb once evidence of an inchoative counterpart is available. This sort of unlearning, however, should be no more problematic than learning that any stream of speech is actually made up of constituents, for example, that *walks* contains two morphemes. This reply does not eliminate the interest of the problem, but it indicates that the present proposal raises no theoretical difficulties we are not already saddled with. The fact that *CAUS* is phonologically zero does mean that a distributional analysis of the segmental stream by the child cannot help in detecting *CAUS*, as it might for *-s*. Perhaps this is why the presence of an alternation between causative and noncausative forms seems to be crucial for detecting certain instances of *CAUS*.

82. Agentive uses of some nominalizations in the *grow* class are better—for example, *the policeman's dispersal of the crowd*—but compare **the rain's dispersal of the crowd* (pointed out by Edit Doron (personal communication)). I have no explanation for this fact.

83. The fact that *-er* does not allow further derivation (as noted by Kiparsky (1982)) might be taken to argue that *-er* is inflectional after all. The meaning expressed by *-er* can be expressed in some other languages (e.g., Russian) by the present participle. I do not know whether such participles should be called inflectional or derivational, since I have only a rule-of-thumb guide to this distinction.

Another possibility, in keeping with Myers's Generalization, might be to posit that *-er* has a silent partner in the form of a succeeding zero affix: for example, *writ-er-∅*. This hypothesis alone would block further derivational affixation. On this view, *-er* might simply serve to bind the external argument of the base verb (perhaps adjectivizing it, as in the Russian participle), with the zero affix functioning as the nominalizer.

84. Noun-forming *-al* is also a group A suffix. This may account for the absence of a causative sense to *arousal*, the only psychological predicate I have found that nominalizes with *-al*.

85. According to Fabb, *-ion* combines only with *-ize*, *-ify*, and *-ate*, among the common suffixes considered in his article.

86. As for the possibility that a simplex causative noun might form the root of words like those in (249), this will be ruled out by the same factors that rule out simplex causative ObjExp verbs, discussed below.

87. There is more structure to these data than meets the eye. Thus, the existence of *Bostonian* does not block the adjectival use of *Boston* in *Boston police*, presumably because *Bostonian culture* is strange. Similarly, although *Rhodesian economy* presumably blocks **Rhodesia economy*, *Zimbabwean economy* does not block *Zimbabwe economy*, probably because of the low familiarity of *Zimbabwean* as an adjective. Of course, there may be other factors (e.g., phonological) at work here, which I have not looked into.

88. I have received mixed reactions to the stars in (256). Beth Levin (personal communication) notes the actual occurrence in text of *Texasness*.

89. Perhaps the special behavior of *-ness* is linked in some fashion to the rarity of adjective-forming zero derivation. Indeed, as noted, Myers argues that there is no adjective-forming zero derivation in English (since, he claims, there is no adjectival inflection). The only examples of which I am aware are those presented above in the text.

90. We are also close to making a substantive claim about the organization of the lexicon (in favor of a phonology-based organization), but I will not explore this possibility here.

91. Though observations by Beth Levin (personal communication) and her students lead me to doubt this somewhat: *containerize*, *Whitemanize*, *upmarketize*, and *nonfirstorderizability* are all attested and do not radically violate my intuitions concerning acceptable word formation.

92. The only empirical reason for distinguishing *-ness* from *-er* and *-able* and invoking the wildcard opacity index is the inability of *-ness* to attach to adjectival forms headed by *SUG*. Thus, if the proposals made here about *SUG* are wrong, it will be possible to eliminate this (slight) complication from the formalism.

93. Beth Levin (personal communication) notes that the majority of *-ize* verbs in the *Oxford Advanced Learner's Dictionary* occur only in a transitive usage, which I would need to analyze as *causativa tantum* (verbs that only occur causativized). This might be an argument against the analysis suggested here, though it is not clear what weight one should attach to such quantitative considerations, since the range of relevant factors is unknown. Levin also notes that verb-forming *-en* displays a more even distribution of transitive and intransitive possibilities.

94. Similarly: *confine*, *conflict*, *conscript*, *consort*, *contract*, *convert*, *convict*, *digest*, *escort*, *export*, *extract*, *ferment*, *import*, *impress*, *increase*, *insult*, *object*, *permit*, *pervert*, *present*, *produce*, *progress*, *project*, *protest*, *rebel*, *record*, *reject*, *regress*, *subject*, *suffix*, *survey*, *torment*, *transfer*, *transport*.

95. Myers makes the proposal (which I will not adopt) that the members of these stress-shifting pairs should not be morphologically related. In support of this

claim, Myers notes first that the semantic relationship between verb and noun in several cases is semantically tenuous (e.g., *objéct* and *óbject*). This argument is weak, since much morphological derivation has this property. By parity of reasoning, one might suggest that the existence of sentence-level idioms argues against phrase structure. Second, Myers notes that there is no blocking effect among pairs like the verb *protést* and the denominal verb *prótest*. However, the very lack of semantic predictability in many of these cases can explain the absence of this blocking effect. Finally, Myers argues that stress-shifting derivation is not productive. In many domains this claim is false. For example, verb-particle combinations quite productively form stress-shifting nouns (see Marchand 1969:378, 384ff.).

- (i) ádd-in, ádd-on, báil-out, bláckout, blást-off, blénd-in, blówout, blówup, breakaway, breákdwn, breák-in, breákout, breákhthrough, breákup, brúsh-down, brúsh-off, brúshup, buildup, búrnout, búst-out, búst-up, búyout, cállback, cách-up, ... méltdown, ... scréw-up, ... touch-up, ... wípeout

Thus, these stress-shifting verb-noun pairs do look like genuine instances of morphological derivation.

96. *John was admiring the painting* is fine, suggesting that *admire* falls under both classes.

97. The argument that English *CAUS* does not attach to adjectives rests on the claim that *CAUS* attaches to the same roots that nominalizers like *-ment* and *-ance* attach to in words like *amusement* and *annoyance*. These nominalizers are deverbal.

98. The discussion in this section was prompted by a conversation with Beth Levin. I am also grateful to Lea Nash for important discussion and Russian data.

99. Except *dégouter* 'disgust', for some reason, as discussed in note 184 below.

100. The *-er* infinitival ending is inflectional, but I ignore this detail for ease of presentation.

101. The two speakers that I consulted disagreed over the acceptability of the animate dative *Ire*. One speaker reported that this example (but not examples with an inanimate dative) improves if a *because*-clause follows; the other reported no improvement. As Lea Nash (personal communication) points out, other verbs, like *zlit* '-sja' 'be unpleasantly mad at', require an animate object. Perhaps animate and inanimate Targets should be thematically distinguished, as Nash suggests.

102. A proposal almost identical to this is advanced by Rothstein (1967), on the basis of a careful study of corresponding facts in Polish (though without the explanation offered below in terms of external and internal arguments). He posits that the reflexive verbs are basic, and the causative verbs derived via an abstract pro-verb *CAUSATIVE*, following Lakoff (1970). The reflexive is deleted in the presence of *CAUSATIVE*.

Rothstein also comes close to anticipating the distinction between T/SM and Causer. One of his arguments acquires special "bite" in the present analysis. He considers a variety of reflexive SubjExp verbs in Polish and compares their Case requirements to those of the corresponding nonreflexive ObjExp verbs.

- (i) a. Jan dziwił się wynikom.
Jan surprised refl results-DAT
'Jan got surprised at the results.'
- b. Wyniki dziwiły Jana.
results surprised Jan-ACC
'The results surprised Jan.'
- (ii) a. Maria cieszyła się z przyjazdu Jana.
Maria gladden refl with arrival of Jan
'Maria became happy at the arrival of Jan.'
- b. Przyjazd Jana cieszyła Marię.
arrival of Jan gladden Maria-ACC
'Jan's arrival made Maria happy.'
- (iii) a. Jan denerwuje się nadchodzącym egzaminem.
Jan be upset refl approaching-INSTR exam-INSTR
'Jan got upset about the upcoming exam.'
- b. Nadchodzący egzamin denerwuje Jana.
approaching-NOM exam upsets Jan-ACC
'The upcoming exam is upsetting Jan.'

Under the analysis proposed here, the variability in the expression of the T/SM argument in (i)–(iii) is expected, given the range of governing lexical items. The lack of variability in the Causer arguments is also expected, given that this argument is external, and a dependent of *CAUS*.

103. I have not investigated whether Fabb's Generalization holds up elsewhere in French or Russian. This, of course, is an important logical gap in the present discussion, but one that must remain unfilled for now. The missing investigations have several prerequisites—most notably the need to resolve the role of the theme vowel in these languages. The theme vowel is a suffix intervening between the stem and the tense/agreement markers of most verbs (*étonn-e-r*, *udivlja-e-t*). If a denominal verb (e.g., *documenter*) is to be regarded as an instance of zero derivation, we would have to suppose that the theme vowel found in the verb is inflectional. Otherwise, the theme vowel could itself be viewed as a potentially denominal verbalizer.

104. English also displays a *reflexivum tantum* here, namely, *pride oneself*. As in Russian, there is no causative.

- (i) John prided himself on his good command of Russian.
- (ii) *His good command of Russian prided John.

105. Examples like (190a) and (192a) displayed Japanese verbs meaning 'be sad' and 'be amused'. The analysis of French and English presented here suggests that these Japanese verbs involve a null reflexive clitic. I do not have any direct evidence in favor of such a hypothesis, but the fact that Japanese allows what looks like object pro-drop might implicate a set of null clitic pronouns for the language. A reflexive clitic would simply be another member of this clitic series.

106. Raising contrasts with control, as the following example (also Rizzi's) shows:

- (i) Gianni_i si_i impone [di PRO_i fare il suo dovere].
 Gianni himself compels to do the his duty
 'Gianni compels himself to do his duty.'

107. This is most likely to be true with the definitions in Chomsky 1986a, under which the clitic, adjoined to Infl or to V, commands everything that this Infl or V c-commands.

108. I use *control* here to refer to a class of phenomena whereby a lexical item requires one of its arguments to bind some other element. Typically, this element is not another argument of the same predicate, as it is in this case.

109. Also *Bill_i's goal was [PRO_i to VP]*, where I assume that the copula allows the infinitival to behave for control purposes as if it were the object of *goal* and hence c-commanded by *Bill*. I will not explore this phenomenon here, however.

110. If my speculations concerning scrambling are correct, it would be impossible to generate a reflexive nonclitic in a more "normal" subject position and satisfy control requirements by scrambling an object to an A-position adjoined to IP, since there are no such positions. Left-dislocated positions are not part of the argument structure of a verb, but are nonetheless θ -positions.

111. If one accepts Chomsky's (1993) arguments that objective Case is assigned to Spec,Agr_O, it is sufficient that there be only one Case position available for subjects outside VP and that Spec,Agr_O be structurally inferior to clitic position.

112. Ruwet gives *die* as a possible counterexample in expressions like *John died a gruesome death*. The cross-linguistic unaccusativity of *die*, however, is not clear, as noted by Rosen (1984).

113. If *it* is in fact limited to subject positions, (305) might be explained in that way.

114. Ruwet (1991:128) suggests that we should not base too much on examples like (307a–b), pointing to parallelism requirements that might play a role in the acceptability of these constructions. Since his discussion of this point is rather inconclusive, I will not take it into account here, except to note its existence.

115. Ruwet (1991:139) discusses examples of this type, crediting them to George Lakoff and Roger Higgins. Unlike Ruwet, I do not think a raising analysis is plausible for these cases. Such an analysis, especially for (308a–b), does not match the semantics, which clearly treats the subjects of *try* and *manage* in the usual fashion. I leave unexplained the differences between the behavior of ambient *it* and agentive controllers in French discussed by Ruwet (pp. 140–141).

116. Note **its attempt to rain*, which continues to show the restriction on possessive ambient *it*, even though the thematic status of *it* is clearer here than in any other case.

117. Ruwet (1991:chap. 3) presents an interesting cross-linguistic survey of weather expressions, which are striking in their variety. He suggests that the facts of meteorological phenomena are such that they do not lend themselves to description in terms of subject and predicate, as our linguistic system requires. Thus, languages resort to various kluges, including, in some languages, expressions that

reproduce some of the content of the verb in its subject (e.g., ‘rain rains’). Although Ruwet seems to stay fairly agnostic concerning the nature of ambient *it*, I would view it as another example of the same thing, wherein a verb of “occurrence of a caused natural phenomenon” takes as one argument a DP that refers to the cause of the phenomenon.

118. Postal (1988) suggests that certain cases of object *it* in idioms like *Melvyn never made it to London*, *He was told not to overdo it* are actually special instances of inherent reflexives—in particular, inherent reflexives that take the verb itself as antecedent. Although this suggestion does not really connect with my proposals here, it inspired the thinking that led to this section.

119. Russian expressions like *menja tošnit* (me-ACC nauseate-3sg ‘I am nauseated’) may be medical instances of this, as might the *it* of *It’s boring in there*.

120. I put aside farmer’s slang that allows *she* in meteorological expressions: for example, *Yup, she sure is rainin’ pretty heavy all right* (Janis Melvold, personal communication). This may involve personification.

121. Unlike what happens in some cross-clausal control relations (e.g., with *try*: **It was tried to leave early*), the absence of the controller in sentence-internal control does not seem to pose problems. There simply is no control in such cases. Thus, I find no implication in the (b) sentences of (i)–(iii) that the actions must be involuntary. The (a) and (b) examples may describe the same events.

- (i) a. Sue blinked her eyes.
b. Her eyes blinked.
- (ii) a. Mary nodded her head.
b. Her head nodded.
- (iii) a. Bill cracked his knuckles.
b. His knuckles cracked.

122. Note also the general incompatibility of T/SM arguments with *by*-phrases, which follows from the analysis. The passive of the causative allows *by*-phrases but not T/SM arguments; the passive of the underlying A-Causer verb allows the T/SM argument, but not a *by*-phrase.

- (i) a. *Sue was quite appalled at the sight by what she’d read.
b. *Bill was very concerned about the price of beans by his teacher.
c. *Harry was dejected about the election results by the analysis in the paper.
d. *Mary was disappointed in Sue by her actions.
e. *John was disgusted with the book by Peter.
f. *Tom was pleased with his success by Sue.

123. One might also use this sort of logic to explain *-ed* adjectives that look as though they are formed from nouns: *talented*, *bearded*, *moneyed*. Conceivably, the nominal roots of these words form zero-derived verbs first. Suppose a verb of this type requires a reflexive clitic. Then the verb will not be usable in English except when first passivized and then adjectivized.

- (i) [[[[talent_N] $\emptyset$ _V] ed_V] $\emptyset$ _A]
- (ii) [[[[beard_N] $\emptyset$ _V] ed_V] $\emptyset$ _A]

An obvious gloss for the null verb here is ‘have’. Perhaps the inalienability of the possessed attribute here (even with *moneyed*, perhaps) in some way contributes to the requirement for a reflexive clitic. I do not know if such reflexive verbs show up overtly in other languages.

124. This usage is strikingly absent in Russian, where **Maša videla-s’* cannot be used to mean ‘Masha saw herself’. The nonclitic reflexive *sebja* must be used instead (*Maša videla sebja*). This might suggest a limitation of the reflexive clitic in Russian to ambient cases (in addition, perhaps, to cases in which it has no semantic force whatsoever). On the other hand, the reflexive clitic is also used to form what is generally described as a passive of imperfective verbs (see Pul’kina and Zakhava-Nekrasova, 1967?:264).

- (i) Proekt sozdaëtsja arxitektorom.
 project create-IMPF-refl architect-INST
 ‘The project is being created by an architect.’

I will not deal with this use of the reflexive clitic here, except to note its relevance for the question of how the instrumental is licensed.

125. This general approach suggests an interesting hypothesis concerning the source of the *by*-phrase. It appears that the *by*-phrase is possible whenever a verb with an external argument undergoes morphology that prevents normal Case assignment to the object. This includes passivization, but also nominalization and adjectivization of all relevant sorts (*-able*). This idea jibes nicely with the proposal hinted at by Fukui and Speas (1986) that the *by*-phrase simply is the VP-internal subject position assigned Case by the verb. This proposal immediately permits (334) and comparable examples with *-able*, as well as allowing *by*-phrases under passivization.

This is close to the suggestion made in *LGB* (chap. 2), without the attempt to derive “dethematization” from suppression of Case. On a theory with VP-internal subjects, this is not necessary. Suppose:

- (i) An affix that suppresses Case either licenses Case to an external argument or suppresses that external argument.

By (335), the semantically noncontentful passive morpheme will only have the option of assigning Case to an external argument, hence the requirement that there *be* an external argument.

This approach requires us to treat examples like (333) as involving control of a *by*-phrase pro or PRO from a Spec,DP that is assigned some θ -role by D, as proposed by Abney (1987). As Abney notes, if the possessive position is thematic, we can immediately understand the absence of expletives in this position (**its appearance that John left*).

126. As has often been reported, (339b) improves when the stranded first object is destressed, with some dialectal variation regarding the acceptability of the structure. I will not account for this possibility here, nor will I try to explain languages like Norwegian, in which the second object appears to be passivizable quite generally.

127. Baker (1993) argues that such cases do not exist and provides a general account of this fact. It appears that these cases may be rare, but *get* certainly seems to be an example of the phenomenon. Syntactic theory is at present ill equipped to explain notions like “rarity.”

128. These considerations therefore militate against accounts of the double object construction that posit a null preposition preceding the first object, but not the second; see, for example, Baker 1988, Den Dikken 1991.

129. Throughout this book, I use the term *command* when I wish to discuss the sorts of relations based on the template $\alpha R\beta$ iff the first X that dominates α also dominates β without specifying *m-command*, *c-command*, or any other specific proposal. This usage is antihistorical, since *command* was introduced by Langacker (1969) for $X = \text{clause}$, but it should cause no confusion here.

130. In section 6.3.2, I will add some nuances to these data, since in certain cases the Goal object of *to* acts as if it c-commands the Theme.

131. Kayne's (1984a:chap. 6) proposal for these cases assigns very different constituencies to double object and *to*-object structures, such that Case marking by *of* must apply over a boundary in the double object case that is not present in the *to*-object case. This seems to be wrong, since (as I note below) all other evidence supports very similar constituent structures for the two constructions. Also see relevant remarks to the same effect in Kayne 1994.

132. The presence of a null element in double object structures is also a feature of the analyses proposed by Kayne (1984a:chap. 9) and Baker (1986:286–87). Thus, my morphological proposals could be “plugged into” these analyses to account for the impossibility of nominalizations of these structures. On the other hand, as noted by Larson (1988), the actual structures assumed by Kayne and Baker do not capture the asymmetries in command relations discussed above. For both authors, double object structures contain a null preposition whose object is the first DP ($V[[e_P] DP] DP$), predicting that if either DP commands the other, it is the second that commands the first.

133. Baker (1988:284–88) makes much the same point, although he assumes that the null preposition introduces Goal, not Theme.

134. (365a) can be explained using the notion “Cascade” developed in chapter 6. *Of* will block affixation of *G*, for reasons that will be clear at that point. On the other hand, the absence of *of* in (365b) will still leave that example unexplained.

Obviously, since the combinatorics of morphemes rest on lexical stipulations, an account can be devised for one or both examples in (365) that “solves” the present problem, but I will refrain from developing such a proposal, in hopes of a more informative account.

135. The view espoused by Jackendoff (1983) and by such investigators as Hale and Keyser (1993) is a species of this proposal, even though it replaces assignment of θ -roles by the predicative morphemes of the lexicon with assignment of θ -roles by more primitive predicates that determine the interpretation of the actual morphemes.

136. In what follows, I attempt to clarify features of Pinker's theory that are somewhat unclear in his presentation. I bear responsibility for any inaccuracies.

137. Verbs that do not describe the manner of causation of motion (e.g., *put*, *carry*, *bring*) allow *in* and *on* just like the verbs of ballistic motion.

138. This is obviously not the full story. The preposition *toward* substitutes for the expected *at-to*. *Under*, *over*, *in back of*, *in front of*, *on top of*, and the like, also behave like prepositions that contain *to*. Perhaps the fact that these latter prepositions are polysyllabic plays a role here. One might suggest that a readjustment rule turns *to* to zero when affixed to a polysyllabic expression.

139. Gropen et al. (1989) include another class, verbs of choosing (*choose*, *pick*, *select*, *favor*, *indicate* ...), among the verbs that might be expected to allow the double object structure. I return to these below.

140. I omit *question*, which simply does not take a Goal argument: **Sue questioned to Mary CP|DP*.

141. *Read* is more complex. Consider:

- (i) Sue read Bill the book about airplanes.
- (ii) Sue read Bill the text of the telegram.
- (iii) #Sue read Bill the tea leaves.

Read appears to allow a double object structure only when its Theme argument names a piece of writing. One might imagine that only in this use of *read* is there an appropriate metaphor of "instantaneous causation of ballistic motion" of the Theme to the Goal. When one reads tea leaves, one is not sending the tea leaves to anyone under any obvious metaphor.

142. Whether or not one fully accepts *instillment* as a nominalization of this use of *instill*, the contrast between (395c) and (395d) seems clear.

143. It is hard to extend the same semantic factors to instances of G_{with} with unaccusative verbs and adjectives (see Kayne 1984a:158), where the semantics are locative rather than directional, though perhaps one can argue that these examples simply involve "state" rather than "change of state" in some fashion.

- (i) a. Palm trees abound in Florida.
- b. Florida abounds with palm trees.
- c. palm trees' abundance in Florida
- d. *Florida's abundance with palm trees
- (ii) a. Strange hoots reverberated in the forest.
- b. The forest reverberated with strange hoots.
- c. the reverberation of strange hoots in the forest
- d. *the reverberation of the forest with strange hoots
- (iii) a. The dust was thick on the table.
- b. The table was thick with dust.
- c. the thickness of the dust on the table
- d. *the thickness of the table with dust

- (iv) a. Mosquitoes were dense in the marsh.
- b. The marsh was dense with mosquitoes.
- c. ?the density of mosquitoes in the marsh
- d. *the density of the marsh with mosquitoes

144. I will suggest an alternative in note 208.

145. This means that syntactic environment can function as an important clue to meaning, as suggested by Gleitman (1990), but cannot completely characterize that meaning.

146. The literature sometimes alleges that the possessed element is “predicated” of the possessor. I do not understand this idea, since (419a) surely does not assert that Sue *is* a book, but, at most, that Sue has a book. This point is also noted by Hoekstra (1988).

147. Of course, one might posit the simultaneous occurrence of a null predicate and a null preposition, if there were reason to do so.

148. Naturally, one might also pursue the question of whether *CAUS* from previous chapters should be identified with *MAKE* in (419b). To avoid this question, I have given the two morphemes different names.

149. I am deliberately not drawing a parallel with the earlier discussion of the morpheme *CAUS*, which will receive a notably different analysis in the next chapter.

150. The judgment in (429b) is denied by Guéron (1986) for the causative sense of *give* discussed in the next chapter. I do not agree with this judgment.

151. Minkoff (1993) observes that there are cases where CFC effects actually arise in this construction. This is the case when the Theme is also an Agent of its own activity, as in the reading of *Sue sent Bill to her relatives* in which Bill travels to Sue’s relatives as a free agent. Thus, **The girls_i sent Bill to each other_i’s relatives*, on this reading. This is a rather startling pattern, which seems to indicate either that clausal status is a possibility for the expressions under consideration, or that agentivity is relevant to this class of phenomena. The latter was already suggested by Chomsky (1972: “Conditions on Rules of Grammar”), with reference to a contrast noted by Yvonne Bordelois, where the contrast between (i) and (428) is already problematic.

(i) John and Mary let the honey drip on each other’s feet.

(ii) *John and Mary let Sue drip honey on each other’s feet.

152. For now, I ignore more complicated possibilities, for example, where coordination unites material in the middle of V’.

(i) Mary bought a book on Friday and a record on Thursday in Paris.

153. Conceivably, a VP shell theory that includes both the V-trace of (442) and a distinct preposition *G* could be developed. Presumably, *G* would affix to V at a stage in the derivation at which *G* is the closest head to V, that is, before V moves to its highest V’ position. This derivation would be necessary if affixation to trace is forbidden. This possibility, however, leads us (in a VP shell system) to expect

other cases in which elements incorporate into verbs that have not fully raised—and some of these cases should involve heads of nonarguments (e.g., some affixal version of *on* in (450)). As far as I know, such cases are not found, modulo the step-by-step movement of *CAUS* discussed in the next chapter.

154. *Refer* and *recommend*, cited by Pinker (1989:119) as members of this class, allow the double object structure much more reluctantly than the verbs in (443), in my judgment.

155. All these verbs contrast with the verbs of choosing discussed briefly in note 139, and grouped by Gropen et al. (1989) among the verbs that might allow the double object structure but do not. These verbs, unlike those we have analyzed with *G* and unlike verbs of future having, do not take any sort of possessor as an argument. The sentence *Bill chose this book for Sue* does not contain any element in its meaning that involves motion of the book to Sue, much less possession of the book by Sue (though the meaning of the sentence is not inconsistent with any of this). By contrast, verbs of future having like *offer* and *promise* contain the notion of possession in their lexical meaning. Verbs of choosing do not fall under any of the semantic classes we have been discussing and thus do not raise any real questions.

156. A common alternative response to binding phenomena with prepositions justifies ignoring them on the grounds that prepositions are quite small. Such a response typically involves the adjective “mere”—for example, calling prepositions “mere Case markers” or “mere realizations” of some more abstract relation. The discussion in this chapter will also constitute an argument against this sort of approach, by affirming in a number of places the relevance of *P* to command calculations. Indeed, this will be my explanation of the *T/SM* restriction.

157. In languages like Italian, this sort of preposition “stranding”, illustrated in (458) is available only with prepositions that are not joined to definite articles in portmanteau fashion (e.g., *nel* ‘in the’). In Spanish, “weightier” prepositions (e.g., *sobre* ‘over’) are easier to accept in these constructions than are lighter prepositions (e.g., *en* ‘in’). I am grateful to Raffaella Zanuttini, Alessandra Giorgi, and Héctor Campos for this information.

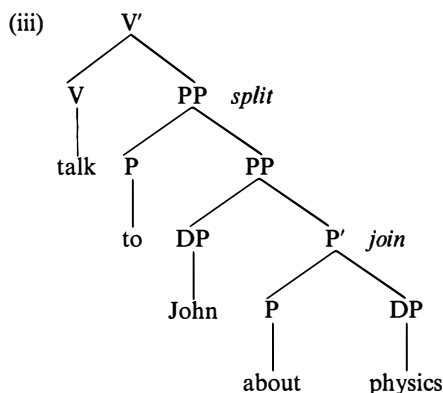
158. There is another factor that can obscure the data here. Conjuncts can share constituents. This is formalized by Muadz (1991), cited by Moltmann (1992), in terms of schemata whereby conjunction happens at nodes called *splitting nodes*, but can be undone at other nodes called *joining nodes*. Consider (i), for example, which looks like a coordination in a traditional structure, where Cascade structures give the wrong constituency.

(i) Sue will talk [to John] and [to Mary] about physics.

We should not propose any extension of Cascade Syntax for this sort of coordination, because other tests suggest that the command relations in Cascade Syntax are the same for these examples as for noncoordinated structures.

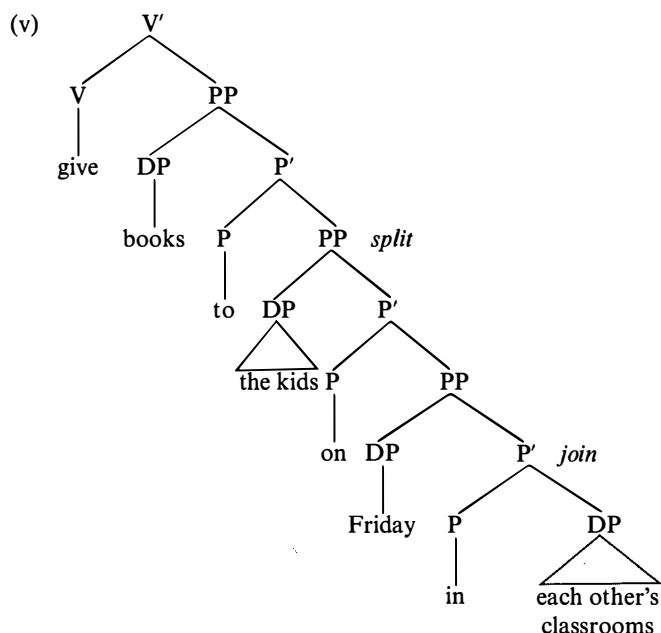
(ii) Sue will talk [to few women] and [to no men] about any scientific subject.

Instead, the “left side” of (i) can be represented as in (iii), where the conjunction splits at the first point indicated and joins again at the second.



A mechanism of this sort is desirable for examples that do not display the simplest sort of constituent coordination in any structural scheme under discussion—either Cascade or traditional.

- (iv) a. Sue will give [some books to John] and [some records to Bill] in the garden on Tuesday.
 b. Mary sells [only rare books to John] and [only common books to Bill] at any of the booklovers' conventions.
 c. Mark introduced [the boys to John] and [the girls to Mary] on each other's birthdays.
 d. I will give books to [the kids on Friday] and [the adults on Saturday] in each other's classrooms.



This elaboration does not deviate too much from Larson's conjecture. The constituents that are relevant (both as splitting nodes and as joining nodes) are still the constituents of Cascade structures. Putting it differently, this proposal allows conjuncts to share structure—but the structure is Cascade structure. Abbott (1976) and Grosu (1976) both point to similar examples as refutation of Bresnan's (1974) and Postal's (1974) claims that "right node raising" provides a test for constituency. If I am correct (in the spirit of Larson's work), then Bresnan and Postal had the right idea, but the wrong constituency. See Moltmann 1992 for a recent general account of coordination.

159. More recent accounts—for example, by Reinhart and Reuland (1993)—do not affect the conclusions that follow. For example, their theory also has a form of Principle C, as part of their Chain Condition. C-command is relevant to their notion of chain.

160. It has always been an open question how to label the nodes I have called $V^{n>0}$. I return to this question in the next chapter.

161. There is an obvious similarity between these notions of semantic relatedness and certain notions of government—in particular, the notion of antecedent government as developed by Lasnik and Saito (1984; also see 1992). Lasnik and Saito (1984) argue at length that antecedent government is blocked by DP or CP. If I am correct, the type of θ -theory developed here might ultimately contribute to one of the goals set by Koster (1978) and developed by Chomsky (1981) but not yet fully accomplished: the unification of various notions of locality. I will not take up this issue here, however.

162. On the other hand, as I discuss elsewhere (Pesetsky, in preparation), if we allow certain instances of semantic relatedness to be established later than D-Structure (Zubizarreta's (1982) "adjunct θ -roles"), we can easily understand paradigms like the following (Postal 1974, Pesetsky 1982):

- (i) I believe John with all my heart to be a genius.
- (ii) *I believe there with all my heart to be only one genius in the world.

Raising from the CP in (i) is possible because *John* bears some semantic relation to *believe* (cf. *believe of John that ...*), whereas no semantic relation is possible between *believe* and the expletive in (ii).

163. Other examples that appear at first glance to present problems can be resolved more simply. Some nice examples are presented by Reinhart (1983:53) as part of an argument that precedence is not a sufficient condition on anaphora. (She also argues that it is not necessary, with which I concur.) Her (ia) is not a problem for Principle C if the structure is something like (ib), even in a Cascade Syntax.

- (i) a. You won't believe who I saw near him_i in Ben_i's car.
- b. ... who_j I saw [_{PP}[_{clause} t_j near him_i] [_P in Ben_i's car]]

Likewise, her (iia) is not a problem if *on her lover's birthday* is semantically related to *fail* rather than *think*. There is thus a contrast with (iib).

- (ii) a. Rosa never fails to think about him_i on her lover_i's birthday.
- b. *Rosa thinks about him_i on her lover_i's birthday.

Fail is an implicative verb (Karttunen 1971, Pesetsky, in preparation) and consequently has the property that temporal adverbials modifying the matrix clause also identify the time of the embedded clause (and conversely). Despite this semantic connection, there is no semantic relatedness in the proposed sense between a matrix occurrence of *on her lover's birthday* and the embedded one. This is, in fact, a nice example of a purely syntactic fact even in a domain where syntactic and semantic facts are closely interrelated. Interestingly, it is also possible that (iii) behaves more like (iia) than (iib). Conceivably, temporal PPs may be related to NegP as well as VP, despite the implicative character of this projection as well, if I am right in grouping (iii) with (iia) rather than (iib).

- (iii) (?)Rosa never thinks about him_i on her lover_i's birthday.

164. In one form or another this observation seems to form part of linguists' lore, and it has recently been discussed by Hale and Keyser (1993).

165. If one can analyze compound expressions like *out of* or *into* in some special way (perhaps in terms of I-selection), one might claim that prepositions do not satisfy lexical requirements via mediated θ -selection. This would solve another difficulty: the noniterability of prepositions, as in **Sue gave the book* [_{PP} *to* [_{PP} *to* [_{PP} *to* [_{DP} *Mary*]]]]. This should be acceptable, since each instance of *to* has as its complement a PP whose head (trivially) selects what *to* itself selects.

166. (477) imposes a limitation on the number of arguments a nonpredicate preposition can take, not a special limitation on the positions that a preposition's arguments may occupy. A preposition certainly may θ -select one internal argument as the specifier of its sister. Indeed, this possibility is at the core of the results reported in this chapter.

167. We can ring the changes on this larger structure as well: *handing of toys to children_i on each other_i's birthdays*, *handing of toys to [children on birthdays]* and *[adults on wedding anniversaries]*, and so on. The structure behaves in every respect as predicted.

168. This formulation allows internal selection in principle to overlook any number of specifier-less nonargument categories, suggesting a possible analysis for aspectual particles and other material that in some languages (e.g., Chinese) may intervene between V and its object.

169. Obviously, I am not advancing a unified theory of minimality here, but I am pointing the way to future work along such lines.

170. The examples in this section include both sentences provided by Oehrle (1976:27ff.) and others modeled on his.

171. Some speakers disagree concerning the acceptability of (503b) and (503d).

172. Compare *Our efforts gave us a new building*, not *Smith's generosity*.

173. Of course, as noted earlier, some causative ObjExp verbs in these languages lack overtly occurring reflexive noncausative counterparts. These gaps might now be due to selection by the noncausative root for *CAUS*, or they might still represent morphological idiosyncrasies. I do not have relevant evidence to offer.

174. When Oehrle's Generalization is violated, the second argument can also serve comfortably as an antecedent. That is, the problem with **Each other's remarks gave a book to John and Mary* is not the binding of the reciprocal.

175. In the previous section, I noted that *CAUS* must be able to appear higher than other adjuncts, so as not to be blocked by them when it moves to the head of its Cascade. One might suppose that *CAUS* is *always* generated higher than locative and temporal expressions (a stronger conclusion than merely allowing it to occur in this position). One would then attribute its position not to the segregation of arguments and adjuncts imposed by θ -selection, but to some U(T)AH principle that places Causer lower than other θ -selected arguments. This sort of θ -selection would presumably be an optional property of the roots like *give* or $\sqrt{\text{annoy}}$, since roots like these can—in principle ($\sqrt{\text{annoy}}$) or in fact (*give*)—occur without *CAUS*. (Instances of overt causative PPs to the right of locatives and temporals would need some other explanation, as noted by Danny Fox (personal communication), for example, rightward movement.)

If this alternative theory about *CAUS* were correct, then one might think that the T/SM restriction should arise from Cascade structures in a different way, as an artifact of the restriction barring more than two internal arguments. The problem with this line of reasoning comes from the presence of *CAUS* with *give* in cases that display Oehrle's Generalization. I know of no reason why *CAUS* should be an argument with Experiencer predicates and an adjunct with *give*. Yet in the cases allowed under Oehrle's Generalization we find two internal arguments (Goal and Theme) coexisting with a Causer inside *V'*. We must conclude that Causer here is not a θ -selected argument.

The case could be made even stronger if a Target or Subject Matter argument could be introduced by an affixal null preposition like *G*. The theory predicts no T/SM effect in such a case, yet such a case would (by hypothesis) display three internal arguments (if Causer were an internal argument). There appears to be no such zero morpheme (independent of the proposed account of T/SM phenomena). I return to this observation in the next chapter.

176. In Chomsky's case, discharge of features is accomplished by raising the more complex form (*V + Infl*) to the simpler one (*Infl*). In our case, the simpler form moves to the more complex (*CAUS* to *V + CAUS*), rather than the other way around as in (527), but this raises no problems that I can determine. This issue was brought to my attention by Shosuke Haraguchi (personal communication).

177. I still assume, with Chomsky, that a Mirror Principle of some sort obtains, so the observations concerning the position of zero morphemes and nonmorphemes that I adopted from Myers keep their old analysis. I leave open whether movement of *G* involves affixation or checking.

178. Technically, this involves replacing the subject Causer with a copy of the lower Causer. This should be nonproblematic, since the subject Causer already is indistinguishable from the lower Causer. If the two were not indistinguishable, the replacement would violate the Thematic Diversity constraint stated in (179). Likewise, perhaps, if *CAUS_P* were to license Case, rendering the two occurrences of Causer dissimilar in Case features. We can thus derive the requirement that *CAUS* not license Case.

179. Actually, one might aim at a more nuanced restriction, which requires non-conflict in θ -features rather than nondistinctness. This would permit the upper occurrence of Causer to bear agentive features that the lower occurrence fails to bear. This approach might provide insight into the reduced acceptability of backward binding with agentive uses of causative Experiencer verbs: *??Each other's relatives deliberately annoyed John and Mary* (Giorgi 1984). Binding might not be evaluable in a “reconstructed” position where the semantics of the phrase when it occupied that position were in some fashion less complete than the semantics of the same phrase in its surface position.

180. If no movement can take place—if, for example, the *by*-phrase is absent—then the object of *CAUS_P* may be stranded without a way of licensing its Case. As it happens, I find that omission of *by* degrades these examples, although I am not secure in these judgments.

(i) *??In those days, Bill was often being frightened when I would come home from work.*

(ii) *??In those days, Bill was again and again being annoyed.*

If these are acceptable, then one would posit a null *by*-phrase, with movement of a null Causer between these positions. Appropriate theory would need to be developed. (Comparable examples with *give* and *teach* need to control for the agentive interpretation, which is difficult.)

181. This is my informal experience in lecturing on this material.

182. Nominalization may also suppress a θ -role. A checking analysis of nominalization might posit [*V* + *N_N*] raising from a clausal constituent to *N* to discharge the strong features of the nominalizer, in the spirit of (364), perhaps with the lower clausal VP there changed to a clausal NP. This will not produce T/SM-type effects.

183. Walinska de Hackbeil (1986) argues that *en-* is actually a prefixal allomorph of the preposition *in-* and not a verbalizer. The actual verbalizer in a form like *embitter*, on her analysis, is a zero morpheme. Note, however, that the form *embitterment* is unique among nominalizations of causative Experiencer predicates in retaining its causative force. This suggests that there is no zero verbalizer, but that *en-* itself may function as a causativizing verbalizer.

184. There are some rogue verbs left, whose behavior the present theory does not shed light on. As (i) shows, for example, the verb *interest*, with a purely causal subject, allows a second object with a reasonable level of acceptability.

- (i) a. ?*The Loom of Language* interested many people in linguistics.
- b. ?Mary's lecture interested me in mathematics.

What is odd is the fact that *interest* appears to behave this way in a number of languages, including French, Dutch, and Finnish (Paul Kiparsky, personal communication). This suggests an explanation rooted in lexical semantics. Yet in English, at least, the near synonyms *concern* and *intrigue* contrast.

- (ii) a. *The book intrigued me with mathematics.
- b. *Mary's lecture concerned me with mathematics.

Interestingly, agentive uses of *interest* (perhaps idiomatic) are better with a prepositional object than purely causal uses like (i).

- (iii) Can I interest you in a cheap copy of *Zero Syntax*?

Excite behaves like *interest*, taking a second argument when agentive. With this second argument, it is considerably worse than *interest* when nonagentive.

- (iv) a. ?Bill really managed to excite him about linguistics.
- b. *Notre Dame cathedral really excited him about architecture.

It may be significant that *interest* is homophonous with the noun *interest*, which might allow the lexical item to enter the system without an A-Causer external argument. French provides similar examples, where *dégoût* and *fatigue* are nouns that plausibly yield the verbs *dégoûter* and *fatiguer* by some sort of zero derivation (but see Ruwet 1993 for problems with this as a general account).

- (v) a. Ce film a dégoûté Pierre du cinéma pour toujours.
this film disgusted Pierre of cinema forever
(Ruwet 1972:181fn., Zubizarreta (1988))
- b. Le spectacle des gens heureux a fatigué Max de sa vie monotone.
the spectacle of happy people tired Max of his monotonous life
(Nicholas Ruwet, personal communication; attrib. to J. P. Boons)

Mere homophony between a noun and an ExpObj verb is insufficient for licensing the T/SM restriction, as is obvious not only from *concern*, but also from noun/verb pairs like *alarm*, *anguish*, *anger*, and *awe* (e.g., **The article angered Bill at the government*, already considered above).

185. The antecedent need not occupy the embedded subject position, though relevant examples are complex: *Those stories about herself made Bill hate Mary* seems, if given a context, no worse than the examples in (549). We therefore should not appeal to any special property whereby the embedded subject raises into the higher clause in accounting for these phenomena (e.g., raising to a matrix object agreement position). I am indebted to Seth Minkoff (personal communication) for this consideration, which arises from his own work on an alternative account.

186. Perhaps one might expect *CAUS_p* to move only as far as *angry*, by analogy with earlier structures, but then the problem would arise again for *Each other's remarks made John and Mary angry at Sue*, which is acceptable. *At* should block raising of *CAUS_p* here.

187. One might argue that the marginal occurrence of *tough*-movement in these periphrastic constructions (and its occurrence in nonperiphrastic causative constructions as well; see Pesetsky 1987a) represents an instance in which two distinct Causers cooccur, perhaps because of the link between them that is established by *tough*-movement.

(i) This book annoys me [to have to read ____].

(ii) ?This book makes me angry [even to look at ____].

If so, then the instance of *CAUS_P* that introduces the *tough*-clause in (ii) should be unable to delete, creating a T/SM effect. Strikingly, this is true.

(iii) *This book makes me angry at its publisher [even to look at ____].

Interestingly, however, the T/SM effect in (iii) can only be analyzed in the manner we have developed if there is no small clause structure in the Cascade representation, so that *at* c-commands *CAUS*.

(iv) makes [me [angry [at [its publisher [*CAUS_P* [even to look at ____]]]]]]]

I will not develop this point further here, but it is a conclusion very much worth noting.

188. I am grateful to Howard Lasnik for pointing me toward the source of these observations, cited by Belletti and Rizzi (1988:316). Burzio's example was *John gave some pictures of each other to the kids*. Here I have picked verbs that allow animates in both Theme and Goal position, to permit construction of a full paradigm with minimally contrasting sentences.

189. Perhaps this specifier position would be occupied by an occurrence of Goal before movement, as in our analysis of periphrastic causatives, or perhaps it is filled only as a consequence of movement, as illustrated in (565).

190. I have left open the nature of the mechanism. Perhaps, in the spirit of Baker, Johnson, and Roberts (1989), that morpheme itself satisfies the external θ -selectional property of the predicate to which it attaches. This mechanism itself is incomplete, however, since it does not tell us why that external role's semantics seemingly disappear from the clause, in contrast to passive.

191. I ignore for the moment the external argument of *give*.

192. Alternatively: If α c-commands β in a Cascade structure, then α precedes β in a Layered structure. I return to questions of linear order when I consider word order variation in section 7.3.

193. The structures of (588) have counterparts with *wh*-movement.

(i) About which pictures of each other did Sue say Mary had spoken to the kids ____?

(ii) *To which kids did each other's friends say Mary had spoken ____ about these pictures?

194. The use of relative clauses rather than information-seeking questions makes (591b) substantially better than its counterpart *wh*-question.

195. In principle, we could mark certain Cascade Syntax P's for adjunct status, and prevent extraction from them as well. A different problem would then arise if it could ever be shown for an example like *a city in which Bill lived in 1986* that the locative started off to the right of the temporal. In Cascade Syntax, the locative P' would be a subconstituent of the temporal P', making this a case of adjunct extraction from an adjunct, which is otherwise excluded by well-known constraints (see, for example, Lasnik and Saito 1984, which follows Huang 1982).

196. The parallel proposal for locatives has a certain precedent in work by Freeze (1992), who motivates a null locative preposition in the derivation of *have*. In his analysis, this preposition, raises to a copula (*have* in English constituting a suppletive form of *be* when it combines with this preposition). I owe this observation to Beth Levin (personal communication).

197. Remember, in any case, that the questions that arise for movement of the Causer do not arise for movement of *CAUS_{aff}*, since the conditions on X⁰-movement definitely hold of Cascade structures.

198. Note that the level to which these adjuncts are attached in Layered Syntax makes no difference for the Cascade Syntax counterpart.

199. *In order to*-clauses, the third of the adverbials in (471), seem to have another life as VP-internal adverbials, given (i), which shows c-command by the object.

- (i) a. We sent none of these boys to West Point in order to please anyone's mother.
- b. We sent only John_i to West Point in order to please his_i mother.

It is thus no surprise that heavy shift is possible around *in order to*-clauses.

- (ii) We sent ____ to West Point, in order to please Bill's mother, [each boy who showed an aptitude for shooting].

With (i), contrast (iii), which contains the adverbials in (612). These never behave like denizens of VP.

- (iii) a. ??We have to fire [only Bill]_i, whether McIntosh likes him_i or not.
- b. *Rosa liked none of the boys, with Ben's mother hanging around at any time.

200. Strictly speaking, the evidence so far has shown that it is a position where these elements *may* occur. We have seen no evidence that other sorts of elements may not occur in such a position. The strengthened claim in (573d) that biuniquely relates VP-adjunction to semantic relatedness is thus motivated by the evidence in this section and is not a fully independent discovery.

201. Of course, we must either abandon Chomsky's (1986a) claim that adjunction to higher VPs is an intermediate step in long movement, or else find some special property of leftward movement that does not entail semantic relatedness. Note that the theory of clause-boundedness presented in the text excludes any sort of long-distance movement to the heavy shift position—even movement that involves intermediate steps.

202. If theories like that of Chomsky (1986a) are correct in positing adjunction to VP as an intermediate step for $\bar{A}$ -movement, adjunction may be assumed to apply to the left margin of VP, which might not entail semantic relatedness. This does not explain the absence of movement (in English) that terminates in that position.

203. I ignore here the “checking” analysis of *G* for the sake of simplicity.

204. Henceforth, I will call the general phenomenon we are discussing *heavy shift*, reserving *heavy DP shift*, *heavy PP shift*, and so on, for special cases of the general phenomenon.

205. Another condition on middles (Jaeggli 1984, cited in Zubizarreta 1987:143) restricts the θ -role of the moved DP to phrases somehow “affected” by the action of the verb.

- (i) a. *Plays like this one enjoy easily.
- b. *Flying fears easily.
- c. *Birds scrutinize easily.
- d. *This sort of fact knows easily.
- e. *Cliffs this big avoid easily.
- f. *Mary sees easily. [on the reading ‘Mary can be easily seen’]
- g. *Gazelles pursue easily.

M. Anderson (1977, 1979) relates parallel facts with passivization in NP to a suggestion that nonaffected arguments are obligatorily introduced by a preposition, at least in NP. If this is quite generally true, then nonaffected arguments of V might be introduced by some null preposition *X*, making the middles in (i) instances of illegal stranding of *X*, or else illegal movement of a phrase headed by *X* to subject position.

206. These examples do not look good when written, but improve when spoken with a healthy pause after the VP-trace.

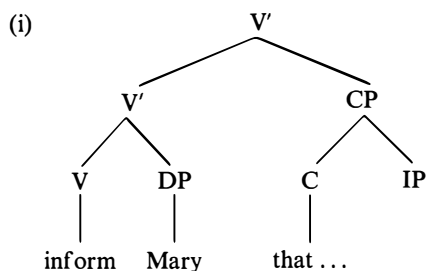
207. The anti-c-command condition on parasitic gaps must follow from a Layered Syntax condition on chains rather than Principle C, given the data noted by Contreras (1984).

(i) Which articles did John file *t* without reading *e*?

(ii) *John filed them_i without reading [Mary’s articles]_i.

C-command in Cascade Syntax accounts for the binding violation in (ii), whereas the c-command relations in Layered Syntax account for the availability of parasitic gaps in (i). This result jibes with the theory in the text of this section.

208. I have assumed, largely on the strength of (570d), that Layered structures are *n*-ary branching, and not strictly binary. Alternatively, we might assume that Layered structures are strictly binary branching, motivating Layered structures in which second objects asymmetrically c-command first objects, as in (i).



We would then need some device like Chomsky's (1986a) distinction between category and segment to prevent fronting of the lower V'. If the text is correct in assigning to Layered structures the licensing conditions for parasitic gaps, then we explain the ability of first objects to license parasitic gaps in second objects, discovered by Engdahl (1983b).

- (ii) a. Who did you inform *t* that we were about to visit *e*?
 b. John, who I persuaded *t* that Mary should talk to *e*, ...

This sort of assumption might also explain the "holism" property of first objects, discussed in section 5.3.3, in connection with examples (401)–(402), repeated here.

- (iii) a. Sue loaded the hay into the wagon.
 b. Sue loaded the wagon with hay.
 (iv) a. Bill smeared the grease onto the axle.
 b. Bill smeared the axle with grease.

I observed, following Anderson (1971), that "in [(iia)], we are likely to understand that all the hay got loaded, but we do not know whether the wagon is full. Likewise, in [(iib)], we are likely to understand that the wagon is full but do not know whether all the hay got loaded. Similarly in [(iv)]. On hearing [(iva)], we naturally understand that all the grease was applied to the axle, but we do not know how much of the axle is covered. On hearing [(ivb)], we might understand the axle to be entirely covered with grease, with some grease left in its container." It is known that adverbs like *completely* seem to be tightly glued to main verbs.

- (v) a. Bill has completely loaded the hay into the wagon.
 b. *Bill completely has loaded the hay into the wagon.

If we understand the scope of *completely* as the minimal Layered Syntax constituent containing the main verb, then we can understand the "holistic" interpretations of (iii) and (iv) as the consequence of a silent form of *completely* applying to the same constituent.

209. We might label *VP* all the Layered Syntax nodes that immediately dominate non- θ -selected arguments of V, to allow them to undergo movement under standard conventions and to allow the non- θ -selected PPs to occupy $\bar{A}$ -positions under standard conventions. This terminological move would eliminate any puzzlement we might have when heavy shift adjoins PPs to nonhighest projections of V.

(i) John arrested ____ on Tuesday [his best friends]_i in each other_i's houses.

Such examples do not otherwise differ from standard ones.

(ii) Sue offended ____ by not honoring ____ [her best friends]_i on each other_i's birthdays.

We might then retain *V* for the constituents discussed in the preceding note.

210. Of five speakers queried, two detected some residual deviance in (645b), though all found the improvement noticeable.

Larson (1989), to whom the suggestion of a c-command explanation is due, presents judgments that disagree with the data on quantifier binding and negative polarity from (643)–(646). Jackendoff (1990), however, appears to share my judgments, as do others whom I have queried.

211. The pronominal examples have alternative interpretations in which the pronouns refer elsewhere, a possibility denied to the anaphors. Thus, the examples may have somewhat different “feels.”

212. If the sequence *G CP* is possible, we can explain the fact that *CP* need not move in passive sentences without recourse to the stipulation that *CP* does not need Case. If we were to adopt Stowell's (1981) idea that apparent *CP* complements to *N* are actually appositive modifiers, and find an extension of this idea to complements of *A*, we could then eliminate the stipulation that *CP* does not need Case. Unfortunately for this suggestion, Stowell's idea does not seem correct for a variety of cases that I discuss in Pesetsky, in preparation.

213. Beth Levin (personal communication) has noted the parallelism between these examples and those where the *CP* is replaced by a *DP*: *Mary suggested an answer to Frank*. Backward binding is, of course, possible from Goal to Theme in such constructions: *I suggested each other_i's ideas to John and Mary*.

In section 6.3.2, I accounted for this sort of backwards binding with the proposal that Goal-phrases introduced by *to* actually start out higher in Cascade Syntax than the Theme, and I further proposed that the Theme at this point in the derivation is introduced by *G*. In other words, the linear sequence at D-Structure is *V to Goal G Theme*. By parity of reasoning, the underlying structure for cases in which the Theme is a *CP* should be the same. This means that the derivations discussed in the text are actually the end of a slightly longer story.

(i) Underlying: *V to Goal G Theme*

Step 1: *G Theme trades places with to Goal*

V G Theme to Goal ... [trace of G Theme may delete]

Step 2: *Incorporation of G into V*

G_i + V t Theme to Goal

Step 3: *Heavy shift*

V [trace of G Theme] to Goal G Theme

The derivation may be somewhat longer than presented in the text, but no new complication is introduced. Step 1 is motivated by the need for *G* to incorporate. In the discussion in section 6.3.2, this was accomplished by moving the Theme out of the phrase introduced by *G*. That will not do when heavy shift needs to apply

later. Consequently, in the examples of the current section, the Theme moves together with *G*, which incorporates in *V* before heavy shift (steps 2 and 3). I am indebted to Danny Fox (personal communication) for this point.

214. At the very least, shift *may* involve a null preposition. We expect *G* to be optionally available in a θ -selected position whenever a verb takes a Theme as its single internal argument. For this reason, *Sue painted the wall* might have the variant analyses in (i) at S-Structure (ignoring the affixation of *G* to *paint*).

- (i) a. Sue painted the wall.
- b. Sue painted [_{PP} *G* the wall].

An unaccusative or passive verb can θ -select Theme but cannot not Case-mark it. If the Theme is directly θ -selected (the “usual” case), the Theme is Caseless in the VP and consequently must move to some Case-marked position. Once again, however, there is no reason to exclude an alternative structure in which θ -selection for Theme is mediated by *G*. Since *G* can also assign (check) Case, the Theme in this configuration need not raise for Case purposes. We thus should find the paradigms in (ii)–(v), with the structures assigned.

- (ii) a. *There were arrested [_{DP} several demonstrators].
- b. [_{DP} Several demonstrators]_i were arrested *t*_i.
- c. There were arrested [_{PP} *G* several demonstrators].
- (iii) a. *There arrived [_{DP} the letter I told you about].
- b. [_{DP} The letter I told you about]_i arrived *t*_i.
- c. There arrived [_{PP} *G* the letter I told you about].
- (iv) a. *There were sent [_{DP} some nasty letters].
- b. [_{DP} Some nasty letters] were sent.
- c. There were sent [_{PP} *G* some nasty letters].
- (v) a. *There were considered brilliant [_{DP} several incoming students].
- b. Several incoming students were considered brilliant.
- c. There were considered brilliant [_{PP} *G* several incoming students].

If these paradigms are correct, the (c) constructions should be possible only with those DPs whose θ -roles can be selected by a null preposition. This is also correct. The Goal arguments in examples like (vi)–(vii) run afoul of Case theory.

- (vi) a. *There were given some men in suits some books about music.
- b. *There were given some books about music some men in suits.
- (vii) a. *There were informed about the children several of the parents.
- b. *There were informed several of the parents about the children.

Curiously, the object of *G* needs to be focused in the (c) examples of (ii)–(iv), even when *G* would seem to be affixable to the main verb. This observation suggests that heavy shift is obligatory for the object of *G* here, a conclusion that is reinforced by the requirement that PP_G be farther to the right than any nonheavy θ -selected argument.

- (viii) a. *There were put several trays on the table on Thursday.
- b. There were put on the table several trays on Thursday.
- c. There were put on the table on Thursday several trays.

- (ix) a. *There was sent a book to him during the conference.
- b. There was sent to him a book during the conference.
- c. There was sent to him during the conference a book.

This suggests that proximity to nominative Case is necessary, in order for *G* to license its object in these constructions; but I will not develop this theme here. The key point for present purposes is the cooccurrence of *G* and heavy shift.

215. If adverbs are generally overlooked, then the impossibility of *V Adv DP* ... would be a consequence, not of an adjacency condition, but of the adverb's inability to function as a sister to *V* in Layered Syntax. In languages like French, of course, *V*-raising over the adverb makes this order possible.

216. In this sense, there may be role for *Agr_O* or *Agr_s* in objective Case assignment, though no requirement that the accusative *DP* occupy a specifier position, as in Chomsky 1993.

217. If movement applies, as suggested by Chomsky (1993), only to satisfy requirements of the moving element, these requirements must not need to be satisfied immediately; instead, it must be possible for additional steps to take place first. This is because if *V* moves first in (684), it must wait until after the *DP* moves to discharge its features. Conversely, if the *DP* moves first, it cannot discharge its features until the *V* moves. I do not know if this consequence is problematic.

218. Emonds (1976:4) does not impose the requirement that the constituents share the same mother node.

219. Strictly speaking, we might suppose that only the trace of Goal is relevant, since only that trace bears the proper configuration to both the verb and the particle. This consideration must be disposed of, if the speculation in the text is correct.

220. One direction concerns clausal Target and Subject Matter arguments. These, though they are not introduced by any overt preposition, nonetheless cannot occur in causative constructions.

- (i) *The book angered me [that anyone could be go stupid].
[cf. *I am angry that anyone could be so stupid.*]

- (ii) *Sue amused me [that she had memorized the poem].
[cf. *I was amused that she had memorized the poem.*]

Interestingly, clausal Target and Subject Matter arguments remain impossible with verb-particle verbs of the same class.

- (iii) *John's remarks pissed me off that the government was so corrupt.
[cf. *I was pissed off that the government was so corrupt.*]
- (iv) *These remarks riled her up that Bill would be so rude.
[cf. *I was really riled up that Bill would be so rude.*]

Thus, the ban on clausal complements appears to have a source different from the T/SM restriction examined in this book.

221. There is surely some relationship between the choice of particles in these constructions and the use of homophonous lexemes (perhaps the same lexical items) as emotive predicates *tout court*, but the present analysis makes nothing of this.

- (i) a. Sue is really *up* about the new play.
- b. Bill is looking very *down* these days.
- c. Arturo was somewhat *off* during the Sibelius today.
- d. Jascha was really *on* during the Chausson *Poème*.

222. A head-final, specifier-initial configuration in Cascade structures would yield bizarre cross-serial dependencies that could not correspond to any Layered structure: for example, $[[DP_1 [DP_2 P_2]] P_1]$, where DP_1 is θ -selected by P_1 and occupies the specifier position of the PP headed by P_2 .

223. Rightward movement is excluded by Kayne (1994), a proposal incompatible with the proposals made here for heavy DP shift. Truckenbrodt (1993) suggests that the effect of rightward V-raising can be achieved by leftward V-raising to some category F, followed by leftward raising of the remnant VP to some higher specifier position.

224. Haider (1993) suggests instead a structure akin to Cascades, but with the verb in the most embedded position. The notion of semantic relatedness is replaced with a notion of extended projection that looks up the tree, much as the system proposed here looks down. Haider argues that such a system captures the alleged fact that argument ordering is freer (“scrambling”) in head-final constituents than in head-initial constituents, as well as an alleged greater freedom of adverb positioning. The former conclusion must be examined carefully, in light of the fact that many head-initial languages (e.g., Russian and Attic Greek) display scrambling, and of the fact that adverb positioning among arguments of V' is freer than generally supposed even in English (Koizumi 1993).

225. I am indebted to Shigeru Miyagawa for these examples and for discussion of the data.

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